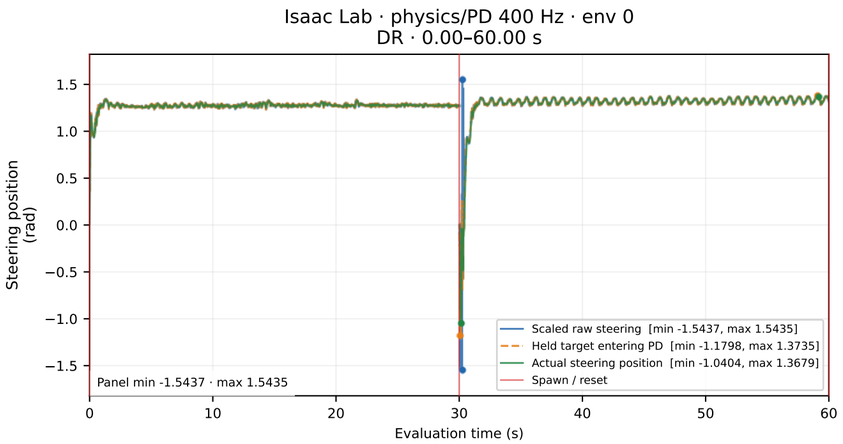
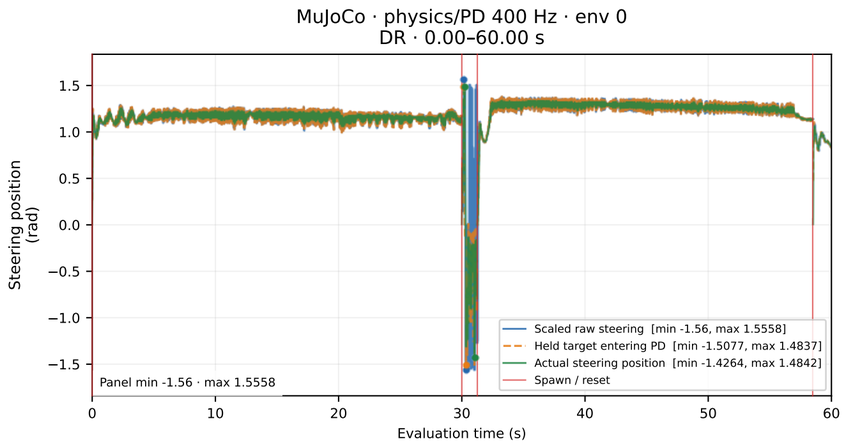
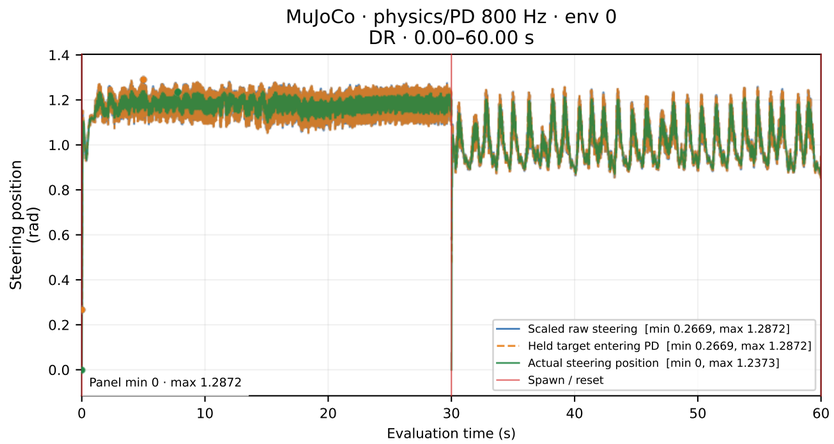
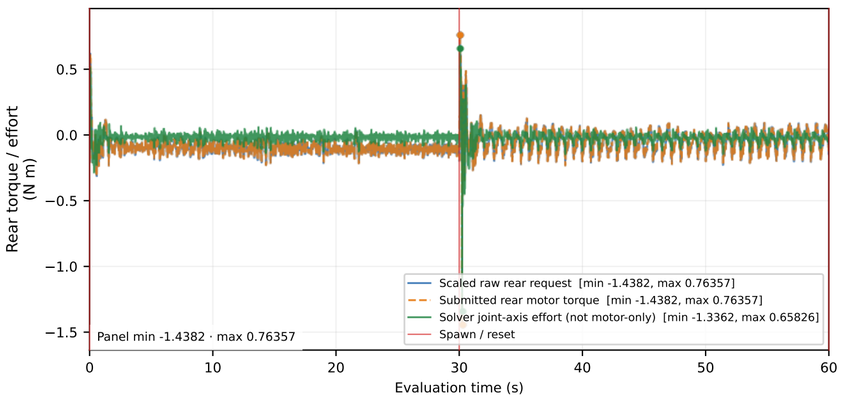
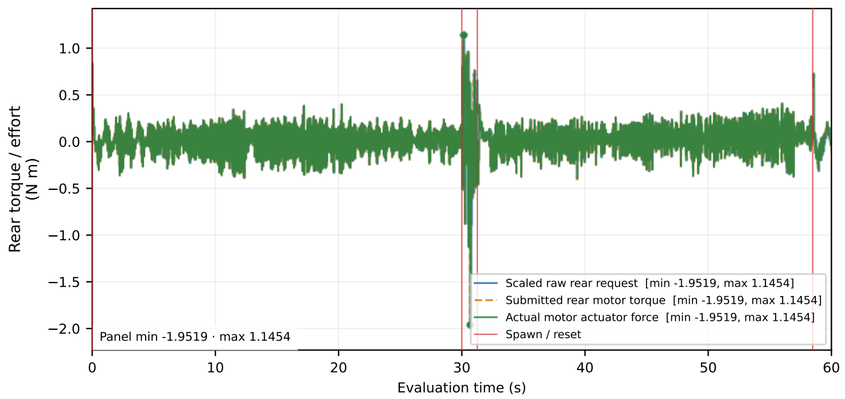
model_2500 · Demand versus actuation · env 0 · full 60 s

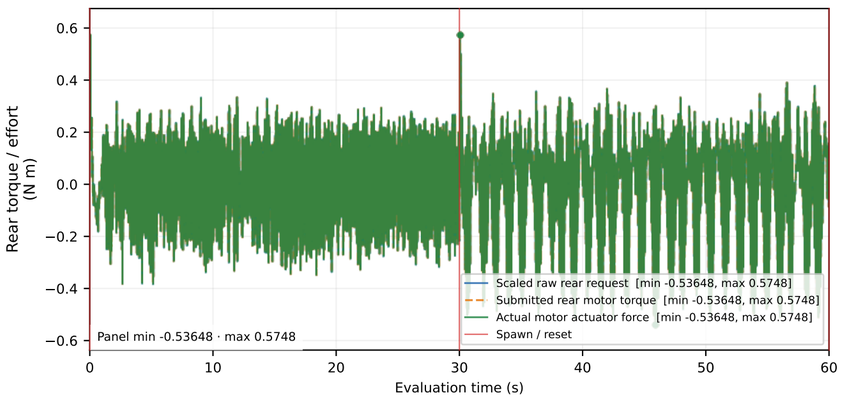












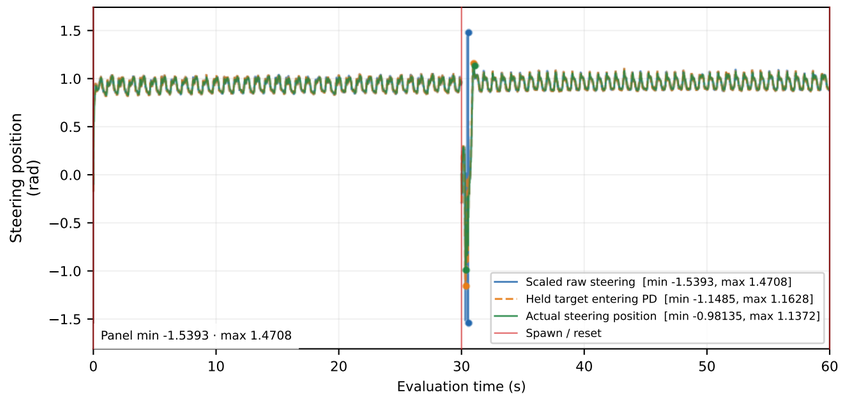
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

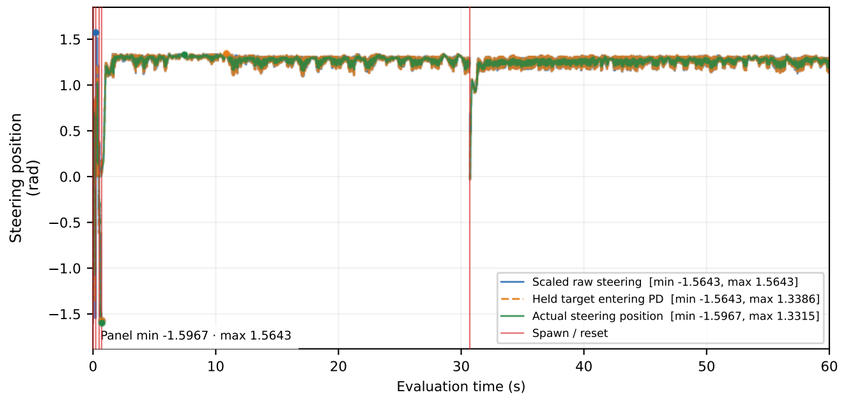
```
SAMPLED DR: env 0; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.04518, 0.804024, 0.959366, 0.816123]
  CoM XYZ mm [4.87718, -3.711, 0.621036]
  mu_s [0.741039, 0.681711, 0.592046, 0.532645]
  mu_d [0.614405, 0.611756, 0.534947, 0.532645]
  Kp [steer,rear] [10.6517, 0]; Kd [0.209184, 0]
  spawn roll 0.0198978 rad; rate 0.00125125 rad/s
  gyro bias XYZ rad/s [-0.000195088, -0.000150685, -2.64955e-05]
  gravity bias YZ [0.00115356, 0.00192213]
E2 [30.00, 60.00) s:
  mass x [1.07174, 1.18127, 1.11706, 0.881481]
  CoM XYZ mm [-0.792429, -4.35209, 3.72319]
  mu_s [0.576972, 0.659445, 0.531594, 0.547743]
  mu_d [0.576972, 0.547619, 0.531594, 0.547743]
  Kp [steer,rear] [9.66427, 0]; Kd [0.207538, 0]
  spawn roll -0.0845213 rad; rate 0.112094 rad/s
  gyro bias XYZ rad/s [-2.89794e-05, -3.58017e-05, 0.000157862]
  gravity bias YZ [-0.0029177, -0.00080667]
Six significant digits shown; full precision: sampled_dr_values.json.
```

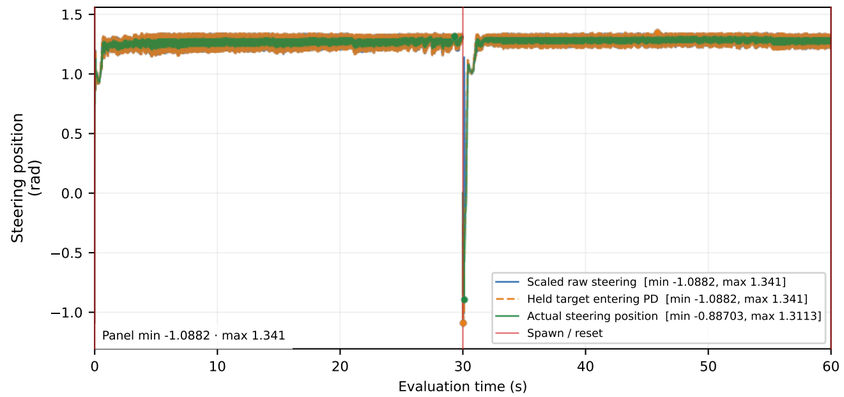
```
SAMPLED DR: env 0; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05389, 1.02143, 1.02368, 0.92158]
  CoM XYZ mm [-4.69182, -0.632826, -2.85415]
  mu_s [0.666375, 0.640016, 0.669571, 0.71155]
  mu_d [0.519145, 0.513141, 0.669571, 0.71155]
  Kp [steer,rear] [9.46788, 0]; Kd [0.191255, 0]
  spawn roll 0.0628041 rad; rate -0.166514 rad/s
  gyro bias XYZ rad/s [0.000138097, -8.54505e-05, 3.17079e-05]
  gravity bias YZ [0.00125118, -0.000618693]
E2 [30.00, 31.26) s:
  mass x [1.02693, 1.08918, 0.965679, 0.927341]
  CoM XYZ mm [1.65679, 1.84224, -2.29571]
  mu_s [0.541939, 0.732187, 0.611239, 0.700521]
  mu_d [0.534359, 0.631664, 0.611239, 0.641329]
  Kp [steer,rear] [10.5106, 0]; Kd [0.202627, 0]
  spawn roll 0.059515 rad; rate -0.0147133 rad/s
  gyro bias XYZ rad/s [-4.78947e-06, 5.43307e-05, 0.000350425]
  gravity bias YZ [-0.000315206, -0.00188852]
E3 [31.26, 58.48) s:
  mass x [0.831615, 1.11129, 0.86509, 1.17322]
  CoM XYZ mm [4.90593, -3.7759, 1.83006]
  mu_s [0.669571, 0.546287, 0.69316, 0.640867]
  mu_d [0.669571, 0.546287, 0.69316, 0.556841]
  Kp [steer,rear] [9.35915, 0]; Kd [0.186665, 0]
  spawn roll -0.0194645 rad; rate -0.0189324 rad/s
  gyro bias XYZ rad/s [9.16669e-05, -2.25439e-05, 8.03725e-05]
  gravity bias YZ [0.00145273, 0.000686634]
E4 [58.48, 60.00) s:
  mass x [1.14858, 0.965632, 0.91743, 0.960669]
  CoM XYZ mm [-4.05657, 3.38234, 4.15215]
  mu_s [0.723429, 0.71008, 0.69316, 0.546287]
  mu_d [0.723429, 0.59371, 0.69316, 0.546287]
  Kp [steer,rear] [10.2283, 0]; Kd [0.182062, 0]
  spawn roll 0.085862 rad; rate 0.244715 rad/s
  gyro bias XYZ rad/s [-8.09711e-05, 0.000283962, 0.000268881]
  gravity bias YZ [-0.000732425, -0.000166156]
Six significant digits shown; full precision: sampled_dr_values.json.
```

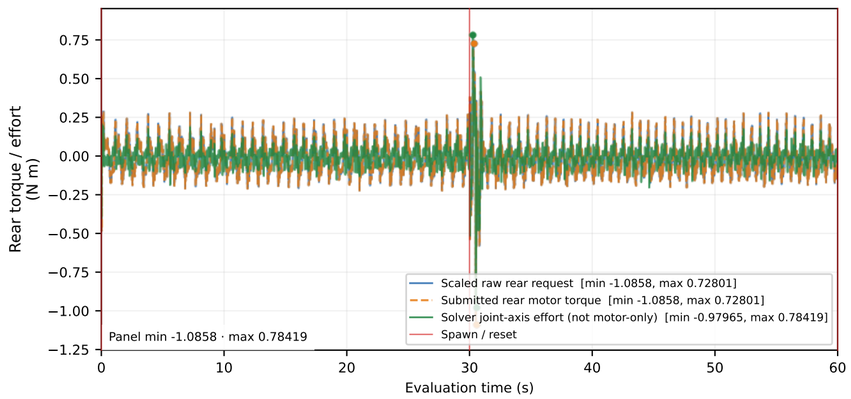
```
SAMPLED DR: env 0; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05389, 1.02143, 1.02368, 0.92158]
  CoM XYZ mm [-4.69182, -0.632826, -2.85415]
  mu_s [0.666375, 0.640016, 0.669571, 0.71155]
  mu_d [0.519145, 0.513141, 0.669571, 0.71155]
  Kp [steer,rear] [9.46788, 0]; Kd [0.191255, 0]
  spawn roll 0.0628041 rad; rate -0.166514 rad/s
  gyro bias XYZ rad/s [0.000138097, -8.54505e-05, 3.17079e-05]
  gravity bias YZ [0.00125118, -0.000618693]
E2 [30.00, 60.00) s:
  mass x [1.02693, 1.08918, 0.965679, 0.927341]
  CoM XYZ mm [1.65679, 1.84224, -2.29571]
  mu_s [0.541939, 0.732187, 0.611239, 0.700521]
  mu_d [0.534359, 0.631664, 0.611239, 0.641329]
  Kp [steer,rear] [10.5106, 0]; Kd [0.202627, 0]
  spawn roll 0.059515 rad; rate -0.0147133 rad/s
  gyro bias XYZ rad/s [-4.78947e-06, 5.43307e-05, 0.000350425]
  gravity bias YZ [-0.000315206, -0.00188852]
Six significant digits shown; full precision: sampled_dr_values.json.
```

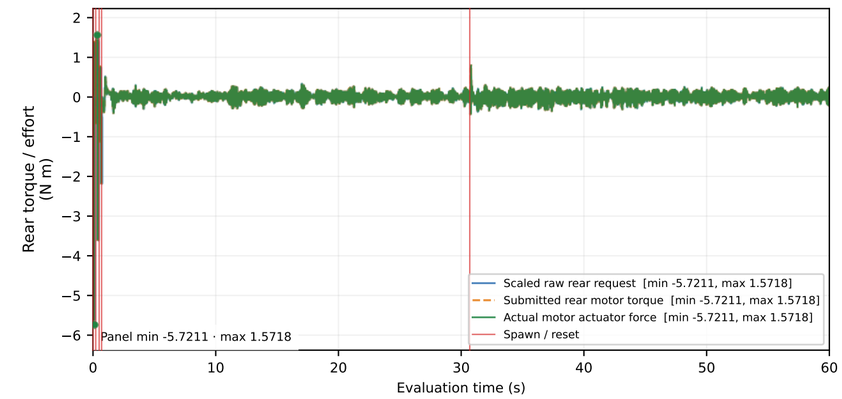
model_2500 · Demand versus actuation · env 1 · full 60 s

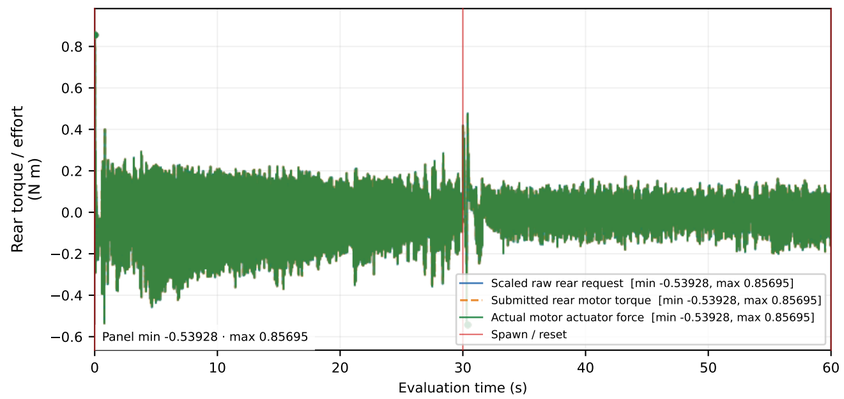












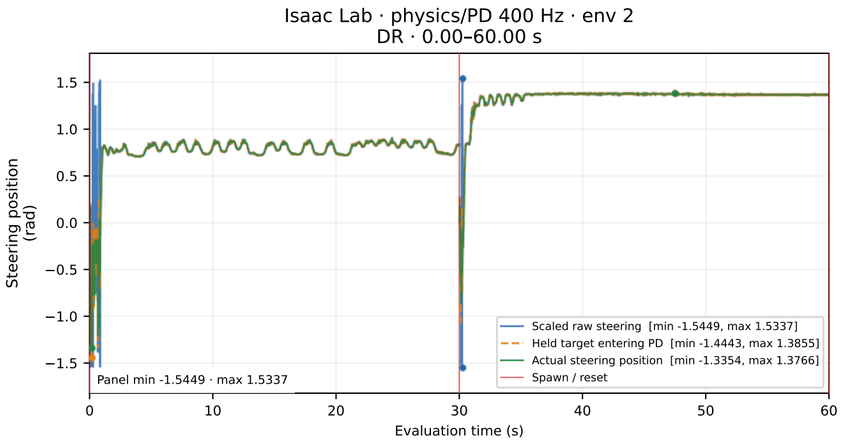
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

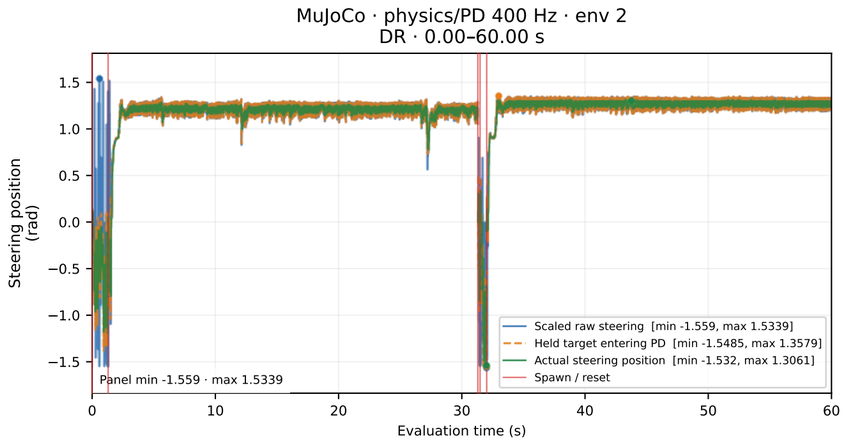
```
SAMPLED DR: env 1; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.862507, 0.992989, 1.0945, 0.962393]
  CoM XYZ mm [0.221074, 2.44478, 0.954762]
  mu_s [0.674963, 0.787792, 0.559809, 0.533455]
  mu_d [0.601294, 0.617134, 0.559809, 0.533455]
  Kp [steer,rear] [9.42558, 0]; Kd [0.184774, 0]
  spawn roll 0.0256746 rad; rate -0.215965 rad/s
  gyro bias XYZ rad/s [-7.03797e-05, 0.000323626, -7.54307e-05]
  gravity bias YZ [0.00224866, -0.00223355]
E2 [30.00, 60.00) s:
  mass x [1.17538, 0.984673, 0.876879, 1.09066]
  CoM XYZ mm [4.70234, 1.68571, 0.744864]
  mu_s [0.533455, 0.602694, 0.693205, 0.579974]
  mu_d [0.533455, 0.602694, 0.693205, 0.579974]
  Kp [steer,rear] [10.2626, 0]; Kd [0.183138, 0]
  spawn roll -0.00394535 rad; rate -0.198856 rad/s
  gyro bias XYZ rad/s [0.00023291, 2.05443e-07, 0.000381584]
  gravity bias YZ [-0.000380348, 0.00116294]
Six significant digits shown; full precision: sampled_dr_values.json.
```

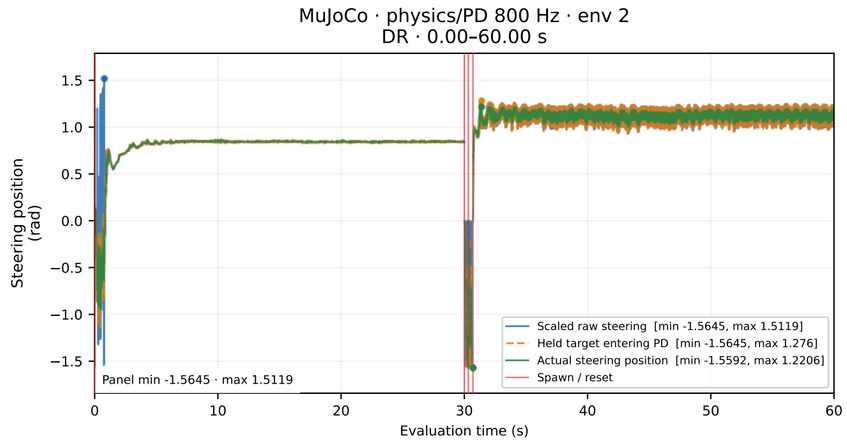
```
SAMPLED DR: env 1; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.22) s:
  mass x [1.06092, 0.81751, 0.808012, 1.13569]
  CoM XYZ mm [0.87143, -2.75295, 2.51792]
  mu_s [0.733515, 0.666375, 0.666375, 0.640016]
  mu_d [0.558392, 0.519145, 0.519145, 0.513141]
  Kp [steer,rear] [9.90206, 0]; Kd [0.215676, 0]
  spawn roll 0.067641 rad; rate 0.0175641 rad/s
  gyro bias XYZ rad/s [-7.13383e-05, -0.000159725, 5.31283e-06]
  gravity bias YZ [-0.00123234, 0.00117109]
E2 [0.22, 0.50) s:
  mass x [1.18392, 1.04402, 1.17318, 0.884353]
  CoM XYZ mm [3.00329, -3.53677, 1.20887]
  mu_s [0.748289, 0.559972, 0.757579, 0.611239]
  mu_d [0.689499, 0.502209, 0.70921, 0.611239]
  Kp [steer,rear] [9.1824, 0]; Kd [0.199162, 0]
  spawn roll 0.0977902 rad; rate -0.0536535 rad/s
  gyro bias XYZ rad/s [-0.000120112, 0.000267945, 2.74423e-05]
  gravity bias YZ [0.00344303, -0.00277117]
E3 [0.50, 0.70) s:
  mass x [0.840627, 1.16411, 1.09729, 0.896648]
  CoM XYZ mm [0.769351, 4.12939, -0.722516]
  mu_s [0.791209, 0.749678, 0.541939, 0.616244]
  mu_d [0.767936, 0.741429, 0.534359, 0.586498]
  Kp [steer,rear] [10.6528, 0]; Kd [0.205113, 0]
  spawn roll -0.0936273 rad; rate 0.000719158 rad/s
  gyro bias XYZ rad/s [0.000257614, -0.000169141, 9.57854e-05]
  gravity bias YZ [0.000545115, 0.00259499]
E4 [0.70, 30.70) s:
  mass x [0.82748, 1.12533, 1.14601, 0.998736]
  CoM XYZ mm [-2.38536, -2.57866, 2.20175]
  mu_s [0.546287, 0.732187, 0.631146, 0.631146]
  mu_d [0.546287, 0.631664, 0.631146, 0.631146]
  Kp [steer,rear] [9.89664, 0]; Kd [0.210172, 0]
  spawn roll -0.033323 rad; rate 0.0957727 rad/s
  gyro bias XYZ rad/s [9.53978e-05, 0.000048847, -0.000220352]
  gravity bias YZ [-0.000144735, 0.00422537]
E5 [30.70, 60.00) s:
  mass x [0.970729, 1.05628, 1.13447, 1.13934]
  CoM XYZ mm [-0.74075, -1.87991, 1.53457]
  mu_s [0.559972, 0.733515, 0.727426, 0.538976]
  mu_d [0.502209, 0.558392, 0.606358, 0.538976]
  Kp [steer,rear] [9.27399, 0]; Kd [0.20469, 0]
  spawn roll 0.043123 rad; rate -0.215016 rad/s
  gyro bias XYZ rad/s [-0.00054727, 0.000248029, -0.000156768]
  gravity bias YZ [-8.41541e-05, -0.00221894]
Six significant digits shown; full precision: sampled_dr_values.json.
```

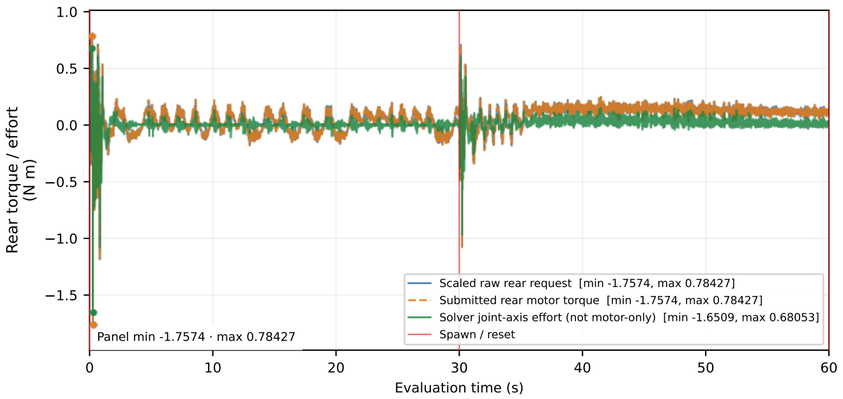
```
SAMPLED DR: env 1; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.06092, 0.81751, 0.808012, 1.13569]
  CoM XYZ mm [0.87143, -2.75295, 2.51792]
  mu_s [0.733515, 0.666375, 0.666375, 0.640016]
  mu_d [0.558392, 0.519145, 0.519145, 0.513141]
  Kp [steer,rear] [9.90206, 0]; Kd [0.215676, 0]
  spawn roll 0.067641 rad; rate 0.0175641 rad/s
  gyro bias XYZ rad/s [-7.13383e-05, -0.000159725, 5.31283e-06]
  gravity bias YZ [-0.00123234, 0.00117109]
E2 [30.00, 60.00) s:
  mass x [0.968901, 1.16355, 0.949248, 1.00463]
  CoM XYZ mm [-2.82922, -3.78885, 2.31511]
  mu_s [0.546287, 0.736077, 0.528253, 0.757579]
  mu_d [0.546287, 0.699455, 0.528253, 0.70921]
  Kp [steer,rear] [9.91541, 0]; Kd [0.217799, 0]
  spawn roll -0.0512449 rad; rate -0.0492599 rad/s
  gyro bias XYZ rad/s [4.82627e-05, -3.74329e-05, 0.000101373]
  gravity bias YZ [0.00136252, 0.00314652]
Six significant digits shown; full precision: sampled_dr_values.json.
```

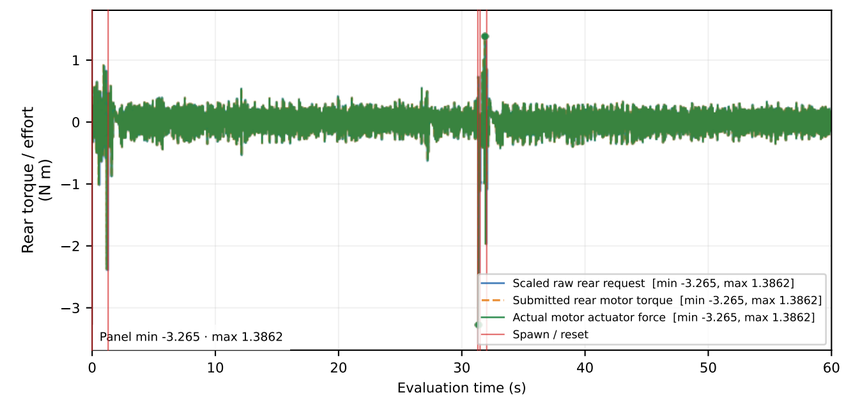
model_2500 · Demand versus actuation · env 2 · full 60 s

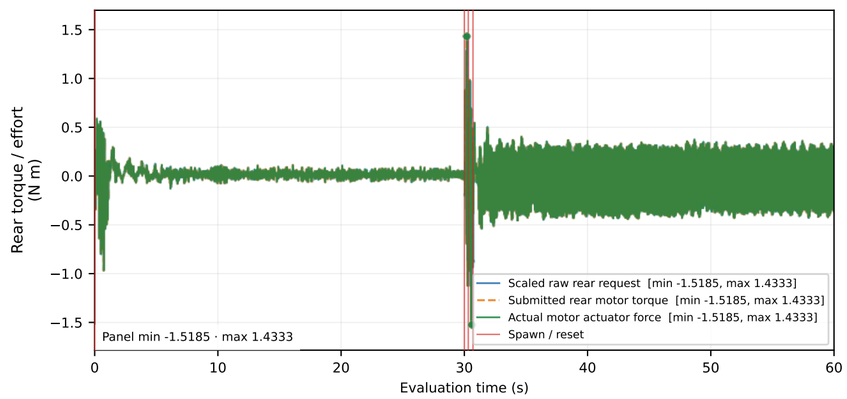












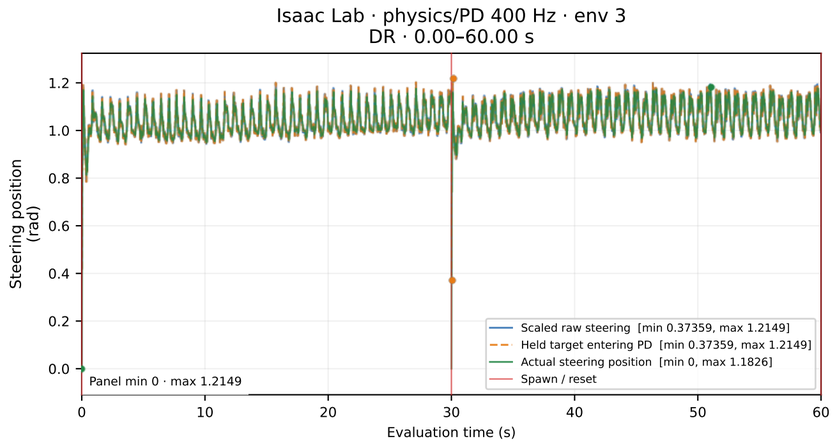
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

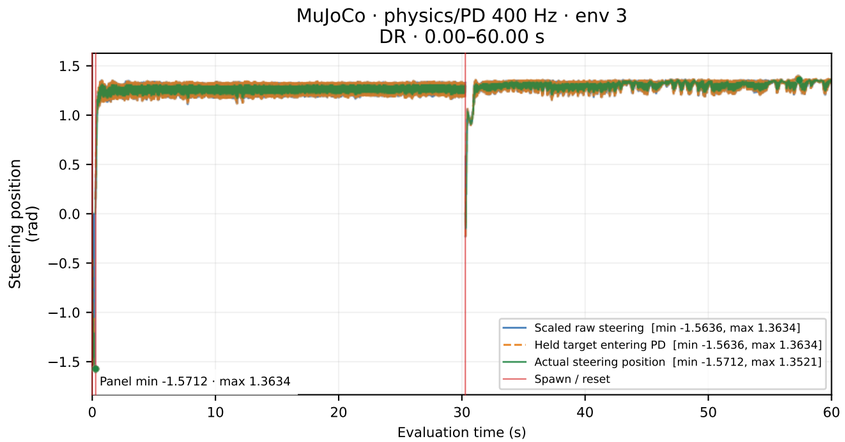
```
SAMPLED DR: env 2; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.00756, 0.914684, 0.896632, 1.16914]
  CoM XYZ mm [4.64708, 3.97917, 2.72967]
  mu_s [0.539956, 0.602694, 0.681711, 0.584424]
  mu_d [0.539956, 0.602694, 0.611756, 0.584424]
  Kp [steer,rear] [10.4584, 0]; Kd [0.182042, 0]
  spawn roll -0.0551139 rad; rate 0.115813 rad/s
  gyro bias XYZ rad/s [-0.000203379, -3.18315e-05, -0.000316606]
  gravity bias YZ [-0.00319071, -0.000254446]
E2 [30.00, 60.00) s:
  mass x [0.957834, 0.884986, 1.1579, 1.16429]
  CoM XYZ mm [-4.14227, -3.11448, -0.689477]
  mu_s [0.584424, 0.584424, 0.547743, 0.764681]
  mu_d [0.584424, 0.584424, 0.547743, 0.764681]
  Kp [steer,rear] [10.343, 0]; Kd [0.219287, 0]
  spawn roll -0.050929 rad; rate 0.070442 rad/s
  gyro bias XYZ rad/s [-0.000212198, -0.000398633, 0.000157541]
  gravity bias YZ [-0.00211123, 0.00306208]
Six significant digits shown; full precision: sampled_dr_values.json.
```

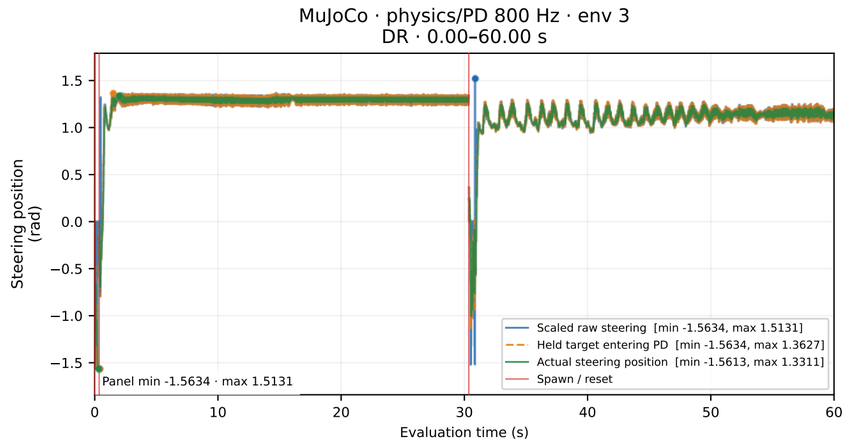
```
SAMPLED DR: env 2; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.30) s:
  mass x [0.849026, 0.903245, 0.962308, 1.18767]
  CoM XYZ mm [-3.37683, 3.57294, -3.36955]
  mu_s [0.728342, 0.727426, 0.733515, 0.71008]
  mu_d [0.728342, 0.606358, 0.558392, 0.59371]
  Kp [steer,rear] [10.2331, 0]; Kd [0.196456, 0]
  spawn roll 0.004772566 rad; rate -0.151419 rad/s
  gyro bias XYZ rad/s [5.35451e-05, 0.000141836, 3.5504e-05]
  gravity bias YZ [-0.00156514, 0.000524658]
E2 [1.30, 31.30) s:
  mass x [0.862338, 1.16887, 1.14804, 1.0074]
  CoM XYZ mm [-1.55679, -1.06343, -4.16265]
  mu_s [0.749678, 0.546287, 0.640867, 0.666375]
  mu_d [0.741429, 0.546287, 0.556841, 0.519145]
  Kp [steer,rear] [9.15738, 0]; Kd [0.211983, 0]
  spawn roll -0.045117 rad; rate 0.104407 rad/s
  gyro bias XYZ rad/s [0.000274867, -4.48659e-05, -0.000265008]
  gravity bias YZ [0.00171783, -0.000824402]
E3 [31.30, 31.48) s:
  mass x [0.955405, 1.09469, 0.992401, 0.887005]
  CoM XYZ mm [-4.13751, -3.91435, 4.85292]
  mu_s [0.633024, 0.546287, 0.640867, 0.732187]
  mu_d [0.568172, 0.546287, 0.556841, 0.631664]
  Kp [steer,rear] [9.8247, 0]; Kd [0.218829, 0]
  spawn roll 0.0975943 rad; rate 0.188861 rad/s
  gyro bias XYZ rad/s [-5.2756e-06, 0.000223486, -5.41635e-05]
  gravity bias YZ [0.00141839, -0.00176458]
E4 [31.48, 32.02) s:
  mass x [0.822835, 0.939556, 1.09824, 0.922163]
  CoM XYZ mm [1.99899, 3.27979, 2.17474]
  mu_s [0.791209, 0.727426, 0.669571, 0.637675]
  mu_d [0.767936, 0.606358, 0.669571, 0.637675]
  Kp [steer,rear] [10.7052, 0]; Kd [0.183295, 0]
  spawn roll -0.00791113 rad; rate 0.138752 rad/s
  gyro bias XYZ rad/s [-0.000304359, 0.000170522, -0.000105578]
  gravity bias YZ [0.00262822, 0.00585244]
E5 [32.02, 60.00) s:
  mass x [0.866334, 0.98371, 1.17728, 1.17219]
  CoM XYZ mm [1.371, -2.09598, -1.37734]
  mu_s [0.541939, 0.633024, 0.538976, 0.597748]
  mu_d [0.534359, 0.568172, 0.538976, 0.597748]
  Kp [steer,rear] [9.16109, 0]; Kd [0.199155, 0]
  spawn roll -0.00517896 rad; rate -0.125551 rad/s
  gyro bias XYZ rad/s [-0.000215667, 4.98964e-05, -0.000202771]
  gravity bias YZ [0.00045478, -0.00260641]
Six significant digits shown; full precision: sampled_dr_values.json.
```

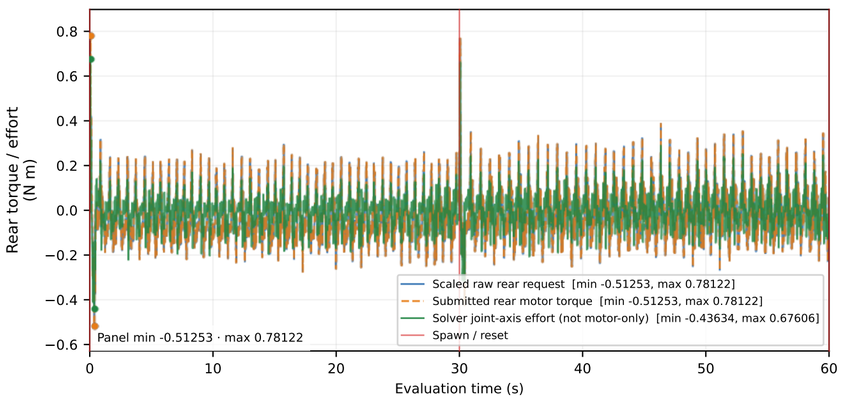
```
SAMPLED DR: env 2; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.849026, 0.903245, 0.962308, 1.18767]
  CoM XYZ mm [-3.37683, 3.57294, -3.36955]
  mu_s [0.728342, 0.727426, 0.733515, 0.71008]
  mu_d [0.728342, 0.606358, 0.558392, 0.59371]
  Kp [steer,rear] [10.2331, 0]; Kd [0.196456, 0]
  spawn roll 0.00477256 rad; rate -0.151419 rad/s
  gyro bias XYZ rad/s [5.35451e-05, 0.000141836, 3.5504e-05]
  gravity bias YZ [-0.00156514, 0.000524658]
E2 [30.00, 30.32) s:
  mass x [1.15833, 0.950965, 1.07119, 0.986383]
  CoM XYZ mm [-2.28574, 4.94326, 0.580657]
  mu_s [0.538976, 0.541939, 0.700521, 0.791209]
  mu_d [0.538976, 0.534359, 0.641329, 0.767936]
  Kp [steer,rear] [9.96591, 0]; Kd [0.207815, 0]
  spawn roll -0.0830041 rad; rate 0.109036 rad/s
  gyro bias XYZ rad/s [-8.98043e-05, 5.78866e-05, 0.000250741]
  gravity bias YZ [-0.000604908, -0.00123248]
E3 [30.32, 30.70) s:
  mass x [0.832338, 0.882538, 1.0699, 1.17094]
  CoM XYZ mm [2.00383, 4.7638, 1.17088]
  mu_s [0.538434, 0.559972, 0.733515, 0.700521]
  mu_d [0.538434, 0.502209, 0.558392, 0.641329]
  Kp [steer,rear] [10.6028, 0]; Kd [0.20534, 0]
  spawn roll -0.0550542 rad; rate 0.058826 rad/s
  gyro bias XYZ rad/s [-1.2178e-05, 0.000148104, 0.000107851]
  gravity bias YZ [-0.00203786, 0.00101602]
E4 [30.70, 60.00) s:
  mass x [1.04999, 0.922194, 1.06322, 0.970641]
  CoM XYZ mm [4.90851, 1.67046, -3.5585]
  mu_s [0.640016, 0.666375, 0.700521, 0.728342]
  mu_d [0.513141, 0.519145, 0.641329, 0.728342]
  Kp [steer,rear] [10.797, 0]; Kd [0.215121, 0]
  spawn roll 0.0438033 rad; rate 0.105346 rad/s
  gyro bias XYZ rad/s [0.000270291, -0.000103782, 2.09693e-05]
  gravity bias YZ [0.0017364, -0.00118816]
Six significant digits shown; full precision: sampled_dr_values.json.
```

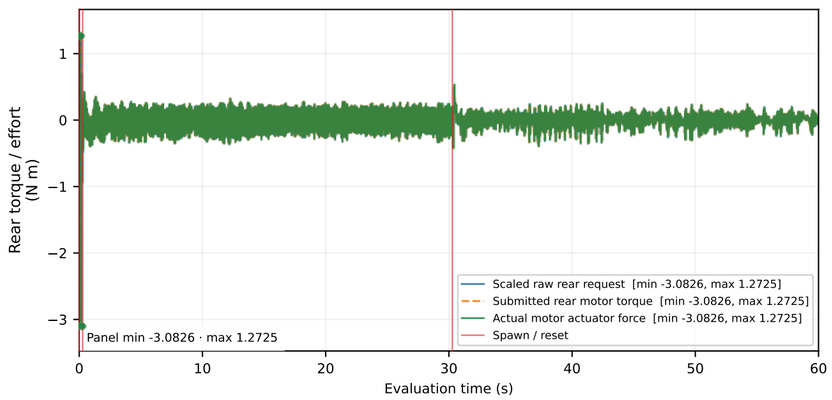
model_2500 · Demand versus actuation · env 3 · full 60 s

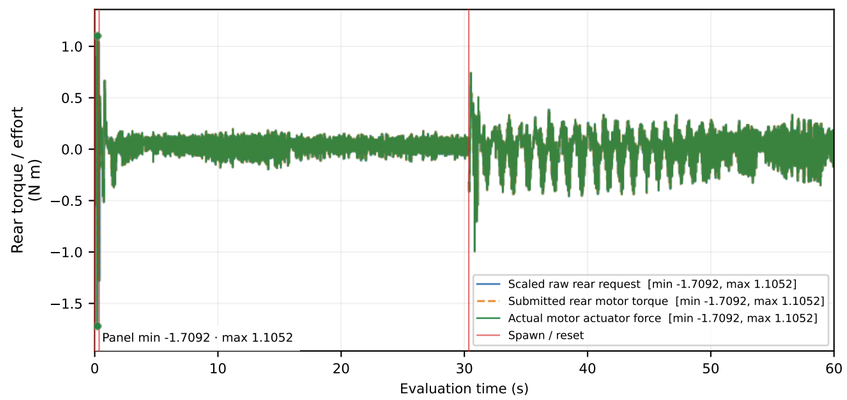












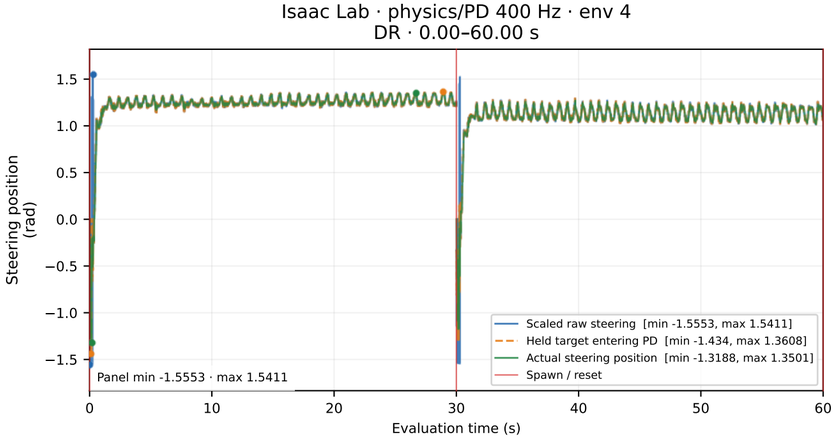
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

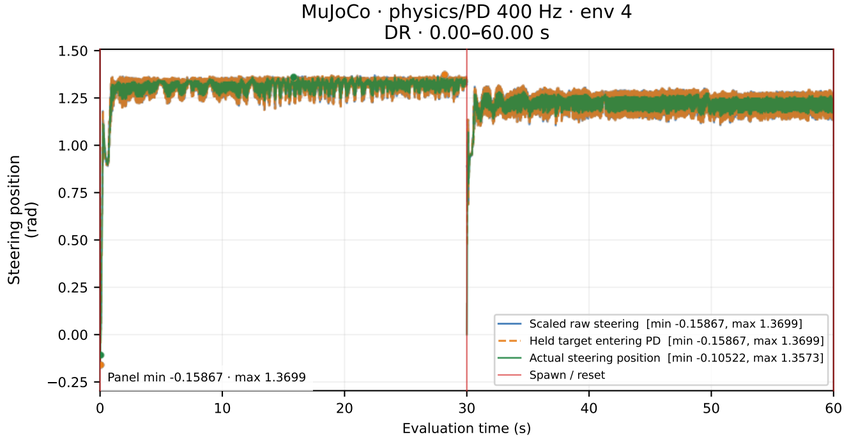
```
SAMPLED DR: env 3; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.13197, 0.813675, 0.955158, 0.832956]
  CoM XYZ mm [1.6811, 0.46216, 0.0711083]
  mu_s [0.713129, 0.632409, 0.602694, 0.774409]
  mu_d [0.713129, 0.589076, 0.602694, 0.561095]
  Kp [steer,rear] [9.6419, 0]; Kd [0.189538, 0]
  spawn roll 0.0698016 rad; rate -0.142062 rad/s
  gyro bias XYZ rad/s [0.000115749, -3.6422e-05, -0.000145327]
  gravity bias YZ [-0.00140168, -0.00330635]
E2 [30.00, 60.00) s:
  mass x [1.01015, 1.06654, 0.832539, 0.908403]
  CoM XYZ mm [-1.31282, 0.536375, 0.871569]
  mu_s [0.539956, 0.539956, 0.579974, 0.632409]
  mu_d [0.539956, 0.539956, 0.579974, 0.589076]
  Kp [steer,rear] [10.686, 0]; Kd [0.201728, 0]
  spawn roll 0.0615653 rad; rate -0.131084 rad/s
  gyro bias XYZ rad/s [-8.79698e-05, -0.000189956, -1.0838e-05]
  gravity bias YZ [-0.00019208, 0.000399016]
Six significant digits shown; full precision: sampled_dr_values.json.
```

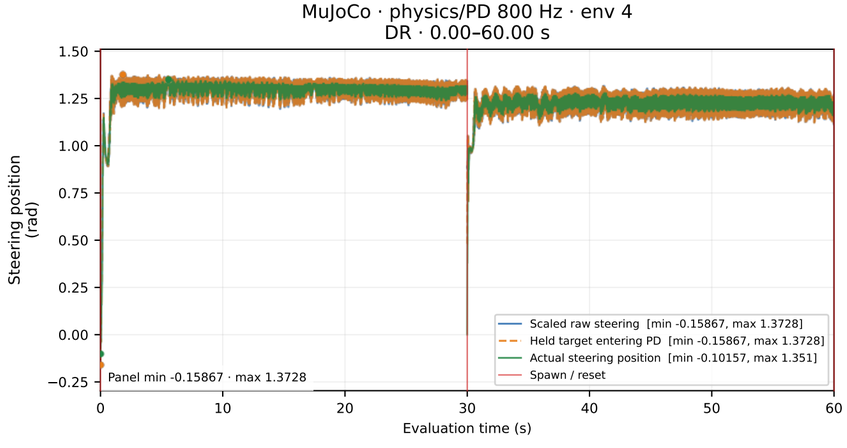
```
SAMPLED DR: env 3; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [1.02925, 1.0114, 1.10546, 1.12468]
  CoM XYZ mm [0.102289, 2.7934, 2.96113]
  mu_s [0.704749, 0.568073, 0.748289, 0.640016]
  mu_d [0.541926, 0.568073, 0.689499, 0.513141]
  Kp [steer,rear] [10.3426, 0]; Kd [0.213603, 0]
  spawn roll -0.0890225 rad; rate -0.19006 rad/s
  gyro bias XYZ rad/s [-4.54431e-05, -0.000198711, -0.000291781]
  gravity bias YZ [0.000127515, 0.00154904]
E2 [0.28, 30.28) s:
  mass x [0.837542, 1.15518, 0.926383, 1.19918]
  CoM XYZ mm [3.94932, -2.13762, 1.29165]
  mu_s [0.546287, 0.541939, 0.538434, 0.732187]
  mu_d [0.546287, 0.534359, 0.538434, 0.631664]
  Kp [steer,rear] [10.1341, 0]; Kd [0.183532, 0]
  spawn roll -0.00720207 rad; rate -0.000312354 rad/s
  gyro bias XYZ rad/s [2.58975e-05, -0.000105769, 0.000105602]
  gravity bias YZ [-0.00163113, -0.00077671]
E3 [30.28, 60.00) s:
  mass x [1.05434, 1.13786, 0.975343, 1.1978]
  CoM XYZ mm [-0.350068, -3.69408, -1.36237]
  mu_s [0.541939, 0.597748, 0.597748, 0.640867]
  mu_d [0.534359, 0.597748, 0.597748, 0.556841]
  Kp [steer,rear] [9.80054, 0]; Kd [0.190214, 0]
  spawn roll -0.000669958 rad; rate -0.187088 rad/s
  gyro bias XYZ rad/s [-3.15961e-05, 5.02063e-05, -0.000199858]
  gravity bias YZ [-0.00370918, 0.00131895]
Six significant digits shown; full precision: sampled_dr_values.json.
```

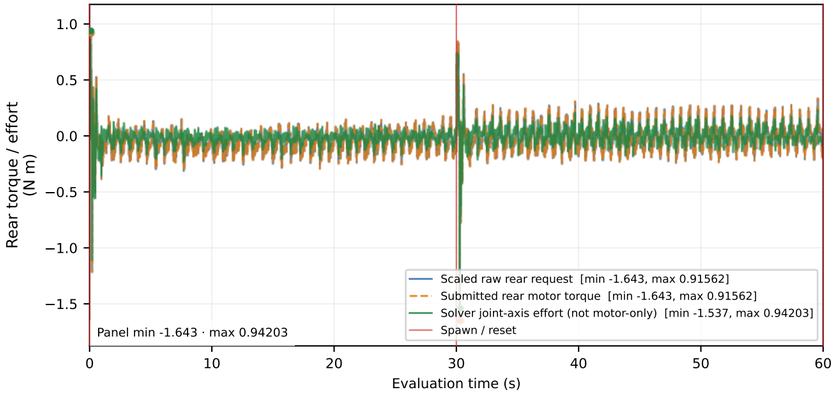
```
SAMPLED DR: env 3; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.36) s:
  mass x [1.02925, 1.0114, 1.10546, 1.12468]
  CoM XYZ mm [0.102289, 2.7934, 2.96113]
  mu_s [0.704749, 0.568073, 0.748289, 0.640016]
  mu_d [0.541926, 0.568073, 0.689499, 0.513141]
  Kp [steer,rear] [10.3426, 0]; Kd [0.213603, 0]
  spawn roll -0.0890225 rad; rate -0.19006 rad/s
  gyro bias XYZ rad/s [-4.54431e-05, -0.000198711, -0.000291781]
  gravity bias YZ [0.000127515, 0.00154904]
E2 [0.36, 30.36) s:
  mass x [1.19907, 1.06352, 1.14454, 1.04925]
  CoM XYZ mm [-0.034423, -4.44887, -0.841293]
  mu_s [0.748289, 0.546287, 0.640867, 0.732187]
  mu_d [0.689499, 0.546287, 0.556841, 0.631664]
  Kp [steer,rear] [10.6007, 0]; Kd [0.21073, 0]
  spawn roll -0.0493498 rad; rate -0.0192491 rad/s
  gyro bias XYZ rad/s [-8.97478e-05, -4.88573e-05, 0.000290481]
  gravity bias YZ [-0.000123604, 0.00476323]
E3 [30.36, 60.00) s:
  mass x [1.04058, 0.989516, 1.13018, 1.10386]
  CoM XYZ mm [-3.10637, 0.0190005, -0.501031]
  mu_s [0.71155, 0.723429, 0.728342, 0.597748]
  mu_d [0.71155, 0.723429, 0.728342, 0.597748]
  Kp [steer,rear] [10.6943, 0]; Kd [0.182485, 0]
  spawn roll -0.0463704 rad; rate 0.205555 rad/s
  gyro bias XYZ rad/s [0.000370802, 0.000462692, -0.000154158]
  gravity bias YZ [0.000957546, -0.00018204]
Six significant digits shown; full precision: sampled_dr_values.json.
```

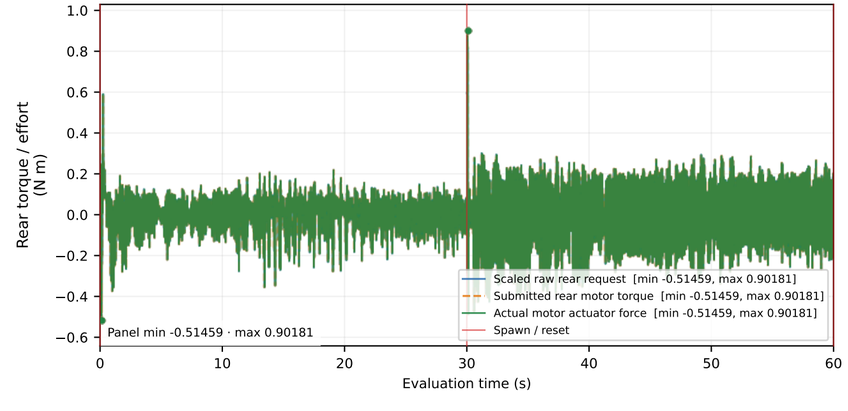
model_2500 · Demand versus actuation · env 4 · full 60 s

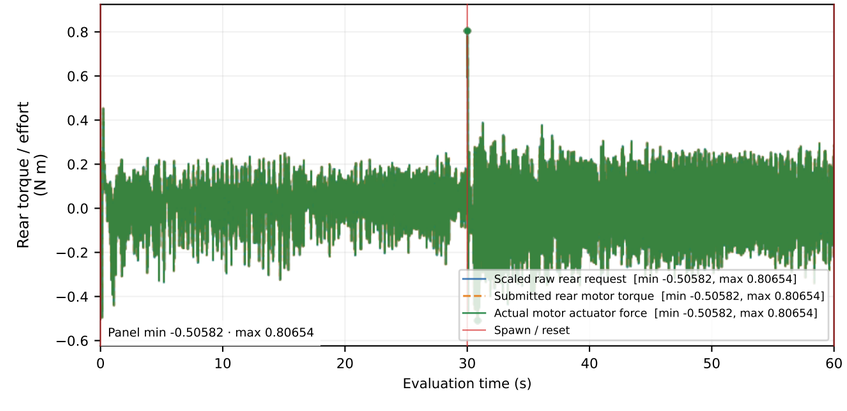












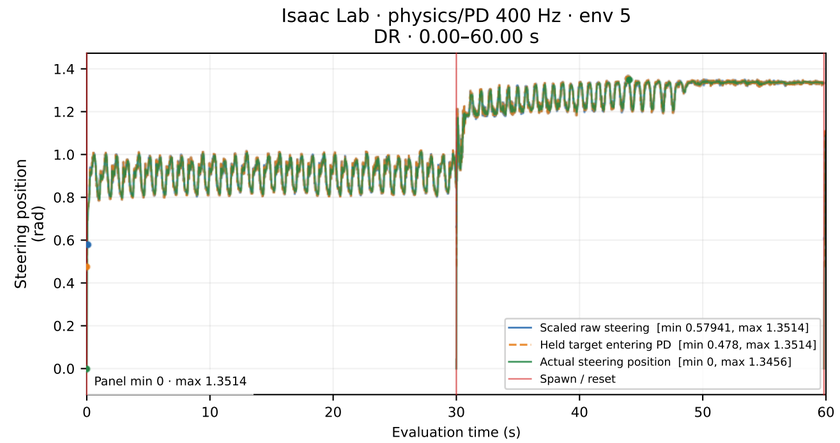
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

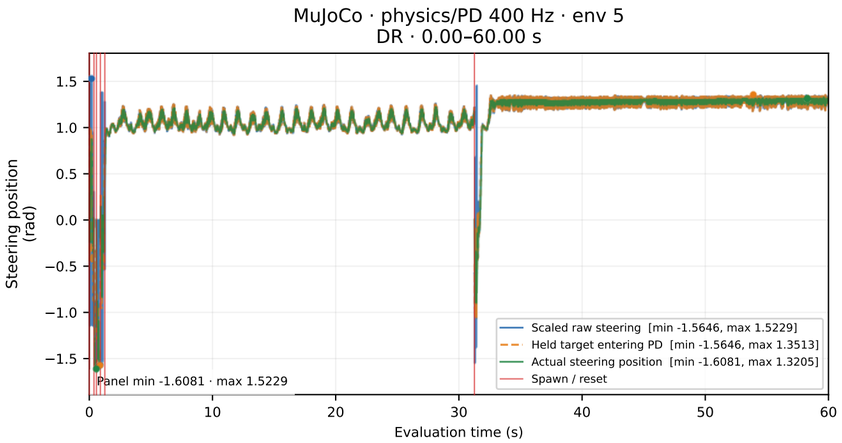
```
SAMPLED DR: env 4; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.10967, 0.911688, 1.00553, 0.882728]
  CoM XYZ mm [0.882208, -2.44468, 3.96283]
  mu_s [0.547743, 0.794338, 0.533455, 0.632409]
  mu_d [0.547743, 0.526209, 0.533455, 0.589076]
  Kp [steer,rear] [10.51, 0]; Kd [0.18806, 0]
  spawn roll -0.0996169 rad; rate -0.165478 rad/s
  gyro bias XYZ rad/s [-0.00025822, 9.37975e-06, 0.000278995]
  gravity bias YZ [0.000311097, 0.00214254]
E2 [30.00, 60.00) s:
  mass x [1.14475, 0.888204, 0.814729, 0.899869]
  CoM XYZ mm [0.76364, -0.353117, -3.2285]
  mu_s [0.726175, 0.760821, 0.579974, 0.547743]
  mu_d [0.558574, 0.670315, 0.579974, 0.547743]
  Kp [steer,rear] [10.4888, 0]; Kd [0.188838, 0]
  spawn roll -0.0724018 rad; rate -0.0422324 rad/s
  gyro bias XYZ rad/s [0.000232035, 0.000363317, -1.80533e-05]
  gravity bias YZ [-0.00186627, -0.000895573]
Six significant digits shown; full precision: sampled_dr_values.json.
```

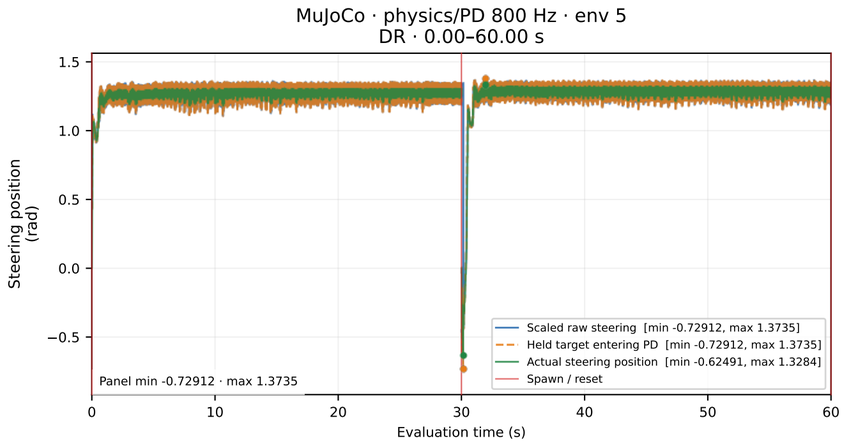
```
SAMPLED DR: env 4; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16224, 0.830891, 0.909028, 1.04874]
  CoM XYZ mm [4.34028, -4.15053, 1.2923]
  mu_s [0.559972, 0.704749, 0.633024, 0.597748]
  mu_d [0.502209, 0.541926, 0.568172, 0.597748]
  Kp [steer,rear] [10.497, 0]; Kd [0.199966, 0]
  spawn roll -0.0219811 rad; rate -0.0869557 rad/s
  gyro bias XYZ rad/s [0.000105059, 6.3236e-05, -7.15947e-05]
  gravity bias YZ [-0.00185749, -0.00210565]
E2 [30.00, 60.00) s:
  mass x [0.818194, 1.13428, 0.985815, 1.09866]
  CoM XYZ mm [-0.129074, -1.07891, 2.27407]
  mu_s [0.559972, 0.757579, 0.541939, 0.568073]
  mu_d [0.502209, 0.70921, 0.534359, 0.568073]
  Kp [steer,rear] [10.0033, 0]; Kd [0.181016, 0]
  spawn roll 0.0661497 rad; rate 0.227347 rad/s
  gyro bias XYZ rad/s [5.09085e-05, 0.00023236, 7.71043e-05]
  gravity bias YZ [7.66479e-05, 0.00122516]
Six significant digits shown; full precision: sampled_dr_values.json.
```

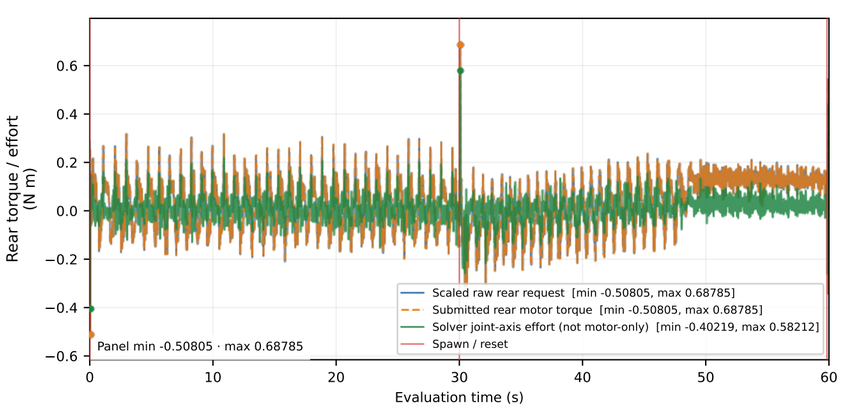
```
SAMPLED DR: env 4; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16224, 0.830891, 0.909028, 1.04874]
  CoM XYZ mm [4.34028, -4.15053, 1.2923]
  mu_s [0.559972, 0.704749, 0.633024, 0.597748]
  mu_d [0.502209, 0.541926, 0.568172, 0.597748]
  Kp [steer,rear] [10.497, 0]; Kd [0.199966, 0]
  spawn roll -0.0219811 rad; rate -0.0869557 rad/s
  gyro bias XYZ rad/s [0.000105059, 6.3236e-05, -7.15947e-05]
  gravity bias YZ [-0.00185749, -0.00210565]
E2 [30.00, 60.00) s:
  mass x [0.818194, 1.13428, 0.985815, 1.09866]
  CoM XYZ mm [-0.129074, -1.07891, 2.27407]
  mu_s [0.559972, 0.757579, 0.541939, 0.568073]
  mu_d [0.502209, 0.70921, 0.534359, 0.568073]
  Kp [steer,rear] [10.0033, 0]; Kd [0.181016, 0]
  spawn roll 0.0661497 rad; rate 0.227347 rad/s
  gyro bias XYZ rad/s [5.09085e-05, 0.00023236, 7.71043e-05]
  gravity bias YZ [7.66479e-05, 0.00122516]
Six significant digits shown; full precision: sampled_dr_values.json.
```

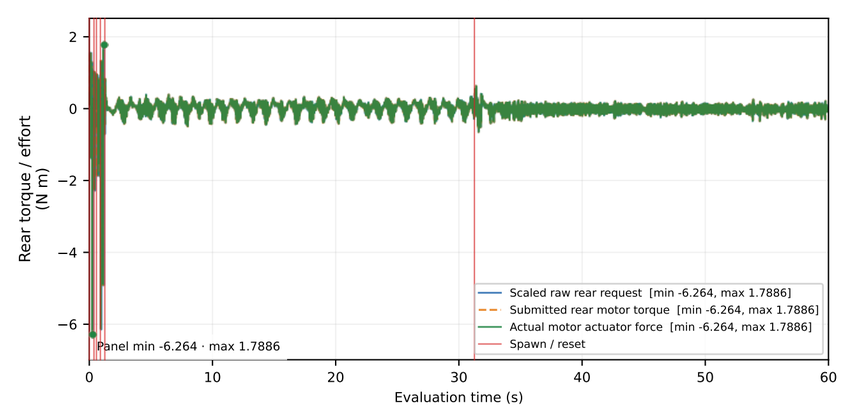
model_2500 · Demand versus actuation · env 5 · full 60 s

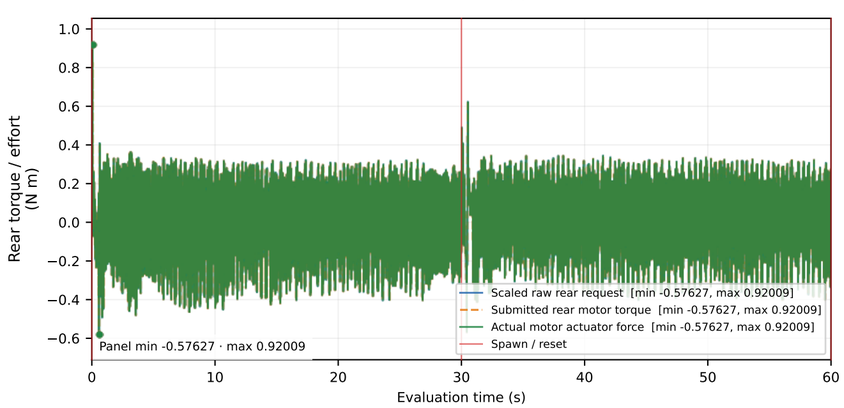












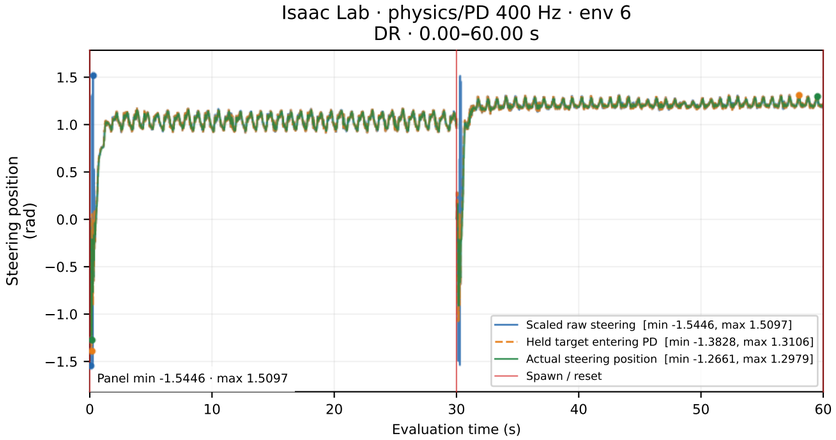
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

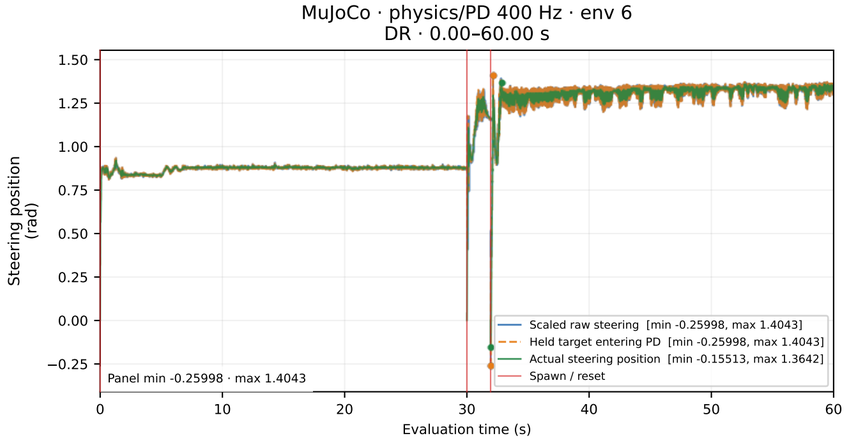
```
SAMPLED DR: env 5; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.802947, 1.15001, 1.18729, 0.848047]
  CoM XYZ mm [-3.13438, 2.23542, -3.2187]
  mu_s [0.659445, 0.584424, 0.673378, 0.592046]
  mu_d [0.547619, 0.584424, 0.673378, 0.534947]
  Kp [steer,rear] [9.42645, 0]; Kd [0.205456, 0]
  spawn roll 0.0161676 rad; rate 0.204907 rad/s
  gyro bias XYZ rad/s [0.000204662, 0.000224783, -8.47435e-05]
  gravity bias YZ [-0.000148007, -0.00267725]
E2 [30.00, 59.84) s:
  mass x [0.881139, 1.13225, 0.949404, 1.08537]
  CoM XYZ mm [0.0505447, -1.46302, 3.60368]
  mu_s [0.647391, 0.741039, 0.647391, 0.539956]
  mu_d [0.647391, 0.614405, 0.647391, 0.539956]
  Kp [steer,rear] [9.62694, 0]; Kd [0.189018, 0]
  spawn roll 0.0462433 rad; rate 0.0846739 rad/s
  gyro bias XYZ rad/s [9.06762e-05, 0.000140041, -6.86844e-05]
  gravity bias YZ [0.00423396, -0.000195554]
E3 [59.84, 60.00) s:
  mass x [1.12477, 1.18441, 0.806715, 0.806501]
  CoM XYZ mm [-2.09889, -3.31393, -0.832587]
  mu_s [0.703711, 0.592046, 0.681711, 0.713129]
  mu_d [0.703711, 0.534947, 0.611756, 0.713129]
  Kp [steer,rear] [9.37835, 0]; Kd [0.192918, 0]
  spawn roll -0.000824928 rad; rate 0.175743 rad/s
  gyro bias XYZ rad/s [7.24127e-05, -0.000395092, 0.000189631]
  gravity bias YZ [-0.00136036, -0.00204384]
Six significant digits shown; full precision: sampled_dr_values.json.
```

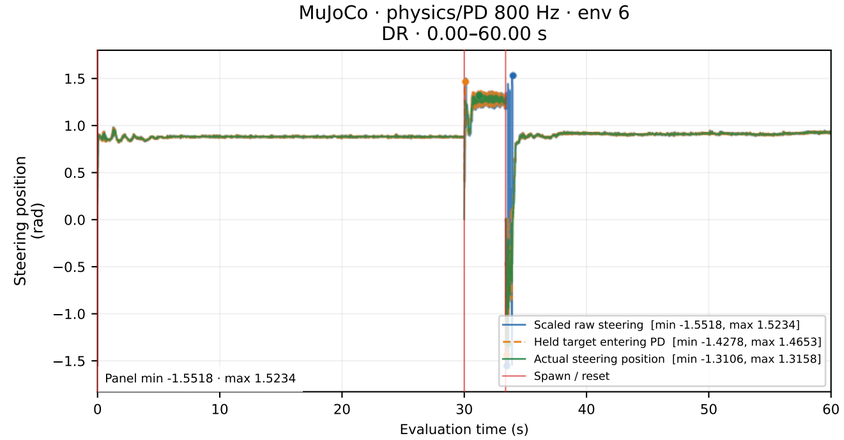
```
SAMPLED DR: env 5; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.38) s:
  mass x [1.09672, 1.10147, 0.986072, 0.84149]
  CoM XYZ mm [4.66852, -1.7925, -2.99668]
  mu_s [0.640016, 0.71155, 0.538434, 0.597748]
  mu_d [0.513141, 0.71155, 0.538434, 0.597748]
  Kp [steer,rear] [9.35848, 0]; Kd [0.211719, 0]
  spawn roll 0.0804349 rad; rate -0.153191 rad/s
  gyro bias XYZ rad/s [0.000176228, 0.000366092, -0.000125907]
  gravity bias YZ [0.00320495, -0.00211962]
E2 [0.38, 0.58) s:
  mass x [1.11491, 1.15873, 0.825652, 0.899697]
  CoM XYZ mm [4.36758, 0.878995, -4.68893]
  mu_s [0.538434, 0.616244, 0.611239, 0.728342]
  mu_d [0.538434, 0.586498, 0.611239, 0.728342]
  Kp [steer,rear] [10.8842, 0]; Kd [0.210061, 0]
  spawn roll -0.0855557 rad; rate -0.247262 rad/s
  gyro bias XYZ rad/s [-0.000168705, -8.27003e-05, 7.35071e-05]
  gravity bias YZ [7.95918e-05, -0.00316247]
E3 [0.58, 0.90) s:
  mass x [0.966119, 0.831289, 1.03341, 1.09674]
  CoM XYZ mm [-0.0965522, 4.67369, -4.64946]
  mu_s [0.727426, 0.538434, 0.538434, 0.528253]
  mu_d [0.606358, 0.538434, 0.538434, 0.528253]
  Kp [steer,rear] [10.4336, 0]; Kd [0.186615, 0]
  spawn roll -0.0877659 rad; rate -0.00385789 rad/s
  gyro bias XYZ rad/s [0.000231287, 0.000186892, 0.000189582]
  gravity bias YZ [-0.00284522, 0.00157655]
E4 [0.90, 1.26) s:
  mass x [0.978842, 1.15451, 1.09598, 0.821619]
  CoM XYZ mm [1.16774, -4.98495, 0.0479427]
  mu_s [0.791209, 0.616244, 0.71155, 0.568073]
  mu_d [0.767936, 0.586498, 0.71155, 0.568073]
  Kp [steer,rear] [10.9289, 0]; Kd [0.209175, 0]
  spawn roll -0.0261108 rad; rate -0.102054 rad/s
  gyro bias XYZ rad/s [-6.47184e-05, 3.55993e-05, -0.000303179]
  gravity bias YZ [0.00180537, 2.05281e-05]
E5 [1.26, 31.26) s:
  mass x [1.11897, 1.17619, 1.12289, 1.14132]
  CoM XYZ mm [-0.575743, 1.12483, -2.3802]
  mu_s [0.541939, 0.71008, 0.71008, 0.733515]
  mu_d [0.534359, 0.59371, 0.59371, 0.558392]
  Kp [steer,rear] [10.6454, 0]; Kd [0.213398, 0]
  spawn roll 0.0508153 rad; rate 0.225338 rad/s
  gyro bias XYZ rad/s [1.53426e-05, -0.000113271, -6.7393e-05]
  gravity bias YZ [0.0019273, 0.00144469]
E6 [31.26, 60.00) s:
  mass x [1.08294, 0.856463, 0.828188, 1.19957]
  CoM XYZ mm [1.13228, -3.49657, 3.12181]
  mu_s [0.631146, 0.616244, 0.568073, 0.728342]
  mu_d [0.631146, 0.586498, 0.568073, 0.728342]
  Kp [steer,rear] [9.07231, 0]; Kd [0.212301, 0]
  spawn roll -0.0509221 rad; rate -0.230631 rad/s
  gyro bias XYZ rad/s [2.57357e-05, -0.000184562, -0.000389171]
  gravity bias YZ [-0.00143281, -0.00148157]
Six significant digits shown; full precision: sampled_dr_values.json.
```

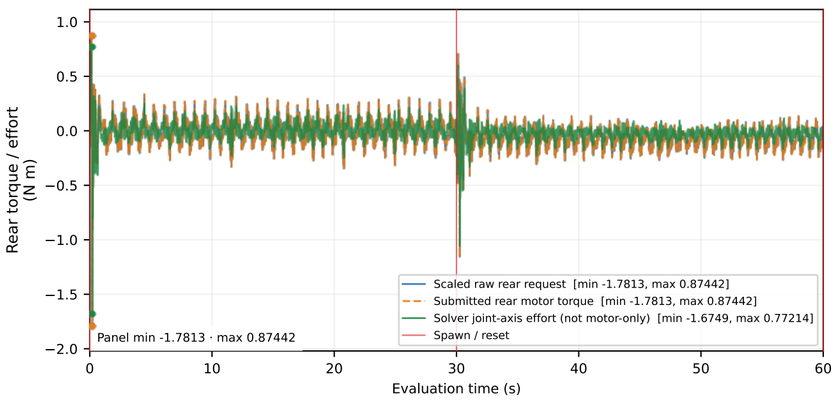
```
SAMPLED DR: env 5; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09672, 1.10147, 0.986072, 0.84149]
  CoM XYZ mm [4.66852, -1.7925, -2.99668]
  mu_s [0.640016, 0.71155, 0.538434, 0.597748]
  mu_d [0.513141, 0.71155, 0.538434, 0.597748]
  Kp [steer,rear] [9.35848, 0]; Kd [0.211719, 0]
  spawn roll 0.0804349 rad; rate -0.153191 rad/s
  gyro bias XYZ rad/s [0.000176228, 0.000366092, -0.000125907]
  gravity bias YZ [0.00320495, -0.00211962]
E2 [30.00, 60.00) s:
  mass x [0.997624, 1.11207, 0.952525, 0.89016]
  CoM XYZ mm [2.79459, -2.94807, -4.33668]
  mu_s [0.640867, 0.666375, 0.700521, 0.616244]
  mu_d [0.556841, 0.519145, 0.641329, 0.586498]
  Kp [steer,rear] [10.059, 0]; Kd [0.186785, 0]
  spawn roll -0.029392 rad; rate -0.22538 rad/s
  gyro bias XYZ rad/s [0.000406189, -0.000215817, 4.80454e-05]
  gravity bias YZ [-0.00327204, -0.00162364]
Six significant digits shown; full precision: sampled_dr_values.json.
```

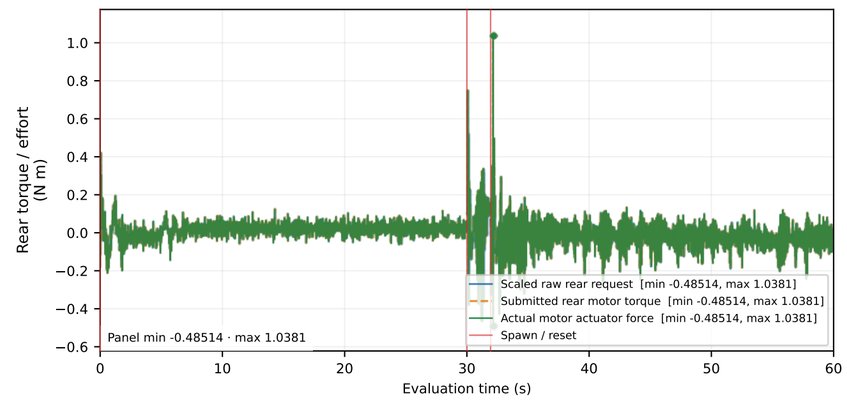
model_2500 · Demand versus actuation · env 6 · full 60 s

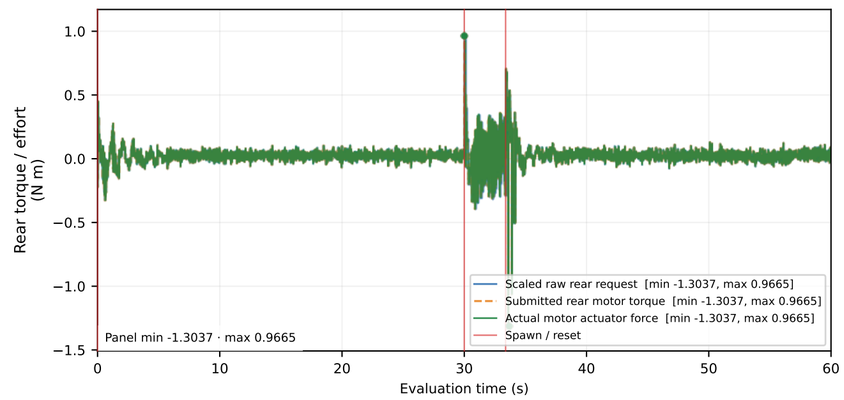












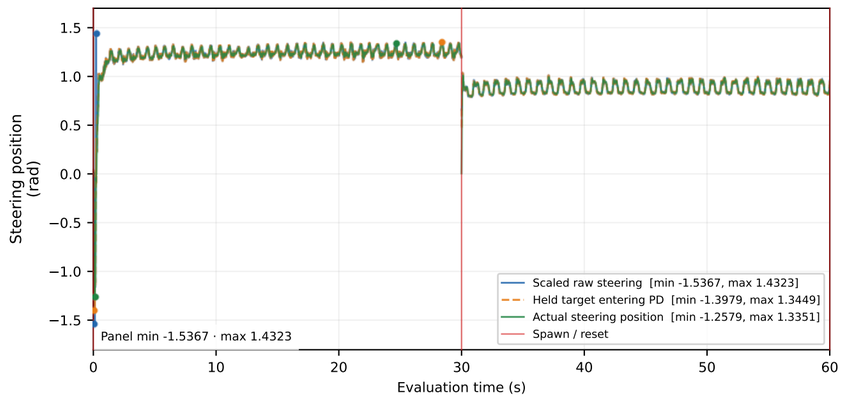
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

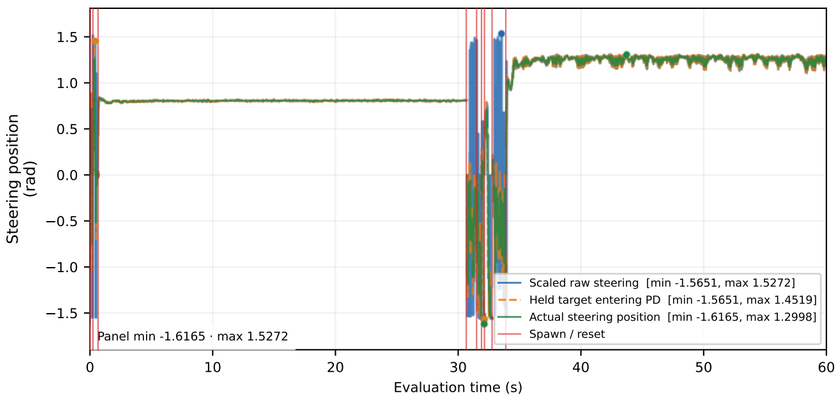
```
SAMPLED DR: env 6; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.878882, 0.980103, 0.820446, 0.995684]
  CoM XYZ mm [4.88032, 1.28894, -0.144839]
  mu_s [0.713129, 0.659445, 0.560535, 0.532645]
  mu_d [0.713129, 0.547619, 0.560535, 0.532645]
  Kp [steer,rear] [10.8556, 0]; Kd [0.211197, 0]
  spawn roll -0.0759049 rad; rate 0.0792792 rad/s
  gyro bias XYZ rad/s [-0.000271052, 0.000281376, 0.00032944]
  gravity bias YZ [0.000762568, -0.00113228]
E2 [30.00, 60.00) s:
  mass x [1.16766, 1.02214, 1.16892, 1.14839]
  CoM XYZ mm [4.2415, -2.19653, 4.4646]
  mu_s [0.547743, 0.628821, 0.579974, 0.592046]
  mu_d [0.547743, 0.628821, 0.579974, 0.534947]
  Kp [steer,rear] [9.40059, 0]; Kd [0.215543, 0]
  spawn roll -0.0541233 rad; rate 0.0767112 rad/s
  gyro bias XYZ rad/s [0.00019548, -1.27438e-05, 0.000412792]
  gravity bias YZ [-0.00398889, -0.000862655]
Six significant digits shown; full precision: sampled_dr_values.json.
```

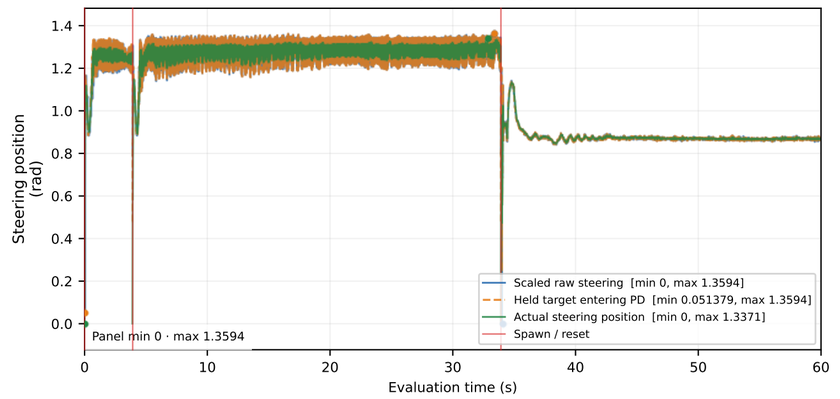
```
SAMPLED DR: env 6; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.95508, 1.03831, 1.00552, 1.0778]
  CoM XYZ mm [1.52497, 3.63117, 0.89843]
  mu_s [0.538434, 0.597748, 0.559972, 0.704749]
  mu_d [0.538434, 0.597748, 0.502209, 0.541926]
  Kp [steer,rear] [9.42108, 0]; Kd [0.183421, 0]
  spawn roll 0.0479951 rad; rate -0.0975409 rad/s
  gyro bias XYZ rad/s [-7.63979e-06, -0.000148874, 0.000268199]
  gravity bias YZ [0.0011309, 0.00172192]
E2 [30.00, 31.94) s:
  mass x [1.0515, 1.19247, 1.19836, 1.13004]
  CoM XYZ mm [4.57824, -4.74443, -1.52432]
  mu_s [0.559972, 0.631146, 0.757579, 0.732187]
  mu_d [0.502209, 0.631146, 0.70921, 0.631664]
  Kp [steer,rear] [9.23497, 0]; Kd [0.200202, 0]
  spawn roll 0.0695217 rad; rate -0.141358 rad/s
  gyro bias XYZ rad/s [-0.00028202, 0.000293422, 0.000191959]
  gravity bias YZ [1.23925e-05, 0.00172226]
E3 [31.94, 60.00) s:
  mass x [1.07042, 0.980495, 0.984247, 1.02319]
  CoM XYZ mm [-0.26196, -4.59484, 2.13633]
  mu_s [0.616244, 0.727426, 0.640016, 0.71155]
  mu_d [0.586498, 0.606358, 0.513141, 0.71155]
  Kp [steer,rear] [9.82772, 0]; Kd [0.219327, 0]
  spawn roll -0.00767592 rad; rate -0.166608 rad/s
  gyro bias XYZ rad/s [8.75006e-06, -0.000253532, 0.000100475]
  gravity bias YZ [-0.0003994, 2.07428e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

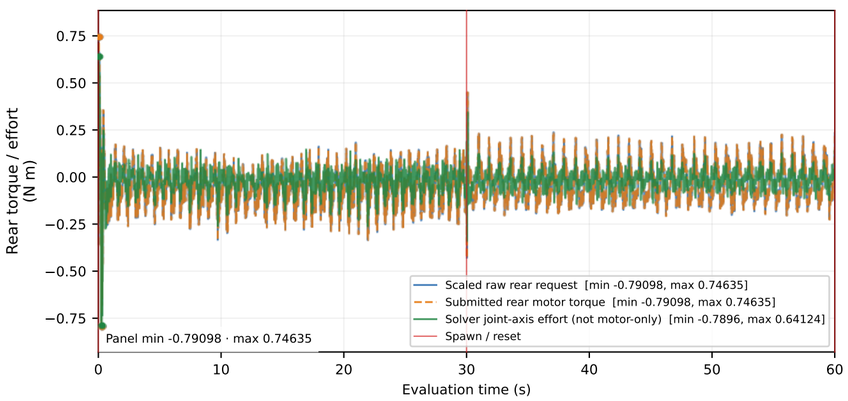
```
SAMPLED DR: env 6; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.95508, 1.03831, 1.00552, 1.0778]
  CoM XYZ mm [1.52497, 3.63117, 0.89843]
  mu_s [0.538434, 0.597748, 0.559972, 0.704749]
  mu_d [0.538434, 0.597748, 0.502209, 0.541926]
  Kp [steer,rear] [9.42108, 0]; Kd [0.183421, 0]
  spawn roll 0.0479951 rad; rate -0.0975409 rad/s
  gyro bias XYZ rad/s [-7.63979e-06, -0.000148874, 0.000268199]
  gravity bias YZ [0.0011309, 0.00172192]
E2 [30.00, 33.38) s:
  mass x [1.0515, 1.19247, 1.19836, 1.13004]
  CoM XYZ mm [4.57824, -4.74443, -1.52432]
  mu_s [0.559972, 0.631146, 0.757579, 0.732187]
  mu_d [0.502209, 0.631146, 0.70921, 0.631664]
  Kp [steer,rear] [9.23497, 0]; Kd [0.200202, 0]
  spawn roll 0.0695217 rad; rate -0.141358 rad/s
  gyro bias XYZ rad/s [-0.00028202, 0.000293422, 0.000191959]
  gravity bias YZ [1.23925e-05, 0.00172226]
E3 [33.38, 60.00) s:
  mass x [1.12701, 0.816186, 1.01265, 1.19488]
  CoM XYZ mm [-0.710368, 3.23515, 2.72403]
  mu_s [0.669571, 0.700521, 0.748289, 0.538976]
  mu_d [0.669571, 0.641329, 0.689499, 0.538976]
  Kp [steer,rear] [10.8693, 0]; Kd [0.190503, 0]
  spawn roll -0.0604354 rad; rate -0.128044 rad/s
  gyro bias XYZ rad/s [5.34909e-05, 0.000369712, 0.00025429]
  gravity bias YZ [0.00398337, 0.00196253]
Six significant digits shown; full precision: sampled_dr_values.json.
```

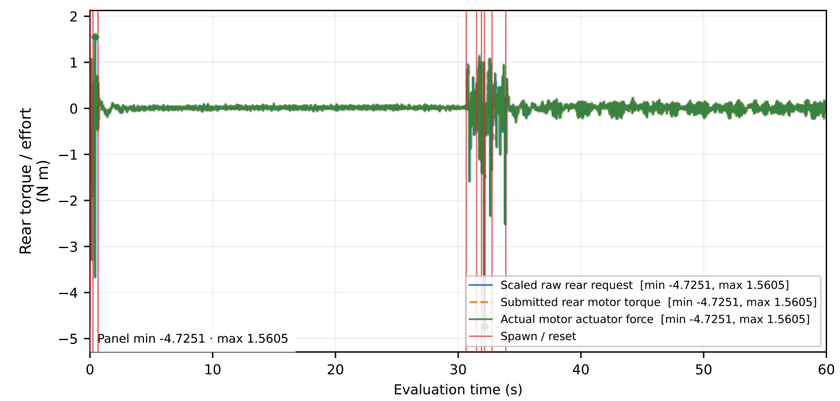
model_2500 · Demand versus actuation · env 7 · full 60 s

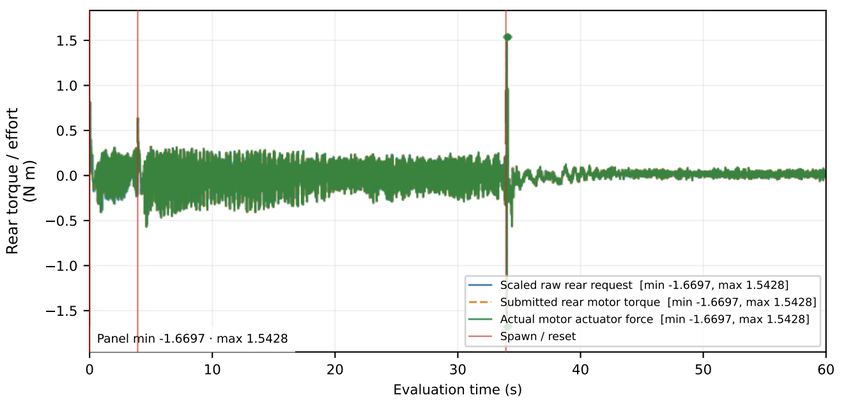












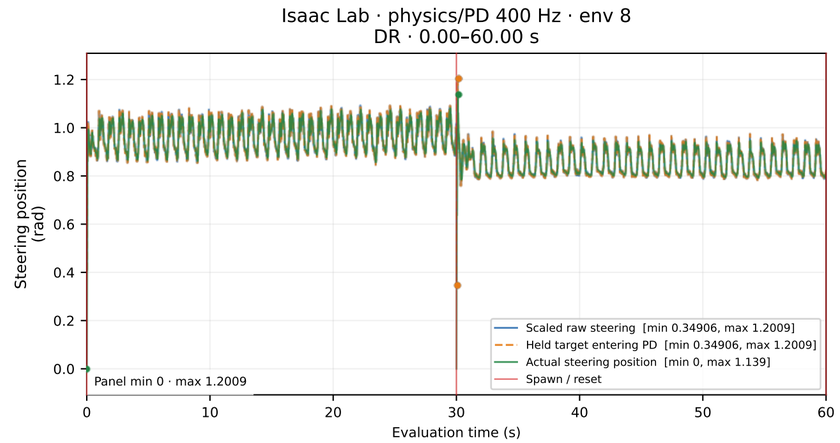
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

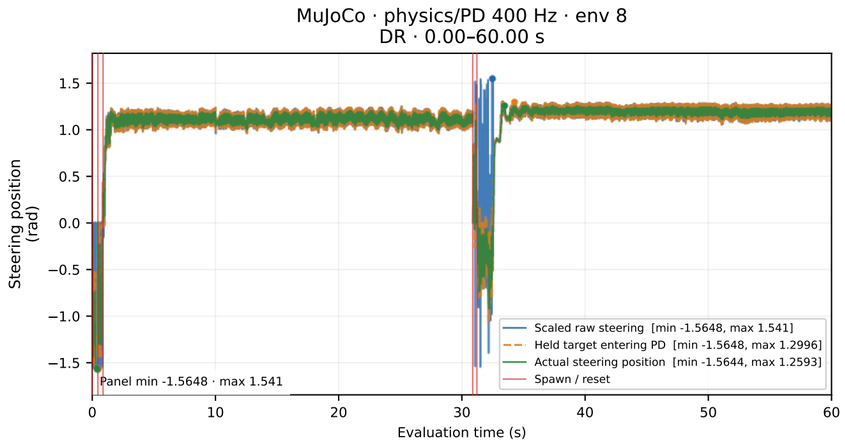
```
SAMPLED DR: env 7; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.17324, 0.952299, 0.82034, 0.906501]
  CoM XYZ mm [2.37444, -2.24125, 4.08924]
  mu_s [0.794338, 0.547743, 0.547743, 0.547743]
  mu_d [0.526209, 0.547743, 0.547743, 0.547743]
  Kp [steer,rear] [9.28848, 0]; Kd [0.203181, 0]
  spawn roll -0.093371 rad; rate 0.204639 rad/s
  gyro bias XYZ rad/s [0.000258929, 0.000124329, 6.37829e-05]
  gravity bias YZ [0.000248948, 0.00114776]
E2 [30.00, 60.00) s:
  mass x [1.12148, 1.1744, 1.19594, 1.05127]
  CoM XYZ mm [4.01582, 3.58083, -0.0416934]
  mu_s [0.681711, 0.760821, 0.647391, 0.681711]
  mu_d [0.611756, 0.670315, 0.647391, 0.611756]
  Kp [steer,rear] [10.4243, 0]; Kd [0.195552, 0]
  spawn roll 0.0374949 rad; rate 0.110923 rad/s
  gyro bias XYZ rad/s [-0.000182822, -9.52629e-05, 0.000164718]
  gravity bias YZ [0.00459708, -0.000579513]
Six significant digits shown; full precision: sampled_dr_values.json.
```

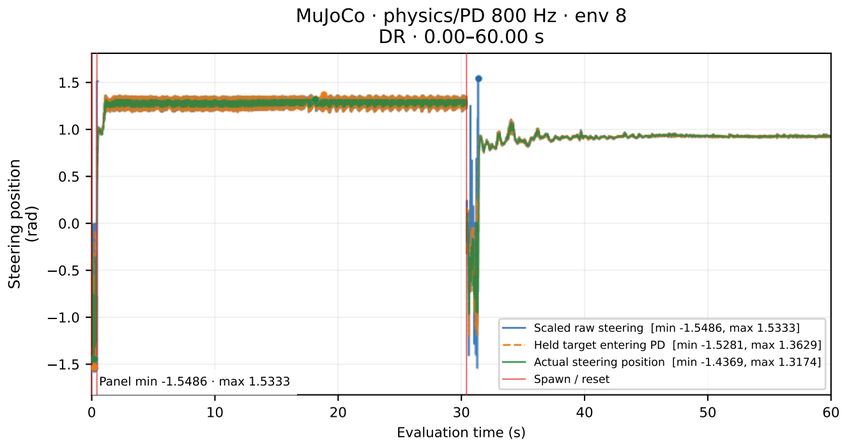
```
SAMPLED DR: env 7; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.24) s:
  mass x [0.945142, 1.03729, 0.95678, 1.04948]
  CoM XYZ mm [1.55815, -4.86414, 2.76336]
  mu_s [0.748289, 0.669571, 0.732187, 0.597748]
  mu_d [0.689499, 0.669571, 0.631664, 0.597748]
  Kp [steer,rear] [9.22333, 0]; Kd [0.205043, 0]
  spawn roll 0.0579807 rad; rate -0.230146 rad/s
  gyro bias XYZ rad/s [0.00021898, 0.000178088, 0.000100494]
  gravity bias YZ [-0.000466871, -0.000566274]
E2 [0.24, 0.66) s:
  mass x [0.815334, 0.846732, 1.04527, 0.935875]
  CoM XYZ mm [-0.394532, -2.53399, 4.82514]
  mu_s [0.727426, 0.704749, 0.541939, 0.541939]
  mu_d [0.606358, 0.541926, 0.534359, 0.534359]
  Kp [steer,rear] [9.38754, 0]; Kd [0.200427, 0]
  spawn roll 0.0472496 rad; rate 0.205943 rad/s
  gyro bias XYZ rad/s [-4.53926e-05, -0.000374891, 0.000302147]
  gravity bias YZ [-0.000488244, -0.000904653]
E3 [0.66, 30.66) s:
  mass x [1.05443, 1.07439, 1.18194, 1.1361]
  CoM XYZ mm [1.26006, 4.69445, 2.3054]
  mu_s [0.637675, 0.568073, 0.748289, 0.732187]
  mu_d [0.637675, 0.568073, 0.689499, 0.631664]
  Kp [steer,rear] [10.13, 0]; Kd [0.180181, 0]
  spawn roll 0.0459918 rad; rate -0.0558392 rad/s
  gyro bias XYZ rad/s [0.000193141, -0.000186143, 3.18235e-05]
  gravity bias YZ [-0.00166348, -0.000338645]
E4 [30.66, 31.50) s:
  mass x [1.01205, 1.11564, 1.17844, 1.01406]
  CoM XYZ mm [2.38007, 4.25924, 1.85643]
  mu_s [0.748289, 0.757579, 0.728342, 0.723429]
  mu_d [0.689499, 0.70921, 0.728342, 0.723429]
  Kp [steer,rear] [9.87057, 0]; Kd [0.191116, 0]
  spawn roll -0.0839131 rad; rate -0.0139192 rad/s
  gyro bias XYZ rad/s [9.68566e-05, 0.00027244, 5.12429e-05]
  gravity bias YZ [0.00121731, 0.000457613]
E5 [31.50, 31.90) s:
  mass x [0.844378, 1.19574, 0.87344, 1.11659]
  CoM XYZ mm [1.87176, 1.99566, -3.27129]
  mu_s [0.728342, 0.640016, 0.69316, 0.69316]
  mu_d [0.728342, 0.513141, 0.69316, 0.69316]
  Kp [steer,rear] [9.49388, 0]; Kd [0.188026, 0]
  spawn roll -0.0894517 rad; rate 0.133838 rad/s
  gyro bias XYZ rad/s [0.000129528, -0.000337925, 1.56767e-06]
  gravity bias YZ [-0.000239414, 0.00301222]
E6 [31.90, 32.14) s:
  mass x [0.967987, 1.12543, 1.13584, 0.841162]
  CoM XYZ mm [-0.510682, 0.130745, -2.72529]
  mu_s [0.546287, 0.559972, 0.71155, 0.727426]
  mu_d [0.546287, 0.502209, 0.71155, 0.606358]
  Kp [steer,rear] [9.57272, 0]; Kd [0.217513, 0]
  spawn roll -0.081501 rad; rate 0.0404057 rad/s
  gyro bias XYZ rad/s [4.89832e-05, -1.75448e-05, 3.42075e-05]
  gravity bias YZ [-0.000777028, -0.00227761]
E7 [32.14, 32.76) s:
  mass x [1.01407, 1.17122, 1.02528, 1.13515]
  CoM XYZ mm [-2.13715, 4.81493, -4.6917]
  mu_s [0.541939, 0.568073, 0.757579, 0.538434]
  mu_d [0.534359, 0.568073, 0.70921, 0.538434]
  Kp [steer,rear] [10.7901, 0]; Kd [0.202416, 0]
  spawn roll -0.0138355 rad; rate 0.236871 rad/s
  gyro bias XYZ rad/s [-0.000234231, -0.000145991, -0.000395346]
  gravity bias YZ [-0.00228167, -0.00111487]
E8 [32.76, 33.88) s:
  mass x [0.983052, 1.13748, 1.1842, 0.97888]
  CoM XYZ mm [-2.31369, 2.93858, 0.875108]
  mu_s [0.640016, 0.71008, 0.559972, 0.528253]
  mu_d [0.513141, 0.59371, 0.502209, 0.528253]
  Kp [steer,rear] [9.09864, 0]; Kd [0.206216, 0]
  spawn roll -0.0149822 rad; rate 0.0646525 rad/s
  gyro bias XYZ rad/s [0.000110128, -0.000243677, -0.000276047]
  gravity bias YZ [0.00222282, 0.000648969]
E9 [33.88, 60.00) s:
  mass x [1.15101, 1.0811, 1.0659, 1.14892]
  CoM XYZ mm [-4.29222, -1.57093, 1.5318]
  mu_s [0.631146, 0.704749, 0.528253, 0.633024]
  mu_d [0.631146, 0.541926, 0.528253, 0.568172]
  Kp [steer,rear] [10.9784, 0]; Kd [0.188898, 0]
  spawn roll 0.0314797 rad; rate -0.0384315 rad/s
  gyro bias XYZ rad/s [-0.000283127, -6.42477e-06, -0.000253066]
  gravity bias YZ [0.00249299, 0.00163119]
Six significant digits shown; full precision: sampled_dr_values.json.
```

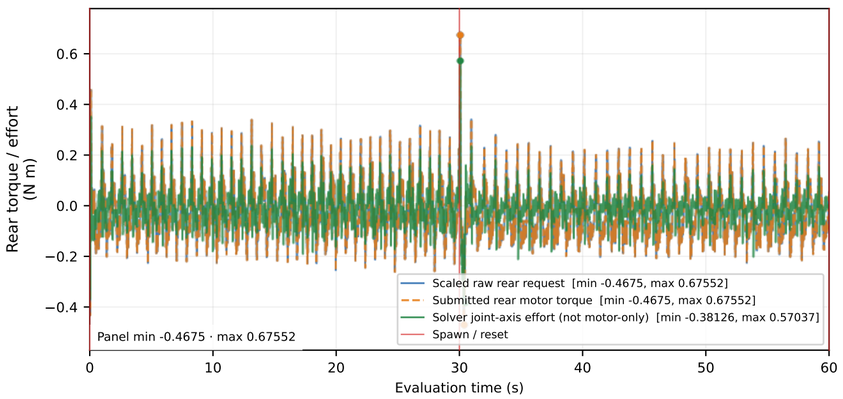
```
SAMPLED DR: env 7; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 3.92) s:
  mass x [0.945142, 1.03729, 0.95678, 1.04948]
  CoM XYZ mm [1.55815, -4.86414, 2.76336]
  mu_s [0.748289, 0.669571, 0.732187, 0.597748]
  mu_d [0.689499, 0.669571, 0.631664, 0.597748]
  Kp [steer,rear] [9.22333, 0]; Kd [0.205043, 0]
  spawn roll 0.0579807 rad; rate -0.230146 rad/s
  gyro bias XYZ rad/s [0.00021898, 0.000178088, 0.000100494]
  gravity bias YZ [-0.000466871, -0.000566274]
E2 [3.92, 33.92) s:
  mass x [1.19354, 0.806697, 1.18383, 1.05628]
  CoM XYZ mm [4.05146, -3.03807, 0.596929]
  mu_s [0.640016, 0.749678, 0.736077, 0.71155]
  mu_d [0.513141, 0.741429, 0.699455, 0.71155]
  Kp [steer,rear] [10.8219, 0]; Kd [0.206327, 0]
  spawn roll 0.0704726 rad; rate 0.0819431 rad/s
  gyro bias XYZ rad/s [-0.000146428, -2.8178e-05, 0.000313634]
  gravity bias YZ [0.00083648, -0.000763304]
E3 [33.92, 60.00) s:
  mass x [0.933955, 1.15257, 0.863114, 0.810173]
  CoM XYZ mm [-0.998784, 3.44166, 4.94422]
  mu_s [0.736077, 0.616244, 0.727426, 0.640016]
  mu_d [0.699455, 0.586498, 0.606358, 0.513141]
  Kp [steer,rear] [10.2999, 0]; Kd [0.21851, 0]
  spawn roll 0.087129 rad; rate 0.0232131 rad/s
  gyro bias XYZ rad/s [0.000176006, -0.00013264, 7.32382e-05]
  gravity bias YZ [-0.000695073, 0.00255973]
Six significant digits shown; full precision: sampled_dr_values.json.
```

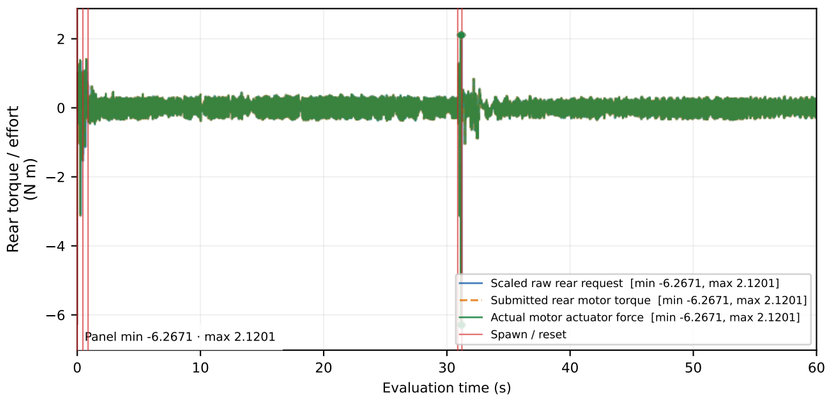
model_2500 · Demand versus actuation · env 8 · full 60 s

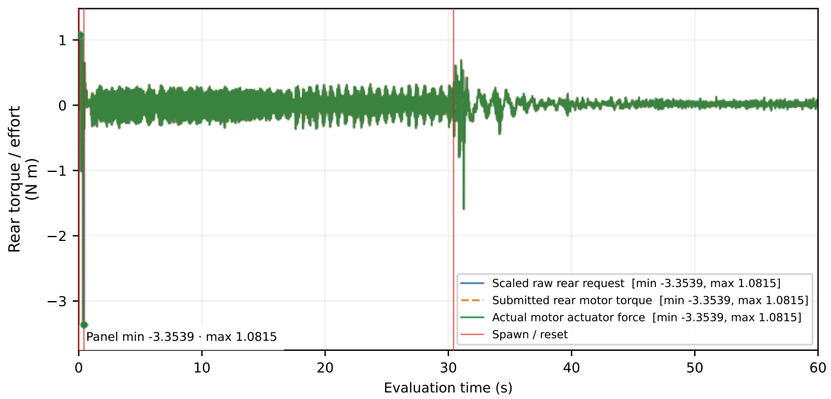












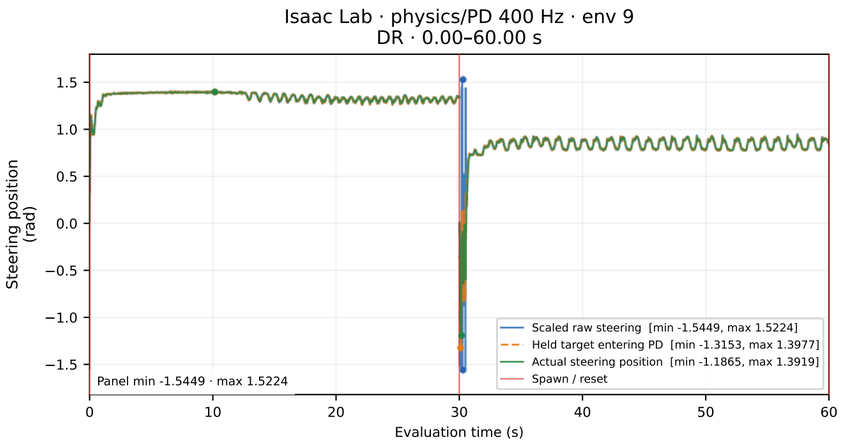
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

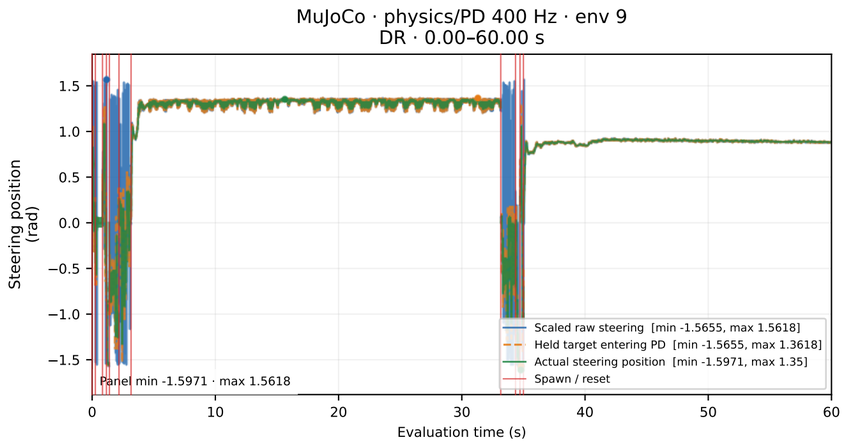
```
SAMPLED DR: env 8; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.05978, 0.906995, 0.869317, 1.14259]
  CoM XYZ mm [1.60898, 1.46585, -1.6959]
  mu_s [0.673378, 0.531594, 0.741039, 0.713129]
  mu_d [0.673378, 0.531594, 0.614405, 0.713129]
  Kp [steer,rear] [9.69437, 0]; Kd [0.203353, 0]
  spawn roll 0.0265545 rad; rate -0.000518277 rad/s
  gyro bias XYZ rad/s [0.000203254, -0.000297841, 0.000206812]
  gravity bias YZ [-0.00221992, 0.000585724]
E2 [30.00, 60.00) s:
  mass x [1.10196, 0.803254, 0.871098, 1.15485]
  CoM XYZ mm [2.80611, 3.00247, 0.461772]
  mu_s [0.533455, 0.751971, 0.741039, 0.576972]
  mu_d [0.533455, 0.541224, 0.614405, 0.576972]
  Kp [steer,rear] [9.27546, 0]; Kd [0.203981, 0]
  spawn roll 0.0704735 rad; rate -0.18156 rad/s
  gyro bias XYZ rad/s [-0.000119344, 0.0002309, 5.98212e-06]
  gravity bias YZ [-0.00216605, -0.00250907]
Six significant digits shown; full precision: sampled_dr_values.json.
```

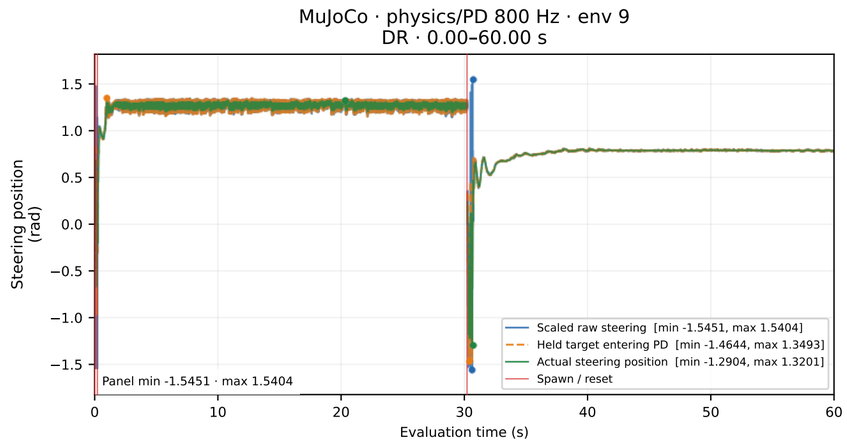
```
SAMPLED DR: env 8; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.46) s:
  mass x [1.11497, 1.13347, 1.01916, 1.18938]
  CoM XYZ mm [-2.63166, 1.46923, -4.34974]
  mu_s [0.700521, 0.640867, 0.732187, 0.733515]
  mu_d [0.641329, 0.556841, 0.631664, 0.558392]
  Kp [steer,rear] [9.53193, 0]; Kd [0.195893, 0]
  spawn roll -0.096984 rad; rate 0.0498493 rad/s
  gyro bias XYZ rad/s [3.95927e-07, -0.000539962, -0.000311445]
  gravity bias YZ [-0.000585585, 0.000518127]
E2 [0.46, 0.88) s:
  mass x [0.971118, 0.848327, 1.18653, 1.02509]
  CoM XYZ mm [-0.876627, 4.23962, 3.52719]
  mu_s [0.733515, 0.704749, 0.546287, 0.528253]
  mu_d [0.558392, 0.541926, 0.546287, 0.528253]
  Kp [steer,rear] [9.90465, 0]; Kd [0.198417, 0]
  spawn roll -0.0590784 rad; rate -0.224446 rad/s
  gyro bias XYZ rad/s [-5.93493e-05, 0.000112362, 0.000246449]
  gravity bias YZ [0.000639852, -0.000516005]
E3 [0.88, 30.88) s:
  mass x [0.860593, 0.937091, 1.18449, 0.958447]
  CoM XYZ mm [-0.61012, 0.384081, -4.92167]
  mu_s [0.611239, 0.71155, 0.757579, 0.538976]
  mu_d [0.611239, 0.71155, 0.70921, 0.538976]
  Kp [steer,rear] [10.827, 0]; Kd [0.191736, 0]
  spawn roll 0.0245912 rad; rate -0.132646 rad/s
  gyro bias XYZ rad/s [0.000117953, -9.70442e-06, 0.00021612]
  gravity bias YZ [-0.00107207, -0.00201265]
E4 [30.88, 31.22) s:
  mass x [1.10668, 1.04798, 0.924135, 1.16744]
  CoM XYZ mm [1.7946, -4.40723, -3.96957]
  mu_s [0.538976, 0.640867, 0.637675, 0.559972]
  mu_d [0.538976, 0.556841, 0.637675, 0.502209]
  Kp [steer,rear] [10.423, 0]; Kd [0.188921, 0]
  spawn roll 0.0447315 rad; rate -0.139914 rad/s
  gyro bias XYZ rad/s [0.000180307, -0.000155625, 6.8454e-05]
  gravity bias YZ [0.00154868, 0.000783916]
E5 [31.22, 60.00) s:
  mass x [0.939216, 0.816572, 0.856842, 1.18614]
  CoM XYZ mm [1.68749, -0.36036, 2.17776]
  mu_s [0.559972, 0.71155, 0.640867, 0.71155]
  mu_d [0.502209, 0.71155, 0.556841, 0.71155]
  Kp [steer,rear] [9.74387, 0]; Kd [0.19657, 0]
  spawn roll -0.0473665 rad; rate 0.161744 rad/s
  gyro bias XYZ rad/s [9.02701e-05, -0.000169255, 0.000344843]
  gravity bias YZ [0.00157456, 0.000492216]
Six significant digits shown; full precision: sampled_dr_values.json.
```

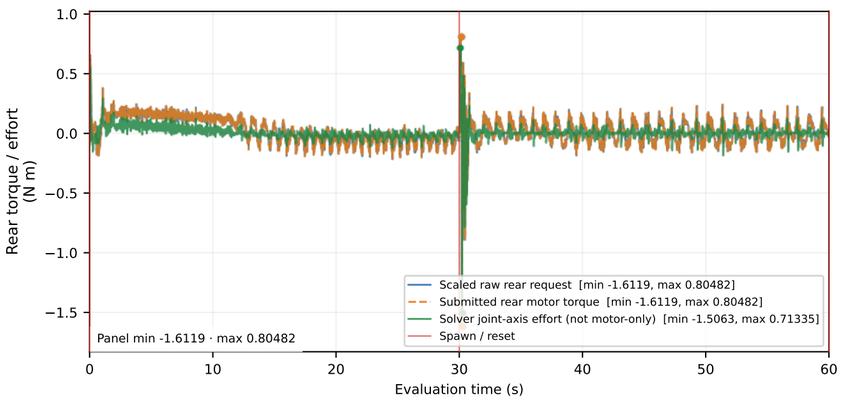
```
SAMPLED DR: env 8; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.42) s:
  mass x [1.11497, 1.13347, 1.01916, 1.18938]
  CoM XYZ mm [-2.63166, 1.46923, -4.34974]
  mu_s [0.700521, 0.640867, 0.732187, 0.733515]
  mu_d [0.641329, 0.556841, 0.631664, 0.558392]
  Kp [steer,rear] [9.53193, 0]; Kd [0.195893, 0]
  spawn roll -0.096984 rad; rate 0.0498493 rad/s
  gyro bias XYZ rad/s [3.95927e-07, -0.000539962, -0.000311445]
  gravity bias YZ [-0.000585585, 0.000518127]
E2 [0.42, 30.42) s:
  mass x [0.838724, 0.932716, 1.01168, 0.832511]
  CoM XYZ mm [-0.979841, -2.97601, -0.845064]
  mu_s [0.597748, 0.71008, 0.559972, 0.541939]
  mu_d [0.597748, 0.59371, 0.502209, 0.534359]
  Kp [steer,rear] [9.30986, 0]; Kd [0.208562, 0]
  spawn roll 0.0204349 rad; rate 0.232107 rad/s
  gyro bias XYZ rad/s [-0.000214265, -8.75294e-05, -0.00010339]
  gravity bias YZ [0.00236122, -0.000522518]
E3 [30.42, 60.00) s:
  mass x [1.19767, 1.13834, 0.819637, 1.1911]
  CoM XYZ mm [0.542863, 3.11743, -1.37944]
  mu_s [0.71008, 0.559972, 0.640016, 0.541939]
  mu_d [0.59371, 0.502209, 0.513141, 0.534359]
  Kp [steer,rear] [9.8392, 0]; Kd [0.199225, 0]
  spawn roll -0.00352456 rad; rate -0.203906 rad/s
  gyro bias XYZ rad/s [0.000280388, -0.000163418, -0.00011797]
  gravity bias YZ [0.00312605, 0.000531834]
Six significant digits shown; full precision: sampled_dr_values.json.
```

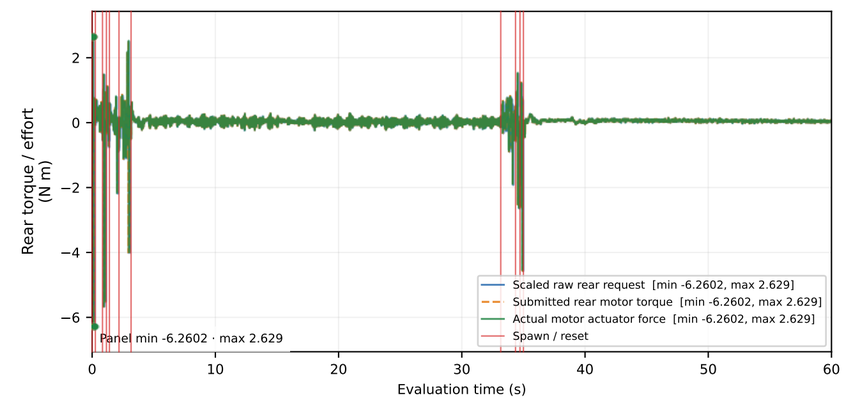
model_2500 · Demand versus actuation · env 9 · full 60 s

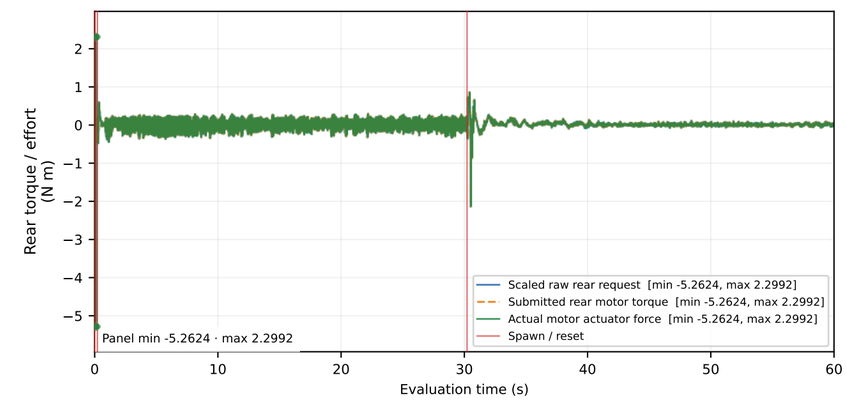












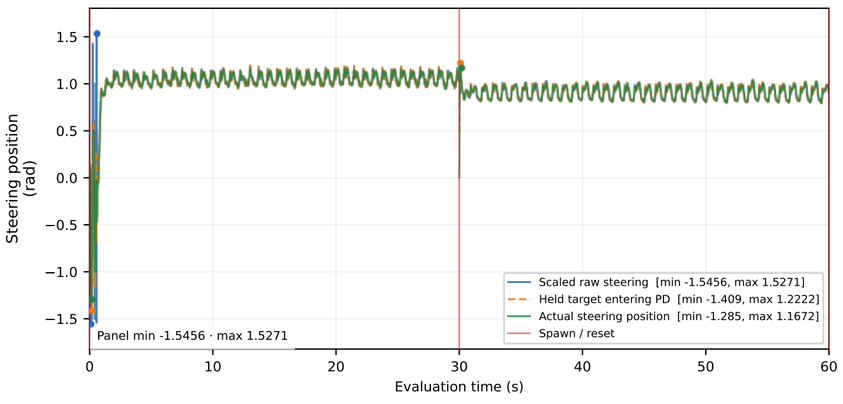
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

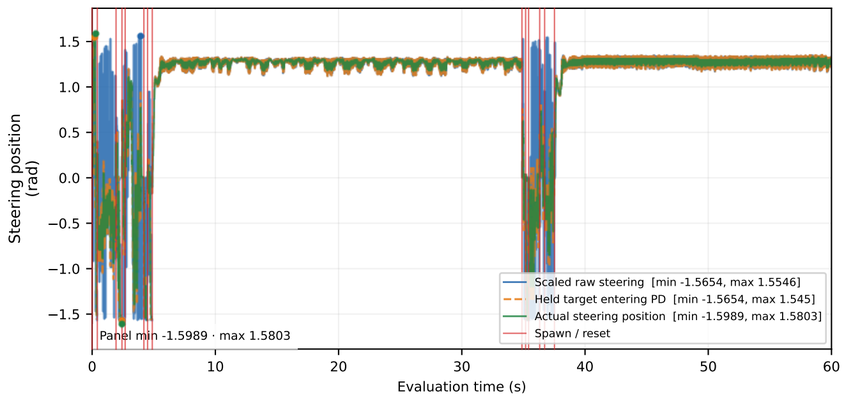
```
SAMPLED DR: env 9; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.910454, 0.883712, 0.82843, 1.10805]
  CoM XYZ mm [-4.77324, -3.57697, -4.34658]
  mu_s [0.673378, 0.764681, 0.794338, 0.579974]
  mu_d [0.673378, 0.764681, 0.526209, 0.579974]
  Kp [steer,rear] [10.9949, 0]; Kd [0.182996, 0]
  spawn roll 0.0113443 rad; rate 0.017793 rad/s
  gyro bias XYZ rad/s [5.89717e-05, -4.01532e-05, 0.000224197]
  gravity bias YZ [-0.00056605, -0.00068148]
E2 [30.00, 60.00) s:
  mass x [0.847115, 0.838555, 0.97014, 0.932349]
  CoM XYZ mm [-0.140801, 3.60915, -0.580743]
  mu_s [0.579974, 0.703711, 0.547743, 0.598343]
  mu_d [0.579974, 0.703711, 0.547743, 0.598343]
  Kp [steer,rear] [9.77015, 0]; Kd [0.193037, 0]
  spawn roll -0.0691044 rad; rate -0.0600907 rad/s
  gyro bias XYZ rad/s [-0.000139313, -0.000349101, -2.96777e-05]
  gravity bias YZ [-0.00147582, -0.00207203]
Six significant digits shown; full precision: sampled_dr_values.json.
```

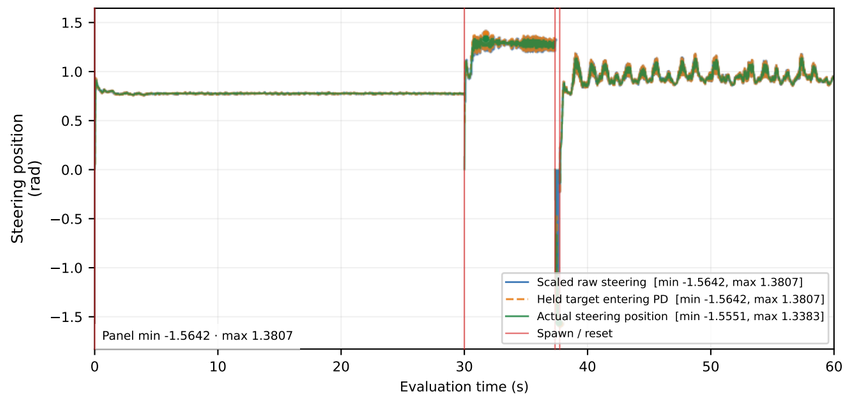
```
SAMPLED DR: env 9; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.26) s:
  mass x [1.17698, 0.922855, 0.901963, 0.889047]
  CoM XYZ mm [2.86204, -2.83292, 3.05727]
  mu_s [0.733515, 0.616244, 0.757579, 0.640016]
  mu_d [0.558392, 0.586498, 0.70921, 0.513141]
  Kp [steer,rear] [9.98367, 0]; Kd [0.211639, 0]
  spawn roll 0.0886575 rad; rate 0.0401016 rad/s
  gyro bias XYZ rad/s [0.000372606, -3.82795e-05, -0.000309129]
  gravity bias YZ [0.00301772, 0.000574019]
E2 [0.26, 0.84) s:
  mass x [1.10292, 0.874408, 1.18535, 0.849379]
  CoM XYZ mm [-0.623202, 0.282927, -2.85789]
  mu_s [0.727426, 0.791209, 0.631146, 0.541939]
  mu_d [0.606358, 0.767936, 0.631146, 0.534359]
  Kp [steer,rear] [10.3967, 0]; Kd [0.215948, 0]
  spawn roll 0.024244 rad; rate -0.0299922 rad/s
  gyro bias XYZ rad/s [4.50181e-05, -0.00023265, 0.000170851]
  gravity bias YZ [0.00123325, -0.00226057]
E3 [0.84, 1.16) s:
  mass x [0.827455, 0.877232, 1.1788, 0.84051]
  CoM XYZ mm [3.9484, -2.10291, -2.05155]
  mu_s [0.568073, 0.791209, 0.541939, 0.704749]
  mu_d [0.568073, 0.767936, 0.534359, 0.541926]
  Kp [steer,rear] [10.7543, 0]; Kd [0.193077, 0]
  spawn roll 0.0680143 rad; rate 0.0817176 rad/s
  gyro bias XYZ rad/s [0.000160368, 2.42672e-06, -0.000305339]
  gravity bias YZ [0.00106168, -0.000653744]
E4 [1.16, 1.40) s:
  mass x [0.950396, 1.01742, 1.18395, 0.902263]
  CoM XYZ mm [-0.61698, 1.3937, 3.67765]
  mu_s [0.749678, 0.700521, 0.528253, 0.616244]
  mu_d [0.741429, 0.641329, 0.528253, 0.586498]
  Kp [steer,rear] [9.00215, 0]; Kd [0.212399, 0]
  spawn roll -0.0832679 rad; rate -0.119692 rad/s
  gyro bias XYZ rad/s [-3.64903e-05, -0.000133754, 0.000118074]
  gravity bias YZ [-0.00157833, -0.00177584]
E5 [1.40, 2.18) s:
  mass x [0.941733, 0.945957, 1.02499, 1.11871]
  CoM XYZ mm [2.34728, 3.86433, -1.70276]
  mu_s [0.568073, 0.791209, 0.631146, 0.723429]
  mu_d [0.568073, 0.767936, 0.631146, 0.723429]
  Kp [steer,rear] [10.9334, 0]; Kd [0.202342, 0]
  spawn roll -0.0297394 rad; rate -0.157282 rad/s
  gyro bias XYZ rad/s [0.000188106, 4.7147e-05, 0.000266679]
  gravity bias YZ [0.00204219, 0.00243793]
E6 [2.18, 3.16) s:
  mass x [0.9028, 0.992604, 0.900615, 0.960603]
  CoM XYZ mm [4.75438, -3.6696, -1.14402]
  mu_s [0.640016, 0.728342, 0.640016, 0.616244]
  mu_d [0.513141, 0.728342, 0.513141, 0.586498]
  Kp [steer,rear] [9.41872, 0]; Kd [0.184475, 0]
  spawn roll -0.0649621 rad; rate 0.141163 rad/s
  gyro bias XYZ rad/s [-1.17404e-05, -0.000343479, -0.00021279]
  gravity bias YZ [-0.00139852, -0.00136185]
E7 [3.16, 33.16) s:
  mass x [0.866846, 1.15696, 1.07471, 0.961656]
  CoM XYZ mm [-3.05226, -3.24011, -4.0631]
  mu_s [0.568073, 0.538976, 0.597748, 0.597748]
  mu_d [0.568073, 0.538976, 0.597748, 0.597748]
  Kp [steer,rear] [9.76421, 0]; Kd [0.189835, 0]
  spawn roll 0.0191086 rad; rate 0.131062 rad/s
  gyro bias XYZ rad/s [2.35736e-05, 2.77649e-05, 0.00015313]
  gravity bias YZ [0.00096978, -0.0013636]
E8 [33.16, 34.36) s:
  mass x [1.13926, 1.08479, 0.942035, 0.904569]
  CoM XYZ mm [3.66064, 4.62531, 0.0420218]
  mu_s [0.71155, 0.727426, 0.633024, 0.669571]
  mu_d [0.71155, 0.606358, 0.568172, 0.669571]
  Kp [steer,rear] [10.7398, 0]; Kd [0.184163, 0]
  spawn roll -0.00481045 rad; rate -0.0372654 rad/s
  gyro bias XYZ rad/s [1.15582e-06, -5.80444e-06, 1.21536e-05]
  gravity bias YZ [-0.000603768, -0.00252194]
E9 [34.36, 34.72) s:
  mass x [1.01226, 1.04554, 1.06772, 0.916796]
  CoM XYZ mm [-4.21703, 2.61234, 3.9977]
  mu_s [0.559972, 0.538976, 0.71008, 0.700521]
  mu_d [0.502209, 0.538976, 0.59371, 0.641329]
  Kp [steer,rear] [9.31866, 0]; Kd [0.20457, 0]
  spawn roll -0.0553254 rad; rate -0.0673743 rad/s
  gyro bias XYZ rad/s [0.000101853, -0.000149137, -0.000188503]
  gravity bias YZ [0.00296788, -0.00027328]
E10 [34.72, 35.00) s:
  mass x [0.877664, 0.904515, 1.14778, 0.974366]
  CoM XYZ mm [-0.144504, -3.22227, 2.01091]
  mu_s [0.538434, 0.669571, 0.736077, 0.528253]
  mu_d [0.538434, 0.669571, 0.699455, 0.528253]
  Kp [steer,rear] [9.30457, 0]; Kd [0.21274, 0]
  spawn roll 0.0433025 rad; rate 0.101667 rad/s
  gyro bias XYZ rad/s [2.90936e-05, -4.72607e-05, -0.000181558]
  gravity bias YZ [-0.000545231, 0.000657348]
E11 [35.00, 60.00) s:
  mass x [1.06552, 0.836028, 1.09686, 0.820407]
  CoM XYZ mm [-4.98273, 2.85038, 1.54199]
  mu_s [0.71008, 0.568073, 0.631146, 0.704749]
  mu_d [0.59371, 0.568073, 0.631146, 0.541926]
  Kp [steer,rear] [9.78271, 0]; Kd [0.183908, 0]
  spawn roll 0.00645888 rad; rate -0.0611585 rad/s
  gyro bias XYZ rad/s [-5.6478e-05, 0.000114369, -0.000147739]
  gravity bias YZ [0.0010745, 0.000864926]
Six significant digits shown; full precision: sampled_dr_values.json.
```

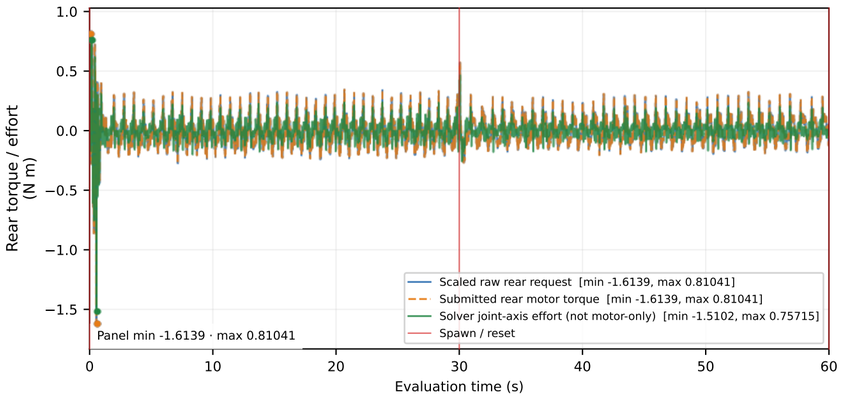
```
SAMPLED DR: env 9; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.22) s:
  mass x [1.17698, 0.922855, 0.901963, 0.889047]
  CoM XYZ mm [2.86204, -2.83292, 3.05727]
  mu_s [0.733515, 0.616244, 0.757579, 0.640016]
  mu_d [0.558392, 0.586498, 0.70921, 0.513141]
  Kp [steer,rear] [9.98367, 0]; Kd [0.211639, 0]
  spawn roll 0.0886575 rad; rate 0.0401016 rad/s
  gyro bias XYZ rad/s [0.000372606, -3.82795e-05, -0.000309129]
  gravity bias YZ [0.00301772, 0.000574019]
E2 [0.22, 30.22) s:
  mass x [1.14099, 0.900238, 0.955974, 1.11702]
  CoM XYZ mm [3.43091, -2.27089, 0.121555]
  mu_s [0.633024, 0.728342, 0.749678, 0.541939]
  mu_d [0.568172, 0.728342, 0.741429, 0.534359]
  Kp [steer,rear] [10.4489, 0]; Kd [0.190282, 0]
  spawn roll -0.000666787 rad; rate -0.218584 rad/s
  gyro bias XYZ rad/s [5.64262e-06, -0.000136915, 8.34335e-06]
  gravity bias YZ [-0.000745782, -0.000370611]
E3 [30.22, 60.00) s:
  mass x [0.843416, 0.994457, 1.07602, 1.15315]
  CoM XYZ mm [0.648931, 4.76041, 3.55979]
  mu_s [0.640016, 0.728342, 0.611239, 0.631146]
  mu_d [0.513141, 0.728342, 0.611239, 0.631146]
  Kp [steer,rear] [9.00674, 0]; Kd [0.206268, 0]
  spawn roll -0.0692728 rad; rate 0.183999 rad/s
  gyro bias XYZ rad/s [-0.000242482, -0.000138084, 1.07059e-05]
  gravity bias YZ [-0.000523423, -0.0036168]
Six significant digits shown; full precision: sampled_dr_values.json.
```

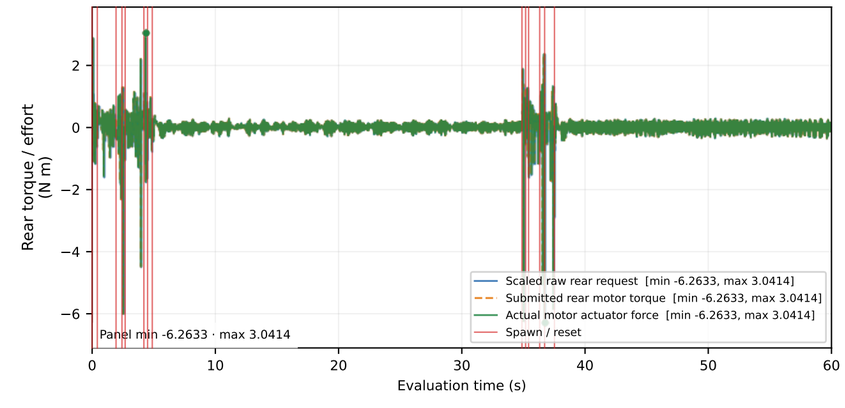
model_2500 · Demand versus actuation · env 10 · full 60 s

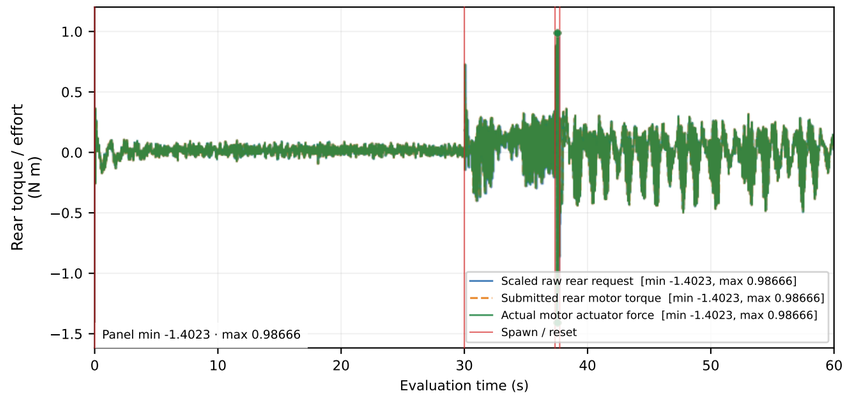












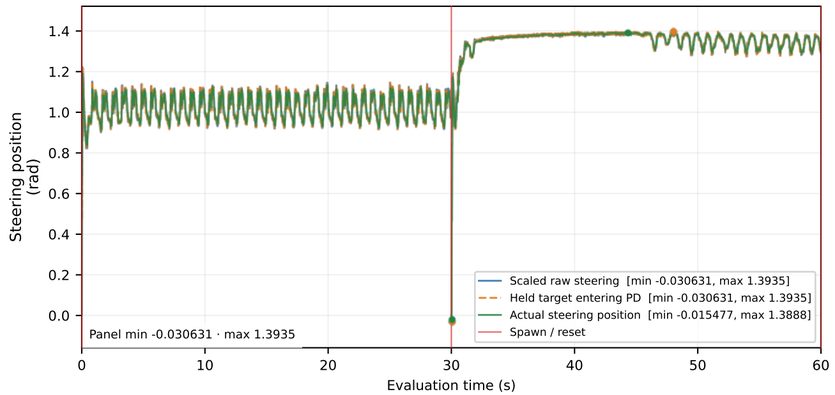
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

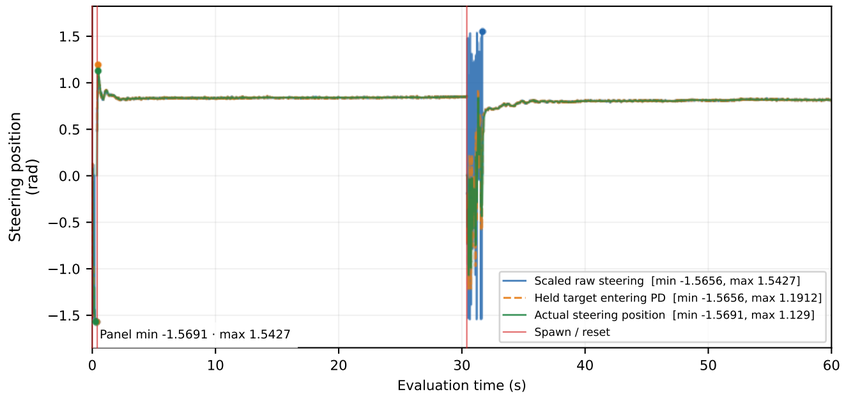
```
SAMPLED DR: env 10; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.044, 0.937751, 1.19408, 0.987669]
  CoM XYZ mm [-0.877663, 0.798646, 3.21598]
  mu_s [0.531594, 0.560535, 0.764681, 0.674963]
  mu_d [0.531594, 0.560535, 0.764681, 0.601294]
  Kp [steer,rear] [10.1063, 0]; Kd [0.216217, 0]
  spawn roll -0.0612238 rad; rate 0.0984062 rad/s
  gyro bias XYZ rad/s [0.00027079, 0.000114682, 0.000188098]
  gravity bias YZ [0.000185434, -0.000602592]
E2 [30.00, 60.00) s:
  mass x [0.897797, 0.967003, 0.894196, 1.05703]
  CoM XYZ mm [0.718489, 2.68479, -3.00688]
  mu_s [0.602694, 0.579974, 0.760821, 0.741039]
  mu_d [0.602694, 0.579974, 0.670315, 0.614405]
  Kp [steer,rear] [10.8715, 0]; Kd [0.202183, 0]
  spawn roll 0.0764369 rad; rate 0.0505117 rad/s
  gyro bias XYZ rad/s [0.000213677, 0.00031753, 0.000137284]
  gravity bias YZ [0.000929145, -0.00170759]
Six significant digits shown; full precision: sampled_dr_values.json.
```

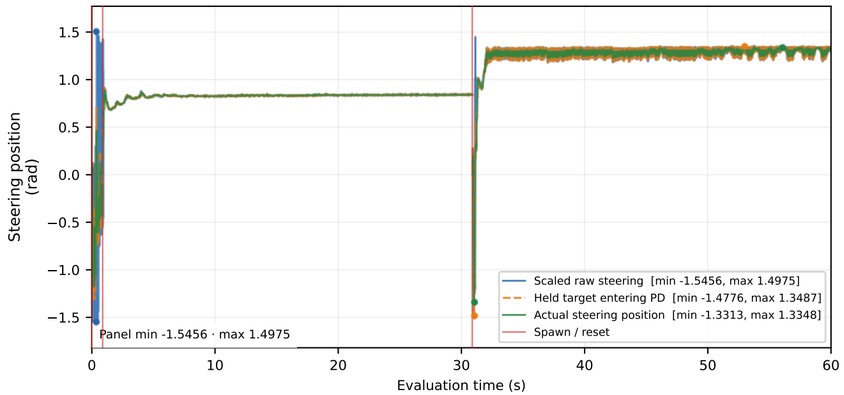
```
SAMPLED DR: env 10; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.42) s:
  mass x [1.04631, 1.02969, 1.00782, 0.802556]
  CoM XYZ mm [0.502924, 4.68338, -1.0194]
  mu_s [0.728342, 0.732187, 0.666375, 0.633024]
  mu_d [0.728342, 0.631664, 0.519145, 0.568172]
  Kp [steer,rear] [10.8929, 0]; Kd [0.198951, 0]
  spawn roll 0.0530206 rad; rate -0.213743 rad/s
  gyro bias XYZ rad/s [-0.000326526, 0.000265956, -0.000136196]
  gravity bias YZ [-0.00207519, -0.000464771]
E2 [0.42, 1.94) s:
  mass x [1.18486, 1.18667, 0.997063, 1.10972]
  CoM XYZ mm [3.44273, 1.65362, -4.23642]
  mu_s [0.631146, 0.69316, 0.546287, 0.69316]
  mu_d [0.631146, 0.69316, 0.546287, 0.69316]
  Kp [steer,rear] [10.2304, 0]; Kd [0.197478, 0]
  spawn roll -0.0587105 rad; rate 0.244661 rad/s
  gyro bias XYZ rad/s [0.000119483, 4.30869e-05, -8.05376e-05]
  gravity bias YZ [-0.00049118, -0.00103959]
E3 [1.94, 2.42) s:
  mass x [0.835614, 0.85745, 0.817507, 0.841165]
  CoM XYZ mm [-2.8191, 2.38297, -2.78244]
  mu_s [0.640016, 0.546287, 0.538434, 0.723429]
  mu_d [0.513141, 0.546287, 0.538434, 0.723429]
  Kp [steer,rear] [10.912, 0]; Kd [0.180721, 0]
  spawn roll 0.00295433 rad; rate 0.140944 rad/s
  gyro bias XYZ rad/s [-2.16887e-05, -5.15281e-05, -0.000142702]
  gravity bias YZ [-0.00417387, 0.0026808]
E4 [2.42, 2.68) s:
  mass x [1.05905, 1.03787, 1.00903, 1.07727]
  CoM XYZ mm [3.46896, -2.57296, 1.83129]
  mu_s [0.700521, 0.704749, 0.791209, 0.559972]
  mu_d [0.641329, 0.541926, 0.767936, 0.502209]
  Kp [steer,rear] [9.35022, 0]; Kd [0.215744, 0]
  spawn roll 0.0582089 rad; rate 0.0912731 rad/s
  gyro bias XYZ rad/s [0.000100303, 5.99272e-05, 0.000117773]
  gravity bias YZ [-0.00461448, 0.000453434]
E5 [2.68, 4.20) s:
  mass x [1.04755, 0.814188, 0.957148, 0.84013]
  CoM XYZ mm [0.49032, -1.96357, -4.73473]
  mu_s [0.791209, 0.71008, 0.616244, 0.538976]
  mu_d [0.767936, 0.59371, 0.586498, 0.538976]
  Kp [steer,rear] [10.3746, 0]; Kd [0.184485, 0]
  spawn roll 0.0022097 rad; rate -0.0359523 rad/s
  gyro bias XYZ rad/s [-3.85463e-05, -0.000203228, -0.000210508]
  gravity bias YZ [0.000717845, 0.00190448]
E6 [4.20, 4.50) s:
  mass x [0.970285, 1.00658, 1.07616, 1.04332]
  CoM XYZ mm [3.39187, -0.295783, 2.0516]
  mu_s [0.723429, 0.69316, 0.791209, 0.727426]
  mu_d [0.723429, 0.69316, 0.767936, 0.606358]
  Kp [steer,rear] [10.9528, 0]; Kd [0.205301, 0]
  spawn roll -0.0915221 rad; rate 0.0388739 rad/s
  gyro bias XYZ rad/s [-0.000200333, 5.84175e-05, 0.000113706]
  gravity bias YZ [-0.00230579, 0.00277618]
E7 [4.50, 4.88) s:
  mass x [0.913092, 1.11761, 0.903278, 0.945502]
  CoM XYZ mm [0.913718, -0.601137, 0.00265632]
  mu_s [0.616244, 0.631146, 0.666375, 0.732187]
  mu_d [0.586498, 0.631146, 0.519145, 0.631664]
  Kp [steer,rear] [9.18786, 0]; Kd [0.20778, 0]
  spawn roll -0.0728771 rad; rate 0.153196 rad/s
  gyro bias XYZ rad/s [6.28892e-06, -0.000165498, -4.58954e-05]
  gravity bias YZ [-0.00158328, -0.00160853]
E8 [4.88, 34.88) s:
  mass x [1.0694, 1.11081, 1.11693, 0.99577]
  CoM XYZ mm [-0.171865, -2.0385, 2.7854]
  mu_s [0.611239, 0.616244, 0.700521, 0.69316]
  mu_d [0.611239, 0.586498, 0.641329, 0.69316]
  Kp [steer,rear] [9.02021, 0]; Kd [0.205425, 0]
  spawn roll -0.00931612 rad; rate -0.125238 rad/s
  gyro bias XYZ rad/s [-5.01276e-05, 0.000197057, -0.000459944]
  gravity bias YZ [0.000572227, 0.00103727]
E9 [34.88, 35.18) s:
  mass x [0.86387, 0.992903, 1.07255, 0.921582]
  CoM XYZ mm [0.823676, 0.0336956, -0.0370925]
  mu_s [0.611239, 0.727426, 0.546287, 0.732187]
  mu_d [0.611239, 0.606358, 0.546287, 0.631664]
  Kp [steer,rear] [10.5863, 0]; Kd [0.205938, 0]
  spawn roll 0.0783838 rad; rate 0.085219 rad/s
  gyro bias XYZ rad/s [-5.24537e-05, -0.000353689, -0.000179775]
  gravity bias YZ [-6.7528e-06, 0.00070289]
E10 [35.18, 35.42) s:
  mass x [1.05469, 1.14715, 0.863331, 1.16766]
  CoM XYZ mm [3.79462, 4.11356, -4.36518]
  mu_s [0.69316, 0.749678, 0.541939, 0.733515]
  mu_d [0.69316, 0.741429, 0.534359, 0.558392]
  Kp [steer,rear] [10.509, 0]; Kd [0.186977, 0]
  spawn roll -0.0988226 rad; rate -0.0518785 rad/s
  gyro bias XYZ rad/s [-0.000270208, -0.000125189, 0.000176881]
  gravity bias YZ [0.000225012, 0.000489124]
E11 [35.42, 36.32) s:
  mass x [1.01652, 0.80575, 1.14129, 1.14338]
  CoM XYZ mm [-2.64045, 4.91107, -4.94833]
  mu_s [0.568073, 0.749678, 0.640867, 0.791209]
  mu_d [0.568073, 0.741429, 0.556841, 0.767936]
  Kp [steer,rear] [9.3717, 0]; Kd [0.185427, 0]
  spawn roll -0.0492544 rad; rate 0.0477335 rad/s
  gyro bias XYZ rad/s [0.000301953, 4.79395e-05, -0.000328948]
  gravity bias YZ [0.000363512, -0.00270865]
E12 [36.32, 36.72) s:
  mass x [1.00043, 0.922407, 1.14794, 0.965364]
  CoM XYZ mm [3.75685, 1.66316, 0.163766]
  mu_s [0.704749, 0.728342, 0.727426, 0.611239]
  mu_d [0.541926, 0.728342, 0.606358, 0.611239]
  Kp [steer,rear] [9.20986, 0]; Kd [0.201713, 0]
  spawn roll 0.0762615 rad; rate 0.0145723 rad/s
  gyro bias XYZ rad/s [-0.000218343, 4.74383e-05, 0.000106282]
  gravity bias YZ [0.0017686, -0.00215085]
E13 [36.72, 37.52) s:
  mass x [0.927922, 1.08205, 1.0375, 0.914039]
  CoM XYZ mm [3.29695, 2.21762, 3.04409]
  mu_s [0.704749, 0.538976, 0.669571, 0.637675]
  mu_d [0.541926, 0.538976, 0.669571, 0.637675]
  Kp [steer,rear] [10.5304, 0]; Kd [0.191862, 0]
  spawn roll -0.0204232 rad; rate -0.0546812 rad/s
  gyro bias XYZ rad/s [0.00014849, -0.00015362, 0.000158029]
  gravity bias YZ [0.000199321, -0.00144308]
E14 [37.52, 60.00) s:
  mass x [1.13732, 0.976452, 0.83567, 1.01326]
  CoM XYZ mm [2.88701, -2.56581, -1.69153]
  mu_s [0.597748, 0.71008, 0.640016, 0.640016]
  mu_d [0.597748, 0.59371, 0.513141, 0.513141]
  Kp [steer,rear] [9.82295, 0]; Kd [0.185461, 0]
  spawn roll 0.0371648 rad; rate 0.0142472 rad/s
  gyro bias XYZ rad/s [0.000117595, -0.000168673, -0.000143787]
  gravity bias YZ [0.000274915, 0.00167859]
Six significant digits shown; full precision: sampled_dr_values.json.
```

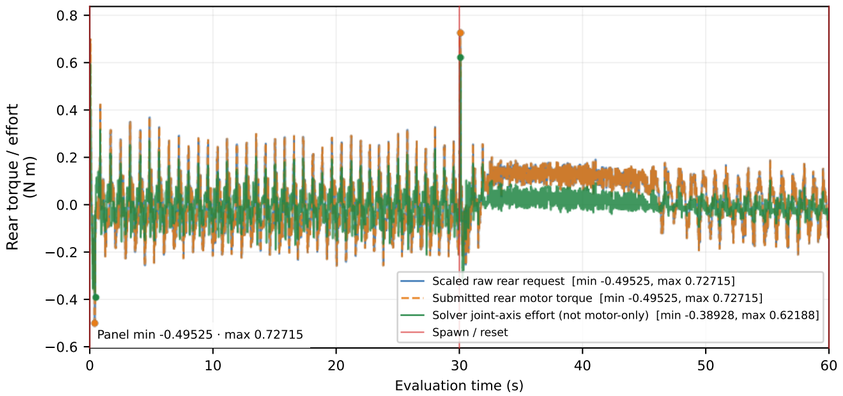
```
SAMPLED DR: env 10; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.04631, 1.02969, 1.00782, 0.802556]
  CoM XYZ mm [0.502924, 4.68338, -1.0194]
  mu_s [0.728342, 0.732187, 0.666375, 0.633024]
  mu_d [0.728342, 0.631664, 0.519145, 0.568172]
  Kp [steer,rear] [10.8929, 0]; Kd [0.198951, 0]
  spawn roll 0.0530206 rad; rate -0.213743 rad/s
  gyro bias XYZ rad/s [-0.000326526, 0.000265956, -0.000136196]
  gravity bias YZ [-0.00207519, -0.000464771]
E2 [30.00, 37.36) s:
  mass x [0.809005, 1.1514, 0.91927, 1.0749]
  CoM XYZ mm [1.54606, -4.9849, -1.7566]
  mu_s [0.631146, 0.727426, 0.538434, 0.611239]
  mu_d [0.631146, 0.606358, 0.538434, 0.611239]
  Kp [steer,rear] [10.5435, 0]; Kd [0.180406, 0]
  spawn roll 0.0331038 rad; rate 0.133716 rad/s
  gyro bias XYZ rad/s [-0.000131766, 0.000104947, -1.18163e-05]
  gravity bias YZ [-0.00112885, 0.000529576]
E3 [37.36, 37.74) s:
  mass x [0.856208, 1.1043, 1.09897, 1.1517]
  CoM XYZ mm [-4.89209, 2.17899, -4.65836]
  mu_s [0.723429, 0.538434, 0.559972, 0.727426]
  mu_d [0.723429, 0.538434, 0.502209, 0.606358]
  Kp [steer,rear] [10.1784, 0]; Kd [0.186521, 0]
  spawn roll -0.0702924 rad; rate -0.141829 rad/s
  gyro bias XYZ rad/s [0.000132492, 2.44563e-05, -1.87751e-06]
  gravity bias YZ [0.00521048, -0.000803314]
E4 [37.74, 60.00) s:
  mass x [1.04979, 1.16416, 1.143, 0.870995]
  CoM XYZ mm [1.27314, 2.53976, -0.629694]
  mu_s [0.546287, 0.757579, 0.640867, 0.757579]
  mu_d [0.546287, 0.70921, 0.556841, 0.70921]
  Kp [steer,rear] [10.2702, 0]; Kd [0.21008, 0]
  spawn roll 0.004211 rad; rate -0.181404 rad/s
  gyro bias XYZ rad/s [-0.000119997, -0.000341104, -0.000298524]
  gravity bias YZ [0.00194827, -0.0028701]
Six significant digits shown; full precision: sampled_dr_values.json.
```

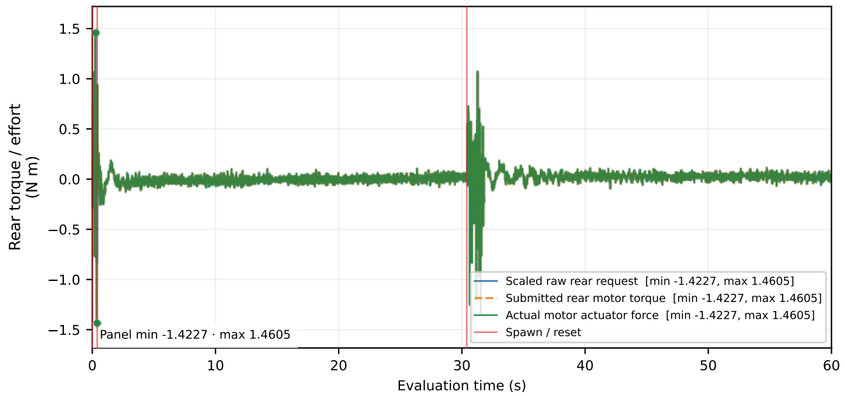
model_2500 · Demand versus actuation · env 11 · full 60 s

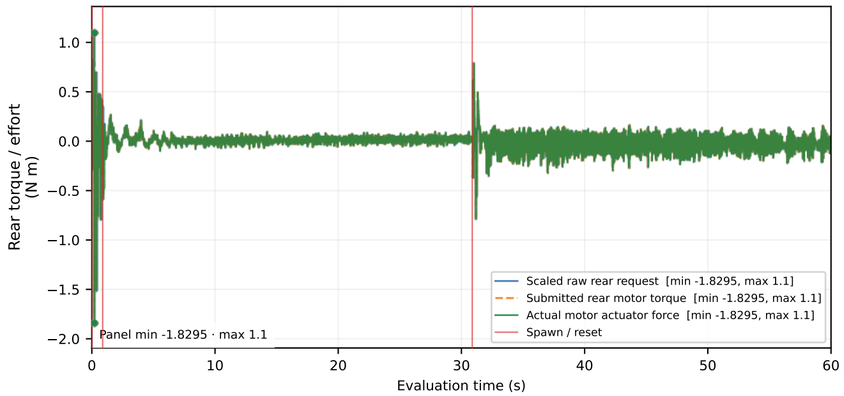












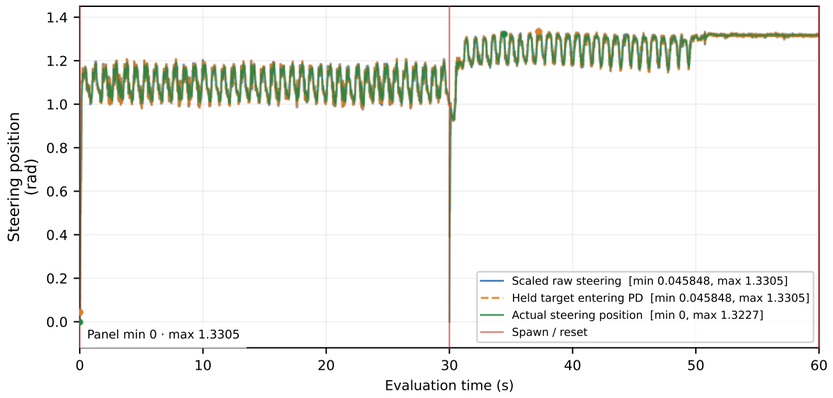
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

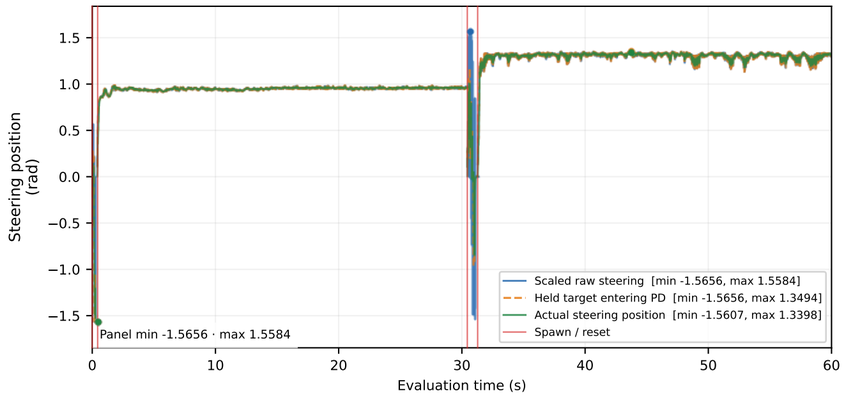
```
SAMPLED DR: env 11; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.16814, 1.08509, 0.842065, 1.12553]
  CoM XYZ mm [1.97017, 0.771385, -4.30979]
  mu_s [0.632409, 0.560535, 0.681711, 0.785466]
  mu_d [0.589076, 0.560535, 0.611756, 0.52258]
  Kp [steer,rear] [9.49828, 0]; Kd [0.189078, 0]
  spawn roll 0.0684242 rad; rate 0.00828874 rad/s
  gyro bias XYZ rad/s [0.000457135, 0.0003747, -0.00031351]
  gravity bias YZ [-0.00394906, 0.00108566]
E2 [30.00, 60.00) s:
  mass x [0.961653, 0.90604, 0.803862, 0.836442]
  CoM XYZ mm [-3.44978, -3.43411, -1.61833]
  mu_s [0.598343, 0.760821, 0.533455, 0.774409]
  mu_d [0.598343, 0.670315, 0.533455, 0.561095]
  Kp [steer,rear] [10.3132, 0]; Kd [0.201799, 0]
  spawn roll 0.0466542 rad; rate -0.224283 rad/s
  gyro bias XYZ rad/s [-6.36372e-05, -0.000145016, 5.87831e-05]
  gravity bias YZ [0.00127278, -0.00158957]
Six significant digits shown; full precision: sampled_dr_values.json.
```

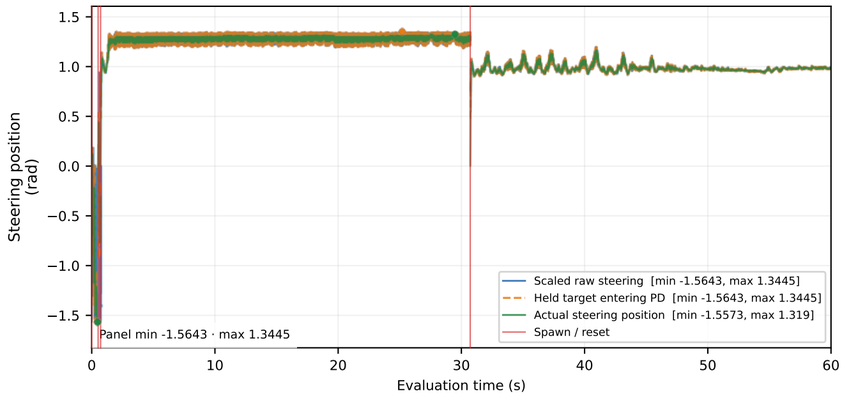
```
SAMPLED DR: env 11; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.40) s:
  mass x [0.804451, 1.08756, 0.932451, 1.17324]
  CoM XYZ mm [-3.95175, 1.66437, 3.19254]
  mu_s [0.69316, 0.749678, 0.616244, 0.669571]
  mu_d [0.69316, 0.741429, 0.586498, 0.669571]
  Kp [steer,rear] [10.5376, 0]; Kd [0.192053, 0]
  spawn roll -0.0839723 rad; rate 0.152834 rad/s
  gyro bias XYZ rad/s [1.32216e-05, -0.000139725, -3.55158e-05]
  gravity bias YZ [0.000531016, -0.00231086]
E2 [0.40, 30.40) s:
  mass x [1.18685, 1.01724, 1.15557, 1.14079]
  CoM XYZ mm [1.51405, 3.86921, -1.64199]
  mu_s [0.546287, 0.631146, 0.611239, 0.727426]
  mu_d [0.546287, 0.631146, 0.611239, 0.606358]
  Kp [steer,rear] [10.0285, 0]; Kd [0.186463, 0]
  spawn roll 0.0861315 rad; rate -0.157324 rad/s
  gyro bias XYZ rad/s [1.12137e-05, -0.000358171, 7.92227e-05]
  gravity bias YZ [-0.000299793, -0.00456061]
E3 [30.40, 60.00) s:
  mass x [0.939248, 0.918164, 1.03497, 0.992503]
  CoM XYZ mm [-3.41698, 3.79475, -2.0653]
  mu_s [0.71155, 0.727426, 0.727426, 0.727426]
  mu_d [0.71155, 0.606358, 0.606358, 0.606358]
  Kp [steer,rear] [9.41325, 0]; Kd [0.201469, 0]
  spawn roll -0.0253665 rad; rate -0.0987004 rad/s
  gyro bias XYZ rad/s [-0.000147457, -7.24679e-05, 0.000179976]
  gravity bias YZ [-0.00232873, -0.000743975]
Six significant digits shown; full precision: sampled_dr_values.json.
```

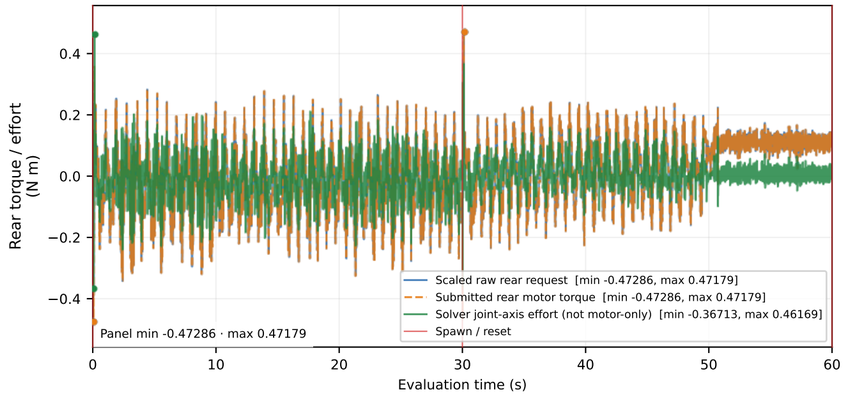
```
SAMPLED DR: env 11; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.88) s:
  mass x [0.804451, 1.08756, 0.932451, 1.17324]
  CoM XYZ mm [-3.95175, 1.66437, 3.19254]
  mu_s [0.69316, 0.749678, 0.616244, 0.669571]
  mu_d [0.69316, 0.741429, 0.586498, 0.669571]
  Kp [steer,rear] [10.5376, 0]; Kd [0.192053, 0]
  spawn roll -0.0839723 rad; rate 0.152834 rad/s
  gyro bias XYZ rad/s [1.32216e-05, -0.000139725, -3.55158e-05]
  gravity bias YZ [0.000531016, -0.00231086]
E2 [0.88, 30.88) s:
  mass x [0.976272, 0.885468, 0.968607, 1.10736]
  CoM XYZ mm [0.554094, 4.28003, 4.62543]
  mu_s [0.749678, 0.568073, 0.700521, 0.538976]
  mu_d [0.741429, 0.568073, 0.641329, 0.538976]
  Kp [steer,rear] [10.8778, 0]; Kd [0.195074, 0]
  spawn roll 0.00723343 rad; rate 0.204298 rad/s
  gyro bias XYZ rad/s [0.000249439, 4.50987e-05, -7.61758e-05]
  gravity bias YZ [0.00256586, -0.00277735]
E3 [30.88, 60.00) s:
  mass x [1.06718, 1.15346, 0.838004, 0.862932]
  CoM XYZ mm [-2.41201, -3.27271, 4.92665]
  mu_s [0.568073, 0.757579, 0.71155, 0.538434]
  mu_d [0.568073, 0.70921, 0.71155, 0.538434]
  Kp [steer,rear] [9.21777, 0]; Kd [0.192868, 0]
  spawn roll -0.0825202 rad; rate 0.213405 rad/s
  gyro bias XYZ rad/s [4.13111e-05, 0.000199249, -0.00012153]
  gravity bias YZ [0.00149148, 0.00141227]
Six significant digits shown; full precision: sampled_dr_values.json.
```

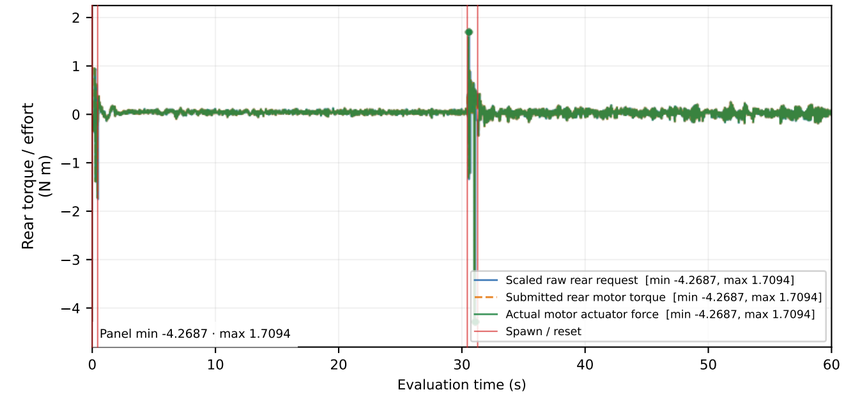
model_2500 · Demand versus actuation · env 12 · full 60 s

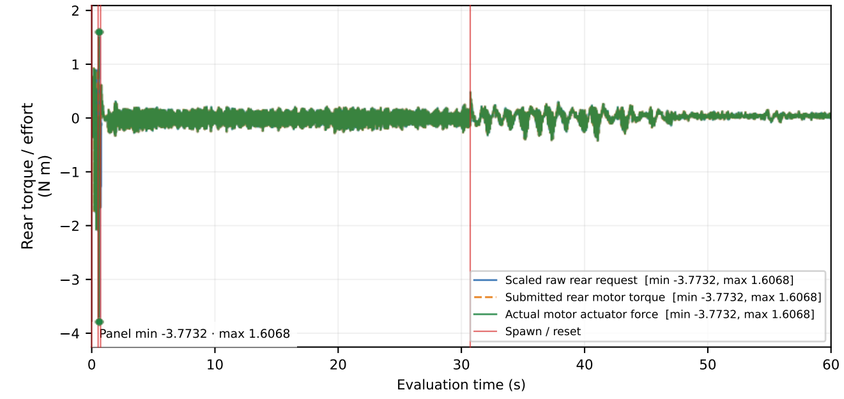












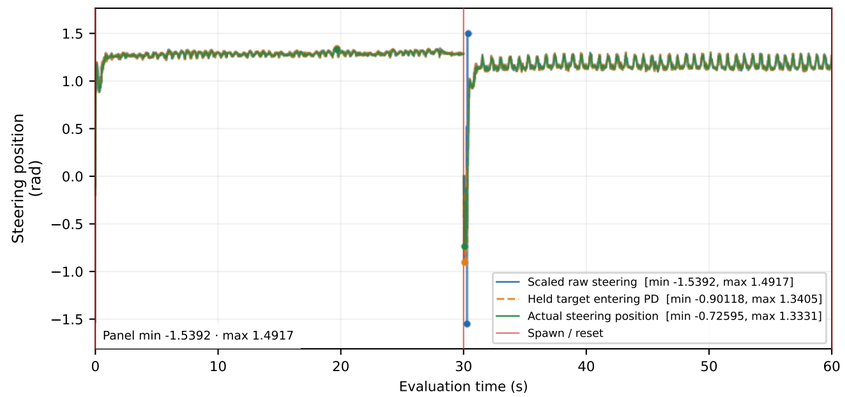
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

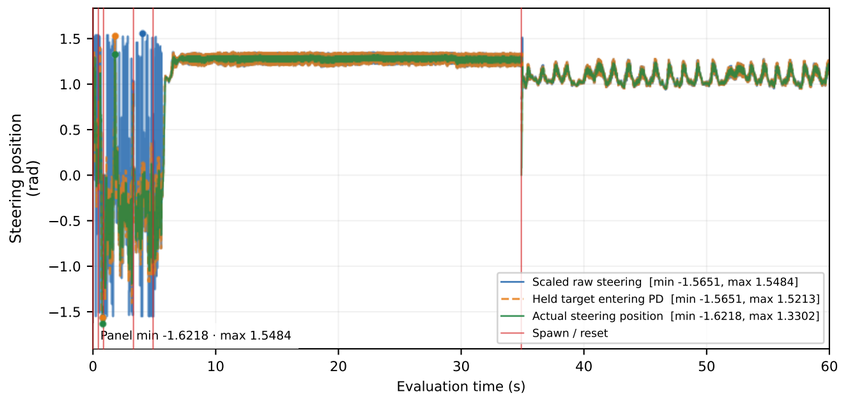
```
SAMPLED DR: env 12; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.843677, 1.04398, 0.870024, 1.13626]
  CoM XYZ mm [2.36219, 0.202864, 1.9197]
  mu_s [0.693205, 0.726175, 0.632409, 0.576972]
  mu_d [0.693205, 0.558574, 0.589076, 0.576972]
  Kp [steer,rear] [10.8603, 0]; Kd [0.20193, 0]
  spawn roll -0.00308955 rad; rate -0.0558987 rad/s
  gyro bias XYZ rad/s [1.09179e-05, -0.00014741, 7.55158e-05]
  gravity bias YZ [-0.00297523, 0.00145696]
E2 [30.00, 60.00) s:
  mass x [0.844162, 0.909308, 0.829776, 1.01903]
  CoM XYZ mm [-2.25903, -1.72129, 2.47356]
  mu_s [0.787792, 0.533455, 0.703711, 0.659445]
  mu_d [0.617134, 0.533455, 0.703711, 0.547619]
  Kp [steer,rear] [10.3056, 0]; Kd [0.210959, 0]
  spawn roll 0.00288594 rad; rate 0.0706386 rad/s
  gyro bias XYZ rad/s [0.000319707, 0.000310775, -0.000162301]
  gravity bias YZ [0.000377707, -0.00287623]
Six significant digits shown; full precision: sampled_dr_values.json.
```

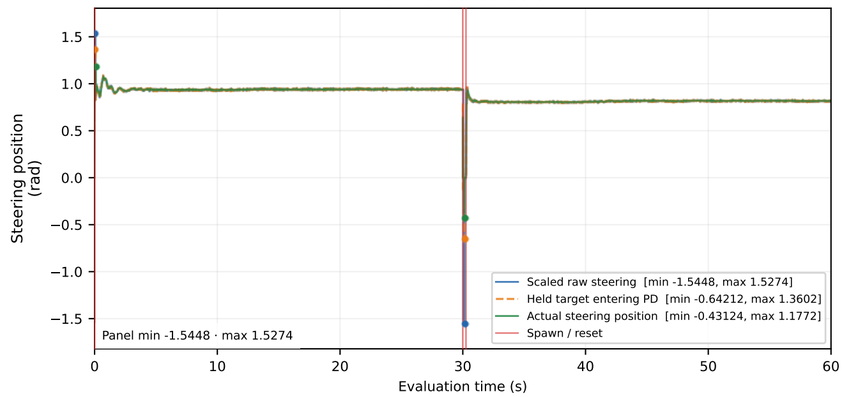
```
SAMPLED DR: env 12; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.44) s:
  mass x [0.888981, 1.00743, 0.831862, 1.03098]
  CoM XYZ mm [-4.1388, -0.447105, 1.67687]
  mu_s [0.733515, 0.538434, 0.704749, 0.541939]
  mu_d [0.558392, 0.538434, 0.541926, 0.534359]
  Kp [steer,rear] [10.8632, 0]; Kd [0.208468, 0]
  spawn roll -0.057985 rad; rate 0.081589 rad/s
  gyro bias XYZ rad/s [-0.00016212, -4.68457e-05, 1.47177e-05]
  gravity bias YZ [-0.00321399, 0.00226372]
E2 [0.44, 30.44) s:
  mass x [1.0346, 1.12616, 1.17533, 1.01379]
  CoM XYZ mm [-4.71101, 1.97284, 1.61729]
  mu_s [0.640016, 0.791209, 0.528253, 0.733515]
  mu_d [0.513141, 0.767936, 0.528253, 0.558392]
  Kp [steer,rear] [10.2496, 0]; Kd [0.217376, 0]
  spawn roll 0.0268486 rad; rate -0.112248 rad/s
  gyro bias XYZ rad/s [5.21292e-05, -0.000253077, -6.43064e-05]
  gravity bias YZ [0.00124444, -0.000723726]
E3 [30.44, 31.28) s:
  mass x [0.895664, 0.861537, 1.01434, 0.801483]
  CoM XYZ mm [-0.735719, 1.39525, -0.138491]
  mu_s [0.616244, 0.633024, 0.568073, 0.640016]
  mu_d [0.586498, 0.568172, 0.568073, 0.513141]
  Kp [steer,rear] [10.0039, 0]; Kd [0.198646, 0]
  spawn roll 0.0119817 rad; rate -0.0276258 rad/s
  gyro bias XYZ rad/s [-0.000283343, 0.000158501, -1.02695e-05]
  gravity bias YZ [0.00112562, 0.00173186]
E4 [31.28, 60.00) s:
  mass x [0.810855, 1.19466, 1.01735, 1.05576]
  CoM XYZ mm [-3.28053, -2.79567, 3.18595]
  mu_s [0.732187, 0.71155, 0.640867, 0.637675]
  mu_d [0.631664, 0.71155, 0.556841, 0.637675]
  Kp [steer,rear] [9.59382, 0]; Kd [0.180522, 0]
  spawn roll -0.0237513 rad; rate 0.0525132 rad/s
  gyro bias XYZ rad/s [-0.000118069, -1.40752e-05, -3.04469e-05]
  gravity bias YZ [-0.00029513, 0.000152421]
Six significant digits shown; full precision: sampled_dr_values.json.
```

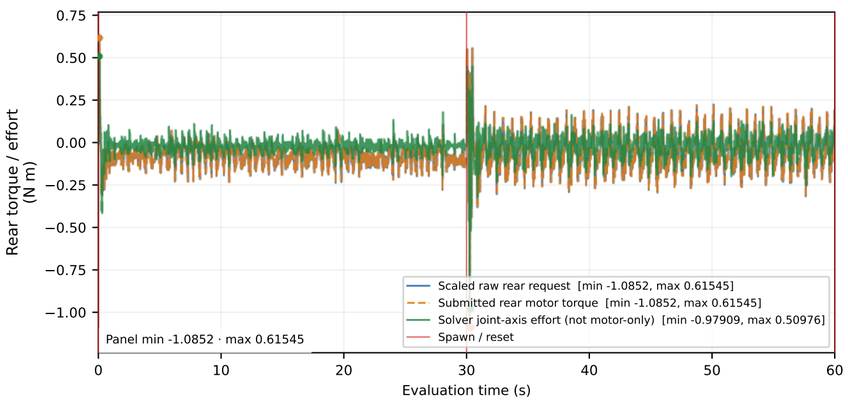
```
SAMPLED DR: env 12; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.52) s:
  mass x [0.888981, 1.00743, 0.831862, 1.03098]
  CoM XYZ mm [-4.1388, -0.447105, 1.67687]
  mu_s [0.733515, 0.538434, 0.704749, 0.541939]
  mu_d [0.558392, 0.538434, 0.541926, 0.534359]
  Kp [steer,rear] [10.8632, 0]; Kd [0.208468, 0]
  spawn roll -0.057985 rad; rate 0.081589 rad/s
  gyro bias XYZ rad/s [-0.00016212, -4.68457e-05, 1.47177e-05]
  gravity bias YZ [-0.00321399, 0.00226372]
E2 [0.52, 0.72) s:
  mass x [0.890015, 1.17945, 1.10153, 1.06853]
  CoM XYZ mm [1.44741, 2.8022, -2.6186]
  mu_s [0.669571, 0.528253, 0.538976, 0.538976]
  mu_d [0.669571, 0.528253, 0.538976, 0.538976]
  Kp [steer,rear] [9.51674, 0]; Kd [0.19014, 0]
  spawn roll 0.0995909 rad; rate 0.23022 rad/s
  gyro bias XYZ rad/s [2.73531e-05, -4.06514e-05, 0.000142005]
  gravity bias YZ [0.000861103, 0.000938522]
E3 [0.72, 30.72) s:
  mass x [0.891332, 0.956026, 1.09053, 0.994289]
  CoM XYZ mm [-1.66715, -3.01877, 2.69457]
  mu_s [0.723429, 0.559972, 0.757579, 0.538434]
  mu_d [0.723429, 0.502209, 0.70921, 0.538434]
  Kp [steer,rear] [9.18502, 0]; Kd [0.215514, 0]
  spawn roll 0.0384913 rad; rate 0.0731196 rad/s
  gyro bias XYZ rad/s [-0.000197728, -0.000124193, -0.000269497]
  gravity bias YZ [0.00145792, -0.00140849]
E4 [30.72, 60.00) s:
  mass x [1.14791, 1.01322, 1.08839, 1.01143]
  CoM XYZ mm [-2.36312, 1.30928, 0.963934]
  mu_s [0.528253, 0.723429, 0.71008, 0.733515]
  mu_d [0.528253, 0.723429, 0.59371, 0.558392]
  Kp [steer,rear] [9.61071, 0]; Kd [0.186415, 0]
  spawn roll 0.0462045 rad; rate 0.218468 rad/s
  gyro bias XYZ rad/s [-0.000102087, -0.000248652, -7.2445e-05]
  gravity bias YZ [0.000117711, 0.000915891]
Six significant digits shown; full precision: sampled_dr_values.json.
```

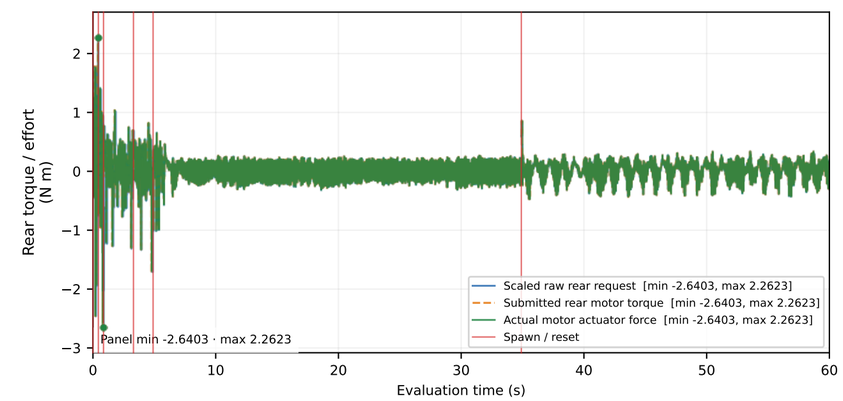
model_2500 · Demand versus actuation · env 13 · full 60 s

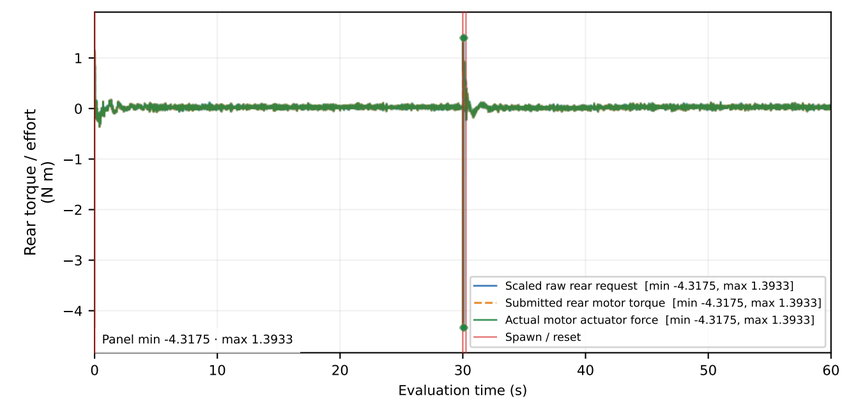












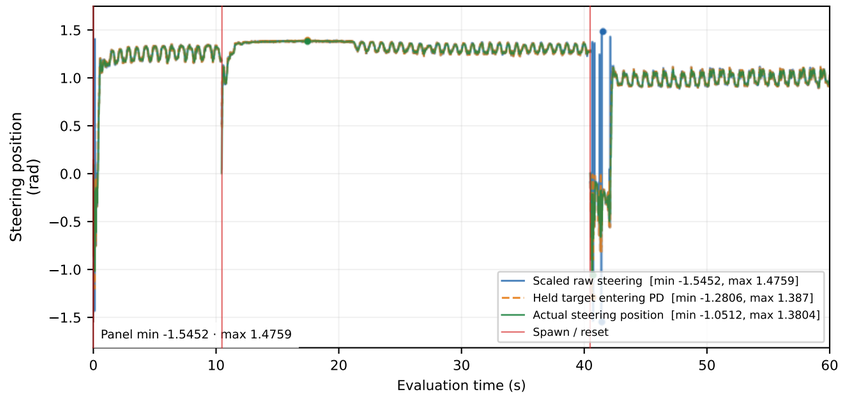
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

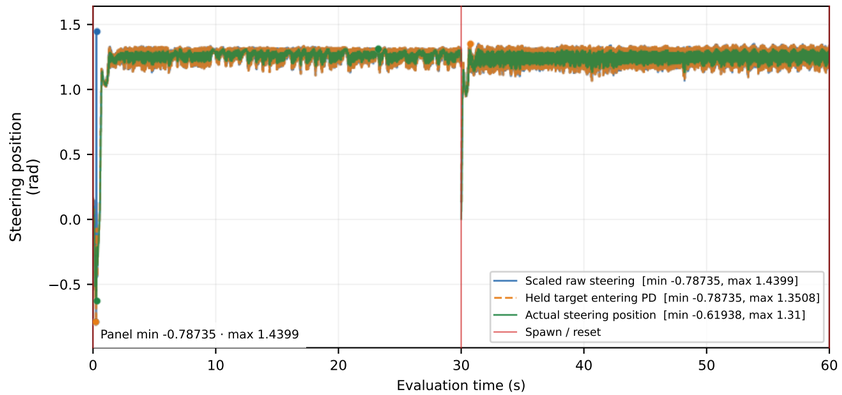
```
SAMPLED DR: env 13; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.14819, 1.09148, 0.901625, 0.866687]
  CoM XYZ mm [-1.49745, -4.85852, 4.10487]
  mu_s [0.576972, 0.764681, 0.703711, 0.760821]
  mu_d [0.576972, 0.764681, 0.703711, 0.670315]
  Kp [steer,rear] [9.5585, 0]; Kd [0.19592, 0]
  spawn roll 0.0150582 rad; rate -0.209008 rad/s
  gyro bias XYZ rad/s [-0.0002458, 0.000257238, 9.44563e-05]
  gravity bias YZ [-0.00107082, -0.000198226]
E2 [30.00, 60.00) s:
  mass x [1.12365, 1.02316, 0.969298, 0.904739]
  CoM XYZ mm [4.35705, -0.931233, 3.05012]
  mu_s [0.598343, 0.628821, 0.713129, 0.659445]
  mu_d [0.598343, 0.628821, 0.713129, 0.547619]
  Kp [steer,rear] [10.1481, 0]; Kd [0.187719, 0]
  spawn roll -0.0219507 rad; rate -0.224828 rad/s
  gyro bias XYZ rad/s [-0.000266752, -4.27344e-05, 0.000211438]
  gravity bias YZ [-0.000458392, 0.00216717]
Six significant digits shown; full precision: sampled_dr_values.json.
```

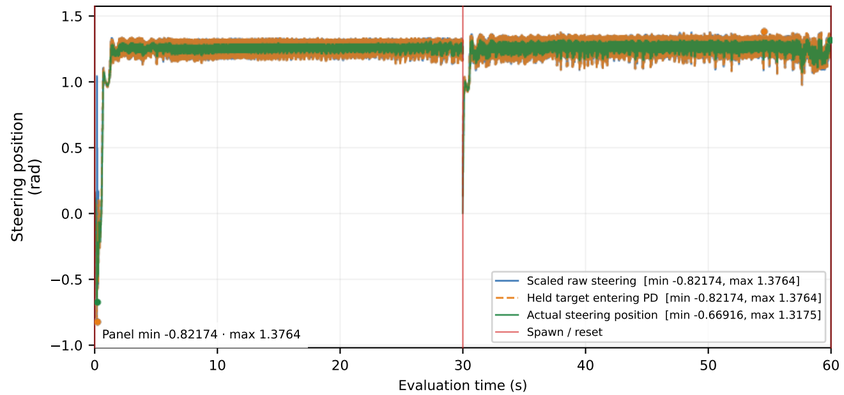
```
SAMPLED DR: env 13; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.44) s:
  mass x [1.1329, 1.14826, 0.887345, 0.892039]
  CoM XYZ mm [-0.072569, 2.53223, 2.04232]
  mu_s [0.728342, 0.700521, 0.559972, 0.528253]
  mu_d [0.728342, 0.641329, 0.502209, 0.528253]
  Kp [steer,rear] [9.98219, 0]; Kd [0.204077, 0]
  spawn roll 0.0961499 rad; rate -0.0648344 rad/s
  gyro bias XYZ rad/s [-8.41829e-05, 9.3231e-05, -0.000287958]
  gravity bias YZ [0.00221957, 8.91091e-05]
E2 [0.44, 0.86) s:
  mass x [1.18048, 1.10671, 0.913639, 0.89274]
  CoM XYZ mm [-0.235183, 4.12978, -3.9335]
  mu_s [0.611239, 0.757579, 0.616244, 0.597748]
  mu_d [0.611239, 0.70921, 0.586498, 0.597748]
  Kp [steer,rear] [9.10023, 0]; Kd [0.215477, 0]
  spawn roll -0.00256985 rad; rate -0.131061 rad/s
  gyro bias XYZ rad/s [7.04588e-05, -3.01761e-05, -0.000176978]
  gravity bias YZ [0.000567873, 0.00103235]
E3 [0.86, 3.30) s:
  mass x [1.16124, 0.824091, 1.11037, 1.07]
  CoM XYZ mm [2.12359, 2.17524, -0.110732]
  mu_s [0.666375, 0.640016, 0.704749, 0.748289]
  mu_d [0.519145, 0.513141, 0.541926, 0.689499]
  Kp [steer,rear] [9.85806, 0]; Kd [0.186709, 0]
  spawn roll -0.0667805 rad; rate -0.0786676 rad/s
  gyro bias XYZ rad/s [0.000465989, 9.39327e-05, 0.000110484]
  gravity bias YZ [0.00476382, -0.00345774]
E4 [3.30, 4.90) s:
  mass x [0.889313, 0.813812, 1.09052, 1.05501]
  CoM XYZ mm [1.75968, 3.00834, -2.88236]
  mu_s [0.640867, 0.597748, 0.559972, 0.71155]
  mu_d [0.556841, 0.597748, 0.502209, 0.71155]
  Kp [steer,rear] [9.80407, 0]; Kd [0.190426, 0]
  spawn roll -0.00749381 rad; rate -0.0256039 rad/s
  gyro bias XYZ rad/s [-4.49526e-05, 0.000359247, 0.000179771]
  gravity bias YZ [-0.00136921, 0.00503738]
E5 [4.90, 34.90) s:
  mass x [1.16635, 0.964278, 0.891603, 1.12673]
  CoM XYZ mm [1.93179, -2.62501, -4.16081]
  mu_s [0.723429, 0.71008, 0.727426, 0.616244]
  mu_d [0.723429, 0.59371, 0.606358, 0.586498]
  Kp [steer,rear] [9.41256, 0]; Kd [0.201394, 0]
  spawn roll -0.0488085 rad; rate -0.0852721 rad/s
  gyro bias XYZ rad/s [-0.000502116, -3.32743e-05, 0.000103514]
  gravity bias YZ [0.000722132, 0.000688708]
E6 [34.90, 60.00) s:
  mass x [0.90303, 1.0851, 0.867758, 1.0553]
  CoM XYZ mm [-2.16255, 0.399354, 1.11696]
  mu_s [0.611239, 0.732187, 0.749678, 0.733515]
  mu_d [0.611239, 0.631664, 0.741429, 0.558392]
  Kp [steer,rear] [9.24214, 0]; Kd [0.181495, 0]
  spawn roll 0.0723589 rad; rate 0.164877 rad/s
  gyro bias XYZ rad/s [-0.000108821, -0.000119695, -2.92104e-05]
  gravity bias YZ [0.000176065, 9.40162e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

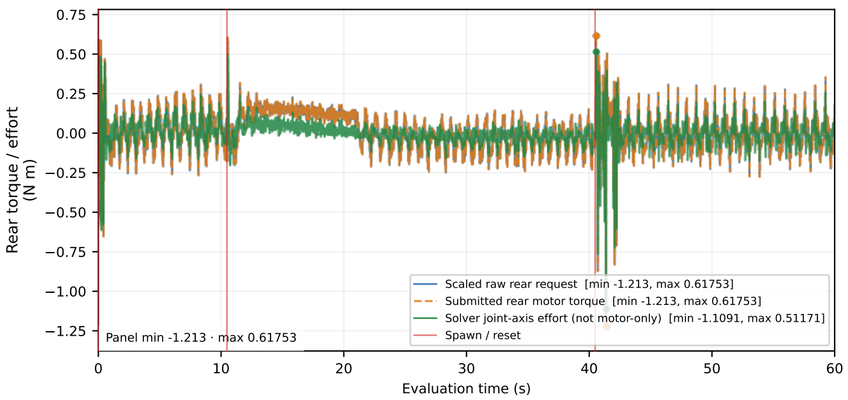
```
SAMPLED DR: env 13; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.1329, 1.14826, 0.887345, 0.892039]
  CoM XYZ mm [-0.072569, 2.53223, 2.04232]
  mu_s [0.728342, 0.700521, 0.559972, 0.528253]
  mu_d [0.728342, 0.641329, 0.502209, 0.528253]
  Kp [steer,rear] [9.98219, 0]; Kd [0.204077, 0]
  spawn roll 0.0961499 rad; rate -0.0648344 rad/s
  gyro bias XYZ rad/s [-8.41829e-05, 9.3231e-05, -0.000287958]
  gravity bias YZ [0.00221957, 8.91091e-05]
E2 [30.00, 30.26) s:
  mass x [1.07626, 0.853924, 0.874087, 0.866441]
  CoM XYZ mm [-1.51051, -2.57644, -2.51735]
  mu_s [0.733515, 0.791209, 0.597748, 0.728342]
  mu_d [0.558392, 0.767936, 0.597748, 0.728342]
  Kp [steer,rear] [10.4238, 0]; Kd [0.186222, 0]
  spawn roll 0.0973156 rad; rate 0.215546 rad/s
  gyro bias XYZ rad/s [-8.64701e-05, -2.65843e-05, 5.33129e-05]
  gravity bias YZ [-0.00274842, -0.00384582]
E3 [30.26, 60.00) s:
  mass x [0.956472, 0.884212, 1.03474, 0.989246]
  CoM XYZ mm [-0.261673, 4.47667, -0.31183]
  mu_s [0.732187, 0.791209, 0.749678, 0.757579]
  mu_d [0.631664, 0.767936, 0.741429, 0.70921]
  Kp [steer,rear] [9.13301, 0]; Kd [0.190991, 0]
  spawn roll 0.0554458 rad; rate -0.237323 rad/s
  gyro bias XYZ rad/s [-0.000265184, 0.000105029, -0.000140464]
  gravity bias YZ [-0.000820023, -0.00188164]
Six significant digits shown; full precision: sampled_dr_values.json.
```

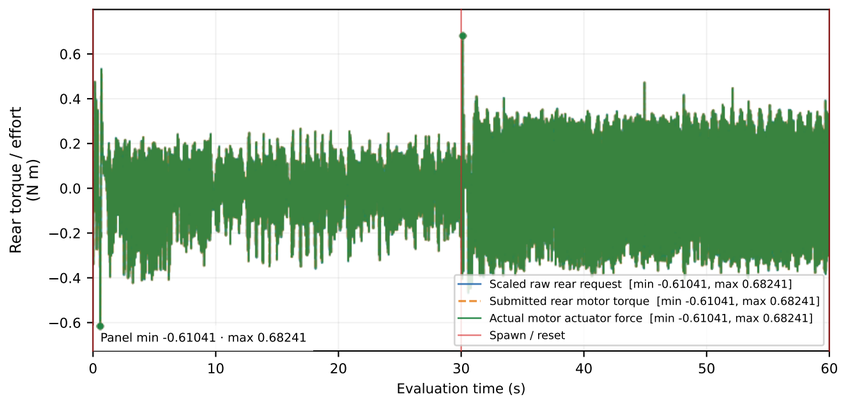
model_2500 · Demand versus actuation · env 14 · full 60 s

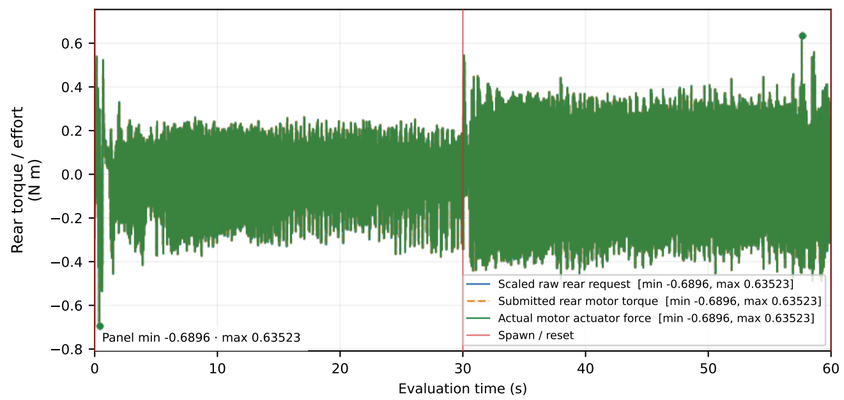












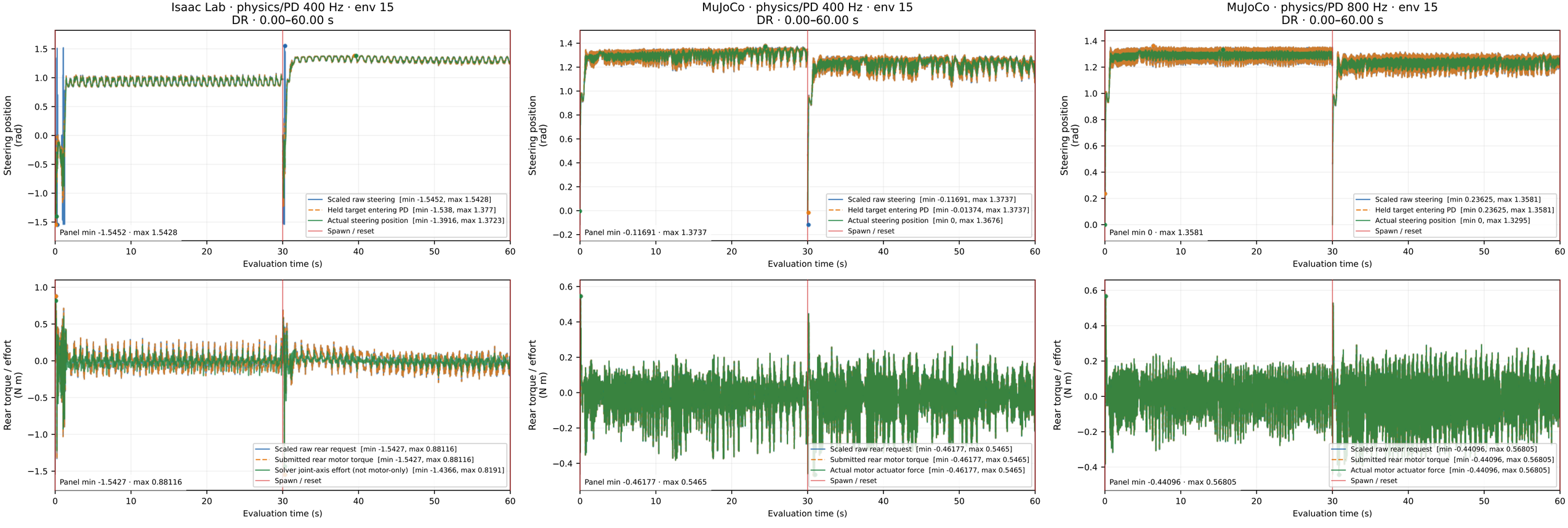
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 14; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 10.48) s:
  mass x [1.0908, 0.959478, 0.907832, 1.12118]
  CoM XYZ mm [-3.98526, -2.2652, 2.011]
  mu_s [0.559809, 0.579974, 0.532645, 0.531594]
  mu_d [0.559809, 0.579974, 0.532645, 0.531594]
  Kp [steer,rear] [10.6855, 0]; Kd [0.196521, 0]
  spawn roll -0.0556784 rad; rate 0.134896 rad/s
  gyro bias XYZ rad/s [0.000404994, 0.000325787, 0.000221077]
  gravity bias YZ [-0.00315518, -0.00475108]
E2 [10.48, 40.48) s:
  mass x [1.12665, 0.974044, 0.963063, 1.15108]
  CoM XYZ mm [-4.75244, -3.39101, -3.00288]
  mu_s [0.592046, 0.741039, 0.547743, 0.681711]
  mu_d [0.534947, 0.614405, 0.547743, 0.611756]
  Kp [steer,rear] [9.77286, 0]; Kd [0.212644, 0]
  spawn roll 0.0134134 rad; rate 0.0812691 rad/s
  gyro bias XYZ rad/s [-9.24104e-05, -0.000422279, -6.17662e-05]
  gravity bias YZ [-0.00234216, 0.000989177]
E3 [40.48, 60.00) s:
  mass x [0.992094, 1.03798, 1.07644, 0.902375]
  CoM XYZ mm [-2.16226, 1.23412, -1.6689]
  mu_s [0.632409, 0.592046, 0.774409, 0.531594]
  mu_d [0.589076, 0.534947, 0.561095, 0.531594]
  Kp [steer,rear] [9.12801, 0]; Kd [0.196552, 0]
  spawn roll -0.0286734 rad; rate -0.0304172 rad/s
  gyro bias XYZ rad/s [-0.000241286, 2.68151e-05, -0.000198635]
  gravity bias YZ [-2.94909e-05, 0.00269875]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 14; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09045, 0.806761, 1.19232, 1.08984]
  CoM XYZ mm [0.442122, -2.6438, 1.06631]
  mu_s [0.611239, 0.791209, 0.669571, 0.538976]
  mu_d [0.611239, 0.767936, 0.669571, 0.538976]
  Kp [steer,rear] [9.52205, 0]; Kd [0.203739, 0]
  spawn roll -0.0427706 rad; rate 0.108207 rad/s
  gyro bias XYZ rad/s [-0.000211617, -0.000269613, 0.000243965]
  gravity bias YZ [-0.00480038, -0.00311316]
E2 [30.00, 60.00) s:
  mass x [0.806949, 1.1354, 0.98674, 0.939223]
  CoM XYZ mm [3.56222, -1.6243, -4.43757]
  mu_s [0.757579, 0.633024, 0.666375, 0.749678]
  mu_d [0.70921, 0.568172, 0.519145, 0.741429]
  Kp [steer,rear] [9.99888, 0]; Kd [0.18346, 0]
  spawn roll 0.0176969 rad; rate -0.075278 rad/s
  gyro bias XYZ rad/s [-6.40627e-05, 0.000153353, 0.000215432]
  gravity bias YZ [-0.00148081, 0.00268728]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 14; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09045, 0.806761, 1.19232, 1.08984]
  CoM XYZ mm [0.442122, -2.6438, 1.06631]
  mu_s [0.611239, 0.791209, 0.669571, 0.538976]
  mu_d [0.611239, 0.767936, 0.669571, 0.538976]
  Kp [steer,rear] [9.52205, 0]; Kd [0.203739, 0]
  spawn roll -0.0427706 rad; rate 0.108207 rad/s
  gyro bias XYZ rad/s [-0.000211617, -0.000269613, 0.000243965]
  gravity bias YZ [-0.00480038, -0.00311316]
E2 [30.00, 60.00) s:
  mass x [0.806949, 1.1354, 0.98674, 0.939223]
  CoM XYZ mm [3.56222, -1.6243, -4.43757]
  mu_s [0.757579, 0.633024, 0.666375, 0.749678]
  mu_d [0.70921, 0.568172, 0.519145, 0.741429]
  Kp [steer,rear] [9.99888, 0]; Kd [0.18346, 0]
  spawn roll 0.0176969 rad; rate -0.075278 rad/s
  gyro bias XYZ rad/s [-6.40627e-05, 0.000153353, 0.000215432]
  gravity bias YZ [-0.00148081, 0.00268728]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 15 · full 60 s



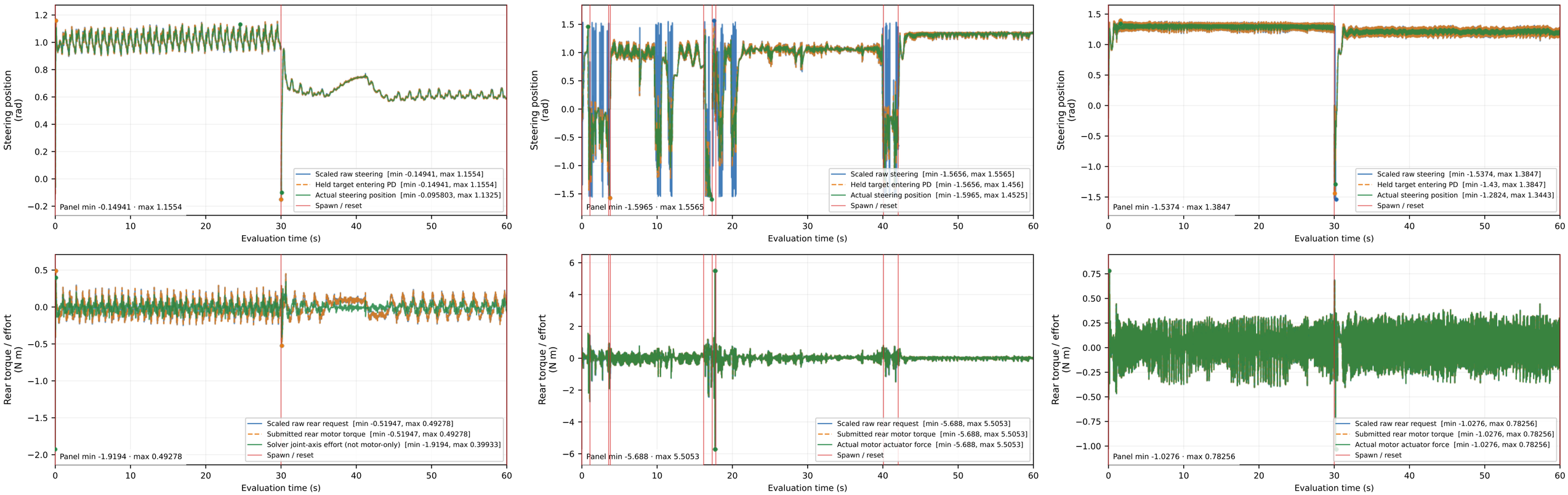
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 15; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.964367, 0.858985, 0.809135, 0.893245]
  CoM XYZ mm [-3.24544, 2.51067, -4.45804]
  mu_s [0.547743, 0.579974, 0.531594, 0.632409]
  mu_d [0.547743, 0.579974, 0.531594, 0.589076]
  Kp [steer,rear] [10.0426, 0]; Kd [0.211879, 0]
  spawn roll -0.0893606 rad; rate -0.0937116 rad/s
  gyro bias XYZ rad/s [-0.000418754, 3.15846e-05, -0.000100589]
  gravity bias YZ [0.00432265, 0.00380675]
E2 [30.00, 60.00) s:
  mass x [1.07244, 1.13643, 1.12039, 1.087]
  CoM XYZ mm [2.09376, -2.7987, -3.77326]
  mu_s [0.559809, 0.774409, 0.532645, 0.560535]
  mu_d [0.559809, 0.561095, 0.532645, 0.560535]
  Kp [steer,rear] [9.93246, 0]; Kd [0.21638, 0]
  spawn roll -0.0822835 rad; rate 0.0203657 rad/s
  gyro bias XYZ rad/s [-5.9777e-05, -0.000493778, 0.000407336]
  gravity bias YZ [-0.00154557, -0.000237761]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 15; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.07367, 0.987325, 1.13649, 0.936128]
  CoM XYZ mm [-1.06178, -3.43558, -0.707819]
  mu_s [0.727426, 0.616244, 0.559972, 0.748289]
  mu_d [0.606358, 0.586498, 0.502209, 0.689499]
  Kp [steer,rear] [9.67415, 0]; Kd [0.199301, 0]
  spawn roll 0.0107642 rad; rate -0.00345246 rad/s
  gyro bias XYZ rad/s [0.000147003, -0.000218396, 6.58599e-05]
  gravity bias YZ [0.00268931, 3.35083e-05]
E2 [30.00, 60.00) s:
  mass x [1.15353, 1.10784, 1.05565, 0.878751]
  CoM XYZ mm [0.324308, -1.41831, 4.63531]
  mu_s [0.669571, 0.71155, 0.631146, 0.732187]
  mu_d [0.669571, 0.71155, 0.631146, 0.631664]
  Kp [steer,rear] [9.99829, 0]; Kd [0.183744, 0]
  spawn roll 0.0191785 rad; rate 0.0552767 rad/s
  gyro bias XYZ rad/s [0.000219153, 0.000119844, -0.000332515]
  gravity bias YZ [0.00203666, -0.00112214]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 15; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.07367, 0.987325, 1.13649, 0.936128]
  CoM XYZ mm [-1.06178, -3.43558, -0.707819]
  mu_s [0.727426, 0.616244, 0.559972, 0.748289]
  mu_d [0.606358, 0.586498, 0.502209, 0.689499]
  Kp [steer,rear] [9.67415, 0]; Kd [0.199301, 0]
  spawn roll 0.0107642 rad; rate -0.00345246 rad/s
  gyro bias XYZ rad/s [0.000147003, -0.000218396, 6.58599e-05]
  gravity bias YZ [0.00268931, 3.35083e-05]
E2 [30.00, 60.00) s:
  mass x [1.15353, 1.10784, 1.05565, 0.878751]
  CoM XYZ mm [0.324308, -1.41831, 4.63531]
  mu_s [0.669571, 0.71155, 0.631146, 0.732187]
  mu_d [0.669571, 0.71155, 0.631146, 0.631664]
  Kp [steer,rear] [9.99829, 0]; Kd [0.183744, 0]
  spawn roll 0.0191785 rad; rate 0.0552767 rad/s
  gyro bias XYZ rad/s [0.000219153, 0.000119844, -0.000332515]
  gravity bias YZ [0.00203666, -0.00112214]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 16 · full 60 s



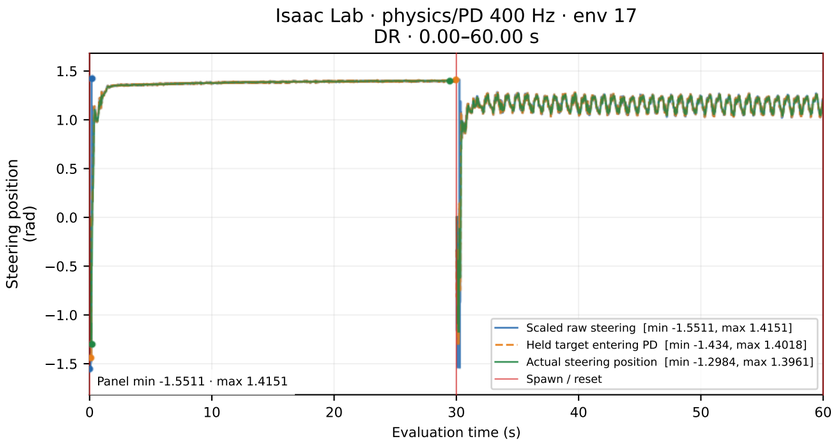
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

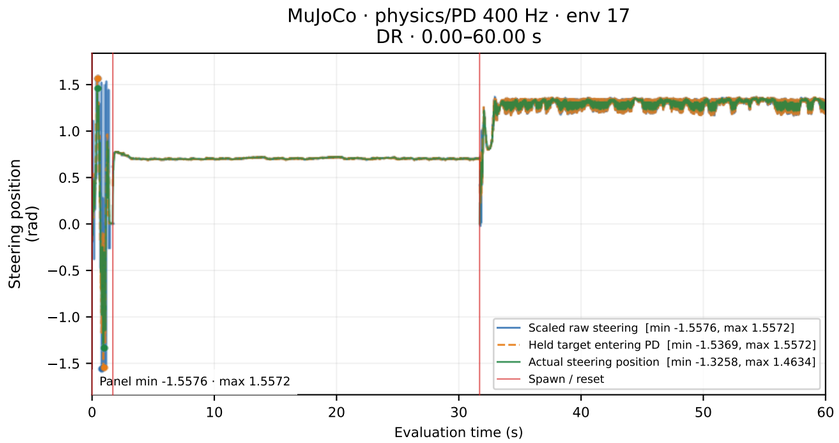
```
SAMPLED DR: env 16; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.891933, 0.959353, 0.992334, 0.924888]
  CoM XYZ mm [4.6868, 1.0248, -3.74794]
  mu_s [0.674963, 0.787792, 0.764681, 0.584424]
  mu_d [0.601294, 0.617134, 0.764681, 0.584424]
  Kp [steer,rear] [9.27673, 0]; Kd [0.181878, 0]
  spawn roll 0.0359022 rad; rate -0.236051 rad/s
  gyro bias XYZ rad/s [-0.000538585, 0.000225163, -0.00014116]
  gravity bias YZ [-0.00284693, 0.000550345]
E2 [30.00, 60.00) s:
  mass x [1.11595, 1.18185, 0.83303, 1.02334]
  CoM XYZ mm [2.93428, 4.99582, 0.202283]
  mu_s [0.579974, 0.760821, 0.764681, 0.673378]
  mu_d [0.579974, 0.670315, 0.764681, 0.673378]
  Kp [steer,rear] [9.60151, 0]; Kd [0.181605, 0]
  spawn roll 0.0172193 rad; rate -0.187324 rad/s
  gyro bias XYZ rad/s [0.000590245, 0.000248146, -0.000151491]
  gravity bias YZ [-0.000822571, -0.00160134]
Six significant digits shown; full precision: sampled_dr_values.json.
```

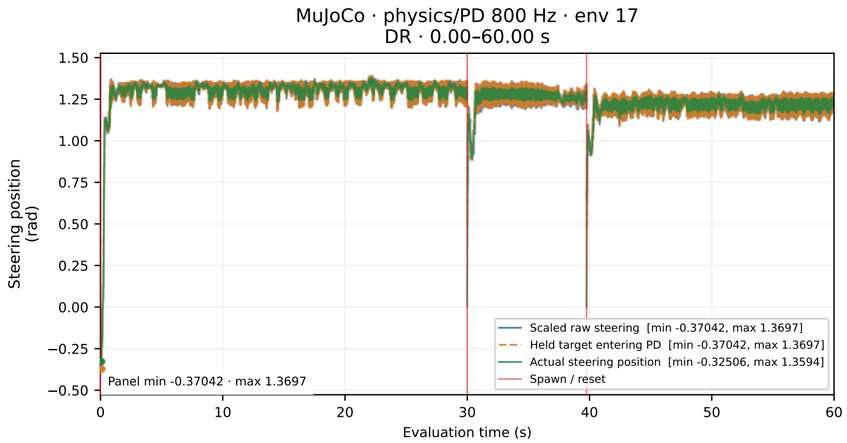
```
SAMPLED DR: env 16; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.08) s:
  mass x [0.923605, 1.15193, 0.897237, 0.883368]
  CoM XYZ mm [3.45813, -3.59427, -1.94667]
  mu_s [0.733515, 0.538434, 0.704749, 0.597748]
  mu_d [0.558392, 0.538434, 0.541926, 0.597748]
  Kp [steer,rear] [10.85, 0]; Kd [0.190802, 0]
  spawn roll -0.00998595 rad; rate -0.146724 rad/s
  gyro bias XYZ rad/s [-0.000146102, -1.28286e-05, -5.60525e-05]
  gravity bias YZ [-0.00196611, 0.000330421]
E2 [1.08, 3.56) s:
  mass x [1.14906, 0.916373, 0.94598, 0.994907]
  CoM XYZ mm [-3.77423, 1.72766, 0.690506]
  mu_s [0.597748, 0.631146, 0.749678, 0.723429]
  mu_d [0.597748, 0.631146, 0.741429, 0.723429]
  Kp [steer,rear] [9.3992, 0]; Kd [0.18431, 0]
  spawn roll -0.0585122 rad; rate -0.069853 rad/s
  gyro bias XYZ rad/s [0.000472192, -0.000347395, 0.000209527]
  gravity bias YZ [0.00233895, 0.000760862]
E3 [3.56, 3.78) s:
  mass x [0.872795, 1.07367, 0.898727, 1.09553]
  CoM XYZ mm [2.65337, 2.57532, -2.65282]
  mu_s [0.528253, 0.640867, 0.71008, 0.559972]
  mu_d [0.528253, 0.556841, 0.59371, 0.502209]
  Kp [steer,rear] [10.8385, 0]; Kd [0.197695, 0]
  spawn roll -0.0884651 rad; rate -0.212415 rad/s
  gyro bias XYZ rad/s [0.0002235, -0.000115595, 0.000163326]
  gravity bias YZ [-0.00208058, -0.00316361]
E4 [3.78, 16.18) s:
  mass x [0.995699, 0.958547, 1.15176, 0.98727]
  CoM XYZ mm [4.87388, 2.07726, -4.03266]
  mu_s [0.723429, 0.640016, 0.757579, 0.640016]
  mu_d [0.723429, 0.513141, 0.70921, 0.513141]
  Kp [steer,rear] [9.97118, 0]; Kd [0.186041, 0]
  spawn roll 0.0429654 rad; rate 0.0871774 rad/s
  gyro bias XYZ rad/s [-0.000137077, 0.00018035, 9.90635e-05]
  gravity bias YZ [0.000820886, -0.00174003]
E5 [16.18, 17.32) s:
  mass x [1.15268, 1.04228, 0.890194, 0.854027]
  CoM XYZ mm [-2.36818, -4.09797, 3.57273]
  mu_s [0.616244, 0.541939, 0.732187, 0.69316]
  mu_d [0.586498, 0.534359, 0.631664, 0.69316]
  Kp [steer,rear] [9.12976, 0]; Kd [0.193339, 0]
  spawn roll 0.0433519 rad; rate -0.12468 rad/s
  gyro bias XYZ rad/s [-2.57608e-05, 1.42294e-05, -8.98742e-05]
  gravity bias YZ [-0.00052632, 0.00179672]
E6 [17.32, 17.80) s:
  mass x [1.14067, 0.926093, 0.909646, 0.951796]
  CoM XYZ mm [1.46079, -3.27898, -3.3874]
  mu_s [0.736077, 0.559972, 0.631146, 0.666375]
  mu_d [0.699455, 0.502209, 0.631146, 0.519145]
  Kp [steer,rear] [9.61725, 0]; Kd [0.210762, 0]
  spawn roll -0.025371 rad; rate 0.172028 rad/s
  gyro bias XYZ rad/s [1.28074e-05, -0.000247523, 6.08333e-05]
  gravity bias YZ [0.00114787, -0.00239577]
E7 [17.80, 40.08) s:
  mass x [1.116, 1.01425, 0.882105, 1.02535]
  CoM XYZ mm [0.112943, 1.39924, -0.747668]
  mu_s [0.723429, 0.616244, 0.528253, 0.704749]
  mu_d [0.723429, 0.586498, 0.528253, 0.541926]
  Kp [steer,rear] [9.60372, 0]; Kd [0.210665, 0]
  spawn roll 0.0702297 rad; rate 0.0838734 rad/s
  gyro bias XYZ rad/s [-0.000403327, -0.000198575, 0.000226331]
  gravity bias YZ [0.00425945, 0.0019684]
E8 [40.08, 42.04) s:
  mass x [0.874414, 1.15853, 1.17124, 1.16893]
  CoM XYZ mm [0.943144, 2.07292, -3.16083]
  mu_s [0.538434, 0.728342, 0.71155, 0.71008]
  mu_d [0.538434, 0.728342, 0.71155, 0.59371]
  Kp [steer,rear] [10.6261, 0]; Kd [0.205094, 0]
  spawn roll -0.0572042 rad; rate -0.121939 rad/s
  gyro bias XYZ rad/s [7.89581e-05, -0.000117877, 0.000182835]
  gravity bias YZ [-0.00179059, 0.00136599]
E9 [42.04, 60.00) s:
  mass x [1.1668, 1.13148, 0.930705, 0.890804]
  CoM XYZ mm [3.5182, -4.51687, 4.93569]
  mu_s [0.640016, 0.71155, 0.637675, 0.546287]
  mu_d [0.513141, 0.71155, 0.637675, 0.546287]
  Kp [steer,rear] [9.47717, 0]; Kd [0.185056, 0]
  spawn roll -0.0297762 rad; rate 0.101724 rad/s
  gyro bias XYZ rad/s [-6.59488e-05, 0.000110389, -4.40996e-05]
  gravity bias YZ [-0.0016465, -0.00234535]
Six significant digits shown; full precision: sampled_dr_values.json.
```

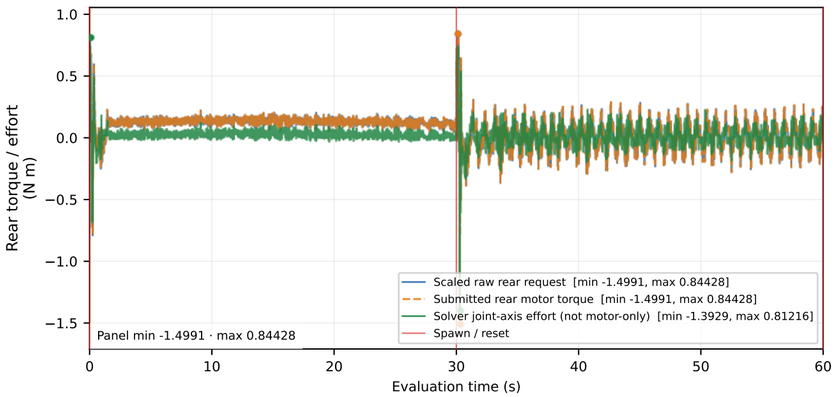
```
SAMPLED DR: env 16; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.923605, 1.15193, 0.897237, 0.883368]
  CoM XYZ mm [3.45813, -3.59427, -1.94667]
  mu_s [0.733515, 0.538434, 0.704749, 0.597748]
  mu_d [0.558392, 0.538434, 0.541926, 0.597748]
  Kp [steer,rear] [10.85, 0]; Kd [0.190802, 0]
  spawn roll -0.00998595 rad; rate -0.146724 rad/s
  gyro bias XYZ rad/s [-0.000146102, -1.28286e-05, -5.60525e-05]
  gravity bias YZ [-0.00196611, 0.000330421]
E2 [30.00, 60.00) s:
  mass x [0.808575, 0.946532, 0.921532, 0.807876]
  CoM XYZ mm [-4.19111, -1.08404, -4.92267]
  mu_s [0.538976, 0.637675, 0.640867, 0.633024]
  mu_d [0.538976, 0.637675, 0.556841, 0.568172]
  Kp [steer,rear] [9.44037, 0]; Kd [0.190149, 0]
  spawn roll -0.051795 rad; rate -0.166744 rad/s
  gyro bias XYZ rad/s [-7.02067e-05, -2.37253e-05, -0.000123674]
  gravity bias YZ [0.000558188, 0.0016771]
Six significant digits shown; full precision: sampled_dr_values.json.
```

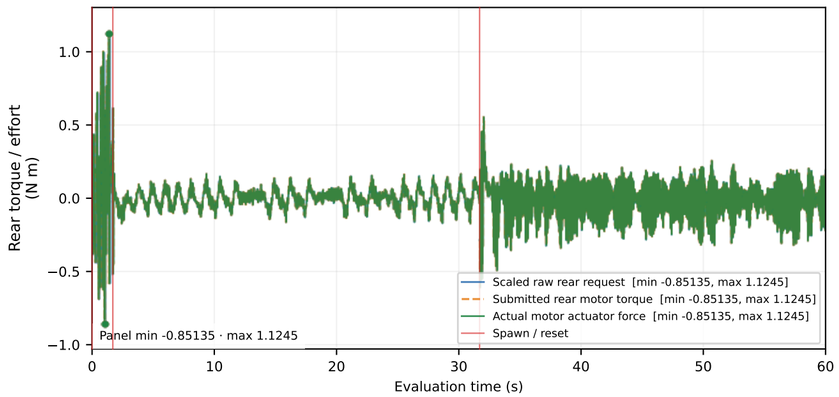
model_2500 · Demand versus actuation · env 17 · full 60 s

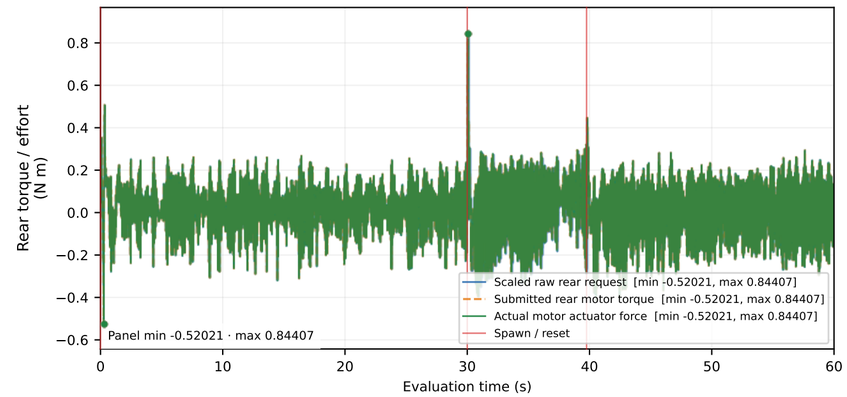












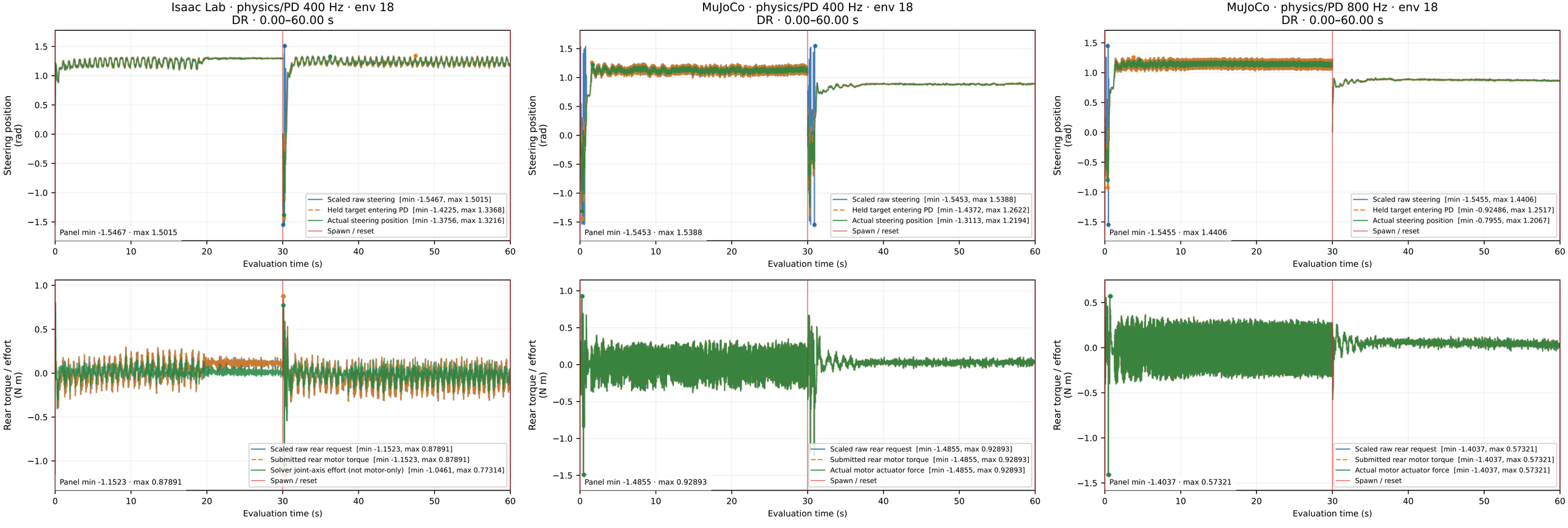
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 17; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.0488, 0.942439, 1.06045, 1.17219]
  CoM XYZ mm [-1.46499, -4.3528, -2.78383]
  mu_s [0.774409, 0.774409, 0.533455, 0.560535]
  mu_d [0.561095, 0.561095, 0.533455, 0.560535]
  Kp [steer,rear] [10.3908, 0]; Kd [0.217746, 0]
  spawn roll -0.0990896 rad; rate -0.0671556 rad/s
  gyro bias XYZ rad/s [-0.000199137, -0.000134969, 1.06368e-05]
  gravity bias YZ [-0.002044, 0.00241515]
E2 [30.00, 60.00) s:
  mass x [0.887591, 1.1975, 1.14532, 0.927279]
  CoM XYZ mm [3.31138, -0.607744, -3.82507]
  mu_s [0.659445, 0.532645, 0.592046, 0.673378]
  mu_d [0.547619, 0.532645, 0.534947, 0.673378]
  Kp [steer,rear] [10.0425, 0]; Kd [0.186075, 0]
  spawn roll -0.085215 rad; rate -0.0130568 rad/s
  gyro bias XYZ rad/s [0.000162379, -0.000128116, 0.000317264]
  gravity bias YZ [-0.00315221, -0.000589016]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 17; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.70) s:
  mass x [1.0351, 1.13043, 0.846778, 0.950972]
  CoM XYZ mm [-3.13738, -3.39717, -4.9418]
  mu_s [0.71155, 0.669571, 0.541939, 0.728342]
  mu_d [0.71155, 0.669571, 0.534359, 0.728342]
  Kp [steer,rear] [10.869, 0]; Kd [0.208257, 0]
  spawn roll -0.0307122 rad; rate -0.0746429 rad/s
  gyro bias XYZ rad/s [8.25675e-06, -0.00013759, 0.000263549]
  gravity bias YZ [-0.00103796, -0.00173413]
E2 [1.70, 31.70) s:
  mass x [1.11575, 0.893922, 1.11038, 1.1095]
  CoM XYZ mm [-0.73171, 4.99729, -4.17946]
  mu_s [0.71008, 0.700521, 0.732187, 0.538976]
  mu_d [0.59371, 0.641329, 0.631664, 0.538976]
  Kp [steer,rear] [10.0259, 0]; Kd [0.217319, 0]
  spawn roll 0.0435569 rad; rate 0.0392872 rad/s
  gyro bias XYZ rad/s [0.000111394, 0.000156834, -0.000151999]
  gravity bias YZ [-0.00336816, -0.00419074]
E3 [31.70, 60.00) s:
  mass x [1.19586, 1.07154, 1.13235, 0.975779]
  CoM XYZ mm [4.70814, -2.75462, 0.661321]
  mu_s [0.640016, 0.631146, 0.541939, 0.749678]
  mu_d [0.513141, 0.631146, 0.534359, 0.741429]
  Kp [steer,rear] [10.1648, 0]; Kd [0.195039, 0]
  spawn roll -0.033512 rad; rate 0.179082 rad/s
  gyro bias XYZ rad/s [-0.000297572, -2.38207e-06, -0.000210845]
  gravity bias YZ [0.00170795, 0.000186021]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 17; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.0351, 1.13043, 0.846778, 0.950972]
  CoM XYZ mm [-3.13738, -3.39717, -4.9418]
  mu_s [0.71155, 0.669571, 0.541939, 0.728342]
  mu_d [0.71155, 0.669571, 0.534359, 0.728342]
  Kp [steer,rear] [10.869, 0]; Kd [0.208257, 0]
  spawn roll -0.0307122 rad; rate -0.0746429 rad/s
  gyro bias XYZ rad/s [8.25675e-06, -0.00013759, 0.000263549]
  gravity bias YZ [-0.00103796, -0.00173413]
E2 [30.00, 39.76) s:
  mass x [1.0484, 0.935286, 0.928736, 1.06293]
  CoM XYZ mm [3.31686, -4.22521, 0.275083]
  mu_s [0.749678, 0.748289, 0.559972, 0.568073]
  mu_d [0.741429, 0.689499, 0.502209, 0.568073]
  Kp [steer,rear] [9.73886, 0]; Kd [0.184541, 0]
  spawn roll 0.0737145 rad; rate 0.0203807 rad/s
  gyro bias XYZ rad/s [-0.000277662, -5.4727e-05, 0.000191924]
  gravity bias YZ [-0.000151406, 0.00473621]
E3 [39.76, 60.00) s:
  mass x [1.10051, 0.929118, 1.16593, 0.84389]
  CoM XYZ mm [-2.34865, -1.86845, 4.7607]
  mu_s [0.568073, 0.541939, 0.727426, 0.732187]
  mu_d [0.568073, 0.534359, 0.606358, 0.631664]
  Kp [steer,rear] [10.7868, 0]; Kd [0.202279, 0]
  spawn roll 0.0280855 rad; rate 0.122722 rad/s
  gyro bias XYZ rad/s [-0.000107871, 2.07815e-05, 0.00015181]
  gravity bias YZ [-0.0024167, 0.000327487]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 18 · full 60 s



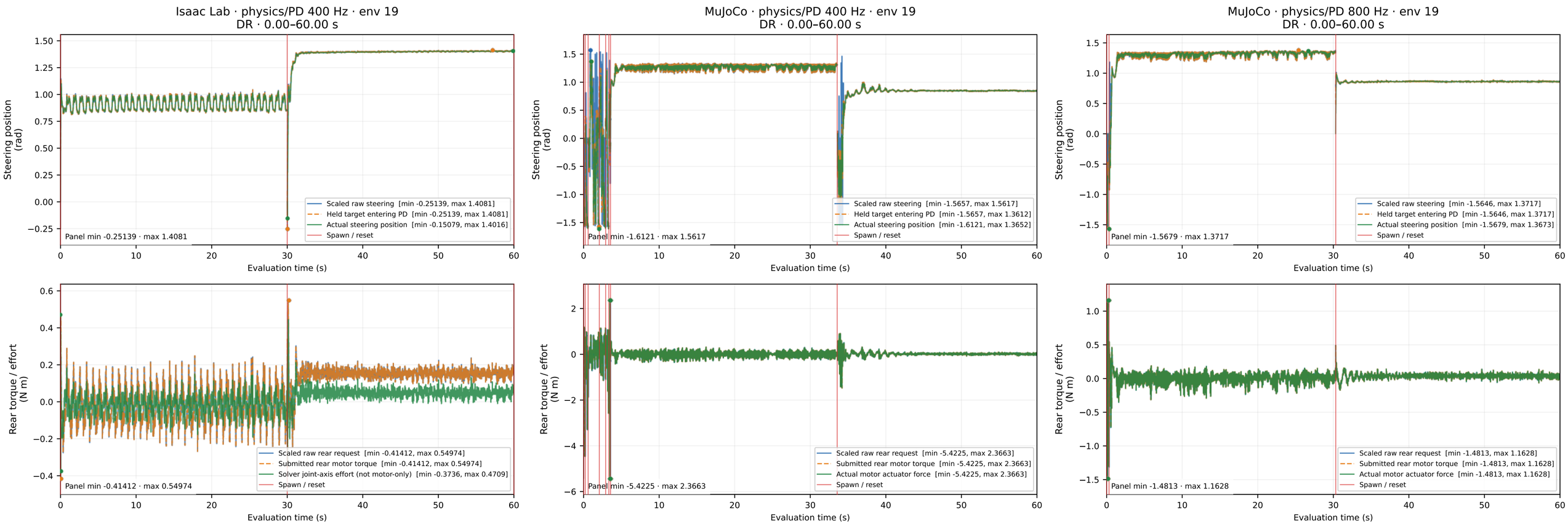
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 18; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.907865, 0.818429, 1.19701, 0.911529]
  CoM XYZ mm [-3.401, -1.14124, 3.49134]
  mu_s [0.760821, 0.576972, 0.674963, 0.632409]
  mu_d [0.670315, 0.576972, 0.601294, 0.589076]
  Kp [steer,rear] [10.1835, 0]; Kd [0.185575, 0]
  spawn roll 0.0792465 rad; rate -0.0982091 rad/s
  gyro bias XYZ rad/s [2.11285e-06, -0.00021168, 0.000204207]
  gravity bias YZ [0.00192337, -0.000994847]
E2 [30.00, 60.00) s:
  mass x [1.11078, 0.987952, 0.801457, 1.18803]
  CoM XYZ mm [3.67439, -1.87362, 0.500634]
  mu_s [0.659445, 0.659445, 0.592046, 0.760821]
  mu_d [0.547619, 0.547619, 0.534947, 0.670315]
  Kp [steer,rear] [9.01972, 0]; Kd [0.200358, 0]
  spawn roll -0.0920569 rad; rate -0.0218368 rad/s
  gyro bias XYZ rad/s [-0.000414288, -0.000261838, 7.58637e-05]
  gravity bias YZ [-0.00230514, 0.00243569]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 18; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.929714, 0.810859, 0.821888, 1.1962]
  CoM XYZ mm [3.6329, 0.505302, 4.68185]
  mu_s [0.637675, 0.669571, 0.757579, 0.733515]
  mu_d [0.637675, 0.669571, 0.70921, 0.558392]
  Kp [steer,rear] [9.63916, 0]; Kd [0.180825, 0]
  spawn roll -0.013749 rad; rate -0.194437 rad/s
  gyro bias XYZ rad/s [6.04374e-05, 0.00035961, 0.000141138]
  gravity bias YZ [-0.000964947, 0.00163817]
E2 [30.00, 60.00) s:
  mass x [1.1533, 1.07496, 0.981384, 1.16623]
  CoM XYZ mm [-4.98669, 2.89448, -0.758341]
  mu_s [0.538976, 0.728342, 0.748289, 0.546287]
  mu_d [0.538976, 0.728342, 0.689499, 0.546287]
  Kp [steer,rear] [10.7239, 0]; Kd [0.20776, 0]
  spawn roll -0.00389257 rad; rate 0.243496 rad/s
  gyro bias XYZ rad/s [0.000289537, 0.000290972, 0.000108021]
  gravity bias YZ [-0.00192329, 0.00103237]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 18; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.929714, 0.810859, 0.821888, 1.1962]
  CoM XYZ mm [3.6329, 0.505302, 4.68185]
  mu_s [0.637675, 0.669571, 0.757579, 0.733515]
  mu_d [0.637675, 0.669571, 0.70921, 0.558392]
  Kp [steer,rear] [9.63916, 0]; Kd [0.180825, 0]
  spawn roll -0.013749 rad; rate -0.194437 rad/s
  gyro bias XYZ rad/s [6.04374e-05, 0.00035961, 0.000141138]
  gravity bias YZ [-0.000964947, 0.00163817]
E2 [30.00, 60.00) s:
  mass x [1.1533, 1.07496, 0.981384, 1.16623]
  CoM XYZ mm [-4.98669, 2.89448, -0.758341]
  mu_s [0.538976, 0.728342, 0.748289, 0.546287]
  mu_d [0.538976, 0.728342, 0.689499, 0.546287]
  Kp [steer,rear] [10.7239, 0]; Kd [0.20776, 0]
  spawn roll -0.00389257 rad; rate 0.243496 rad/s
  gyro bias XYZ rad/s [0.000289537, 0.000290972, 0.000108021]
  gravity bias YZ [-0.00192329, 0.00103237]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 19 · full 60 s



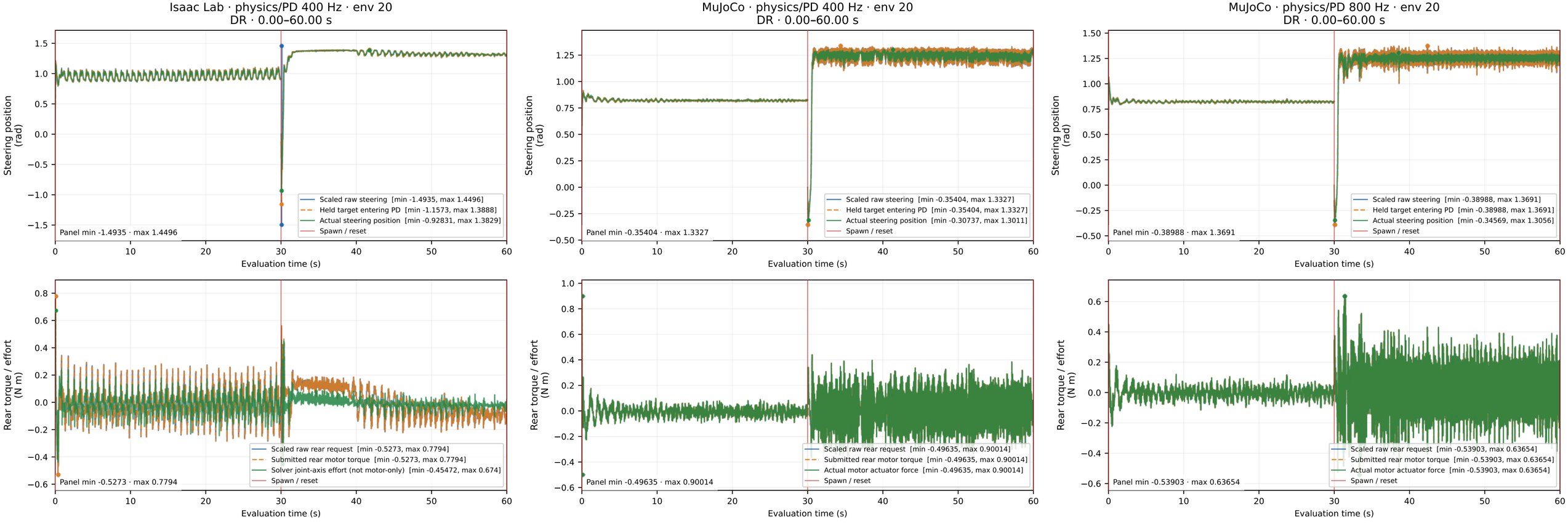
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 19; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.00429, 1.0378, 1.04659, 0.931652]
  CoM XYZ mm [2.20187, 2.93466, 3.4705]
  mu_s [0.598343, 0.703711, 0.576972, 0.764681]
  mu_d [0.598343, 0.703711, 0.576972, 0.764681]
  Kp [steer,rear] [10.741, 0]; Kd [0.205689, 0]
  spawn roll 0.0489928 rad; rate 0.230745 rad/s
  gyro bias XYZ rad/s [9.13323e-05, -7.36136e-05, -7.71893e-05]
  gravity bias YZ [0.00412858, 0.00170572]
E2 [30.00, 60.00) s:
  mass x [1.1418, 1.11129, 0.848825, 0.839723]
  CoM XYZ mm [-4.78759, -4.57539, -3.77113]
  mu_s [0.673378, 0.751971, 0.539956, 0.726175]
  mu_d [0.673378, 0.541224, 0.539956, 0.558574]
  Kp [steer,rear] [10.0199, 0]; Kd [0.219382, 0]
  spawn roll -0.00345242 rad; rate -0.178608 rad/s
  gyro bias XYZ rad/s [-0.00048902, -0.000157415, -0.000121871]
  gravity bias YZ [-0.0010147, 0.00292473]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 19; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.22) s:
  mass x [0.94975, 1.10877, 1.06846, 1.08366]
  CoM XYZ mm [-0.662472, 4.97269, 4.98093]
  mu_s [0.748289, 0.700521, 0.71155, 0.69316]
  mu_d [0.689499, 0.641329, 0.71155, 0.69316]
  Kp [steer,rear] [10.3709, 0]; Kd [0.204186, 0]
  spawn roll -0.0960328 rad; rate -0.238798 rad/s
  gyro bias XYZ rad/s [-0.000101158, 0.000235751, 0.000149081]
  gravity bias YZ [0.00322361, 0.00163551]
E2 [0.22, 0.60) s:
  mass x [1.16152, 1.09921, 0.911187, 0.870989]
  CoM XYZ mm [0.152926, 2.03583, -4.97262]
  mu_s [0.748289, 0.640867, 0.748289, 0.727426]
  mu_d [0.689499, 0.556841, 0.689499, 0.606358]
  Kp [steer,rear] [9.58862, 0]; Kd [0.205841, 0]
  spawn roll -0.0643534 rad; rate 0.0627738 rad/s
  gyro bias XYZ rad/s [0.000155158, -0.000431857, 0.000147435]
  gravity bias YZ [0.00131533, -0.0046838]
E3 [0.60, 2.08) s:
  mass x [0.859996, 0.850183, 1.05983, 0.888393]
  CoM XYZ mm [-4.74363, -4.76717, 4.30503]
  mu_s [0.749678, 0.700521, 0.748289, 0.757579]
  mu_d [0.741429, 0.641329, 0.689499, 0.70921]
  Kp [steer,rear] [10.3475, 0]; Kd [0.218515, 0]
  spawn roll -0.0411368 rad; rate 0.0241542 rad/s
  gyro bias XYZ rad/s [-1.22991e-05, -0.000179141, -3.93248e-06]
  gravity bias YZ [-0.00078878, 0.00196354]
E4 [2.08, 2.92) s:
  mass x [0.946687, 0.815144, 0.806943, 0.830702]
  CoM XYZ mm [0.369905, 0.356829, -3.35087]
  mu_s [0.791209, 0.666375, 0.666375, 0.666375]
  mu_d [0.767936, 0.519145, 0.519145, 0.519145]
  Kp [steer,rear] [9.02024, 0]; Kd [0.208556, 0]
  spawn roll -0.0149148 rad; rate 0.222933 rad/s
  gyro bias XYZ rad/s [0.000306687, -0.000102122, -3.7636e-05]
  gravity bias YZ [0.000331526, 0.000600647]
E5 [2.92, 3.34) s:
  mass x [1.02102, 0.991683, 0.990575, 0.808083]
  CoM XYZ mm [-1.19474, -2.0852, 2.67834]
  mu_s [0.611239, 0.749678, 0.528253, 0.616244]
  mu_d [0.611239, 0.741429, 0.528253, 0.586498]
  Kp [steer,rear] [10.9107, 0]; Kd [0.210466, 0]
  spawn roll -0.0600638 rad; rate 0.24504 rad/s
  gyro bias XYZ rad/s [-0.000331161, 0.000435587, -0.000245418]
  gravity bias YZ [0.00255999, 0.00126006]
E6 [3.34, 3.56) s:
  mass x [1.19377, 1.0168, 1.1573, 0.863469]
  CoM XYZ mm [2.70902, -3.40868, 1.48517]
  mu_s [0.616244, 0.631146, 0.640867, 0.69316]
  mu_d [0.586498, 0.631146, 0.556841, 0.69316]
  Kp [steer,rear] [9.02462, 0]; Kd [0.186443, 0]
  spawn roll 0.0919497 rad; rate 0.0209312 rad/s
  gyro bias XYZ rad/s [3.47464e-05, -6.53378e-05, -3.27807e-05]
  gravity bias YZ [-0.00232706, 0.00199089]
E7 [3.56, 33.56) s:
  mass x [0.96629, 1.15956, 1.12709, 1.14772]
  CoM XYZ mm [-1.28262, -1.86038, -1.8101]
  mu_s [0.538434, 0.736077, 0.538976, 0.71155]
  mu_d [0.538434, 0.699455, 0.538976, 0.71155]
  Kp [steer,rear] [10.4069, 0]; Kd [0.190008, 0]
  spawn roll 0.0358482 rad; rate -0.0162534 rad/s
  gyro bias XYZ rad/s [-2.00516e-05, -0.000129817, -0.000108534]
  gravity bias YZ [0.00326072, -0.00332642]
E8 [33.56, 60.00) s:
  mass x [0.936235, 1.02765, 0.988345, 1.15599]
  CoM XYZ mm [0.0885807, 3.90616, -3.79963]
  mu_s [0.528253, 0.611239, 0.568073, 0.559972]
  mu_d [0.528253, 0.611239, 0.568073, 0.502209]
  Kp [steer,rear] [10.8696, 0]; Kd [0.195431, 0]
  spawn roll -0.0335909 rad; rate 0.0919542 rad/s
  gyro bias XYZ rad/s [-0.000503023, 2.49212e-05, 0.000124039]
  gravity bias YZ [0.00275775, 0.00143926]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 19; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.32) s:
  mass x [0.94975, 1.10877, 1.06846, 1.08366]
  CoM XYZ mm [-0.662472, 4.97269, 4.98093]
  mu_s [0.748289, 0.700521, 0.71155, 0.69316]
  mu_d [0.689499, 0.641329, 0.71155, 0.69316]
  Kp [steer,rear] [10.3709, 0]; Kd [0.204186, 0]
  spawn roll -0.0960328 rad; rate -0.238798 rad/s
  gyro bias XYZ rad/s [-0.000101158, 0.000235751, 0.000149081]
  gravity bias YZ [0.00322361, 0.00163551]
E2 [0.32, 30.32) s:
  mass x [1.11003, 0.829622, 0.950311, 0.915982]
  CoM XYZ mm [-3.24813, -3.4527, 1.06287]
  mu_s [0.791209, 0.538976, 0.791209, 0.568073]
  mu_d [0.767936, 0.538976, 0.767936, 0.568073]
  Kp [steer,rear] [10.5427, 0]; Kd [0.19121, 0]
  spawn roll -0.0509555 rad; rate 0.0191255 rad/s
  gyro bias XYZ rad/s [-4.50528e-05, 6.58198e-05, -0.000161347]
  gravity bias YZ [0.00208725, -0.00132221]
E3 [30.32, 60.00) s:
  mass x [0.991628, 0.905446, 0.945454, 0.992245]
  CoM XYZ mm [-1.76291, 3.41365, -2.11742]
  mu_s [0.700521, 0.669571, 0.727426, 0.71155]
  mu_d [0.641329, 0.669571, 0.606358, 0.71155]
  Kp [steer,rear] [9.00862, 0]; Kd [0.188959, 0]
  spawn roll 0.0561029 rad; rate 0.159022 rad/s
  gyro bias XYZ rad/s [-0.000149479, 0.000295165, 7.51162e-05]
  gravity bias YZ [-0.000705262, -0.000130155]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 20 · full 60 s



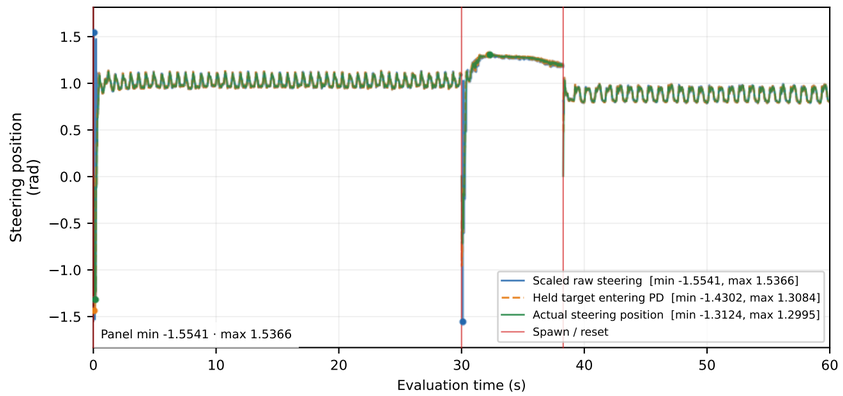
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

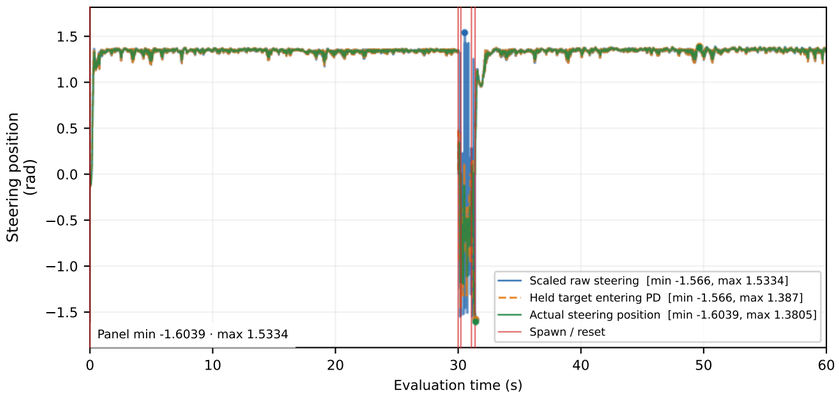
```
SAMPLED DR: env 20; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.12145, 0.998445, 1.07913, 1.05785]
  CoM XYZ mm [-1.91383, 1.48481, 4.09499]
  mu_s [0.576972, 0.760821, 0.713129, 0.602694]
  mu_d [0.576972, 0.670315, 0.713129, 0.602694]
  Kp [steer,rear] [9.81716, 0]; Kd [0.218603, 0]
  spawn roll 0.0737982 rad; rate 0.0786717 rad/s
  gyro bias XYZ rad/s [0.000357197, -0.000176057, -0.000173052]
  gravity bias YZ [-0.000308403, 0.000634569]
E2 [30.00, 60.00) s:
  mass x [0.892234, 0.823505, 0.947098, 0.964512]
  CoM XYZ mm [-0.711992, -3.47873, -4.83014]
  mu_s [0.576972, 0.726175, 0.547743, 0.785466]
  mu_d [0.576972, 0.558574, 0.547743, 0.52258]
  Kp [steer,rear] [9.22803, 0]; Kd [0.188455, 0]
  spawn roll -0.0530459 rad; rate 0.0152707 rad/s
  gyro bias XYZ rad/s [8.27099e-05, 0.000162087, 9.96022e-05]
  gravity bias YZ [0.00105849, -0.00303754]
Six significant digits shown; full precision: sampled_dr_values.json.
```

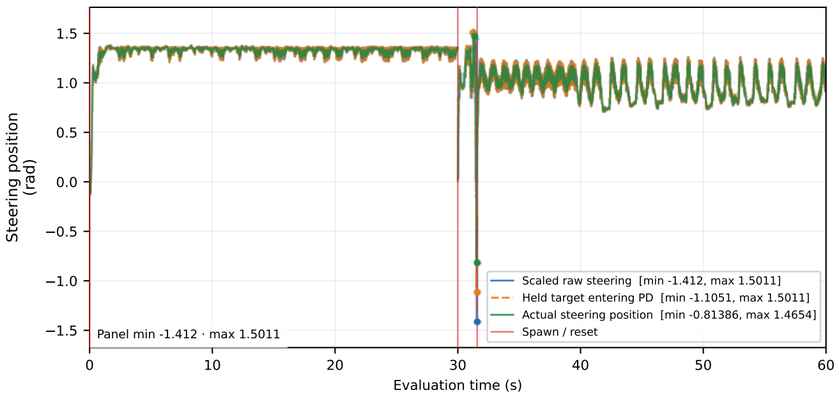
```
SAMPLED DR: env 20; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.04373, 1.09911, 1.11228, 0.915142]
  CoM XYZ mm [1.93236, 4.26073, 0.13635]
  mu_s [0.541939, 0.71155, 0.727426, 0.736077]
  mu_d [0.534359, 0.71155, 0.606358, 0.699455]
  Kp [steer,rear] [9.47149, 0]; Kd [0.180846, 0]
  spawn roll 0.0663592 rad; rate 0.0689231 rad/s
  gyro bias XYZ rad/s [-7.44315e-05, 0.000269738, -0.000209089]
  gravity bias YZ [-0.000127877, 0.000950033]
E2 [30.00, 60.00) s:
  mass x [0.834144, 1.07257, 0.865316, 0.977668]
  CoM XYZ mm [-2.13668, -1.82604, -4.49653]
  mu_s [0.723429, 0.538434, 0.637675, 0.611239]
  mu_d [0.723429, 0.538434, 0.637675, 0.611239]
  Kp [steer,rear] [9.62454, 0]; Kd [0.191168, 0]
  spawn roll -0.0238987 rad; rate -0.10103 rad/s
  gyro bias XYZ rad/s [-3.95046e-05, 0.000102812, 0.000123087]
  gravity bias YZ [-2.13084e-05, 3.84637e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

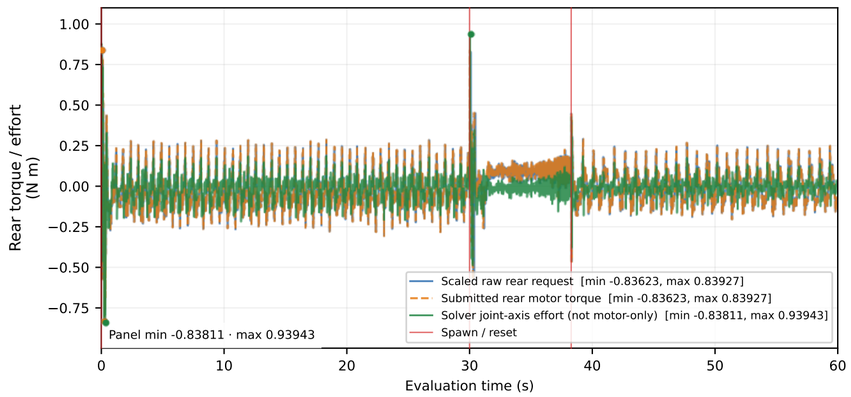
```
SAMPLED DR: env 20; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.04373, 1.09911, 1.11228, 0.915142]
  CoM XYZ mm [1.93236, 4.26073, 0.13635]
  mu_s [0.541939, 0.71155, 0.727426, 0.736077]
  mu_d [0.534359, 0.71155, 0.606358, 0.699455]
  Kp [steer,rear] [9.47149, 0]; Kd [0.180846, 0]
  spawn roll 0.0663592 rad; rate 0.0689231 rad/s
  gyro bias XYZ rad/s [-7.44315e-05, 0.000269738, -0.000209089]
  gravity bias YZ [-0.000127877, 0.000950033]
E2 [30.00, 60.00) s:
  mass x [0.834144, 1.07257, 0.865316, 0.977668]
  CoM XYZ mm [-2.13668, -1.82604, -4.49653]
  mu_s [0.723429, 0.538434, 0.637675, 0.611239]
  mu_d [0.723429, 0.538434, 0.637675, 0.611239]
  Kp [steer,rear] [9.62454, 0]; Kd [0.191168, 0]
  spawn roll -0.0238987 rad; rate -0.10103 rad/s
  gyro bias XYZ rad/s [-3.95046e-05, 0.000102812, 0.000123087]
  gravity bias YZ [-2.13084e-05, 3.84637e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

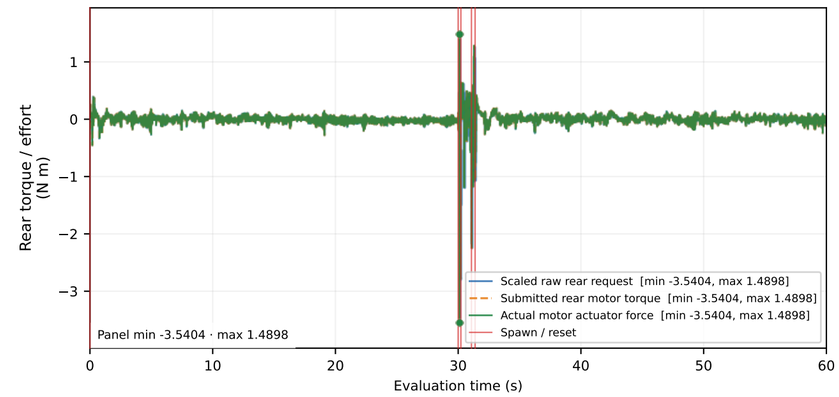
model_2500 · Demand versus actuation · env 21 · full 60 s

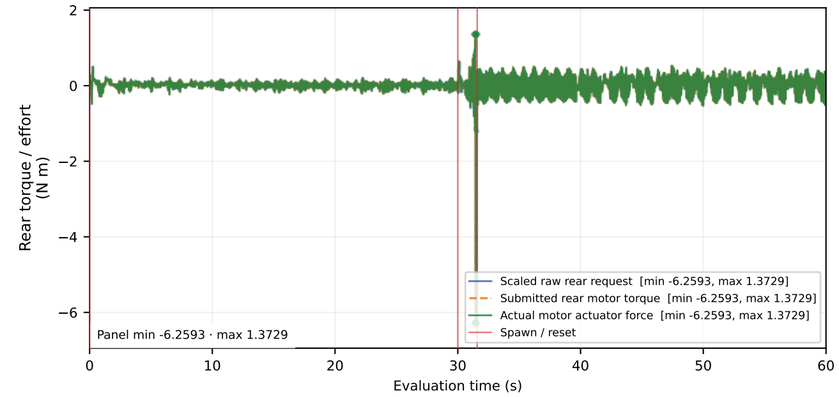












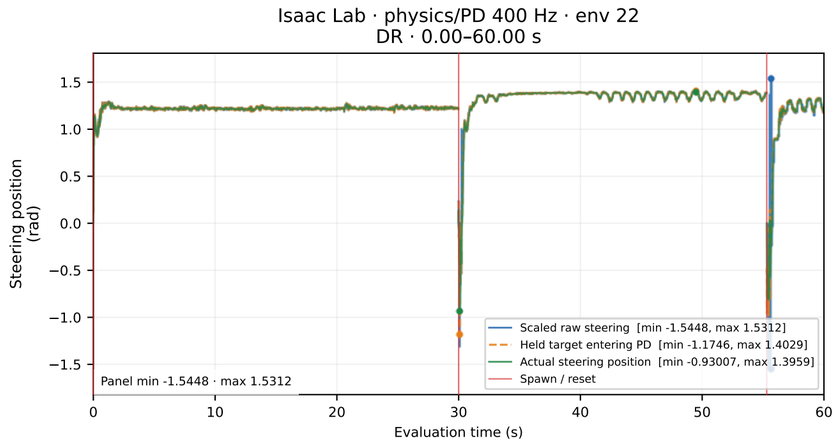
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

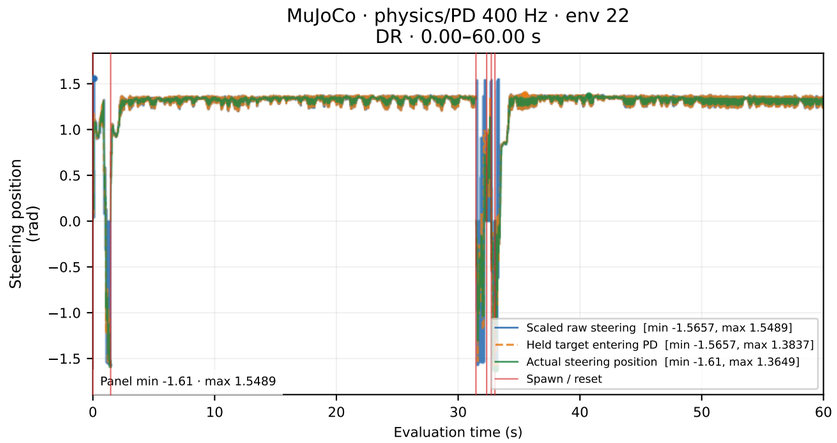
```
SAMPLED DR: env 21; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.16234, 0.931657, 0.874102, 0.941904]
  CoM XYZ mm [-0.778958, -0.191033, 4.88482]
  mu_s [0.576972, 0.764681, 0.760821, 0.532645]
  mu_d [0.576972, 0.764681, 0.670315, 0.532645]
  Kp [steer,rear] [10.558, 0]; Kd [0.190695, 0]
  spawn roll -0.0385937 rad; rate -0.201642 rad/s
  gyro bias XYZ rad/s [-0.000537161, -9.48584e-06, -0.000165546]
  gravity bias YZ [5.74883e-05, -0.000606045]
E2 [30.00, 38.28) s:
  mass x [1.07496, 1.12446, 0.947485, 1.11741]
  CoM XYZ mm [0.648886, -4.50441, 1.39657]
  mu_s [0.681711, 0.659445, 0.647391, 0.751971]
  mu_d [0.611756, 0.547619, 0.647391, 0.541224]
  Kp [steer,rear] [10.4772, 0]; Kd [0.21541, 0]
  spawn roll -0.0491127 rad; rate -0.225306 rad/s
  gyro bias XYZ rad/s [-9.14914e-05, 4.75103e-05, 3.46783e-05]
  gravity bias YZ [0.00205013, 0.00204799]
E3 [38.28, 60.00) s:
  mass x [0.925078, 0.848264, 1.02357, 1.1917]
  CoM XYZ mm [2.09817, 3.03209, -4.31511]
  mu_s [0.785466, 0.559809, 0.713129, 0.726175]
  mu_d [0.52258, 0.559809, 0.713129, 0.558574]
  Kp [steer,rear] [10.7923, 0]; Kd [0.180827, 0]
  spawn roll 0.0395315 rad; rate 0.215036 rad/s
  gyro bias XYZ rad/s [0.000233514, -4.39318e-05, 0.000166626]
  gravity bias YZ [-0.000890945, -0.00300513]
Six significant digits shown; full precision: sampled_dr_values.json.
```

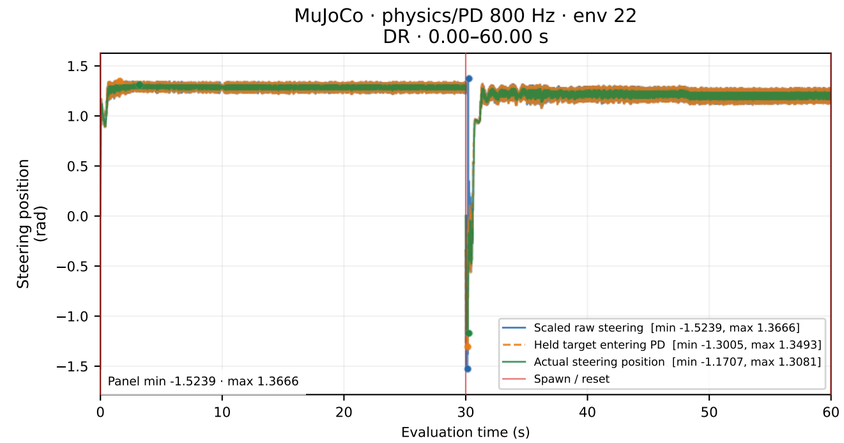
```
SAMPLED DR: env 21; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.02403, 1.1969, 0.878675, 1.00212]
  CoM XYZ mm [-2.99621, -3.27672, -1.3238]
  mu_s [0.538434, 0.748289, 0.727426, 0.640016]
  mu_d [0.538434, 0.689499, 0.606358, 0.513141]
  Kp [steer,rear] [10.979, 0]; Kd [0.199518, 0]
  spawn roll -0.0234954 rad; rate -0.0591556 rad/s
  gyro bias XYZ rad/s [-0.000390077, -5.04864e-05, -1.20618e-05]
  gravity bias YZ [0.00233172, -0.000743702]
E2 [30.00, 30.22) s:
  mass x [0.857328, 1.19008, 0.849359, 0.847076]
  CoM XYZ mm [2.84863, -3.96132, -1.49141]
  mu_s [0.631146, 0.538434, 0.749678, 0.700521]
  mu_d [0.631146, 0.538434, 0.741429, 0.641329]
  Kp [steer,rear] [9.09774, 0]; Kd [0.189967, 0]
  spawn roll 0.0231506 rad; rate -0.123269 rad/s
  gyro bias XYZ rad/s [7.24468e-05, -0.00027306, 0.000115335]
  gravity bias YZ [0.00107675, -0.00217639]
E3 [30.22, 31.08) s:
  mass x [1.151, 1.08475, 0.979946, 1.11803]
  CoM XYZ mm [-1.63771, 4.35002, 3.89226]
  mu_s [0.640016, 0.631146, 0.538434, 0.749678]
  mu_d [0.513141, 0.631146, 0.538434, 0.741429]
  Kp [steer,rear] [10.3954, 0]; Kd [0.209618, 0]
  spawn roll -0.0395964 rad; rate -0.196297 rad/s
  gyro bias XYZ rad/s [-0.000107801, -0.000248744, -0.00030249]
  gravity bias YZ [0.00149914, -0.00151463]
E4 [31.08, 31.38) s:
  mass x [1.02905, 1.06181, 1.19, 0.8162]
  CoM XYZ mm [4.10381, 4.61135, -0.571818]
  mu_s [0.71155, 0.559972, 0.631146, 0.541939]
  mu_d [0.71155, 0.502209, 0.631146, 0.534359]
  Kp [steer,rear] [9.43782, 0]; Kd [0.200375, 0]
  spawn roll -0.0668429 rad; rate 0.139502 rad/s
  gyro bias XYZ rad/s [-9.08241e-05, 1.6222e-05, -0.000179747]
  gravity bias YZ [-8.45339e-05, 0.000930439]
E5 [31.38, 60.00) s:
  mass x [1.10165, 0.913317, 0.839364, 1.16837]
  CoM XYZ mm [-3.28761, -3.97198, -0.854517]
  mu_s [0.597748, 0.616244, 0.538976, 0.538976]
  mu_d [0.597748, 0.586498, 0.538976, 0.538976]
  Kp [steer,rear] [10.3977, 0]; Kd [0.184072, 0]
  spawn roll -0.0282585 rad; rate 0.221166 rad/s
  gyro bias XYZ rad/s [-0.000123733, -3.5891e-05, 0.00017009]
  gravity bias YZ [-0.00179086, 0.000860961]
Six significant digits shown; full precision: sampled_dr_values.json.
```

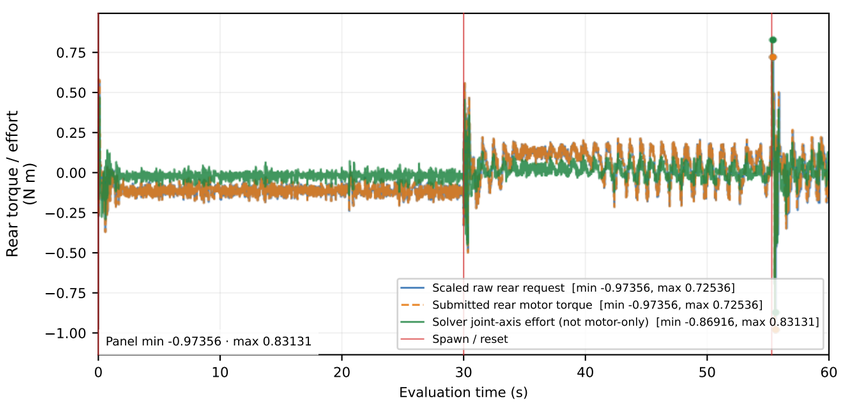
```
SAMPLED DR: env 21; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.02403, 1.1969, 0.878675, 1.00212]
  CoM XYZ mm [-2.99621, -3.27672, -1.3238]
  mu_s [0.538434, 0.748289, 0.727426, 0.640016]
  mu_d [0.538434, 0.689499, 0.606358, 0.513141]
  Kp [steer,rear] [10.979, 0]; Kd [0.199518, 0]
  spawn roll -0.0234954 rad; rate -0.0591556 rad/s
  gyro bias XYZ rad/s [-0.000390077, -5.04864e-05, -1.20618e-05]
  gravity bias YZ [0.00233172, -0.000743702]
E2 [30.00, 31.58) s:
  mass x [0.857328, 1.19008, 0.849359, 0.847076]
  CoM XYZ mm [2.84863, -3.96132, -1.49141]
  mu_s [0.631146, 0.538434, 0.749678, 0.700521]
  mu_d [0.631146, 0.538434, 0.741429, 0.641329]
  Kp [steer,rear] [9.09774, 0]; Kd [0.189967, 0]
  spawn roll 0.0231506 rad; rate -0.123269 rad/s
  gyro bias XYZ rad/s [7.24468e-05, -0.00027306, 0.000115335]
  gravity bias YZ [0.00107675, -0.00217639]
E3 [31.58, 60.00) s:
  mass x [0.824134, 0.974825, 1.1904, 0.828339]
  CoM XYZ mm [2.50539, 2.62759, -4.6295]
  mu_s [0.704749, 0.727426, 0.69316, 0.559972]
  mu_d [0.541926, 0.606358, 0.69316, 0.502209]
  Kp [steer,rear] [10.9743, 0]; Kd [0.184708, 0]
  spawn roll 0.0205726 rad; rate 0.0219324 rad/s
  gyro bias XYZ rad/s [-0.000251298, 0.000132695, 5.19054e-05]
  gravity bias YZ [0.000527718, -0.00169607]
Six significant digits shown; full precision: sampled_dr_values.json.
```

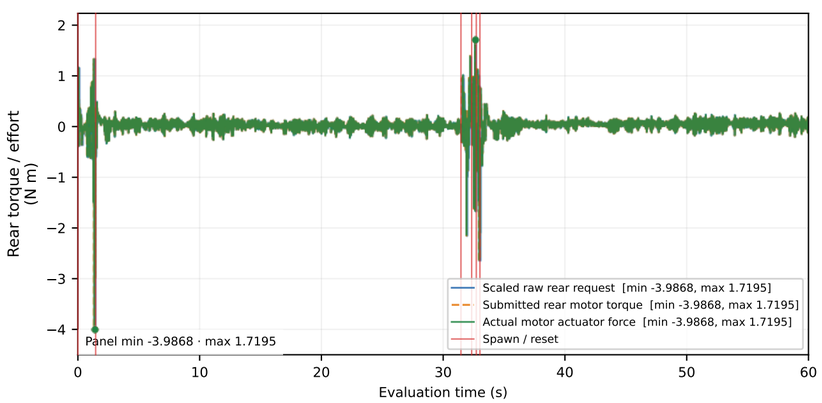
model_2500 · Demand versus actuation · env 22 · full 60 s

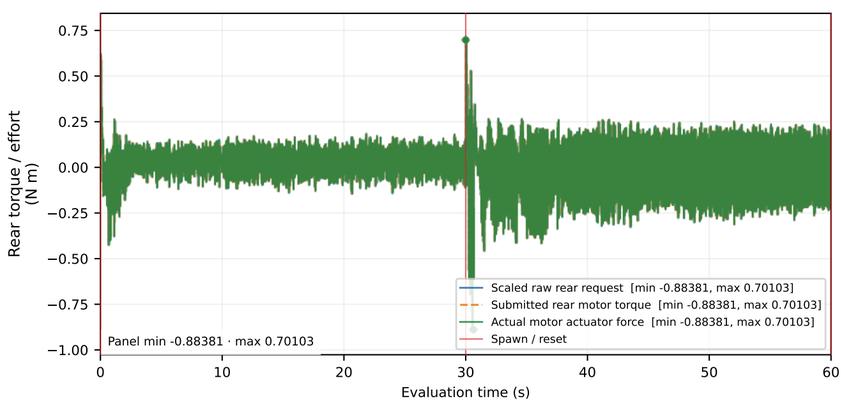












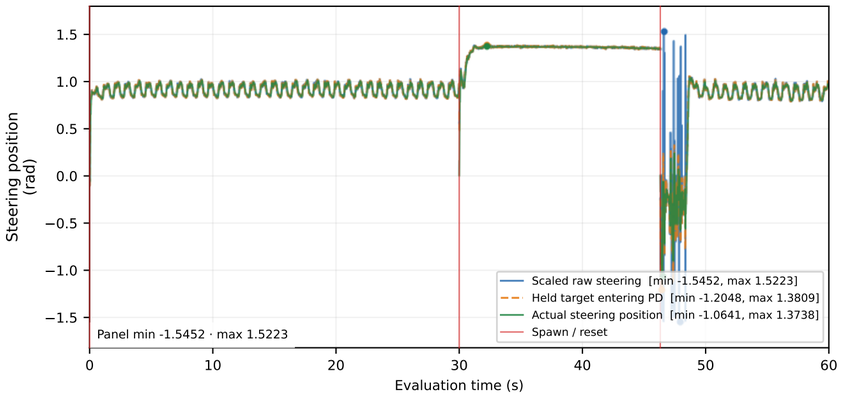
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

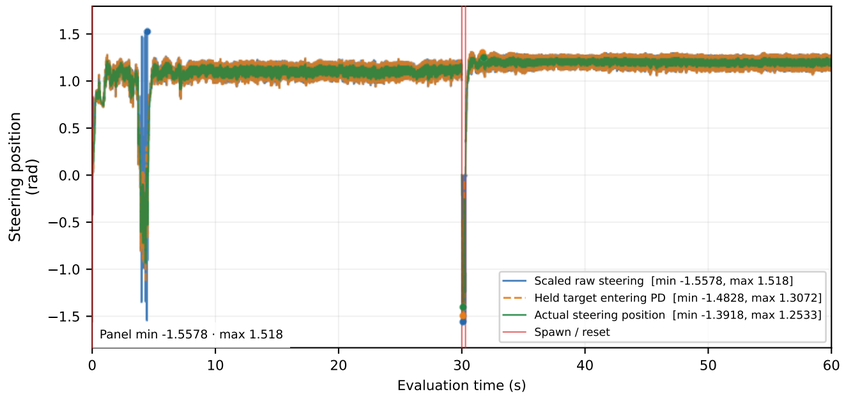
```
SAMPLED DR: env 22; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.09989, 0.913112, 1.11555, 1.16651]
  CoM XYZ mm [3.35515, -2.84438, 2.06447]
  mu_s [0.774409, 0.681711, 0.602694, 0.579974]
  mu_d [0.561095, 0.611756, 0.602694, 0.579974]
  Kp [steer,rear] [9.75244, 0]; Kd [0.205282, 0]
  spawn roll 0.0130913 rad; rate -0.061652 rad/s
  gyro bias XYZ rad/s [0.000563281, -0.000617255, 0.000169907]
  gravity bias YZ [0.0011242, 6.85766e-05]
E2 [30.00, 55.30) s:
  mass x [1.19175, 1.10707, 1.14586, 0.845834]
  CoM XYZ mm [-2.34434, -3.65275, -1.65352]
  mu_s [0.774409, 0.659445, 0.532645, 0.533455]
  mu_d [0.561095, 0.547619, 0.532645, 0.533455]
  Kp [steer,rear] [10.0599, 0]; Kd [0.215464, 0]
  spawn roll -0.0596136 rad; rate 0.198514 rad/s
  gyro bias XYZ rad/s [-4.66532e-05, -9.9831e-05, -0.000144437]
  gravity bias YZ [0.00113814, 0.00285899]
E3 [55.30, 60.00) s:
  mass x [0.861296, 0.981169, 1.11128, 1.19447]
  CoM XYZ mm [-4.05623, -2.17658, -2.20144]
  mu_s [0.785466, 0.794338, 0.598343, 0.547743]
  mu_d [0.52258, 0.526209, 0.598343, 0.547743]
  Kp [steer,rear] [9.01218, 0]; Kd [0.195789, 0]
  spawn roll -0.0596743 rad; rate -0.183025 rad/s
  gyro bias XYZ rad/s [0.000484342, -0.000219137, -0.000125366]
  gravity bias YZ [-0.00415205, 0.00169506]
Six significant digits shown; full precision: sampled_dr_values.json.
```

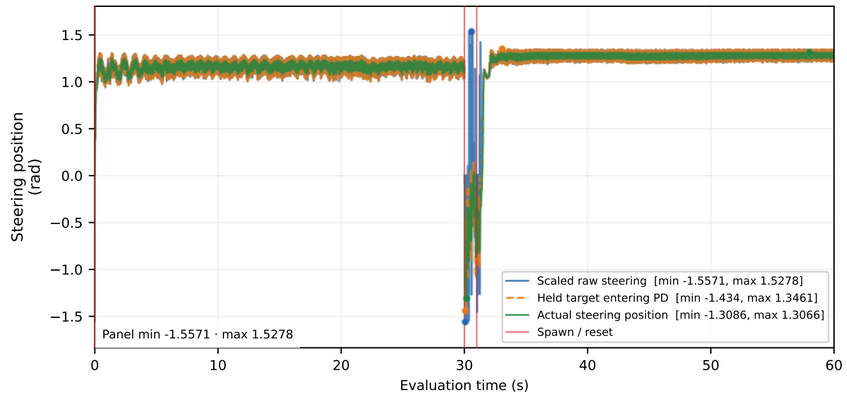
```
SAMPLED DR: env 22; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.46) s:
  mass x [1.1774, 1.00934, 0.826284, 0.927424]
  CoM XYZ mm [3.58417, -4.15441, 4.85365]
  mu_s [0.704749, 0.631146, 0.736077, 0.538434]
  mu_d [0.541926, 0.631146, 0.699455, 0.538434]
  Kp [steer,rear] [9.06763, 0]; Kd [0.202224, 0]
  spawn roll 0.0374538 rad; rate 0.0326712 rad/s
  gyro bias XYZ rad/s [-0.000140744, -0.000192308, 0.000153964]
  gravity bias YZ [-0.000118989, 0.00169111]
E2 [1.46, 31.46) s:
  mass x [0.836772, 1.01964, 1.09406, 1.12996]
  CoM XYZ mm [-2.6505, -3.9608, -3.50786]
  mu_s [0.727426, 0.757579, 0.733515, 0.568073]
  mu_d [0.606358, 0.70921, 0.558392, 0.568073]
  Kp [steer,rear] [9.24874, 0]; Kd [0.218677, 0]
  spawn roll -0.0268408 rad; rate 0.194205 rad/s
  gyro bias XYZ rad/s [-6.24926e-05, 7.9838e-05, -0.000160236]
  gravity bias YZ [-0.00274159, -0.000387806]
E3 [31.46, 32.34) s:
  mass x [0.890461, 1.13429, 0.828668, 1.07681]
  CoM XYZ mm [4.4716, -2.65451, -3.88171]
  mu_s [0.700521, 0.69316, 0.749678, 0.723429]
  mu_d [0.641329, 0.69316, 0.741429, 0.723429]
  Kp [steer,rear] [10.2565, 0]; Kd [0.217698, 0]
  spawn roll -0.0679824 rad; rate -0.0581301 rad/s
  gyro bias XYZ rad/s [1.34444e-05, 1.75308e-06, -1.70683e-05]
  gravity bias YZ [-0.000319099, -0.000625235]
E4 [32.34, 32.72) s:
  mass x [0.875501, 1.04263, 0.881064, 1.04699]
  CoM XYZ mm [-1.33345, -4.39802, 3.62697]
  mu_s [0.633024, 0.538434, 0.757579, 0.704749]
  mu_d [0.568172, 0.538434, 0.70921, 0.541926]
  Kp [steer,rear] [9.25831, 0]; Kd [0.198036, 0]
  spawn roll 0.0501151 rad; rate 0.207299 rad/s
  gyro bias XYZ rad/s [-0.000203632, 0.000105006, 0.000199391]
  gravity bias YZ [0.00161353, 0.00366085]
E5 [32.72, 33.02) s:
  mass x [1.14551, 1.11546, 0.866295, 0.830485]
  CoM XYZ mm [0.0241074, 3.01149, 0.832467]
  mu_s [0.733515, 0.559972, 0.633024, 0.736077]
  mu_d [0.558392, 0.502209, 0.568172, 0.699455]
  Kp [steer,rear] [9.22348, 0]; Kd [0.193362, 0]
  spawn roll -0.0414557 rad; rate -0.127543 rad/s
  gyro bias XYZ rad/s [-0.000168263, 0.000126677, -0.000223143]
  gravity bias YZ [-0.00219662, 0.000639729]
E6 [33.02, 60.00) s:
  mass x [0.873044, 0.892759, 1.0757, 0.971105]
  CoM XYZ mm [1.30932, -4.6614, -3.35964]
  mu_s [0.69316, 0.700521, 0.723429, 0.736077]
  mu_d [0.69316, 0.641329, 0.723429, 0.699455]
  Kp [steer,rear] [10.374, 0]; Kd [0.18768, 0]
  spawn roll -0.0733841 rad; rate -0.114182 rad/s
  gyro bias XYZ rad/s [0.000219676, 0.000425842, 0.000271977]
  gravity bias YZ [-0.00149297, -0.00327126]
Six significant digits shown; full precision: sampled_dr_values.json.
```

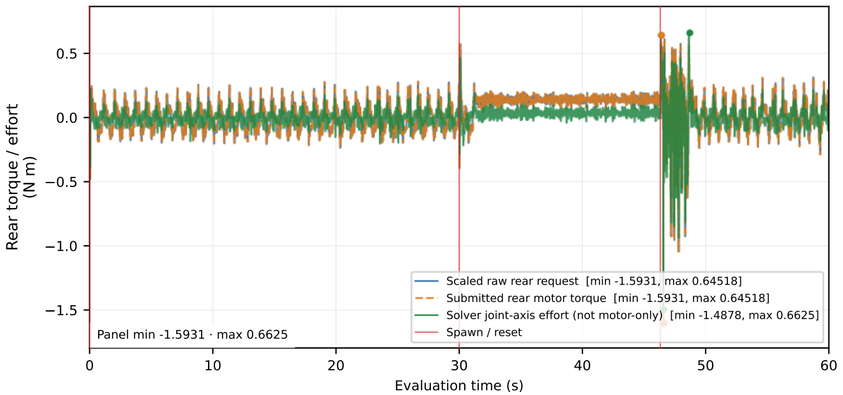
```
SAMPLED DR: env 22; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.1774, 1.00934, 0.826284, 0.927424]
  CoM XYZ mm [3.58417, -4.15441, 4.85365]
  mu_s [0.704749, 0.631146, 0.736077, 0.538434]
  mu_d [0.541926, 0.631146, 0.699455, 0.538434]
  Kp [steer,rear] [9.06763, 0]; Kd [0.202224, 0]
  spawn roll 0.0374538 rad; rate 0.0326712 rad/s
  gyro bias XYZ rad/s [-0.000140744, -0.000192308, 0.000153964]
  gravity bias YZ [-0.000118989, 0.00169111]
E2 [30.00, 60.00) s:
  mass x [1.10383, 1.04039, 1.03433, 1.08747]
  CoM XYZ mm [0.44504, -1.15206, 3.22108]
  mu_s [0.749678, 0.748289, 0.748289, 0.749678]
  mu_d [0.741429, 0.689499, 0.689499, 0.741429]
  Kp [steer,rear] [10.2047, 0]; Kd [0.202195, 0]
  spawn roll -0.069652 rad; rate -0.0482418 rad/s
  gyro bias XYZ rad/s [0.000186282, 2.88473e-05, -0.000391111]
  gravity bias YZ [0.0013609, -0.00096558]
Six significant digits shown; full precision: sampled_dr_values.json.
```

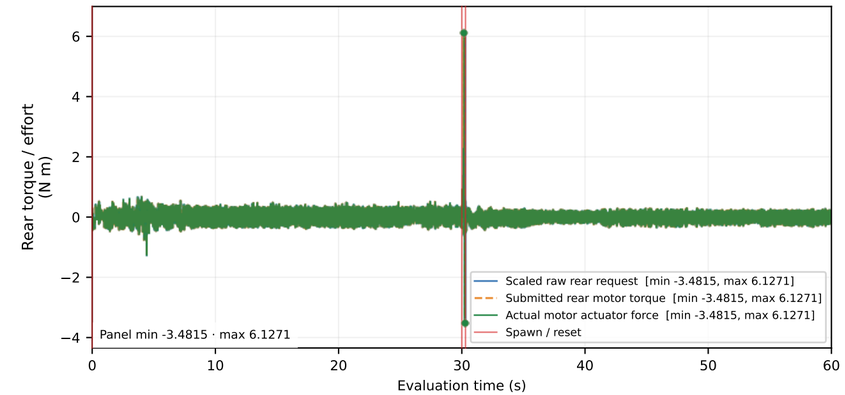
model_2500 · Demand versus actuation · env 23 · full 60 s

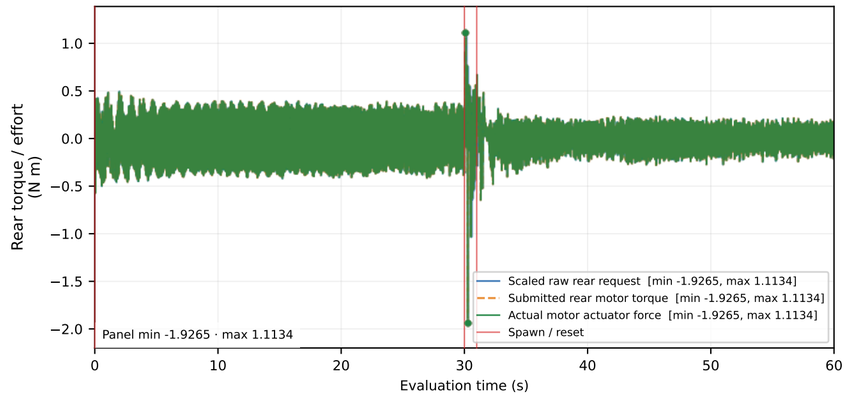












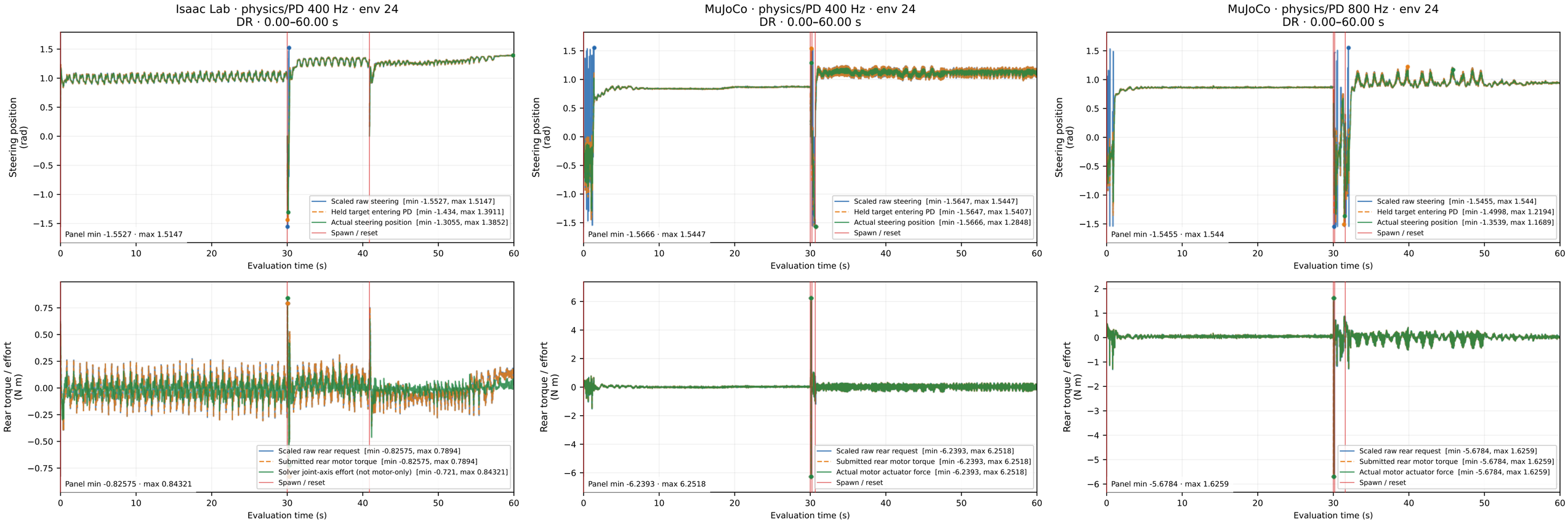
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 23; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.902276, 1.07803, 1.00334, 0.809615]
  CoM XYZ mm [3.325, 2.94543, -3.45263]
  mu_s [0.703711, 0.531594, 0.713129, 0.713129]
  mu_d [0.703711, 0.531594, 0.713129, 0.713129]
  Kp [steer,rear] [9.33442, 0]; Kd [0.190002, 0]
  spawn roll 0.0350462 rad; rate -0.216377 rad/s
  gyro bias XYZ rad/s [0.00011978, -9.29031e-05, 0.000219528]
  gravity bias YZ [0.00326947, 0.00403598]
E2 [30.00, 46.32) s:
  mass x [0.960775, 1.19447, 1.09524, 0.929989]
  CoM XYZ mm [1.72573, -4.97271, -3.64296]
  mu_s [0.539956, 0.787792, 0.647391, 0.584424]
  mu_d [0.539956, 0.617134, 0.647391, 0.584424]
  Kp [steer,rear] [10.5864, 0]; Kd [0.219723, 0]
  spawn roll 0.00635613 rad; rate 0.130233 rad/s
  gyro bias XYZ rad/s [0.000137682, 0.000124554, 0.000185817]
  gravity bias YZ [-9.12121e-06, -0.0011008]
E3 [46.32, 60.00) s:
  mass x [1.07932, 0.846165, 1.19697, 1.03142]
  CoM XYZ mm [-0.91359, 2.53814, -1.48316]
  mu_s [0.602694, 0.726175, 0.560535, 0.659445]
  mu_d [0.602694, 0.558574, 0.560535, 0.547619]
  Kp [steer,rear] [9.03699, 0]; Kd [0.194668, 0]
  spawn roll -0.0463249 rad; rate -0.0693574 rad/s
  gyro bias XYZ rad/s [-0.000236129, 4.4613e-05, -6.95118e-05]
  gravity bias YZ [-0.00167178, -0.000764033]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 23; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.818957, 1.00729, 0.949943, 0.891471]
  CoM XYZ mm [4.41723, 0.945015, -2.21149]
  mu_s [0.568073, 0.757579, 0.559972, 0.538434]
  mu_d [0.568073, 0.70921, 0.502209, 0.538434]
  Kp [steer,rear] [9.0213, 0]; Kd [0.181029, 0]
  spawn roll 0.0108942 rad; rate 0.249432 rad/s
  gyro bias XYZ rad/s [0.000368914, -1.73389e-05, 0.000229485]
  gravity bias YZ [-0.000507162, -0.00263101]
E2 [30.00, 30.30) s:
  mass x [1.09747, 1.09297, 0.987042, 0.984079]
  CoM XYZ mm [4.1462, 4.50607, -3.18988]
  mu_s [0.538976, 0.640867, 0.791209, 0.757579]
  mu_d [0.538976, 0.556841, 0.767936, 0.70921]
  Kp [steer,rear] [9.91003, 0]; Kd [0.187922, 0]
  spawn roll -0.0730179 rad; rate -0.233558 rad/s
  gyro bias XYZ rad/s [9.01731e-06, 0.0001623, 6.279e-05]
  gravity bias YZ [0.00214148, -0.00207834]
E3 [30.30, 60.00) s:
  mass x [0.939329, 1.19123, 0.965854, 1.16404]
  CoM XYZ mm [3.06904, -0.920233, 2.4337]
  mu_s [0.723429, 0.568073, 0.71155, 0.748289]
  mu_d [0.723429, 0.568073, 0.71155, 0.689499]
  Kp [steer,rear] [9.03876, 0]; Kd [0.218117, 0]
  spawn roll -0.00209098 rad; rate 0.0701048 rad/s
  gyro bias XYZ rad/s [-0.000131661, 0.000121779, -0.000318406]
  gravity bias YZ [-0.00128338, 0.000232989]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 23; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.818957, 1.00729, 0.949943, 0.891471]
  CoM XYZ mm [4.41723, 0.945015, -2.21149]
  mu_s [0.568073, 0.757579, 0.559972, 0.538434]
  mu_d [0.568073, 0.70921, 0.502209, 0.538434]
  Kp [steer,rear] [9.0213, 0]; Kd [0.181029, 0]
  spawn roll 0.0108942 rad; rate 0.249432 rad/s
  gyro bias XYZ rad/s [0.000368914, -1.73389e-05, 0.000229485]
  gravity bias YZ [-0.000507162, -0.00263101]
E2 [30.00, 31.00) s:
  mass x [1.09747, 1.09297, 0.987042, 0.984079]
  CoM XYZ mm [4.1462, 4.50607, -3.18988]
  mu_s [0.538976, 0.640867, 0.791209, 0.757579]
  mu_d [0.538976, 0.556841, 0.767936, 0.70921]
  Kp [steer,rear] [9.91003, 0]; Kd [0.187922, 0]
  spawn roll -0.0730179 rad; rate -0.233558 rad/s
  gyro bias XYZ rad/s [9.01731e-06, 0.0001623, 6.279e-05]
  gravity bias YZ [0.00214148, -0.00207834]
E3 [31.00, 60.00) s:
  mass x [0.918573, 0.834252, 0.82717, 1.00434]
  CoM XYZ mm [1.36049, -2.84484, 2.19587]
  mu_s [0.611239, 0.640867, 0.69316, 0.559972]
  mu_d [0.611239, 0.556841, 0.69316, 0.502209]
  Kp [steer,rear] [10.1131, 0]; Kd [0.189644, 0]
  spawn roll -0.0488518 rad; rate -0.186283 rad/s
  gyro bias XYZ rad/s [-0.000249822, 0.000127333, 2.32506e-06]
  gravity bias YZ [0.00168369, -0.00131681]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 24 · full 60 s



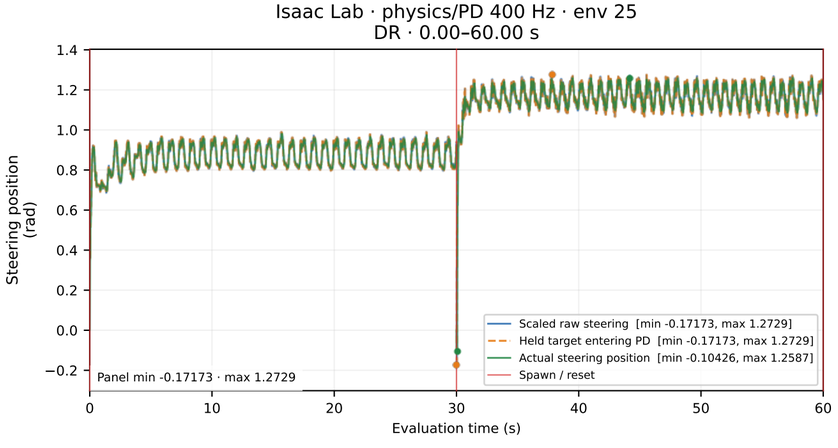
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

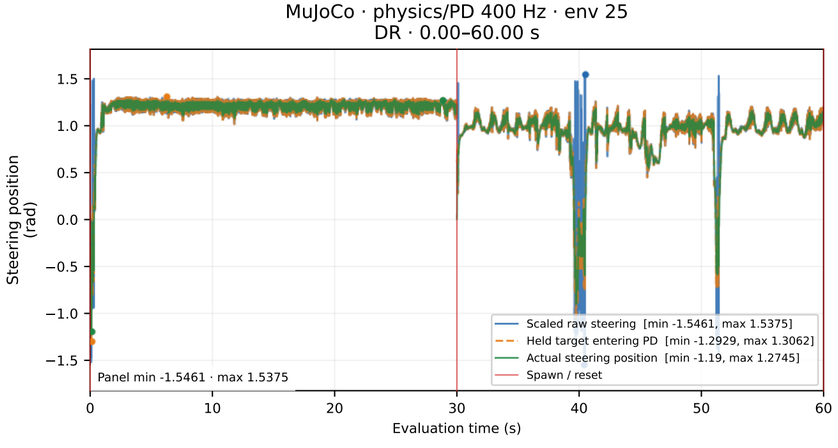
```
SAMPLED DR: env 24; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.02376, 1.12788, 0.816827, 1.00857]
  CoM XYZ mm [4.92334, 1.35443, -0.119776]
  mu_s [0.532645, 0.559809, 0.547743, 0.673378]
  mu_d [0.532645, 0.559809, 0.547743, 0.673378]
  Kp [steer,rear] [9.66888, 0]; Kd [0.192899, 0]
  spawn roll 0.0707435 rad; rate 0.193241 rad/s
  gyro bias XYZ rad/s [0.000166648, 0.000478024, 0.00031368]
  gravity bias YZ [3.62278e-05, 0.005236]
E2 [30.00, 40.88) s:
  mass x [1.05178, 0.919355, 0.940272, 1.03161]
  CoM XYZ mm [-4.1834, -1.93626, -2.92845]
  mu_s [0.547743, 0.592046, 0.632409, 0.576972]
  mu_d [0.547743, 0.534947, 0.589076, 0.576972]
  Kp [steer,rear] [10.494, 0]; Kd [0.192404, 0]
  spawn roll -0.0747468 rad; rate -0.16735 rad/s
  gyro bias XYZ rad/s [-3.86446e-05, -0.000305564, -0.000323233]
  gravity bias YZ [0.00139849, -0.000392816]
E3 [40.88, 60.00) s:
  mass x [0.966667, 0.967923, 1.07066, 1.15713]
  CoM XYZ mm [-4.86504, -4.15821, 3.49082]
  mu_s [0.794338, 0.632409, 0.703711, 0.602694]
  mu_d [0.526209, 0.589076, 0.703711, 0.602694]
  Kp [steer,rear] [9.02357, 0]; Kd [0.183519, 0]
  spawn roll 0.0516136 rad; rate -0.164886 rad/s
  gyro bias XYZ rad/s [-5.06925e-05, -6.31963e-05, 0.000165454]
  gravity bias YZ [0.00173651, 0.000749239]
Six significant digits shown; full precision: sampled_dr_values.json.
```

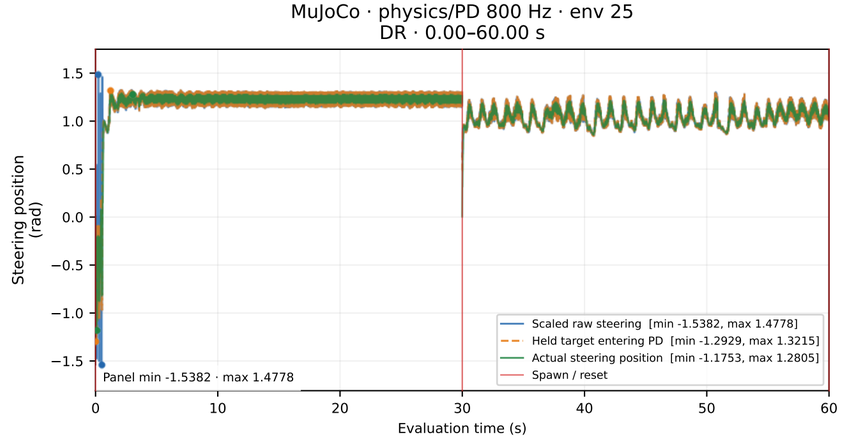
```
SAMPLED DR: env 24; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16949, 1.07713, 0.829174, 1.0497]
  CoM XYZ mm [-0.274809, 3.49019, -3.97172]
  mu_s [0.736077, 0.546287, 0.71155, 0.546287]
  mu_d [0.699455, 0.546287, 0.71155, 0.546287]
  Kp [steer,rear] [10.8176, 0]; Kd [0.201189, 0]
  spawn roll -0.00614922 rad; rate -0.118431 rad/s
  gyro bias XYZ rad/s [0.000194581, -0.000155248, 0.000214357]
  gravity bias YZ [-0.000313656, -0.000110502]
E2 [30.00, 30.22) s:
  mass x [1.11045, 1.01966, 1.1954, 0.860601]
  CoM XYZ mm [4.11354, -2.76793, 2.64046]
  mu_s [0.541939, 0.640867, 0.637675, 0.559972]
  mu_d [0.534359, 0.556841, 0.637675, 0.502209]
  Kp [steer,rear] [9.4654, 0]; Kd [0.219437, 0]
  spawn roll 0.0981903 rad; rate -0.215889 rad/s
  gyro bias XYZ rad/s [0.000261415, 0.000154779, 0.000156588]
  gravity bias YZ [0.000304513, 0.000957138]
E3 [30.22, 30.66) s:
  mass x [1.03549, 1.0774, 0.809523, 0.982319]
  CoM XYZ mm [-2.63449, 0.558577, -3.9431]
  mu_s [0.728342, 0.568073, 0.616244, 0.640016]
  mu_d [0.728342, 0.568073, 0.586498, 0.513141]
  Kp [steer,rear] [9.28898, 0]; Kd [0.1849, 0]
  spawn roll -0.066071 rad; rate 0.0949529 rad/s
  gyro bias XYZ rad/s [-0.000316964, -0.000304982, -2.27981e-05]
  gravity bias YZ [-0.000705733, -0.00386866]
E4 [30.66, 60.00) s:
  mass x [1.00793, 0.966922, 0.803254, 1.08329]
  CoM XYZ mm [-3.46147, -0.00834489, -3.56986]
  mu_s [0.736077, 0.728342, 0.597748, 0.728342]
  mu_d [0.699455, 0.728342, 0.597748, 0.728342]
  Kp [steer,rear] [10.033, 0]; Kd [0.205122, 0]
  spawn roll -0.0216071 rad; rate 0.230526 rad/s
  gyro bias XYZ rad/s [0.00025827, 0.000150042, -1.7466e-05]
  gravity bias YZ [-0.00113656, 0.00450769]
Six significant digits shown; full precision: sampled_dr_values.json.
```

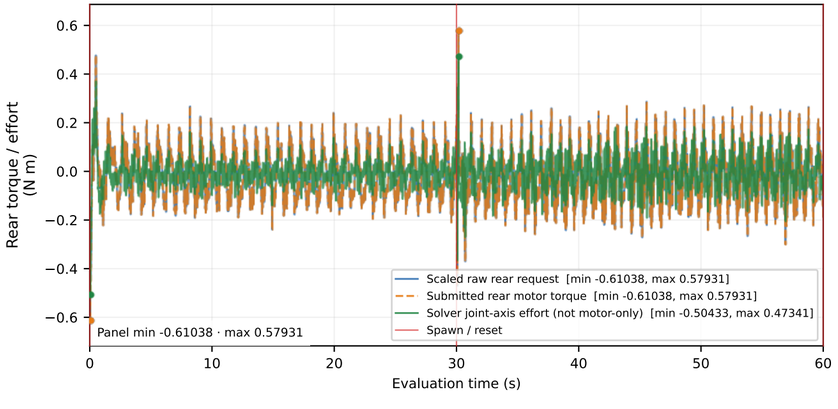
```
SAMPLED DR: env 24; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16949, 1.07713, 0.829174, 1.0497]
  CoM XYZ mm [-0.274809, 3.49019, -3.97172]
  mu_s [0.736077, 0.546287, 0.71155, 0.546287]
  mu_d [0.699455, 0.546287, 0.71155, 0.546287]
  Kp [steer,rear] [10.8176, 0]; Kd [0.201189, 0]
  spawn roll -0.00614922 rad; rate -0.118431 rad/s
  gyro bias XYZ rad/s [0.000194581, -0.000155248, 0.000214357]
  gravity bias YZ [-0.000313656, -0.000110502]
E2 [30.00, 30.18) s:
  mass x [1.11045, 1.01966, 1.1954, 0.860601]
  CoM XYZ mm [4.11354, -2.76793, 2.64046]
  mu_s [0.541939, 0.640867, 0.637675, 0.559972]
  mu_d [0.534359, 0.556841, 0.637675, 0.502209]
  Kp [steer,rear] [9.4654, 0]; Kd [0.219437, 0]
  spawn roll 0.0981903 rad; rate -0.215889 rad/s
  gyro bias XYZ rad/s [0.000261415, 0.000154779, 0.000156588]
  gravity bias YZ [0.000304513, 0.000957138]
E3 [30.18, 31.58) s:
  mass x [1.1719, 1.00104, 1.09403, 1.01886]
  CoM XYZ mm [-2.9091, 3.2673, 1.6565]
  mu_s [0.791209, 0.728342, 0.637675, 0.528253]
  mu_d [0.767936, 0.728342, 0.637675, 0.528253]
  Kp [steer,rear] [10.0026, 0]; Kd [0.199406, 0]
  spawn roll -0.0513004 rad; rate 0.145146 rad/s
  gyro bias XYZ rad/s [-7.91602e-05, 0.000116882, 3.57925e-06]
  gravity bias YZ [0.000482736, 0.00138094]
E4 [31.58, 60.00) s:
  mass x [0.875317, 1.13192, 1.16229, 1.14476]
  CoM XYZ mm [-1.50508, 2.1826, 4.04661]
  mu_s [0.71008, 0.723429, 0.546287, 0.640016]
  mu_d [0.59371, 0.723429, 0.546287, 0.513141]
  Kp [steer,rear] [10.7536, 0]; Kd [0.215146, 0]
  spawn roll -0.0486227 rad; rate 0.215478 rad/s
  gyro bias XYZ rad/s [7.06148e-05, 5.34391e-06, 0.000158052]
  gravity bias YZ [0.00137741, -0.000929329]
Six significant digits shown; full precision: sampled_dr_values.json.
```

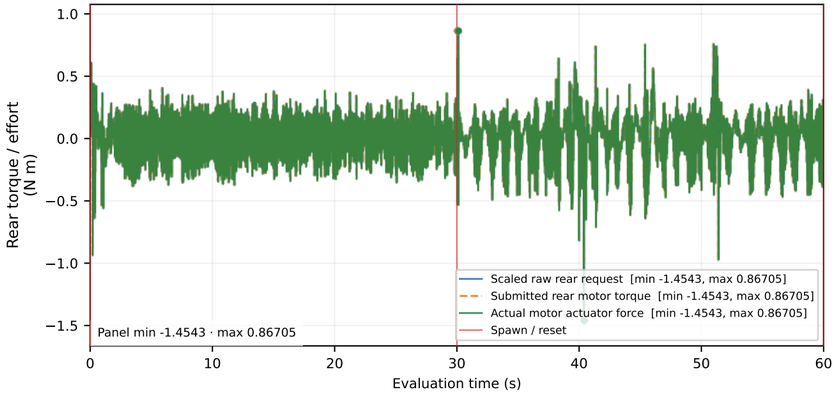
model_2500 · Demand versus actuation · env 25 · full 60 s

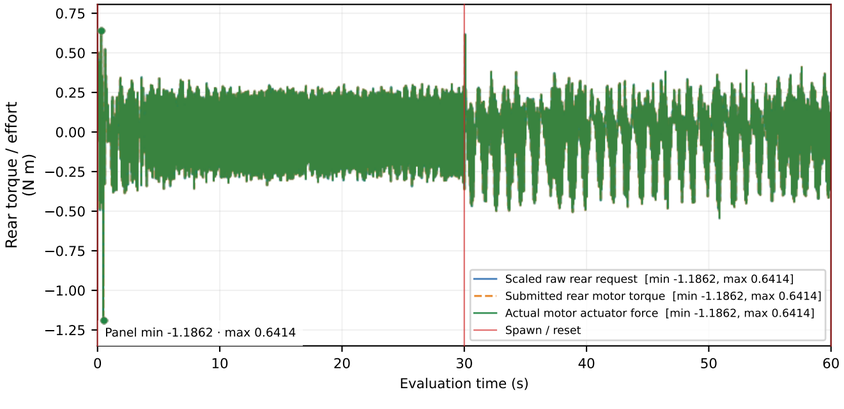












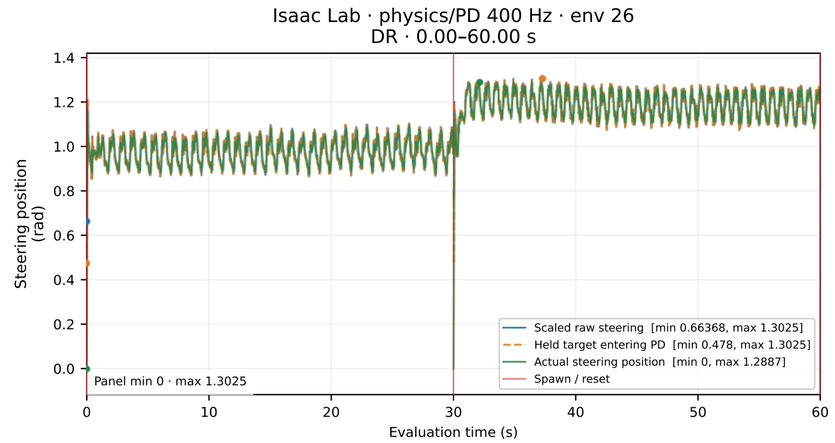
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

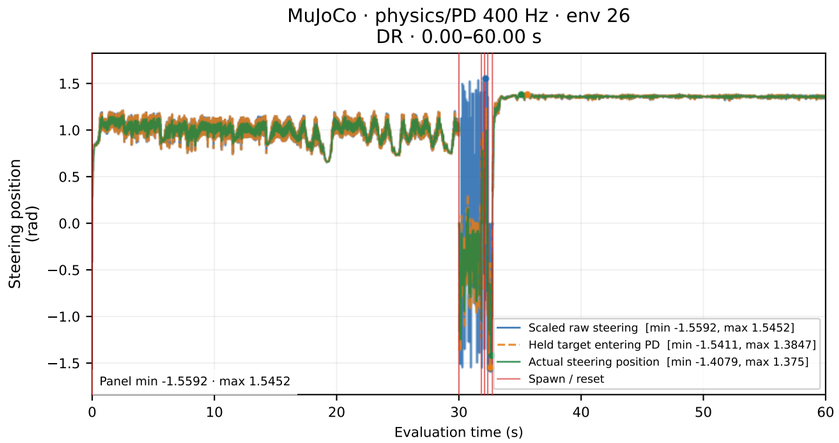
```
SAMPLED DR: env 25; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.875616, 1.17975, 0.903825, 0.929668]
  CoM XYZ mm [1.81167, 3.34438, -0.377148]
  mu_s [0.693205, 0.681711, 0.751971, 0.579974]
  mu_d [0.693205, 0.611756, 0.541224, 0.579974]
  Kp [steer,rear] [9.83792, 0]; Kd [0.197306, 0]
  spawn roll -0.0205786 rad; rate 0.173932 rad/s
  gyro bias XYZ rad/s [9.18702e-05, -4.10651e-05, -0.000261445]
  gravity bias YZ [0.00265045, 0.00207929]
E2 [30.00, 60.00) s:
  mass x [0.831212, 1.14263, 0.900257, 0.944867]
  CoM XYZ mm [0.970975, -0.505716, 1.01537]
  mu_s [0.531594, 0.741039, 0.647391, 0.741039]
  mu_d [0.531594, 0.614405, 0.647391, 0.614405]
  Kp [steer,rear] [9.08753, 0]; Kd [0.190724, 0]
  spawn roll 0.0144805 rad; rate -0.187457 rad/s
  gyro bias XYZ rad/s [-0.000286282, 0.000117795, -0.000100435]
  gravity bias YZ [0.000908414, -0.00157245]
Six significant digits shown; full precision: sampled_dr_values.json.
```

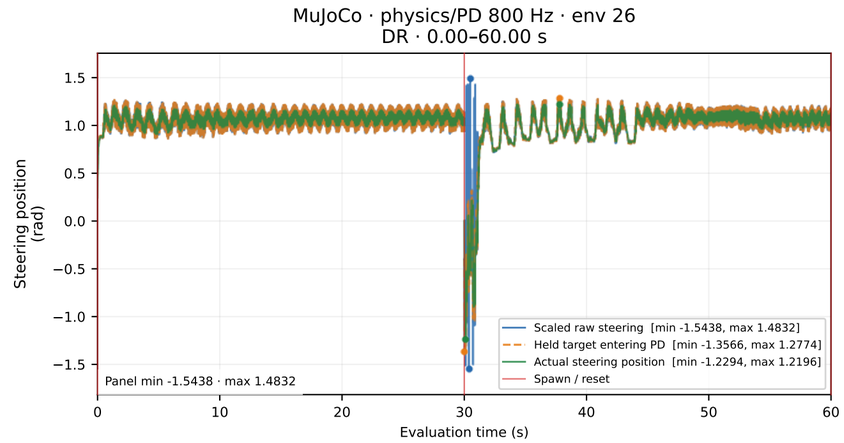
```
SAMPLED DR: env 25; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.965733, 0.850775, 1.00559, 0.918897]
  CoM XYZ mm [0.98037, -0.687294, -1.91706]
  mu_s [0.728342, 0.791209, 0.640016, 0.528253]
  mu_d [0.728342, 0.767936, 0.513141, 0.528253]
  Kp [steer,rear] [10.8245, 0]; Kd [0.180481, 0]
  spawn roll -0.0544291 rad; rate -0.0819372 rad/s
  gyro bias XYZ rad/s [-0.000260812, -0.000319122, -0.000211122]
  gravity bias YZ [0.00357963, -5.96259e-05]
E2 [30.00, 60.00) s:
  mass x [1.06177, 0.820528, 0.83899, 0.997132]
  CoM XYZ mm [1.44562, 1.0066, 0.635074]
  mu_s [0.546287, 0.616244, 0.631146, 0.669571]
  mu_d [0.546287, 0.586498, 0.631146, 0.669571]
  Kp [steer,rear] [9.37293, 0]; Kd [0.200151, 0]
  spawn roll 0.0338803 rad; rate 0.0723154 rad/s
  gyro bias XYZ rad/s [-0.000207155, 0.000191967, -0.000169671]
  gravity bias YZ [-0.00415054, 0.00403475]
Six significant digits shown; full precision: sampled_dr_values.json.
```

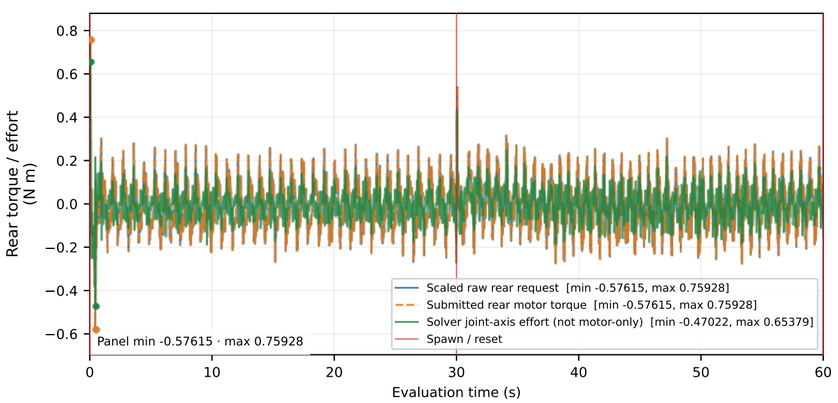
```
SAMPLED DR: env 25; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.965733, 0.850775, 1.00559, 0.918897]
  CoM XYZ mm [0.98037, -0.687294, -1.91706]
  mu_s [0.728342, 0.791209, 0.640016, 0.528253]
  mu_d [0.728342, 0.767936, 0.513141, 0.528253]
  Kp [steer,rear] [10.8245, 0]; Kd [0.180481, 0]
  spawn roll -0.0544291 rad; rate -0.0819372 rad/s
  gyro bias XYZ rad/s [-0.000260812, -0.000319122, -0.000211122]
  gravity bias YZ [0.00357963, -5.96259e-05]
E2 [30.00, 60.00) s:
  mass x [1.06177, 0.820528, 0.83899, 0.997132]
  CoM XYZ mm [1.44562, 1.0066, 0.635074]
  mu_s [0.546287, 0.616244, 0.631146, 0.669571]
  mu_d [0.546287, 0.586498, 0.631146, 0.669571]
  Kp [steer,rear] [9.37293, 0]; Kd [0.200151, 0]
  spawn roll 0.0338803 rad; rate 0.0723154 rad/s
  gyro bias XYZ rad/s [-0.000207155, 0.000191967, -0.000169671]
  gravity bias YZ [-0.00415054, 0.00403475]
Six significant digits shown; full precision: sampled_dr_values.json.
```

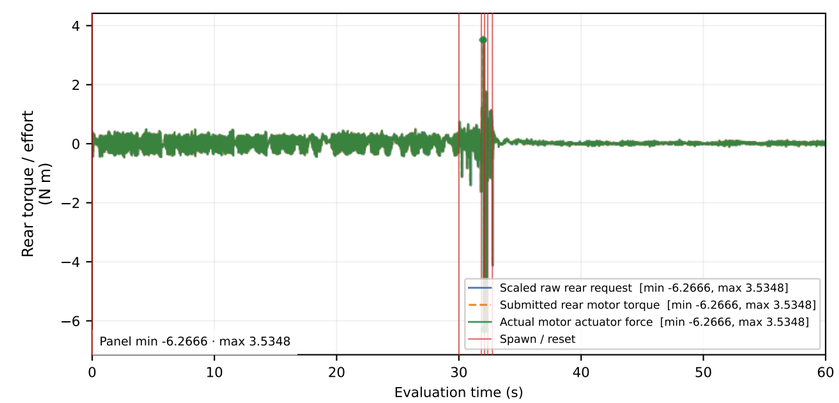
model_2500 · Demand versus actuation · env 26 · full 60 s

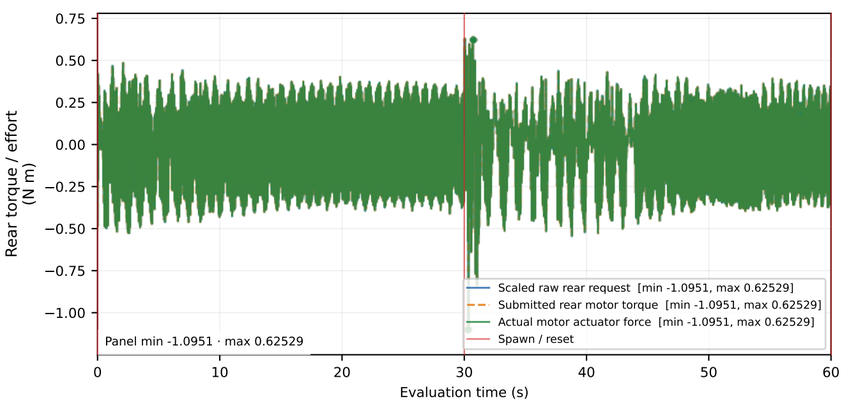












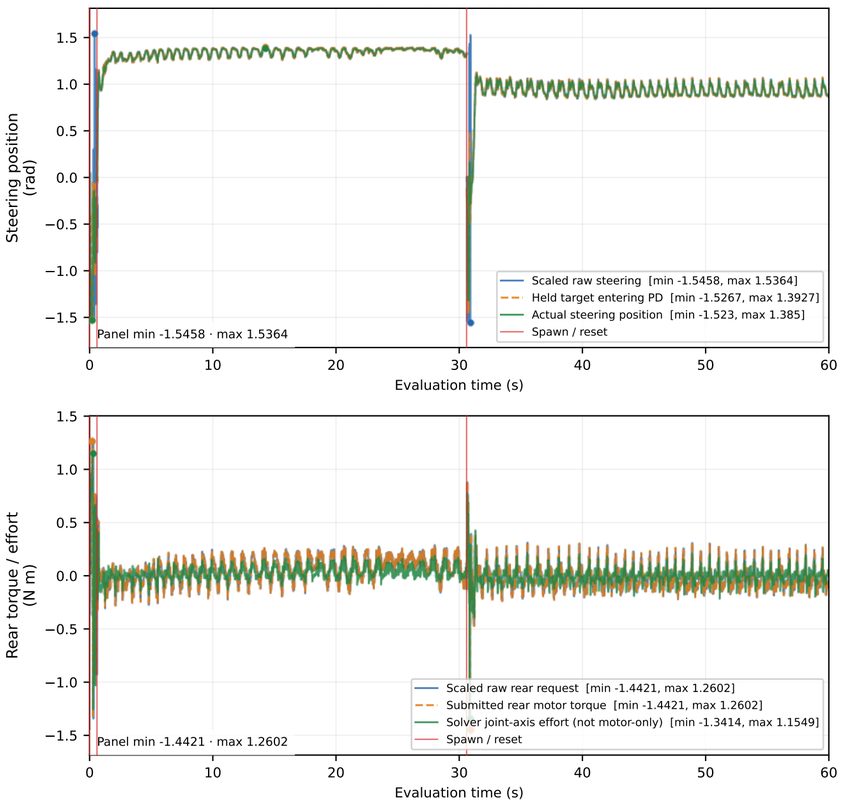
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

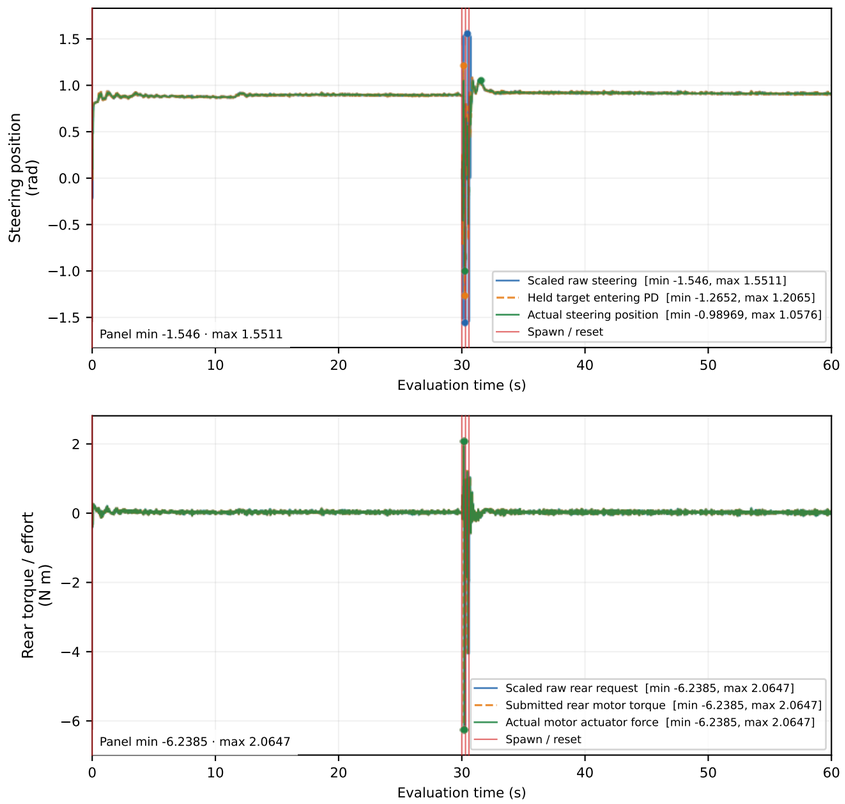
```
SAMPLED DR: env 26; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.915996, 0.906212, 1.03858, 1.1616]
  CoM XYZ mm [3.93371, 1.73644, -2.09413]
  mu_s [0.726175, 0.647391, 0.741039, 0.751971]
  mu_d [0.558574, 0.647391, 0.614405, 0.541224]
  Kp [steer,rear] [9.22716, 0]; Kd [0.204445, 0]
  spawn roll 0.0750011 rad; rate -0.0215228 rad/s
  gyro bias XYZ rad/s [0.000132369, -0.000562396, 1.03834e-05]
  gravity bias YZ [-0.000777999, 0.00171696]
E2 [30.00, 60.00) s:
  mass x [0.877824, 1.01535, 1.1085, 1.17473]
  CoM XYZ mm [-4.86545, -1.24921, -4.78084]
  mu_s [0.628821, 0.559809, 0.659445, 0.559809]
  mu_d [0.628821, 0.559809, 0.547619, 0.559809]
  Kp [steer,rear] [10.4904, 0]; Kd [0.211355, 0]
  spawn roll 0.0364383 rad; rate 0.174227 rad/s
  gyro bias XYZ rad/s [0.000113855, 0.000254793, -0.000328278]
  gravity bias YZ [-0.00144561, 0.00106124]
Six significant digits shown; full precision: sampled_dr_values.json.
```

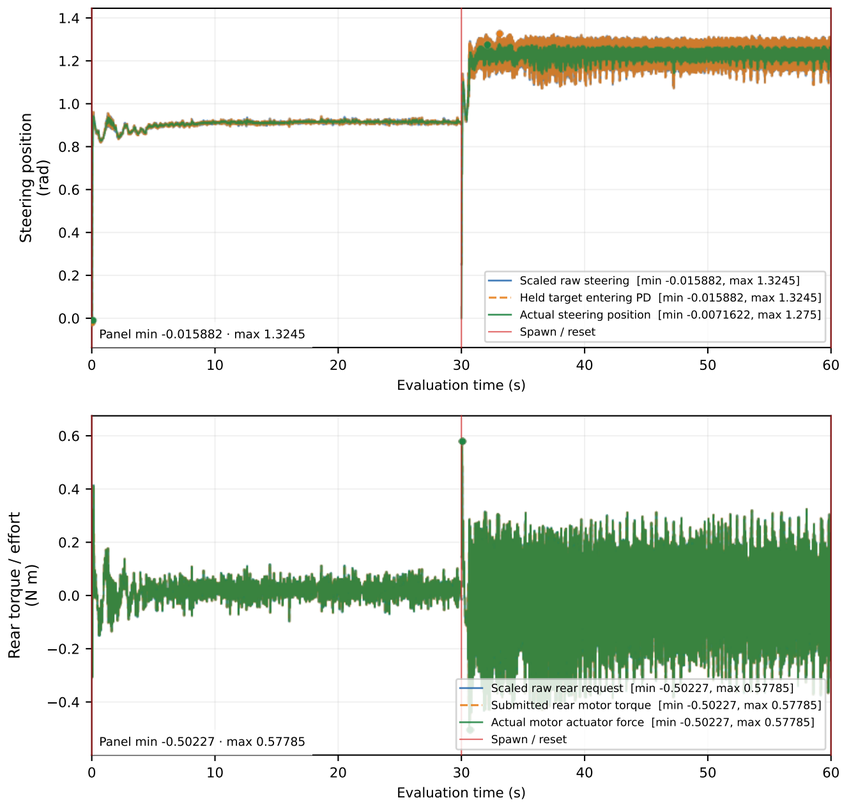
```
SAMPLED DR: env 26; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.993747, 1.13933, 1.06941, 1.0505]
  CoM XYZ mm [4.75818, 1.9279, -3.69763]
  mu_s [0.723429, 0.597748, 0.538976, 0.568073]
  mu_d [0.723429, 0.597748, 0.538976, 0.568073]
  Kp [steer,rear] [9.35545, 0]; Kd [0.209963, 0]
  spawn roll 0.0290254 rad; rate 0.209869 rad/s
  gyro bias XYZ rad/s [0.00013998, 0.000133799, -1.10574e-05]
  gravity bias YZ [-0.00157627, 0.00273748]
E2 [30.00, 31.84) s:
  mass x [0.885949, 1.08233, 0.917855, 1.03555]
  CoM XYZ mm [-1.94472, 1.7385, -4.32292]
  mu_s [0.669571, 0.669571, 0.757579, 0.748289]
  mu_d [0.669571, 0.669571, 0.70921, 0.689499]
  Kp [steer,rear] [10.2197, 0]; Kd [0.18084, 0]
  spawn roll -0.0491084 rad; rate -0.133235 rad/s
  gyro bias XYZ rad/s [2.47219e-05, -0.000191236, 2.73441e-05]
  gravity bias YZ [-0.0014935, 0.00221651]
E3 [31.84, 32.10) s:
  mass x [0.942888, 0.971694, 1.11327, 1.06906]
  CoM XYZ mm [0.537742, 0.536609, 0.630142]
  mu_s [0.611239, 0.631146, 0.728342, 0.528253]
  mu_d [0.611239, 0.631146, 0.728342, 0.528253]
  Kp [steer,rear] [9.96441, 0]; Kd [0.192504, 0]
  spawn roll 0.0934041 rad; rate -0.144448 rad/s
  gyro bias XYZ rad/s [0.000218819, -0.000101322, -0.000225819]
  gravity bias YZ [-0.000608424, -0.00229416]
E4 [32.10, 32.36) s:
  mass x [1.12464, 0.953291, 1.0491, 0.924926]
  CoM XYZ mm [0.0679545, -0.031636, 3.96472]
  mu_s [0.704749, 0.736077, 0.749678, 0.666375]
  mu_d [0.541926, 0.699455, 0.741429, 0.519145]
  Kp [steer,rear] [9.76565, 0]; Kd [0.200747, 0]
  spawn roll 0.0978547 rad; rate 0.198396 rad/s
  gyro bias XYZ rad/s [-1.95181e-05, 0.000111424, -0.00014902]
  gravity bias YZ [-0.00139232, -0.000730083]
E5 [32.36, 32.74) s:
  mass x [0.922887, 1.11895, 0.973031, 1.1487]
  CoM XYZ mm [4.99285, 2.89677, 3.46991]
  mu_s [0.633024, 0.631146, 0.748289, 0.732187]
  mu_d [0.568172, 0.631146, 0.689499, 0.631664]
  Kp [steer,rear] [10.1915, 0]; Kd [0.217585, 0]
  spawn roll -0.0739943 rad; rate -0.0990864 rad/s
  gyro bias XYZ rad/s [-0.000252783, -1.43901e-05, 0.000141257]
  gravity bias YZ [-0.000742643, -0.00254135]
E6 [32.74, 60.00) s:
  mass x [0.818999, 1.19359, 0.905628, 1.02562]
  CoM XYZ mm [-4.64729, -3.71685, -0.274436]
  mu_s [0.704749, 0.640867, 0.736077, 0.791209]
  mu_d [0.541926, 0.556841, 0.699455, 0.767936]
  Kp [steer,rear] [9.04605, 0]; Kd [0.1897, 0]
  spawn roll -0.0316723 rad; rate 0.158515 rad/s
  gyro bias XYZ rad/s [-6.09537e-05, -2.27372e-06, 0.000226784]
  gravity bias YZ [0.003926, -0.00216295]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 26; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.993747, 1.13933, 1.06941, 1.0505]
  CoM XYZ mm [4.75818, 1.9279, -3.69763]
  mu_s [0.723429, 0.597748, 0.538976, 0.568073]
  mu_d [0.723429, 0.597748, 0.538976, 0.568073]
  Kp [steer,rear] [9.35545, 0]; Kd [0.209963, 0]
  spawn roll 0.0290254 rad; rate 0.209869 rad/s
  gyro bias XYZ rad/s [0.00013998, 0.000133799, -1.10574e-05]
  gravity bias YZ [-0.00157627, 0.00273748]
E2 [30.00, 60.00) s:
  mass x [0.885949, 1.08233, 0.917855, 1.03555]
  CoM XYZ mm [-1.94472, 1.7385, -4.32292]
  mu_s [0.669571, 0.669571, 0.757579, 0.748289]
  mu_d [0.669571, 0.669571, 0.70921, 0.689499]
  Kp [steer,rear] [10.2197, 0]; Kd [0.18084, 0]
  spawn roll -0.0491084 rad; rate -0.133235 rad/s
  gyro bias XYZ rad/s [2.47219e-05, -0.000191236, 2.73441e-05]
  gravity bias YZ [-0.0014935, 0.00221651]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 27 · full 60 s







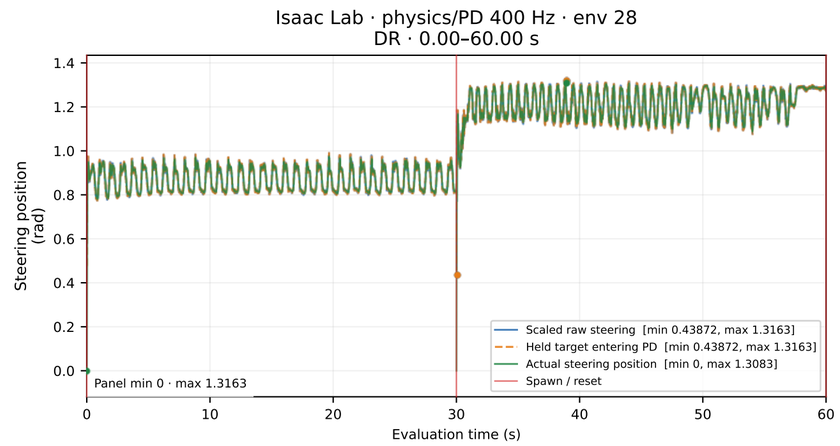
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

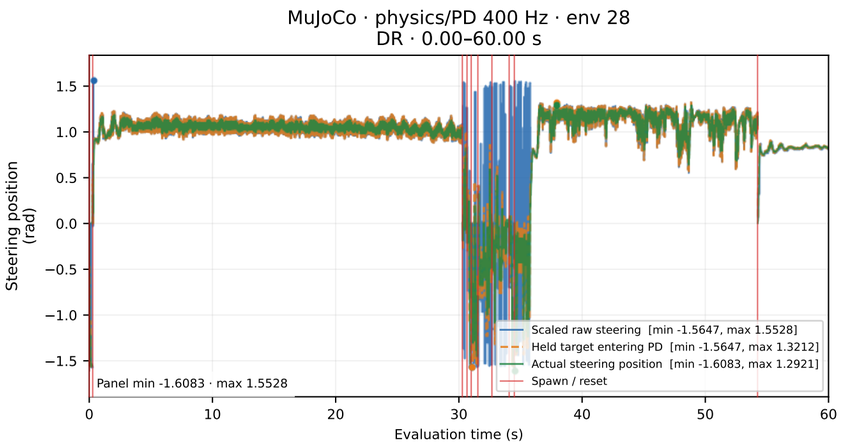
```
SAMPLED DR: env 27; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 0.60) s:
  mass x [1.14218, 1.17407, 0.880879, 0.815383]
  CoM XYZ mm [-3.5619, 4.63646, 2.62785]
  mu_s [0.539956, 0.533455, 0.760821, 0.774409]
  mu_d [0.539956, 0.533455, 0.670315, 0.561095]
  Kp [steer,rear] [10.978, 0]; Kd [0.180798, 0]
  spawn roll -0.0902905 rad; rate 0.206057 rad/s
  gyro bias XYZ rad/s [-9.91288e-05, -3.06844e-06, -2.71612e-05]
  gravity bias YZ [0.0023112, 0.00116169]
E2 [0.60, 30.60) s:
  mass x [1.09414, 0.88537, 0.903026, 0.999329]
  CoM XYZ mm [-1.14301, -3.40579, 4.10023]
  mu_s [0.751971, 0.751971, 0.751971, 0.547743]
  mu_d [0.541224, 0.541224, 0.541224, 0.547743]
  Kp [steer,rear] [9.60229, 0]; Kd [0.183213, 0]
  spawn roll 0.00356567 rad; rate -0.112306 rad/s
  gyro bias XYZ rad/s [6.14946e-06, 0.000212763, -0.000417311]
  gravity bias YZ [-0.00135359, 0.00155985]
E3 [30.60, 60.00) s:
  mass x [1.00297, 0.822399, 1.13368, 1.1895]
  CoM XYZ mm [-3.1565, 1.60713, 3.7401]
  mu_s [0.632409, 0.787792, 0.598343, 0.673378]
  mu_d [0.589076, 0.617134, 0.598343, 0.673378]
  Kp [steer,rear] [10.8456, 0]; Kd [0.217123, 0]
  spawn roll -0.0781943 rad; rate 0.0748008 rad/s
  gyro bias XYZ rad/s [0.000252128, -9.52991e-05, -5.25896e-05]
  gravity bias YZ [-0.000183114, -0.00352336]
Six significant digits shown; full precision: sampled_dr_values.json.
```

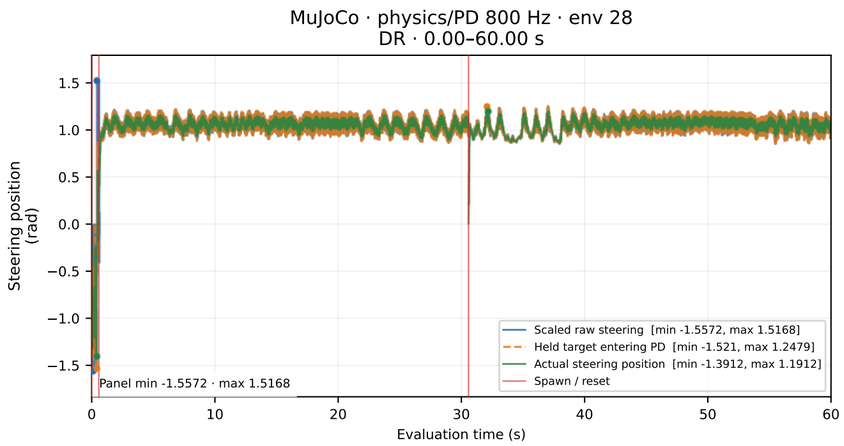
```
SAMPLED DR: env 27; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.03215, 1.01074, 1.14598, 1.0939]
  CoM XYZ mm [-0.234152, 3.16696, 0.446348]
  mu_s [0.559972, 0.640016, 0.559972, 0.732187]
  mu_d [0.502209, 0.513141, 0.502209, 0.631664]
  Kp [steer,rear] [10.3569, 0]; Kd [0.208237, 0]
  spawn roll 0.0467654 rad; rate -0.215901 rad/s
  gyro bias XYZ rad/s [0.000142147, -3.23553e-05, -8.01081e-05]
  gravity bias YZ [0.00221784, 0.000939233]
E2 [30.00, 30.30) s:
  mass x [1.0161, 0.989662, 0.917133, 0.918384]
  CoM XYZ mm [3.42184, -0.646235, 1.00044]
  mu_s [0.666375, 0.723429, 0.727426, 0.597748]
  mu_d [0.519145, 0.723429, 0.606358, 0.597748]
  Kp [steer,rear] [9.98354, 0]; Kd [0.218973, 0]
  spawn roll 0.0708753 rad; rate -0.194923 rad/s
  gyro bias XYZ rad/s [6.70637e-05, 2.76081e-05, 0.000159323]
  gravity bias YZ [0.00398194, 0.000394695]
E3 [30.30, 30.58) s:
  mass x [0.964189, 1.19385, 1.16629, 1.00441]
  CoM XYZ mm [3.59202, -1.0898, -0.457886]
  mu_s [0.700521, 0.528253, 0.538976, 0.611239]
  mu_d [0.641329, 0.528253, 0.538976, 0.611239]
  Kp [steer,rear] [9.55067, 0]; Kd [0.214054, 0]
  spawn roll 0.0634218 rad; rate 0.0615261 rad/s
  gyro bias XYZ rad/s [0.000118771, -0.000156696, -7.15329e-05]
  gravity bias YZ [-0.00242644, -0.000123051]
E4 [30.58, 60.00) s:
  mass x [0.846888, 0.844771, 1.08345, 1.06525]
  CoM XYZ mm [-4.91461, 2.57639, -2.21418]
  mu_s [0.640867, 0.559972, 0.69316, 0.640016]
  mu_d [0.556841, 0.502209, 0.69316, 0.513141]
  Kp [steer,rear] [10.5368, 0]; Kd [0.184188, 0]
  spawn roll 0.0444598 rad; rate 0.101519 rad/s
  gyro bias XYZ rad/s [-0.000230281, -0.000219928, -0.000218251]
  gravity bias YZ [0.00296641, -0.00362709]
Six significant digits shown; full precision: sampled_dr_values.json.
```

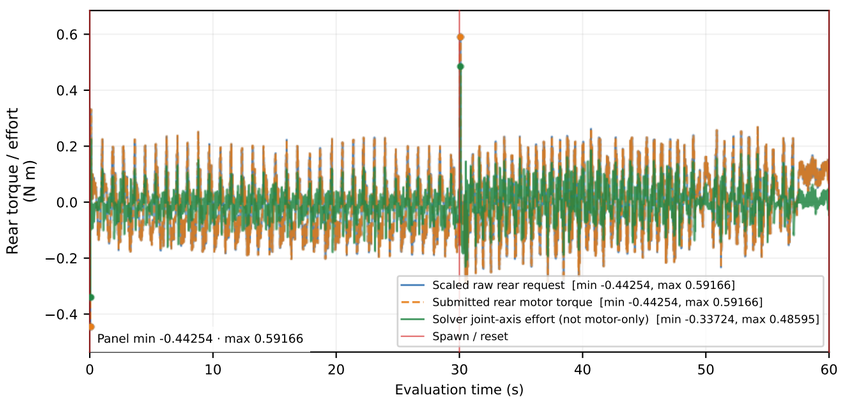
```
SAMPLED DR: env 27; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.03215, 1.01074, 1.14598, 1.0939]
  CoM XYZ mm [-0.234152, 3.16696, 0.446348]
  mu_s [0.559972, 0.640016, 0.559972, 0.732187]
  mu_d [0.502209, 0.513141, 0.502209, 0.631664]
  Kp [steer,rear] [10.3569, 0]; Kd [0.208237, 0]
  spawn roll 0.0467654 rad; rate -0.215901 rad/s
  gyro bias XYZ rad/s [0.000142147, -3.23553e-05, -8.01081e-05]
  gravity bias YZ [0.00221784, 0.000939233]
E2 [30.00, 60.00) s:
  mass x [1.0161, 0.989662, 0.917133, 0.918384]
  CoM XYZ mm [3.42184, -0.646235, 1.00044]
  mu_s [0.666375, 0.723429, 0.727426, 0.597748]
  mu_d [0.519145, 0.723429, 0.606358, 0.597748]
  Kp [steer,rear] [9.98354, 0]; Kd [0.218973, 0]
  spawn roll 0.0708753 rad; rate -0.194923 rad/s
  gyro bias XYZ rad/s [6.70637e-05, 2.76081e-05, 0.000159323]
  gravity bias YZ [0.00398194, 0.000394695]
Six significant digits shown; full precision: sampled_dr_values.json.
```

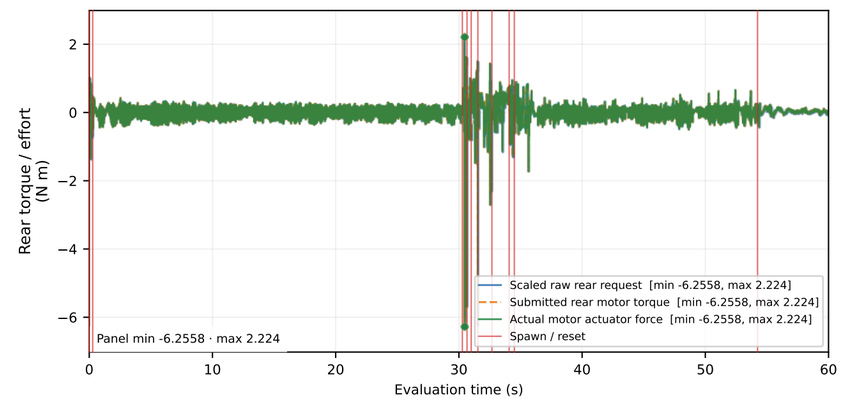
model_2500 · Demand versus actuation · env 28 · full 60 s

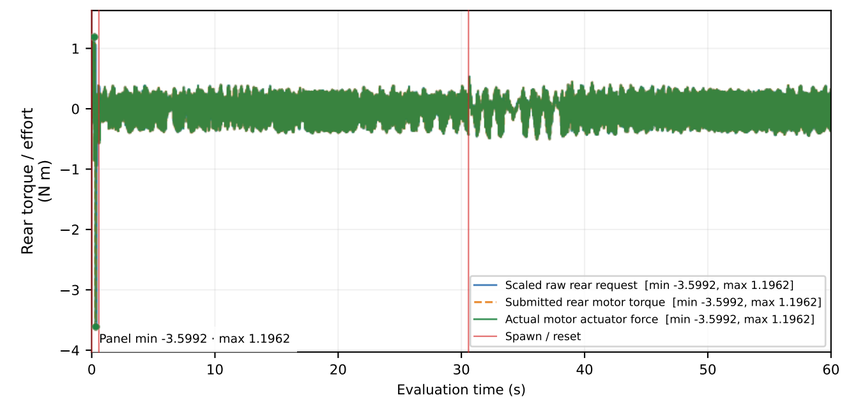












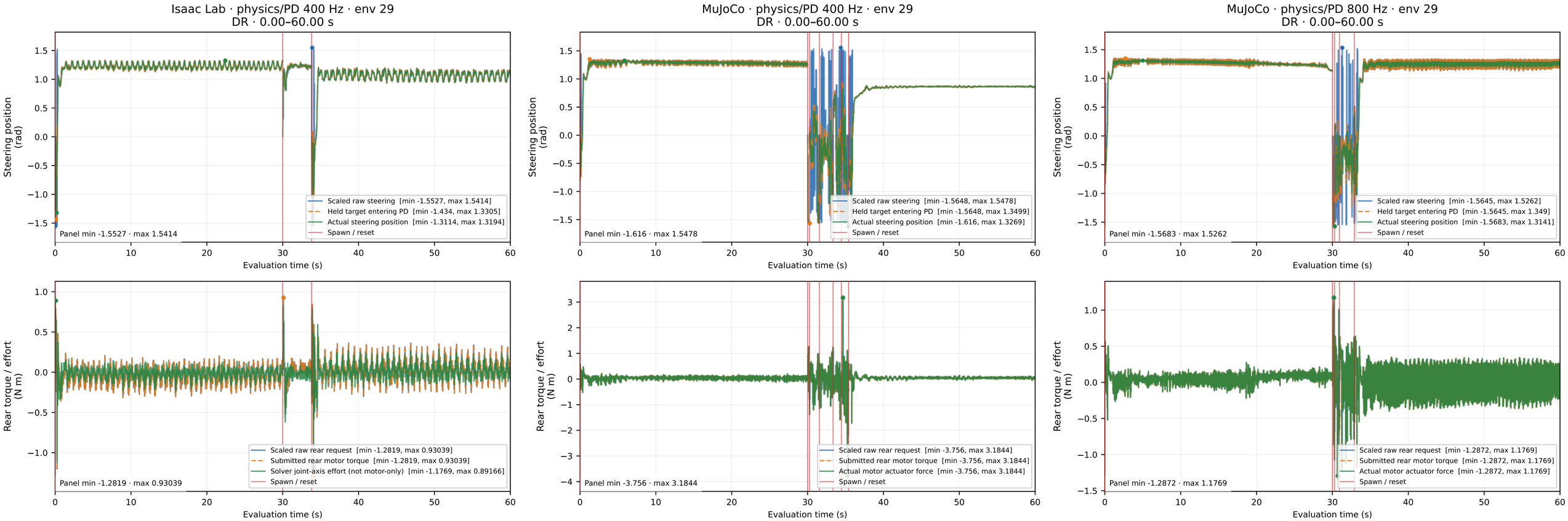
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 28; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.10236, 1.12912, 1.19072, 1.15239]
  CoM XYZ mm [-0.636846, 3.37844, 3.57158]
  mu_s [0.579974, 0.673378, 0.713129, 0.785466]
  mu_d [0.579974, 0.673378, 0.713129, 0.52258]
  Kp [steer,rear] [10.2666, 0]; Kd [0.203469, 0]
  spawn roll 0.0384418 rad; rate 0.0541661 rad/s
  gyro bias XYZ rad/s [-0.000126931, -6.078e-05, 0.000257724]
  gravity bias YZ [0.00306666, -0.000674733]
E2 [30.00, 60.00) s:
  mass x [1.11974, 0.873552, 1.01957, 1.08793]
  CoM XYZ mm [-1.75007, -0.848766, 0.974908]
  mu_s [0.673378, 0.726175, 0.531594, 0.751971]
  mu_d [0.673378, 0.558574, 0.531594, 0.541224]
  Kp [steer,rear] [10.9783, 0]; Kd [0.200438, 0]
  spawn roll 0.0327392 rad; rate 0.00453961 rad/s
  gyro bias XYZ rad/s [0.000201461, 9.5479e-05, 3.07736e-05]
  gravity bias YZ [0.00220735, 0.00233325]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 28; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [0.952294, 1.08952, 0.928876, 1.18609]
  CoM XYZ mm [2.40347, -0.41802, -2.92448]
  mu_s [0.568073, 0.732187, 0.538434, 0.559972]
  mu_d [0.568073, 0.631664, 0.538434, 0.502209]
  Kp [steer,rear] [9.46634, 0]; Kd [0.193779, 0]
  spawn roll -0.0959019 rad; rate -0.208002 rad/s
  gyro bias XYZ rad/s [-8.36518e-05, 0.00020827, -0.000112668]
  gravity bias YZ [0.00304337, -0.00101055]
E2 [0.28, 30.28) s:
  mass x [0.93304, 1.00889, 1.11889, 1.08516]
  CoM XYZ mm [3.50917, 1.95555, 4.919]
  mu_s [0.631146, 0.528253, 0.700521, 0.732187]
  mu_d [0.631146, 0.528253, 0.641329, 0.631664]
  Kp [steer,rear] [10.2706, 0]; Kd [0.208555, 0]
  spawn roll 0.0571224 rad; rate -0.242375 rad/s
  gyro bias XYZ rad/s [0.000175128, -0.000326341, -0.000202794]
  gravity bias YZ [0.00267025, 0.000785849]
E3 [30.28, 30.66) s:
  mass x [1.04399, 0.869427, 0.823365, 0.934411]
  CoM XYZ mm [4.46926, -1.76103, -2.60666]
  mu_s [0.728342, 0.538434, 0.568073, 0.633024]
  mu_d [0.728342, 0.538434, 0.568073, 0.568172]
  Kp [steer,rear] [10.8384, 0]; Kd [0.193374, 0]
  spawn roll 0.00969489 rad; rate -0.226411 rad/s
  gyro bias XYZ rad/s [0.00011068, 0.000119604, 0.000314661]
  gravity bias YZ [-0.00140637, -0.00214619]
E4 [30.66, 31.00) s:
  mass x [1.19968, 1.01752, 1.0602, 0.913877]
  CoM XYZ mm [-0.991653, -1.08678, 4.77046]
  mu_s [0.559972, 0.71155, 0.541939, 0.633024]
  mu_d [0.502209, 0.71155, 0.534359, 0.568172]
  Kp [steer,rear] [9.41789, 0]; Kd [0.218262, 0]
  spawn roll -0.0938252 rad; rate 0.043656 rad/s
  gyro bias XYZ rad/s [-8.90731e-05, -3.84677e-05, 0.000531459]
  gravity bias YZ [-0.000366339, -0.0024708]
E5 [31.00, 31.54) s:
  mass x [1.06646, 0.984685, 1.00894, 1.10895]
  CoM XYZ mm [-2.39553, 4.82211, 1.03344]
  mu_s [0.749678, 0.666375, 0.559972, 0.538976]
  mu_d [0.741429, 0.519145, 0.502209, 0.538976]
  Kp [steer,rear] [9.37931, 0]; Kd [0.191366, 0]
  spawn roll -0.0513507 rad; rate 0.0101969 rad/s
  gyro bias XYZ rad/s [-0.000205965, 2.3688e-05, 0.00025687]
  gravity bias YZ [-2.8465e-06, -0.000470191]
E6 [31.54, 32.68) s:
  mass x [0.972611, 0.832211, 0.87056, 1.18059]
  CoM XYZ mm [4.07078, -1.95287, 1.45444]
  mu_s [0.640016, 0.733515, 0.640016, 0.69316]
  mu_d [0.513141, 0.558392, 0.513141, 0.69316]
  Kp [steer,rear] [9.46859, 0]; Kd [0.201181, 0]
  spawn roll -0.0680625 rad; rate 0.236171 rad/s
  gyro bias XYZ rad/s [0.000200285, 0.000310343, 0.000162168]
  gravity bias YZ [-0.00164235, -0.00253692]
E7 [32.68, 34.08) s:
  mass x [0.972775, 1.04356, 1.14376, 0.884581]
  CoM XYZ mm [-0.413933, 0.0959689, 2.696]
  mu_s [0.704749, 0.640016, 0.71008, 0.616244]
  mu_d [0.541926, 0.513141, 0.59371, 0.586498]
  Kp [steer,rear] [9.20953, 0]; Kd [0.197529, 0]
  spawn roll -0.0263583 rad; rate -0.0092295 rad/s
  gyro bias XYZ rad/s [-0.000194922, -0.000416148, -0.000107549]
  gravity bias YZ [-0.00426185, 3.11676e-05]
E8 [34.08, 34.50) s:
  mass x [1.18303, 0.986336, 1.03512, 0.807493]
  CoM XYZ mm [-0.762478, 2.35782, -0.110923]
  mu_s [0.568073, 0.541939, 0.736077, 0.69316]
  mu_d [0.568073, 0.534359, 0.699455, 0.69316]
  Kp [steer,rear] [10.2904, 0]; Kd [0.213506, 0]
  spawn roll -0.0150451 rad; rate -0.24368 rad/s
  gyro bias XYZ rad/s [0.000322825, -0.000121041, 4.10367e-05]
  gravity bias YZ [0.00235073, -0.00210662]
E9 [34.50, 54.24) s:
  mass x [1.06933, 1.11779, 0.925928, 0.881714]
  CoM XYZ mm [2.58632, -0.582327, -4.11222]
  mu_s [0.616244, 0.732187, 0.538976, 0.538976]
  mu_d [0.586498, 0.631664, 0.538976, 0.538976]
  Kp [steer,rear] [10.128, 0]; Kd [0.182089, 0]
  spawn roll -0.0443302 rad; rate -0.0679515 rad/s
  gyro bias XYZ rad/s [-0.000133908, -0.000311232, 2.11184e-05]
  gravity bias YZ [0.00115997, 0.000949268]
E10 [54.24, 60.00) s:
  mass x [0.857086, 1.16586, 1.1047, 0.989514]
  CoM XYZ mm [-1.1716, 4.02087, -3.10204]
  mu_s [0.727426, 0.71008, 0.791209, 0.640867]
  mu_d [0.606358, 0.59371, 0.767936, 0.556841]
  Kp [steer,rear] [9.13905, 0]; Kd [0.208661, 0]
  spawn roll 0.0237928 rad; rate -0.110249 rad/s
  gyro bias XYZ rad/s [0.000181704, -5.97139e-06, -3.66438e-05]
  gravity bias YZ [0.00189326, -0.000104177]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 28; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.58) s:
  mass x [0.952294, 1.08952, 0.928876, 1.18609]
  CoM XYZ mm [2.40347, -0.41802, -2.92448]
  mu_s [0.568073, 0.732187, 0.538434, 0.559972]
  mu_d [0.568073, 0.631664, 0.538434, 0.502209]
  Kp [steer,rear] [9.46634, 0]; Kd [0.193779, 0]
  spawn roll -0.0959019 rad; rate -0.208002 rad/s
  gyro bias XYZ rad/s [-8.36518e-05, 0.00020827, -0.000112668]
  gravity bias YZ [0.00304337, -0.00101055]
E2 [0.58, 30.58) s:
  mass x [0.937336, 0.889866, 1.1976, 0.989658]
  CoM XYZ mm [-0.185031, 1.64127, -0.30178]
  mu_s [0.637675, 0.723429, 0.757579, 0.757579]
  mu_d [0.637675, 0.723429, 0.70921, 0.70921]
  Kp [steer,rear] [10.4582, 0]; Kd [0.215546, 0]
  spawn roll -0.00614233 rad; rate 0.123039 rad/s
  gyro bias XYZ rad/s [4.75919e-05, -1.56271e-05, -0.000574203]
  gravity bias YZ [-0.00245388, -0.00177541]
E3 [30.58, 60.00) s:
  mass x [1.01946, 0.881359, 1.05001, 0.98384]
  CoM XYZ mm [1.85371, 2.16592, -3.81601]
  mu_s [0.723429, 0.749678, 0.736077, 0.727426]
  mu_d [0.723429, 0.741429, 0.699455, 0.606358]
  Kp [steer,rear] [9.88319, 0]; Kd [0.207719, 0]
  spawn roll 0.0637601 rad; rate -0.196215 rad/s
  gyro bias XYZ rad/s [5.14055e-06, -0.000353386, -0.000115272]
  gravity bias YZ [-0.000504244, -0.000612158]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 29 · full 60 s



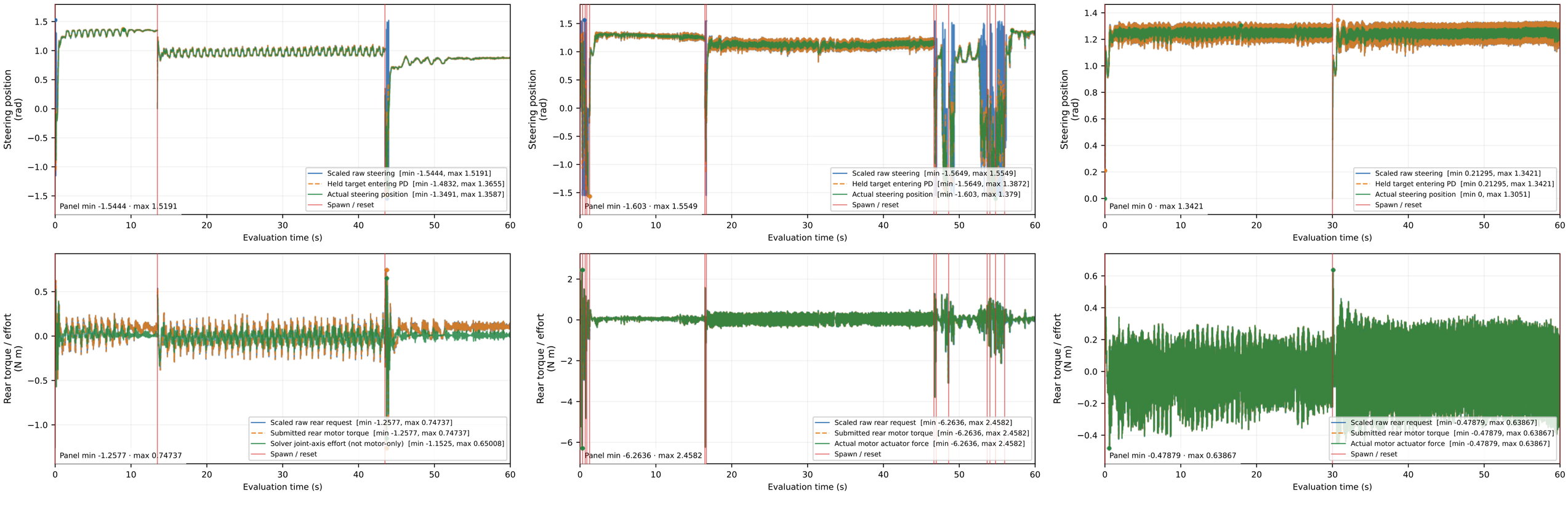
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 29; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.04223, 0.986278, 1.12749, 1.0329]
  CoM XYZ mm [4.67661, -1.2452, 4.37889]
  mu_s [0.741039, 0.539956, 0.673378, 0.751971]
  mu_d [0.614405, 0.539956, 0.673378, 0.541224]
  Kp [steer,rear] [10.2652, 0]; Kd [0.185641, 0]
  spawn roll -0.0778912 rad; rate -0.12325 rad/s
  gyro bias XYZ rad/s [-0.000114268, 0.000298355, 0.00044557]
  gravity bias YZ [0.00208657, 0.00182313]
E2 [30.00, 33.80) s:
  mass x [1.08136, 1.17486, 1.0021, 0.847269]
  CoM XYZ mm [-0.536606, -4.60852, 4.68202]
  mu_s [0.726175, 0.560535, 0.531594, 0.531594]
  mu_d [0.558574, 0.560535, 0.531594, 0.531594]
  Kp [steer,rear] [10.49, 0]; Kd [0.203965, 0]
  spawn roll 0.0716322 rad; rate -0.183393 rad/s
  gyro bias XYZ rad/s [0.000145528, 0.000137405, -0.000204785]
  gravity bias YZ [0.000279636, -0.00157764]
E3 [33.80, 60.00) s:
  mass x [1.06325, 1.00468, 0.822012, 1.06386]
  CoM XYZ mm [-3.90828, 0.396512, -4.04747]
  mu_s [0.751971, 0.751971, 0.547743, 0.787792]
  mu_d [0.541224, 0.541224, 0.547743, 0.617134]
  Kp [steer,rear] [10.9457, 0]; Kd [0.214372, 0]
  spawn roll -0.0390346 rad; rate -0.167397 rad/s
  gyro bias XYZ rad/s [0.000233973, -2.20054e-05, -0.000181076]
  gravity bias YZ [-0.00162685, 0.00190149]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 29; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.901365, 1.01102, 1.10439, 1.0539]
  CoM XYZ mm [-2.95828, -4.5423, 0.231654]
  mu_s [0.723429, 0.69316, 0.633024, 0.538976]
  mu_d [0.723429, 0.69316, 0.568172, 0.538976]
  Kp [steer,rear] [9.07765, 0]; Kd [0.195074, 0]
  spawn roll -0.0440389 rad; rate -0.170196 rad/s
  gyro bias XYZ rad/s [7.55189e-05, 6.51789e-05, -0.000228312]
  gravity bias YZ [0.00108292, -0.000849417]
E2 [30.00, 30.26) s:
  mass x [1.15712, 1.14701, 0.925754, 1.00218]
  CoM XYZ mm [0.654529, 3.45013, 4.30141]
  mu_s [0.538434, 0.728342, 0.546287, 0.538976]
  mu_d [0.538434, 0.728342, 0.546287, 0.538976]
  Kp [steer,rear] [10.7335, 0]; Kd [0.186154, 0]
  spawn roll -0.0988358 rad; rate 0.0124952 rad/s
  gyro bias XYZ rad/s [0.000154901, 0.000115283, -5.695e-05]
  gravity bias YZ [-0.00325704, 5.60172e-05]
E3 [30.26, 31.56) s:
  mass x [1.12305, 1.17015, 1.15517, 0.865258]
  CoM XYZ mm [-1.57713, -0.299899, 0.111842]
  mu_s [0.749678, 0.791209, 0.669571, 0.728342]
  mu_d [0.741429, 0.767936, 0.669571, 0.728342]
  Kp [steer,rear] [10.6883, 0]; Kd [0.18963, 0]
  spawn roll -0.00759973 rad; rate -0.150361 rad/s
  gyro bias XYZ rad/s [0.00016499, -0.00012605, 0.000268673]
  gravity bias YZ [0.00414226, 0.00181435]
E4 [31.56, 33.36) s:
  mass x [0.994299, 0.90087, 1.0446, 1.10651]
  CoM XYZ mm [-4.55593, 0.216405, -0.69874]
  mu_s [0.541939, 0.727426, 0.631146, 0.528253]
  mu_d [0.534359, 0.606358, 0.631146, 0.528253]
  Kp [steer,rear] [9.10137, 0]; Kd [0.216598, 0]
  spawn roll -0.0415815 rad; rate -0.00592296 rad/s
  gyro bias XYZ rad/s [0.000175236, 0.000150931, 0.000124915]
  gravity bias YZ [0.000199283, -0.00204661]
E5 [33.36, 34.46) s:
  mass x [0.81369, 0.857251, 1.10322, 1.14334]
  CoM XYZ mm [0.838735, 3.11839, -3.82533]
  mu_s [0.541939, 0.538976, 0.728342, 0.640867]
  mu_d [0.534359, 0.538976, 0.728342, 0.556841]
  Kp [steer,rear] [9.12319, 0]; Kd [0.201365, 0]
  spawn roll -0.0193479 rad; rate 0.139358 rad/s
  gyro bias XYZ rad/s [0.000185854, -5.29748e-05, 7.57583e-05]
  gravity bias YZ [-0.00245777, 0.00206656]
E6 [34.46, 35.40) s:
  mass x [1.14994, 1.0652, 0.887675, 0.832326]
  CoM XYZ mm [4.13229, 4.01106, -0.34915]
  mu_s [0.757579, 0.546287, 0.69316, 0.631146]
  mu_d [0.70921, 0.546287, 0.69316, 0.631146]
  Kp [steer,rear] [10.9448, 0]; Kd [0.205048, 0]
  spawn roll 0.0030687 rad; rate -0.118402 rad/s
  gyro bias XYZ rad/s [7.88943e-05, -0.000186644, 0.000384477]
  gravity bias YZ [-0.00249443, 0.00376283]
E7 [35.40, 60.00) s:
  mass x [1.17704, 0.998073, 0.97533, 0.980104]
  CoM XYZ mm [-0.150428, 3.38569, -3.16444]
  mu_s [0.791209, 0.637675, 0.538976, 0.633024]
  mu_d [0.767936, 0.637675, 0.538976, 0.568172]
  Kp [steer,rear] [10.07, 0]; Kd [0.180639, 0]
  spawn roll -0.0282665 rad; rate -0.128308 rad/s
  gyro bias XYZ rad/s [9.44895e-05, -8.33654e-05, -6.65546e-05]
  gravity bias YZ [8.47903e-05, -0.000443589]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 29; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.901365, 1.01102, 1.10439, 1.0539]
  CoM XYZ mm [-2.95828, -4.5423, 0.231654]
  mu_s [0.723429, 0.69316, 0.633024, 0.538976]
  mu_d [0.723429, 0.69316, 0.568172, 0.538976]
  Kp [steer,rear] [9.07765, 0]; Kd [0.195074, 0]
  spawn roll -0.0440389 rad; rate -0.170196 rad/s
  gyro bias XYZ rad/s [7.55189e-05, 6.51789e-05, -0.000228312]
  gravity bias YZ [0.00108292, -0.000849417]
E2 [30.00, 30.28) s:
  mass x [1.15712, 1.14701, 0.925754, 1.00218]
  CoM XYZ mm [0.654529, 3.45013, 4.30141]
  mu_s [0.538434, 0.728342, 0.546287, 0.538976]
  mu_d [0.538434, 0.728342, 0.546287, 0.538976]
  Kp [steer,rear] [10.7335, 0]; Kd [0.186154, 0]
  spawn roll -0.0988358 rad; rate 0.0124952 rad/s
  gyro bias XYZ rad/s [0.000154901, 0.000115283, -5.695e-05]
  gravity bias YZ [-0.00325704, 5.60172e-05]
E3 [30.28, 30.92) s:
  mass x [0.896302, 0.960689, 1.12332, 0.967739]
  CoM XYZ mm [-2.92105, -2.56118, 1.92169]
  mu_s [0.791209, 0.546287, 0.528253, 0.69316]
  mu_d [0.767936, 0.546287, 0.528253, 0.69316]
  Kp [steer,rear] [10.9657, 0]; Kd [0.201706, 0]
  spawn roll -0.0789953 rad; rate 0.205389 rad/s
  gyro bias XYZ rad/s [-8.82892e-05, -9.22037e-05, -3.24904e-05]
  gravity bias YZ [-0.000913649, 0.00295795]
E4 [30.92, 32.88) s:
  mass x [1.19163, 0.952337, 0.964596, 0.953458]
  CoM XYZ mm [-4.88524, 2.50624, 0.287979]
  mu_s [0.611239, 0.71155, 0.791209, 0.616244]
  mu_d [0.611239, 0.71155, 0.767936, 0.586498]
  Kp [steer,rear] [10.1973, 0]; Kd [0.21061, 0]
  spawn roll -0.0341768 rad; rate -0.0950615 rad/s
  gyro bias XYZ rad/s [-4.88644e-05, 8.31145e-05, -0.000226066]
  gravity bias YZ [0.000706437, 0.000516435]
E5 [32.88, 60.00) s:
  mass x [1.04432, 1.13414, 1.1779, 1.14262]
  CoM XYZ mm [4.7274, -1.36799, -0.577485]
  mu_s [0.748289, 0.728342, 0.748289, 0.616244]
  mu_d [0.689499, 0.728342, 0.689499, 0.586498]
  Kp [steer,rear] [9.3829, 0]; Kd [0.196605, 0]
  spawn roll -0.0652417 rad; rate 0.233495 rad/s
  gyro bias XYZ rad/s [-0.000249581, 6.77096e-05, 4.06295e-05]
  gravity bias YZ [0.000499155, -0.00600322]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 30 · full 60 s



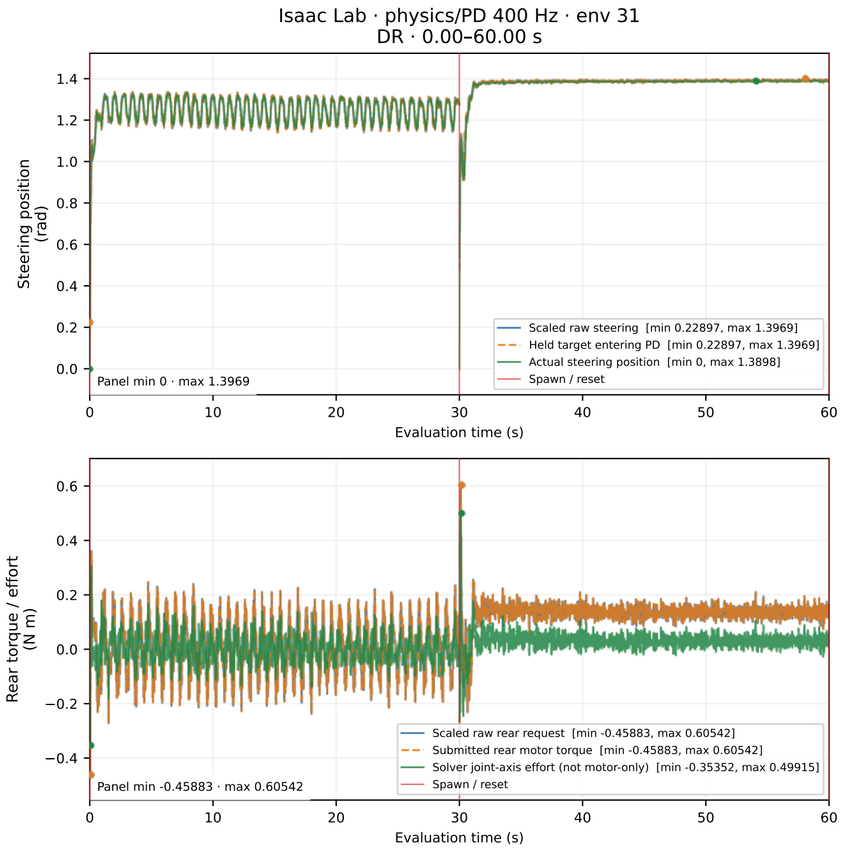
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

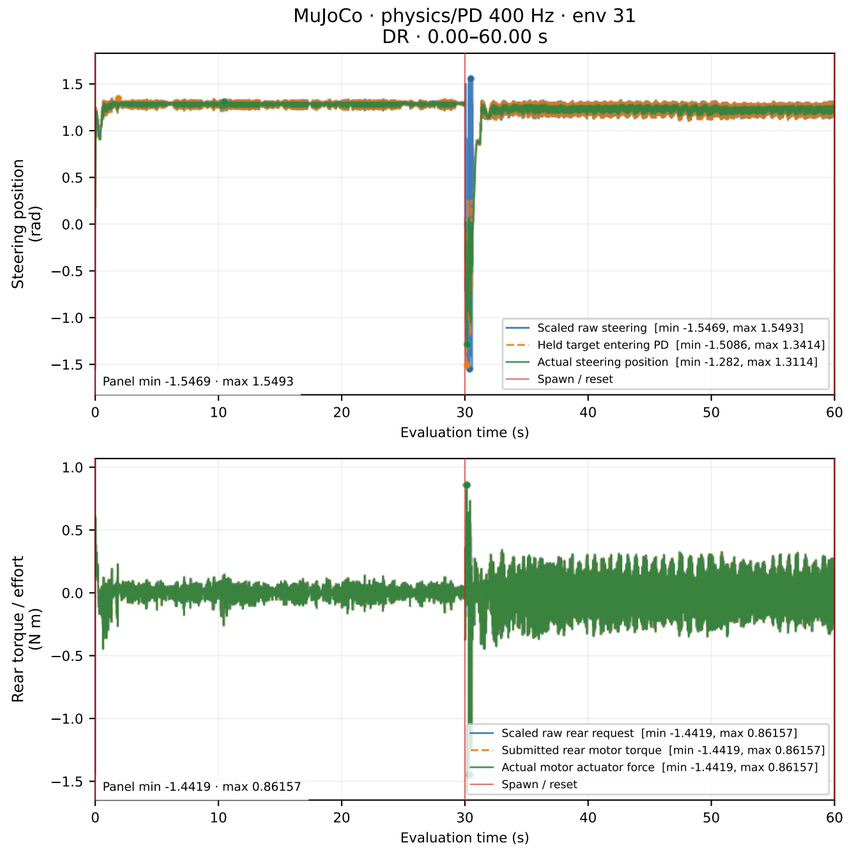
```
SAMPLED DR: env 30; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 13.48) s:
  mass x [0.924668, 1.07947, 1.12819, 1.14008]
  CoM XYZ mm [-3.09983, -2.81616, 0.847831]
  mu_s [0.579974, 0.703711, 0.693205, 0.598343]
  mu_d [0.579974, 0.703711, 0.693205, 0.598343]
  Kp [steer,rear] [10.1628, 0]; Kd [0.185653, 0]
  spawn roll -0.0300911 rad; rate -0.232015 rad/s
  gyro bias XYZ rad/s [0.000215005, 0.00012291, -0.000360078]
  gravity bias YZ [-0.0029152, -0.000581331]
E2 [13.48, 43.48) s:
  mass x [0.997325, 1.1252, 1.01393, 0.844822]
  CoM XYZ mm [4.16946, 1.41413, 4.34159]
  mu_s [0.628821, 0.751971, 0.741039, 0.785466]
  mu_d [0.628821, 0.541224, 0.614405, 0.52258]
  Kp [steer,rear] [9.06119, 0]; Kd [0.18606, 0]
  spawn roll 0.0747708 rad; rate 0.100169 rad/s
  gyro bias XYZ rad/s [-0.000290741, 4.18535e-05, 0.000158674]
  gravity bias YZ [-0.000980066, 0.00217898]
E3 [43.48, 60.00) s:
  mass x [0.831246, 0.886655, 0.923206, 0.906771]
  CoM XYZ mm [2.95128, 4.11288, 2.91291]
  mu_s [0.547743, 0.532645, 0.794338, 0.592046]
  mu_d [0.547743, 0.532645, 0.526209, 0.534947]
  Kp [steer,rear] [10.178, 0]; Kd [0.193007, 0]
  spawn roll -0.0606137 rad; rate 0.22692 rad/s
  gyro bias XYZ rad/s [6.64626e-06, -4.15746e-06, 1.02434e-05]
  gravity bias YZ [0.000547827, 0.0021663]
Six significant digits shown; full precision: sampled_dr_values.json.
```

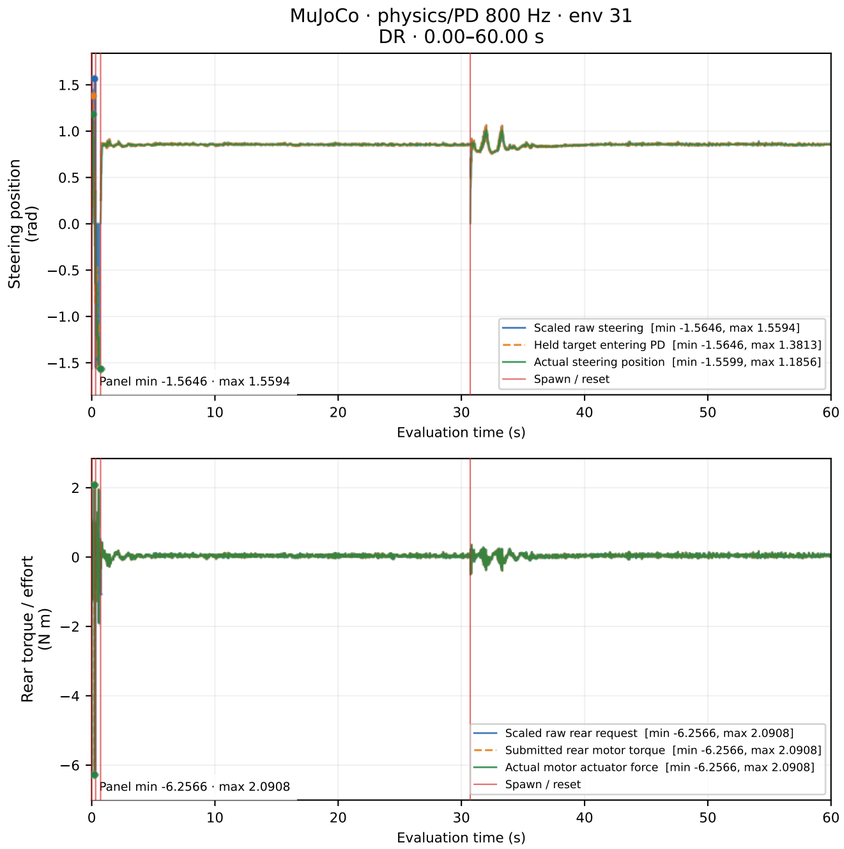
```
SAMPLED DR: env 30; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.34) s:
  mass x [1.13513, 0.80267, 0.892579, 0.845694]
  CoM XYZ mm [2.15583, -1.89545, 0.733078]
  mu_s [0.559972, 0.757579, 0.640867, 0.748289]
  mu_d [0.502209, 0.70921, 0.556841, 0.689499]
  Kp [steer,rear] [10.9072, 0]; Kd [0.193931, 0]
  spawn roll 0.0574297 rad; rate -0.165354 rad/s
  gyro bias XYZ rad/s [0.00015069, -1.76586e-05, -0.000395626]
  gravity bias YZ [0.000517843, 0.000246497]
E2 [0.34, 0.68) s:
  mass x [0.814908, 0.930503, 1.00509, 1.18996]
  CoM XYZ mm [-4.98003, -4.36759, -1.88294]
  mu_s [0.71008, 0.669571, 0.748289, 0.528253]
  mu_d [0.59371, 0.669571, 0.689499, 0.528253]
  Kp [steer,rear] [10.0369, 0]; Kd [0.208819, 0]
  spawn roll 0.0361603 rad; rate -0.237526 rad/s
  gyro bias XYZ rad/s [0.00016799, 2.68466e-05, 8.12228e-05]
  gravity bias YZ [-0.00577595, -0.000572445]
E3 [0.68, 0.90) s:
  mass x [1.14454, 1.19582, 0.826954, 0.940737]
  CoM XYZ mm [2.94838, -4.74773, 2.04879]
  mu_s [0.631146, 0.541939, 0.541939, 0.704749]
  mu_d [0.631146, 0.534359, 0.534359, 0.541926]
  Kp [steer,rear] [10.2926, 0]; Kd [0.21078, 0]
  spawn roll 0.0200943 rad; rate 0.184764 rad/s
  gyro bias XYZ rad/s [-0.000151875, 9.23402e-05, 0.000155635]
  gravity bias YZ [0.00014534, -0.000752591]
E4 [0.90, 1.26) s:
  mass x [1.0366, 1.17037, 0.903821, 0.866983]
  CoM XYZ mm [-3.62126, 4.25613, -1.90264]
  mu_s [0.71008, 0.541939, 0.728342, 0.704749]
  mu_d [0.59371, 0.534359, 0.728342, 0.541926]
  Kp [steer,rear] [10.7814, 0]; Kd [0.202091, 0]
  spawn roll -0.0368995 rad; rate -0.137435 rad/s
  gyro bias XYZ rad/s [0.000173283, -2.58349e-06, 0.000304482]
  gravity bias YZ [-0.000570498, -0.00117011]
E5 [1.26, 16.46) s:
  mass x [0.942509, 0.814669, 0.926681, 1.19858]
  CoM XYZ mm [-1.89303, -4.76264, -0.849581]
  mu_s [0.538434, 0.633024, 0.69316, 0.757579]
  mu_d [0.538434, 0.568172, 0.69316, 0.70921]
  Kp [steer,rear] [9.83469, 0]; Kd [0.211493, 0]
  spawn roll 0.0108082 rad; rate -0.218889 rad/s
  gyro bias XYZ rad/s [0.000160507, -0.000274877, -0.00013085]
  gravity bias YZ [0.00150107, -0.00064928]
E6 [16.46, 16.66) s:
  mass x [0.946566, 1.17225, 1.09063, 0.804103]
  CoM XYZ mm [-0.174654, 0.0803397, 3.76314]
  mu_s [0.728342, 0.736077, 0.546287, 0.611239]
  mu_d [0.728342, 0.699455, 0.546287, 0.611239]
  Kp [steer,rear] [10.9993, 0]; Kd [0.195934, 0]
  spawn roll 0.0770678 rad; rate -0.00949378 rad/s
  gyro bias XYZ rad/s [-4.24715e-05, -6.68252e-05, -0.000254964]
  gravity bias YZ [-0.00118173, -0.0016484]
E7 [16.66, 46.66) s:
  mass x [0.810637, 1.03981, 1.14723, 1.1206]
  CoM XYZ mm [2.22889, 0.925924, -4.88346]
  mu_s [0.757579, 0.538434, 0.748289, 0.559972]
  mu_d [0.70921, 0.538434, 0.689499, 0.502209]
  Kp [steer,rear] [9.69393, 0]; Kd [0.214131, 0]
  spawn roll 0.0192156 rad; rate -0.227771 rad/s
  gyro bias XYZ rad/s [-9.3961e-05, -3.74562e-05, 0.000290037]
  gravity bias YZ [0.00207558, 0.000515313]
E8 [46.66, 46.96) s:
  mass x [0.91235, 1.1217, 1.09581, 0.858619]
  CoM XYZ mm [-1.50039, 3.80481, -0.620426]
  mu_s [0.757579, 0.736077, 0.749678, 0.728342]
  mu_d [0.70921, 0.699455, 0.741429, 0.728342]
  Kp [steer,rear] [10.1318, 0]; Kd [0.200717, 0]
  spawn roll 0.0444284 rad; rate 0.0910835 rad/s
  gyro bias XYZ rad/s [0.000170838, -0.000171349, -3.40807e-06]
  gravity bias YZ [-0.0000961515, -0.00179085]
E9 [46.96, 48.60) s:
  mass x [1.1734, 1.15025, 0.876941, 0.993433]
  CoM XYZ mm [4.87998, 1.67559, -2.34641]
  mu_s [0.71008, 0.597748, 0.757579, 0.748289]
  mu_d [0.59371, 0.597748, 0.70921, 0.689499]
  Kp [steer,rear] [10.3798, 0]; Kd [0.213412, 0]
  spawn roll 0.0482838 rad; rate 0.158738 rad/s
  gyro bias XYZ rad/s [0.000146746, -0.000149573, -5.23109e-05]
  gravity bias YZ [0.000241813, 0.000491757]
E10 [48.60, 53.68) s:
  mass x [0.812724, 0.999431, 0.939454, 1.03354]
  CoM XYZ mm [-4.6654, 2.52297, -1.23633]
  mu_s [0.748289, 0.733515, 0.723429, 0.71008]
  mu_d [0.689499, 0.558392, 0.723429, 0.59371]
  Kp [steer,rear] [9.70706, 0]; Kd [0.193021, 0]
  spawn roll -0.0399049 rad; rate 0.215487 rad/s
  gyro bias XYZ rad/s [6.37738e-05, 0.00011428, 9.97606e-05]
  gravity bias YZ [0.00241983, 0.00194708]
E11 [53.68, 54.04) s:
  mass x [1.02559, 1.08955, 0.975661, 1.13058]
  CoM XYZ mm [2.08939, 4.20951, 3.9372]
  mu_s [0.611239, 0.541939, 0.568073, 0.611239]
  mu_d [0.611239, 0.534359, 0.568073, 0.611239]
  Kp [steer,rear] [9.0622, 0]; Kd [0.209563, 0]
  spawn roll -0.0816654 rad; rate 0.226339 rad/s
  gyro bias XYZ rad/s [0.00010862, 0.000208024, -0.000175451]
  gravity bias YZ [0.000335151, 0.000128454]
E12 [54.04, 54.78) s:
  mass x [0.867411, 1.11808, 0.961108, 0.899199]
  CoM XYZ mm [-2.40154, 2.61398, 3.00352]
  mu_s [0.727426, 0.71008, 0.71155, 0.732187]
  mu_d [0.606358, 0.59371, 0.71155, 0.631664]
  Kp [steer,rear] [10.2915, 0]; Kd [0.216056, 0]
  spawn roll -0.0422033 rad; rate 0.0399885 rad/s
  gyro bias XYZ rad/s [0.000379283, 0.000219853, 0.000175484]
  gravity bias YZ [0.00150931, 0.000822312]
E13 [54.78, 55.98) s:
  mass x [0.971327, 1.06309, 0.882956, 1.04843]
  CoM XYZ mm [-1.44661, 0.67016, 3.7355]
  mu_s [0.723429, 0.749678, 0.71155, 0.71155]
  mu_d [0.723429, 0.741429, 0.71155, 0.71155]
  Kp [steer,rear] [9.38777, 0]; Kd [0.207596, 0]
  spawn roll -0.0713152 rad; rate -0.0630836 rad/s
  gyro bias XYZ rad/s [-1.12214e-05, 5.21473e-05, -0.000534602]
  gravity bias YZ [0.000144944, 0.000659295]
E14 [55.98, 60.00) s:
  mass x [0.800622, 1.15836, 1.0868, 0.876023]
  CoM XYZ mm [-2.21413, -3.54437, -1.20594]
  mu_s [0.700521, 0.640016, 0.546287, 0.723429]
  mu_d [0.641329, 0.513141, 0.546287, 0.723429]
  Kp [steer,rear] [9.76497, 0]; Kd [0.19418, 0]
  spawn roll 0.0248617 rad; rate -0.214199 rad/s
  gyro bias XYZ rad/s [-0.000122645, 6.29751e-05, 0.000443132]
  gravity bias YZ [0.00233486, -0.00106821]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 30; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.13513, 0.80267, 0.892579, 0.845694]
  CoM XYZ mm [2.15583, -1.89545, 0.733078]
  mu_s [0.559972, 0.757579, 0.640867, 0.748289]
  mu_d [0.502209, 0.70921, 0.556841, 0.689499]
  Kp [steer,rear] [10.9072, 0]; Kd [0.193931, 0]
  spawn roll 0.0574297 rad; rate -0.165354 rad/s
  gyro bias XYZ rad/s [0.00015069, -1.76586e-05, -0.000395626]
  gravity bias YZ [0.000517843, 0.000246497]
E2 [30.00, 60.00) s:
  mass x [0.856053, 1.04344, 1.00687, 1.18416]
  CoM XYZ mm [2.97977, -0.942431, -4.27524]
  mu_s [0.791209, 0.71155, 0.704749, 0.597748]
  mu_d [0.767936, 0.71155, 0.541926, 0.597748]
  Kp [steer,rear] [9.25988, 0]; Kd [0.186814, 0]
  spawn roll 0.0430583 rad; rate -0.0280241 rad/s
  gyro bias XYZ rad/s [0.000535818, 0.000184314, 0.000122627]
  gravity bias YZ [-0.000818223, 0.000831351]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 31 · full 60 s







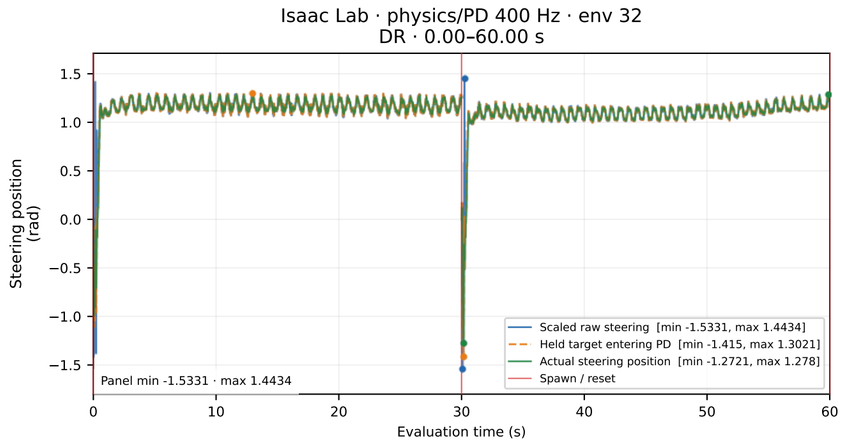
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

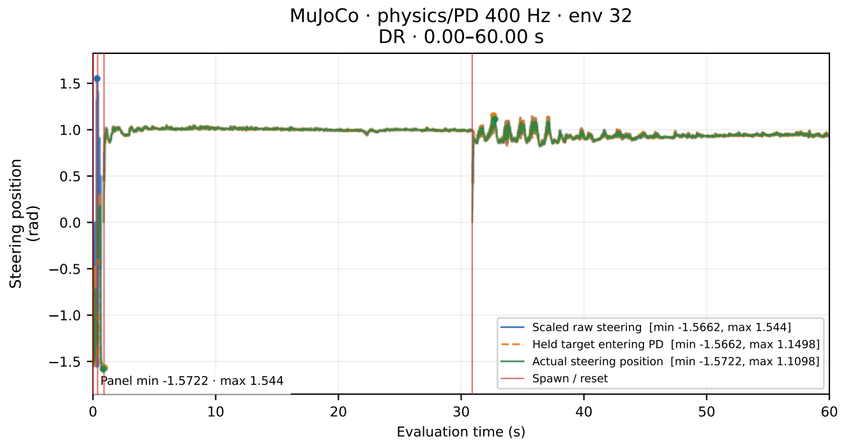
```
SAMPLED DR: env 31; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.888913, 0.920628, 1.16873, 0.999925]
  CoM XYZ mm [-2.61347, -1.67952, -0.69344]
  mu_s [0.547743, 0.787792, 0.559809, 0.673378]
  mu_d [0.547743, 0.617134, 0.559809, 0.673378]
  Kp [steer,rear] [10.5455, 0]; Kd [0.198474, 0]
  spawn roll -0.00862077 rad; rate 0.0478292 rad/s
  gyro bias XYZ rad/s [3.97985e-05, -0.000110208, -1.53622e-05]
  gravity bias YZ [0.000798645, 0.00177435]
E2 [30.00, 60.00) s:
  mass x [1.16364, 0.887446, 1.15393, 1.06103]
  CoM XYZ mm [-0.166044, -4.54881, -4.72042]
  mu_s [0.764681, 0.787792, 0.547743, 0.584424]
  mu_d [0.764681, 0.617134, 0.547743, 0.584424]
  Kp [steer,rear] [9.07667, 0]; Kd [0.215742, 0]
  spawn roll 0.0105771 rad; rate 0.105873 rad/s
  gyro bias XYZ rad/s [1.70873e-05, -9.48296e-05, -6.3715e-05]
  gravity bias YZ [1.16744e-05, -0.00196036]
Six significant digits shown; full precision: sampled_dr_values.json.
```

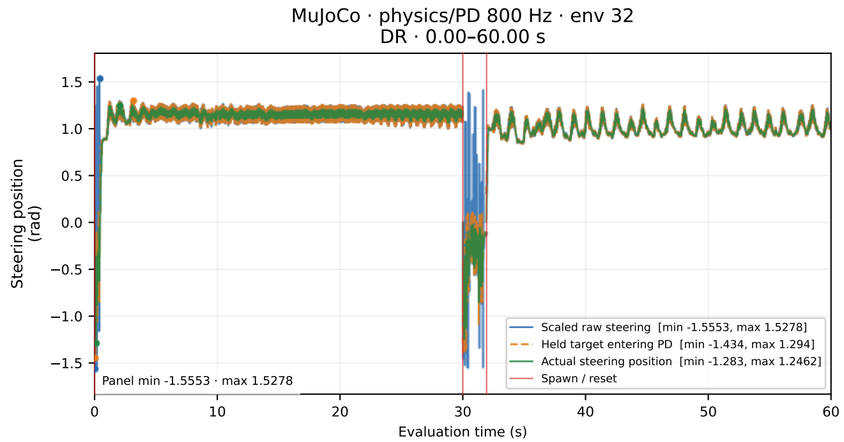
```
SAMPLED DR: env 31; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.997054, 0.921347, 0.907427, 1.19406]
  CoM XYZ mm [-2.80109, -4.67992, 3.54697]
  mu_s [0.749678, 0.611239, 0.736077, 0.736077]
  mu_d [0.741429, 0.611239, 0.699455, 0.699455]
  Kp [steer,rear] [10.4465, 0]; Kd [0.184146, 0]
  spawn roll 0.0612062 rad; rate -0.192569 rad/s
  gyro bias XYZ rad/s [-0.000168277, 0.000281201, 0.000184886]
  gravity bias YZ [-0.000333241, 0.00361406]
E2 [30.00, 60.00) s:
  mass x [0.964597, 0.977022, 1.08373, 1.10108]
  CoM XYZ mm [2.24418, -0.39583, -3.62629]
  mu_s [0.748289, 0.71008, 0.791209, 0.597748]
  mu_d [0.689499, 0.59371, 0.767936, 0.597748]
  Kp [steer,rear] [9.78566, 0]; Kd [0.194591, 0]
  spawn roll -0.095277 rad; rate 0.228682 rad/s
  gyro bias XYZ rad/s [-1.55867e-05, -0.000193466, -4.36437e-05]
  gravity bias YZ [0.00434751, 0.0011049]
Six significant digits shown; full precision: sampled_dr_values.json.
```

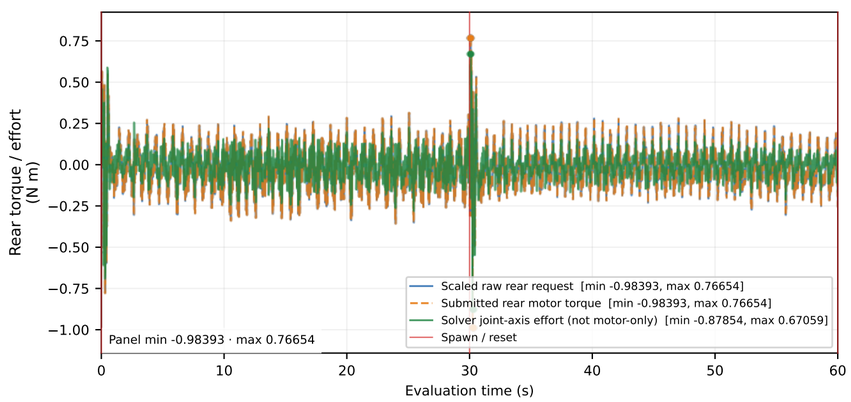
```
SAMPLED DR: env 31; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.32) s:
  mass x [0.997054, 0.921347, 0.907427, 1.19406]
  CoM XYZ mm [-2.80109, -4.67992, 3.54697]
  mu_s [0.749678, 0.611239, 0.736077, 0.736077]
  mu_d [0.741429, 0.611239, 0.699455, 0.699455]
  Kp [steer,rear] [10.4465, 0]; Kd [0.184146, 0]
  spawn roll 0.0612062 rad; rate -0.192569 rad/s
  gyro bias XYZ rad/s [-0.000168277, 0.000281201, 0.000184886]
  gravity bias YZ [-0.000333241, 0.00361406]
E2 [0.32, 0.72) s:
  mass x [0.966366, 1.19041, 0.936999, 1.10457]
  CoM XYZ mm [-4.39216, 2.49833, 3.69328]
  mu_s [0.749678, 0.704749, 0.640867, 0.69316]
  mu_d [0.741429, 0.541926, 0.556841, 0.69316]
  Kp [steer,rear] [10.7143, 0]; Kd [0.19111, 0]
  spawn roll -0.0976437 rad; rate -0.134834 rad/s
  gyro bias XYZ rad/s [0.000126475, -0.000147112, -0.00012958]
  gravity bias YZ [0.000159554, 0.00158182]
E3 [0.72, 30.72) s:
  mass x [0.900998, 0.800325, 1.1386, 1.04068]
  CoM XYZ mm [1.69883, 3.85388, 4.13658]
  mu_s [0.71155, 0.637675, 0.732187, 0.640867]
  mu_d [0.71155, 0.637675, 0.631664, 0.556841]
  Kp [steer,rear] [9.83752, 0]; Kd [0.215052, 0]
  spawn roll 0.0430423 rad; rate -0.109067 rad/s
  gyro bias XYZ rad/s [-0.000180353, 0.000203191, 0.00028134]
  gravity bias YZ [0.000324342, -0.000905106]
E4 [30.72, 60.00) s:
  mass x [0.957543, 0.910966, 0.804324, 0.859663]
  CoM XYZ mm [4.23565, 3.55379, -3.90607]
  mu_s [0.568073, 0.538976, 0.757579, 0.538976]
  mu_d [0.568073, 0.538976, 0.70921, 0.538976]
  Kp [steer,rear] [9.0268, 0]; Kd [0.216424, 0]
  spawn roll 0.0149589 rad; rate 0.184369 rad/s
  gyro bias XYZ rad/s [-0.000210905, -0.000187521, -0.000211918]
  gravity bias YZ [-0.00353316, -0.00337261]
Six significant digits shown; full precision: sampled_dr_values.json.
```

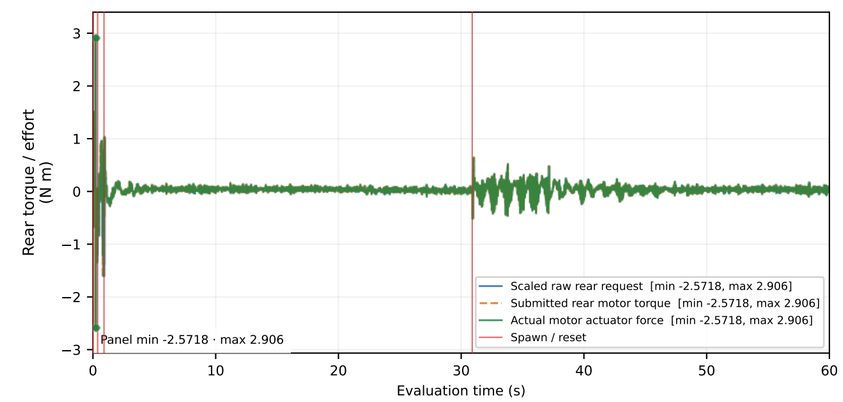
model_2500 · Demand versus actuation · env 32 · full 60 s

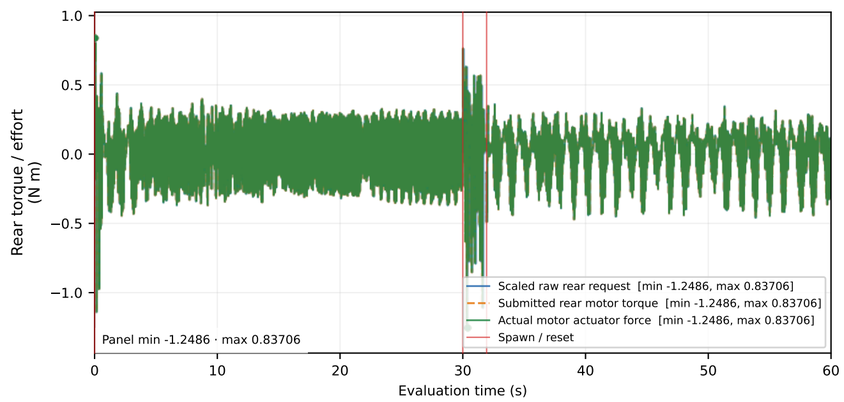












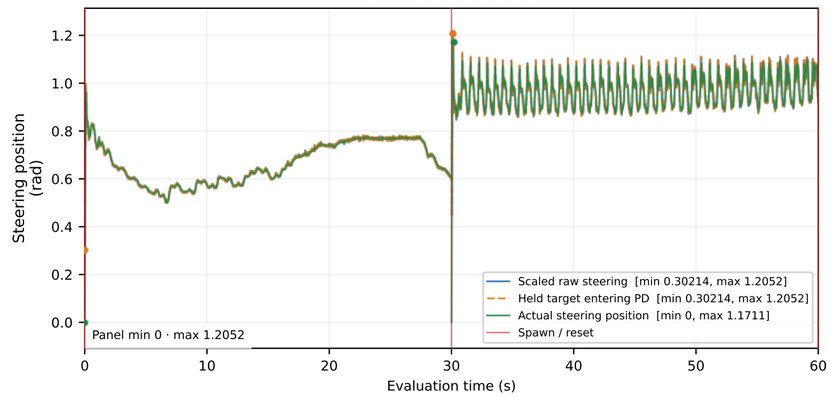
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

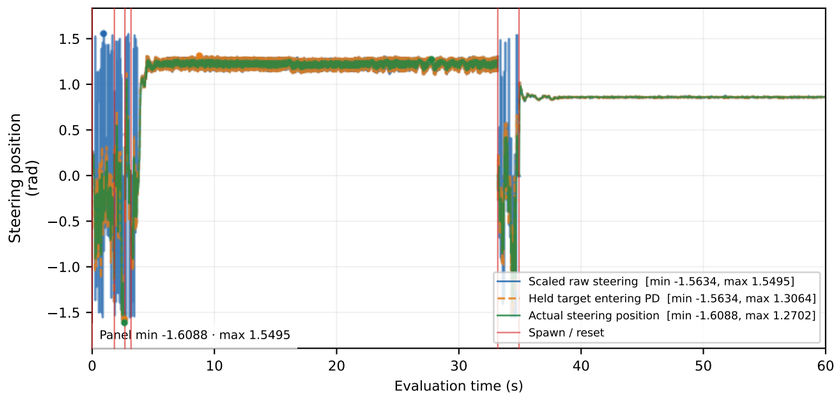
```
SAMPLED DR: env 32; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.15354, 0.994755, 1.13212, 0.895552]
  CoM XYZ mm [0.827014, -0.963841, -4.22029]
  mu_s [0.533455, 0.794338, 0.659445, 0.628821]
  mu_d [0.533455, 0.526209, 0.547619, 0.628821]
  Kp [steer,rear] [9.07548, 0]; Kd [0.198498, 0]
  spawn roll -0.0426314 rad; rate -0.19918 rad/s
  gyro bias XYZ rad/s [-0.000229271, 0.00035585, 0.000120208]
  gravity bias YZ [-0.00213508, -0.0010165]
E2 [30.00, 60.00) s:
  mass x [1.11867, 1.17444, 0.927634, 0.801768]
  CoM XYZ mm [2.46421, -1.17602, 4.30979]
  mu_s [0.741039, 0.764681, 0.559809, 0.592046]
  mu_d [0.614405, 0.764681, 0.559809, 0.534947]
  Kp [steer,rear] [9.73381, 0]; Kd [0.202134, 0]
  spawn roll -0.0633899 rad; rate 0.130295 rad/s
  gyro bias XYZ rad/s [-0.000184547, -0.000195907, 0.000120958]
  gravity bias YZ [0.000194793, -8.68671e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

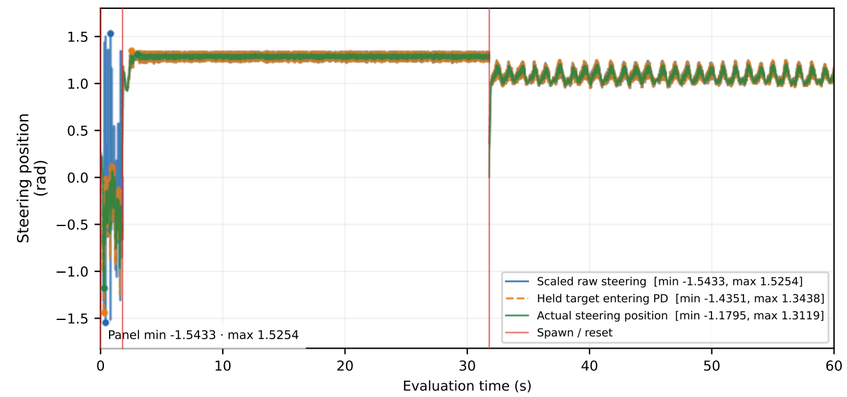
```
SAMPLED DR: env 32; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.38) s:
  mass x [1.15775, 1.00829, 1.07056, 1.06768]
  CoM XYZ mm [0.648381, 0.245347, -4.96027]
  mu_s [0.669571, 0.616244, 0.723429, 0.669571]
  mu_d [0.669571, 0.586498, 0.723429, 0.669571]
  Kp [steer,rear] [10.5561, 0]; Kd [0.218481, 0]
  spawn roll -0.0797969 rad; rate -0.148618 rad/s
  gyro bias XYZ rad/s [0.000199814, -6.26873e-05, 1.47468e-05]
  gravity bias YZ [0.000709582, 0.000497162]
E2 [0.38, 0.90) s:
  mass x [1.16707, 0.822879, 1.17134, 0.853268]
  CoM XYZ mm [-4.69558, -2.80143, 0.637254]
  mu_s [0.541939, 0.637675, 0.568073, 0.546287]
  mu_d [0.534359, 0.637675, 0.568073, 0.546287]
  Kp [steer,rear] [10.1914, 0]; Kd [0.19981, 0]
  spawn roll -0.04112596 rad; rate -0.0897145 rad/s
  gyro bias XYZ rad/s [0.00037647, 7.1709e-05, -3.5109e-05]
  gravity bias YZ [0.00389114, -0.00101151]
E3 [0.90, 30.90) s:
  mass x [1.17554, 1.16043, 0.862716, 0.958754]
  CoM XYZ mm [1.08892, 1.02467, 3.20079]
  mu_s [0.723429, 0.757579, 0.69316, 0.568073]
  mu_d [0.723429, 0.70921, 0.69316, 0.568073]
  Kp [steer,rear] [10.1718, 0]; Kd [0.19536, 0]
  spawn roll 0.0820112 rad; rate 0.0157743 rad/s
  gyro bias XYZ rad/s [-9.90621e-05, -0.00010471, 0.000284694]
  gravity bias YZ [0.00247652, 0.00237624]
E4 [30.90, 60.00) s:
  mass x [0.876745, 0.875549, 0.953014, 0.9418]
  CoM XYZ mm [-2.51835, 2.13953, -3.58757]
  mu_s [0.791209, 0.640867, 0.669571, 0.704749]
  mu_d [0.767936, 0.556841, 0.669571, 0.541926]
  Kp [steer,rear] [10.1907, 0]; Kd [0.194124, 0]
  spawn roll 0.0385587 rad; rate 0.0718184 rad/s
  gyro bias XYZ rad/s [-0.000119636, 0.000344695, 5.95735e-05]
  gravity bias YZ [0.000369942, 0.000484901]
Six significant digits shown; full precision: sampled_dr_values.json.
```

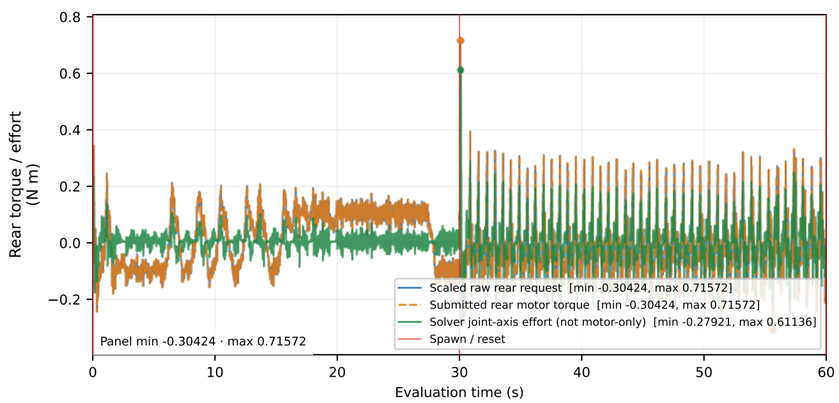
```
SAMPLED DR: env 32; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.15775, 1.00829, 1.07056, 1.06768]
  CoM XYZ mm [0.648381, 0.245347, -4.96027]
  mu_s [0.669571, 0.616244, 0.723429, 0.669571]
  mu_d [0.669571, 0.586498, 0.723429, 0.669571]
  Kp [steer,rear] [10.5561, 0]; Kd [0.218481, 0]
  spawn roll -0.0797969 rad; rate -0.148618 rad/s
  gyro bias XYZ rad/s [0.000199814, -6.26873e-05, 1.47468e-05]
  gravity bias YZ [0.000709582, 0.000497162]
E2 [30.00, 31.94) s:
  mass x [0.844711, 1.06078, 1.07912, 1.0883]
  CoM XYZ mm [-0.750851, 2.08231, 3.34438]
  mu_s [0.611239, 0.791209, 0.538976, 0.748289]
  mu_d [0.611239, 0.767936, 0.538976, 0.689499]
  Kp [steer,rear] [10.4035, 0]; Kd [0.215361, 0]
  spawn roll -0.0623382 rad; rate -0.11613 rad/s
  gyro bias XYZ rad/s [-6.18384e-05, -0.000112417, 0.000212459]
  gravity bias YZ [-0.000220083, -0.00215042]
E3 [31.94, 60.00) s:
  mass x [0.918723, 0.858997, 1.01245, 1.1257]
  CoM XYZ mm [-3.37096, 1.56511, -1.90493]
  mu_s [0.727426, 0.69316, 0.71008, 0.732187]
  mu_d [0.606358, 0.69316, 0.59371, 0.631664]
  Kp [steer,rear] [10.2083, 0]; Kd [0.199927, 0]
  spawn roll 0.00198396 rad; rate 0.0446447 rad/s
  gyro bias XYZ rad/s [-0.000102384, -0.000182165, 0.000137707]
  gravity bias YZ [0.00167768, -0.00296171]
Six significant digits shown; full precision: sampled_dr_values.json.
```

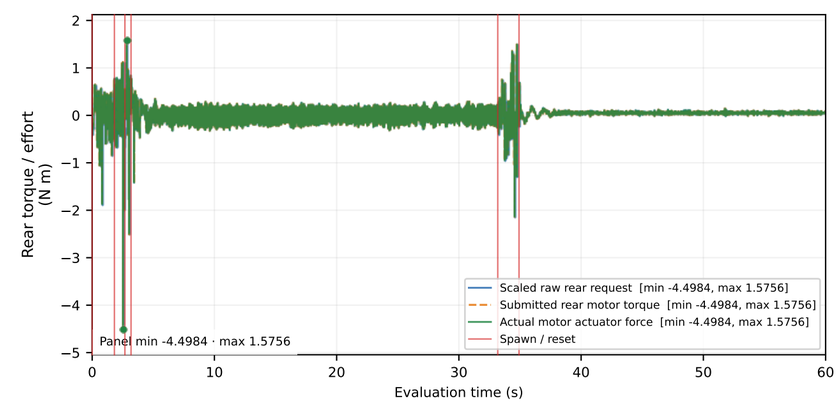
model_2500 · Demand versus actuation · env 33 · full 60 s

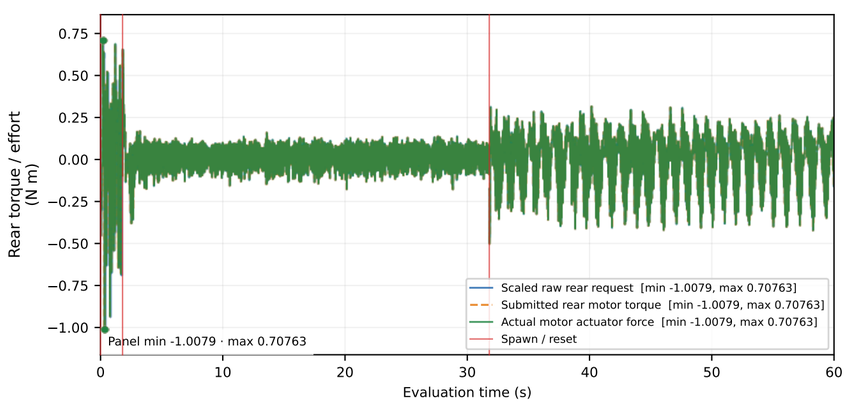












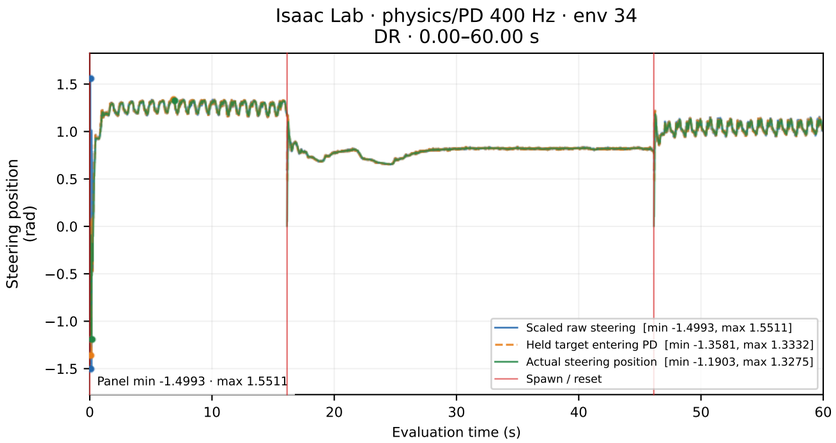
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

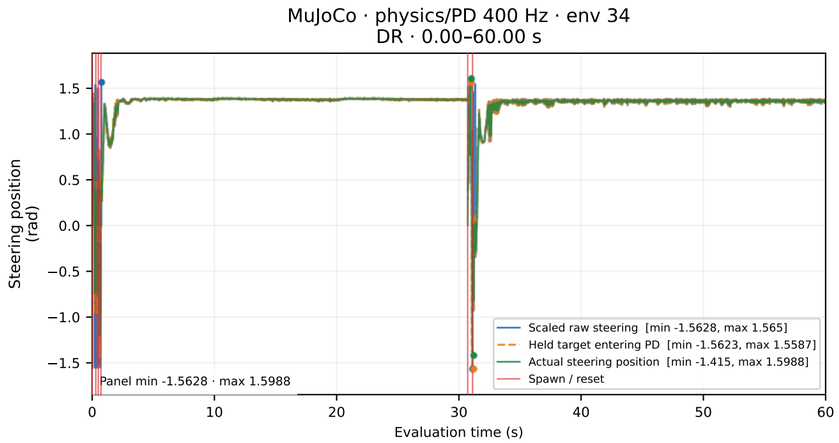
```
SAMPLED DR: env 33; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.918787, 0.909582, 0.860474, 0.816054]
  CoM XYZ mm [-1.41342, 4.90098, 1.82721]
  mu_s [0.547743, 0.547743, 0.674963, 0.674963]
  mu_d [0.547743, 0.547743, 0.601294, 0.601294]
  Kp [steer,rear] [9.07892, 0]; Kd [0.187439, 0]
  spawn roll 0.0479791 rad; rate -0.107307 rad/s
  gyro bias XYZ rad/s [5.93858e-05, 0.000281976, 0.000140135]
  gravity bias YZ [-0.000156539, 0.00012416]
E2 [30.00, 60.00) s:
  mass x [1.18798, 0.981603, 0.839946, 1.0543]
  CoM XYZ mm [-2.37614, 1.46068, 1.5182]
  mu_s [0.532645, 0.751971, 0.547743, 0.632409]
  mu_d [0.532645, 0.541224, 0.547743, 0.589076]
  Kp [steer,rear] [9.99212, 0]; Kd [0.192678, 0]
  spawn roll 0.0498407 rad; rate -0.0544084 rad/s
  gyro bias XYZ rad/s [0.000215835, 0.000132055, -0.000148476]
  gravity bias YZ [0.00101097, -0.00239106]
Six significant digits shown; full precision: sampled_dr_values.json.
```

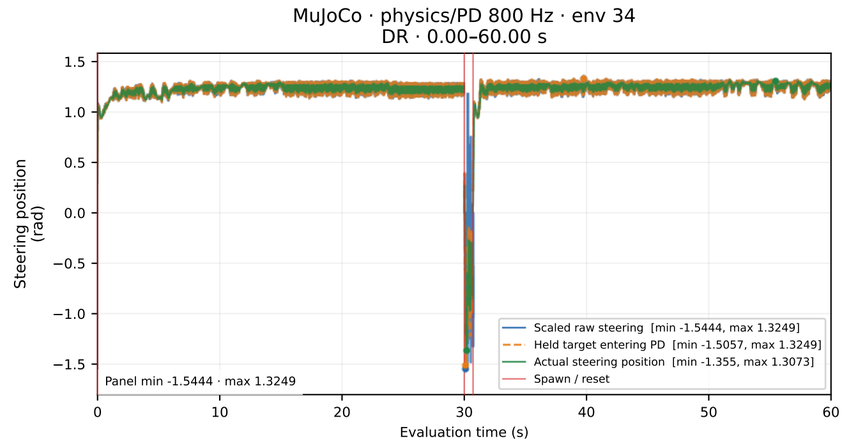
```
SAMPLED DR: env 33; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.82) s:
  mass x [0.885317, 0.861252, 1.0657, 1.07373]
  CoM XYZ mm [4.33164, 2.5149, 4.26862]
  mu_s [0.704749, 0.700521, 0.749678, 0.71155]
  mu_d [0.541926, 0.641329, 0.741429, 0.71155]
  Kp [steer,rear] [9.0252, 0]; Kd [0.212922, 0]
  spawn roll -0.0197365 rad; rate 0.0835653 rad/s
  gyro bias XYZ rad/s [-3.26169e-05, 0.000319795, -0.000258576]
  gravity bias YZ [-0.00102961, -0.00178678]
E2 [1.82, 2.68) s:
  mass x [0.977003, 0.802712, 0.882463, 0.834067]
  CoM XYZ mm [-2.07259, -2.23026, -0.33118]
  mu_s [0.611239, 0.71008, 0.704749, 0.538976]
  mu_d [0.611239, 0.59371, 0.541926, 0.538976]
  Kp [steer,rear] [10.9342, 0]; Kd [0.192036, 0]
  spawn roll -0.0112658 rad; rate 0.0655895 rad/s
  gyro bias XYZ rad/s [0.000292643, -0.000192386, 0.000102021]
  gravity bias YZ [-0.00345355, 0.000951618]
E3 [2.68, 3.18) s:
  mass x [0.99145, 1.09962, 0.923818, 0.832726]
  CoM XYZ mm [2.22311, 1.26657, -4.71255]
  mu_s [0.757579, 0.640016, 0.669571, 0.568073]
  mu_d [0.70921, 0.513141, 0.669571, 0.568073]
  Kp [steer,rear] [9.43703, 0]; Kd [0.212337, 0]
  spawn roll 0.0622928 rad; rate -0.0996375 rad/s
  gyro bias XYZ rad/s [0.000216561, 0.000179249, 0.000152826]
  gravity bias YZ [-0.000605763, -0.00251901]
E4 [3.18, 33.18) s:
  mass x [0.951043, 1.06067, 1.1946, 1.08842]
  CoM XYZ mm [-2.4262, -1.6443, 0.533732]
  mu_s [0.757579, 0.728342, 0.71008, 0.597748]
  mu_d [0.70921, 0.728342, 0.59371, 0.597748]
  Kp [steer,rear] [9.39824, 0]; Kd [0.181749, 0]
  spawn roll -0.0559239 rad; rate -0.19513 rad/s
  gyro bias XYZ rad/s [0.000296256, -0.000421684, 1.61355e-05]
  gravity bias YZ [-0.00111529, 0.000311467]
E5 [33.18, 34.92) s:
  mass x [0.958945, 1.1996, 0.885858, 1.13036]
  CoM XYZ mm [2.48551, -0.882421, 4.8983]
  mu_s [0.704749, 0.538434, 0.546287, 0.568073]
  mu_d [0.541926, 0.538434, 0.546287, 0.568073]
  Kp [steer,rear] [10.4711, 0]; Kd [0.201411, 0]
  spawn roll -0.00859318 rad; rate -0.165972 rad/s
  gyro bias XYZ rad/s [-0.000297346, -0.000132116, 1.4391e-05]
  gravity bias YZ [0.000481221, 0.00282605]
E6 [34.92, 60.00) s:
  mass x [1.15527, 1.01585, 1.04687, 1.03184]
  CoM XYZ mm [-4.60195, 3.53239, -1.15752]
  mu_s [0.700521, 0.568073, 0.616244, 0.528253]
  mu_d [0.641329, 0.568073, 0.586498, 0.528253]
  Kp [steer,rear] [10.6587, 0]; Kd [0.207751, 0]
  spawn roll 0.0648386 rad; rate 0.202985 rad/s
  gyro bias XYZ rad/s [-0.000350997, -3.76622e-05, -2.16671e-05]
  gravity bias YZ [0.00101979, 0.000359558]
Six significant digits shown; full precision: sampled_dr_values.json.
```

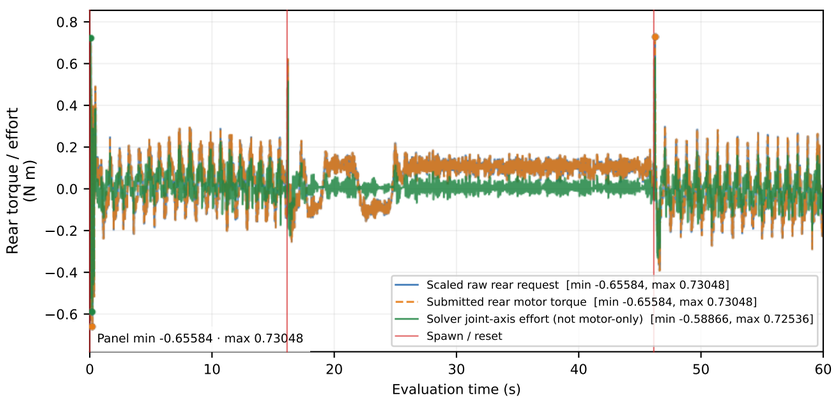
```
SAMPLED DR: env 33; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.80) s:
  mass x [0.885317, 0.861252, 1.0657, 1.07373]
  CoM XYZ mm [4.33164, 2.5149, 4.26862]
  mu_s [0.704749, 0.700521, 0.749678, 0.71155]
  mu_d [0.541926, 0.641329, 0.741429, 0.71155]
  Kp [steer,rear] [9.0252, 0]; Kd [0.212922, 0]
  spawn roll -0.0197365 rad; rate 0.0835653 rad/s
  gyro bias XYZ rad/s [-3.26169e-05, 0.000319795, -0.000258576]
  gravity bias YZ [-0.00102961, -0.00178678]
E2 [1.80, 31.80) s:
  mass x [1.07268, 1.1171, 1.18264, 1.04602]
  CoM XYZ mm [-2.91, -4.40165, 2.16322]
  mu_s [0.733515, 0.666375, 0.640867, 0.704749]
  mu_d [0.558392, 0.519145, 0.556841, 0.541926]
  Kp [steer,rear] [9.84797, 0]; Kd [0.1977, 0]
  spawn roll 0.0330717 rad; rate 0.0440185 rad/s
  gyro bias XYZ rad/s [0.00028135, 0.000405887, 4.42249e-05]
  gravity bias YZ [-0.000279101, 0.00258951]
E3 [31.80, 60.00) s:
  mass x [1.04684, 1.16331, 0.83554, 0.81731]
  CoM XYZ mm [-2.19189, 0.128327, 3.89774]
  mu_s [0.611239, 0.611239, 0.640016, 0.732187]
  mu_d [0.611239, 0.611239, 0.513141, 0.631664]
  Kp [steer,rear] [9.59445, 0]; Kd [0.201467, 0]
  spawn roll -0.00351131 rad; rate 0.0643324 rad/s
  gyro bias XYZ rad/s [3.65111e-05, -5.71696e-06, 0.000277037]
  gravity bias YZ [-0.00317783, 0.00182454]
Six significant digits shown; full precision: sampled_dr_values.json.
```

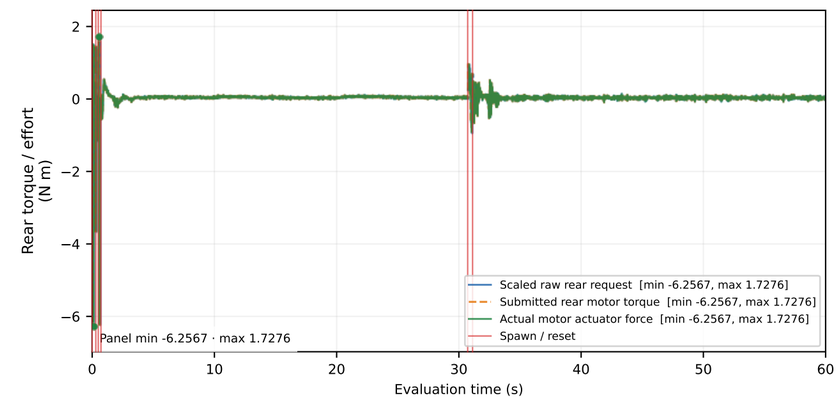
model_2500 · Demand versus actuation · env 34 · full 60 s

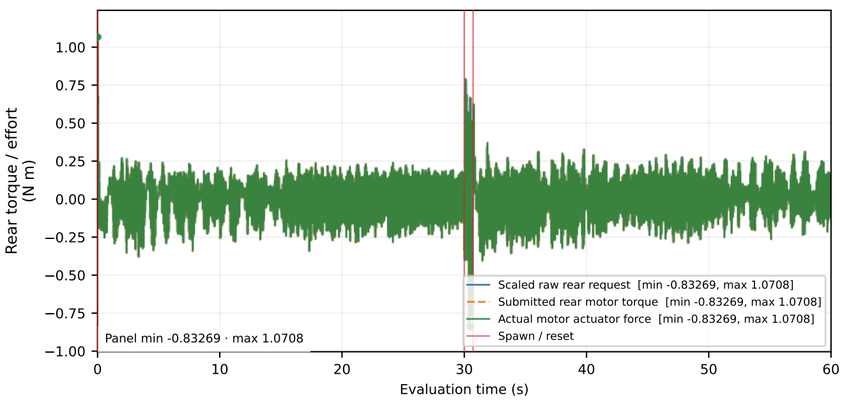












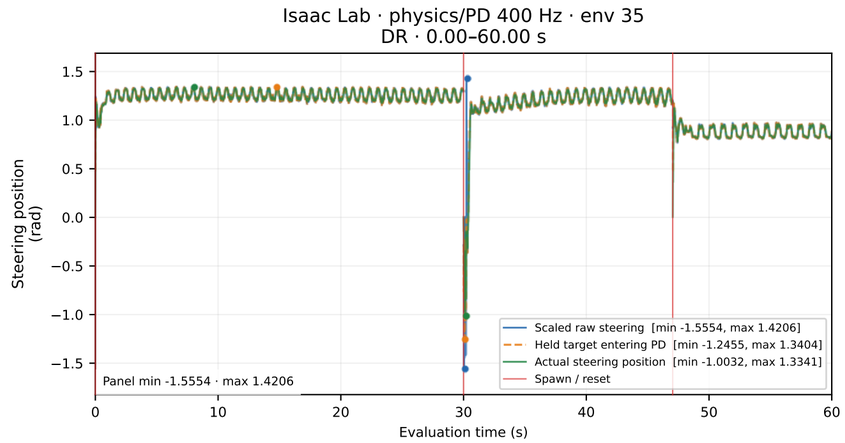
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

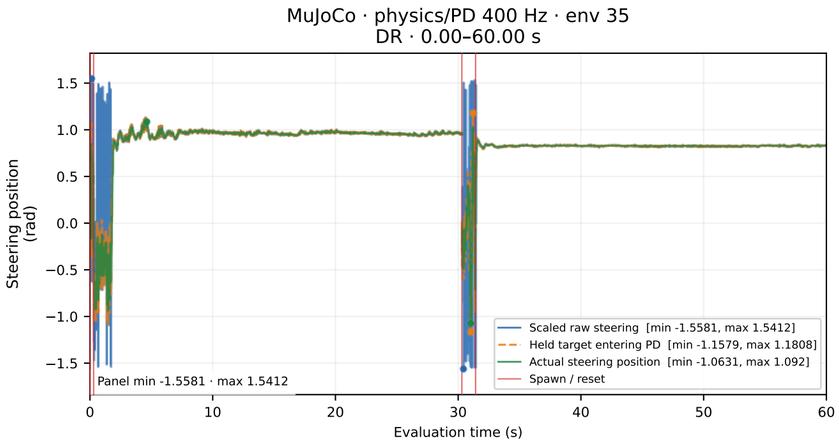
```
SAMPLED DR: env 34; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 16.14) s:
  mass x [0.805827, 1.08268, 0.968777, 1.1818]
  CoM XYZ mm [-4.69837, -2.09229, 0.019148]
  mu_s [0.647391, 0.533455, 0.533455, 0.785466]
  mu_d [0.647391, 0.533455, 0.533455, 0.52258]
  Kp [steer,rear] [9.28387, 0]; Kd [0.215265, 0]
  spawn roll -0.0397326 rad; rate -0.186314 rad/s
  gyro bias XYZ rad/s [-0.000172667, 0.000245817, -0.000265783]
  gravity bias YZ [-0.00288783, -0.000176845]
E2 [16.14, 46.14) s:
  mass x [0.825074, 0.83507, 1.14831, 1.11127]
  CoM XYZ mm [-1.40844, 4.53725, 1.21087]
  mu_s [0.693205, 0.681711, 0.693205, 0.760821]
  mu_d [0.693205, 0.611756, 0.693205, 0.670315]
  Kp [steer,rear] [10.0339, 0]; Kd [0.215451, 0]
  spawn roll 0.0688019 rad; rate -0.0408712 rad/s
  gyro bias XYZ rad/s [-0.000203892, -0.00045729, 0.000102071]
  gravity bias YZ [8.0981e-05, 0.00307239]
E3 [46.14, 60.00) s:
  mass x [1.08489, 1.15281, 1.16129, 1.07297]
  CoM XYZ mm [4.08788, 0.834204, -3.63411]
  mu_s [0.659445, 0.674963, 0.751971, 0.693205]
  mu_d [0.547619, 0.601294, 0.541224, 0.693205]
  Kp [steer,rear] [9.46826, 0]; Kd [0.188964, 0]
  spawn roll 0.0654882 rad; rate -0.035815 rad/s
  gyro bias XYZ rad/s [0.000109645, 8.30663e-05, -0.000334825]
  gravity bias YZ [-0.00258604, -0.000351566]
Six significant digits shown; full precision: sampled_dr_values.json.
```

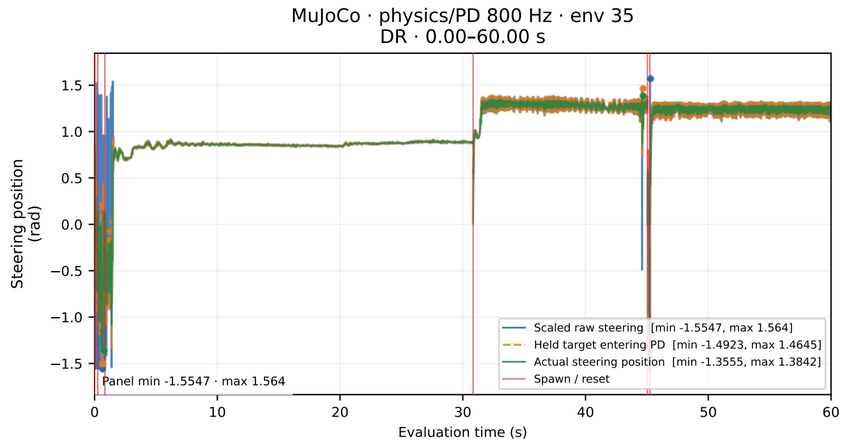
```
SAMPLED DR: env 34; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.30) s:
  mass x [1.12253, 1.19421, 1.07296, 1.08835]
  CoM XYZ mm [-4.21886, -1.91108, 0.617315]
  mu_s [0.666375, 0.700521, 0.637675, 0.633024]
  mu_d [0.519145, 0.641329, 0.637675, 0.568172]
  Kp [steer,rear] [9.69958, 0]; Kd [0.182268, 0]
  spawn roll 0.0739992 rad; rate 0.209117 rad/s
  gyro bias XYZ rad/s [-0.000150689, 0.00024818, -7.63476e-06]
  gravity bias YZ [-0.000274781, -0.000871272]
E2 [0.30, 0.50) s:
  mass x [0.980226, 0.867885, 1.18425, 1.17197]
  CoM XYZ mm [-4.6277, 2.63561, -1.75512]
  mu_s [0.568073, 0.541939, 0.538434, 0.611239]
  mu_d [0.568073, 0.534359, 0.538434, 0.611239]
  Kp [steer,rear] [10.1143, 0]; Kd [0.20075, 0]
  spawn roll 0.0905334 rad; rate 0.248604 rad/s
  gyro bias XYZ rad/s [8.45795e-05, 0.000149024, -4.7732e-05]
  gravity bias YZ [0.00152551, -0.00321815]
E3 [0.50, 0.72) s:
  mass x [0.990933, 0.881057, 1.10269, 1.13758]
  CoM XYZ mm [-0.711965, 1.16799, -3.54662]
  mu_s [0.640867, 0.538434, 0.611239, 0.597748]
  mu_d [0.556841, 0.538434, 0.611239, 0.597748]
  Kp [steer,rear] [10.3673, 0]; Kd [0.201971, 0]
  spawn roll 0.0713888 rad; rate -0.092217 rad/s
  gyro bias XYZ rad/s [-0.000371305, 1.68319e-05, 0.000262002]
  gravity bias YZ [0.00232817, 0.00141465]
E4 [0.72, 30.72) s:
  mass x [1.07908, 1.014, 0.901382, 1.13736]
  CoM XYZ mm [-0.410973, -4.45773, 4.44689]
  mu_s [0.633024, 0.71008, 0.791209, 0.528253]
  mu_d [0.568172, 0.59371, 0.767936, 0.528253]
  Kp [steer,rear] [10.0404, 0]; Kd [0.21151, 0]
  spawn roll -0.0604781 rad; rate 0.24403 rad/s
  gyro bias XYZ rad/s [-8.95758e-07, 0.000374733, -4.32786e-05]
  gravity bias YZ [0.00513053, -0.00115521]
E5 [30.72, 31.12) s:
  mass x [0.994516, 1.11325, 0.884855, 0.872714]
  CoM XYZ mm [-2.46737, 1.57546, 0.486792]
  mu_s [0.568073, 0.700521, 0.666375, 0.631146]
  mu_d [0.568073, 0.641329, 0.519145, 0.631146]
  Kp [steer,rear] [9.88636, 0]; Kd [0.209558, 0]
  spawn roll 0.0833926 rad; rate -0.00718131 rad/s
  gyro bias XYZ rad/s [-0.000106101, -0.000147011, 0.000211347]
  gravity bias YZ [-0.00190917, -0.0003670 94]
E6 [31.12, 60.00) s:
  mass x [0.804439, 1.19971, 1.02108, 0.989968]
  CoM XYZ mm [2.5749, -4.23626, 1.23556]
  mu_s [0.733515, 0.640867, 0.669571, 0.541939]
  mu_d [0.558392, 0.556841, 0.669571, 0.534359]
  Kp [steer,rear] [10.9656, 0]; Kd [0.199435, 0]
  spawn roll -0.0371786 rad; rate -0.178639 rad/s
  gyro bias XYZ rad/s [2.13248e-05, -0.00025547, 0.000179823]
  gravity bias YZ [0.00265764, -0.00570382]
Six significant digits shown; full precision: sampled_dr_values.json.
```

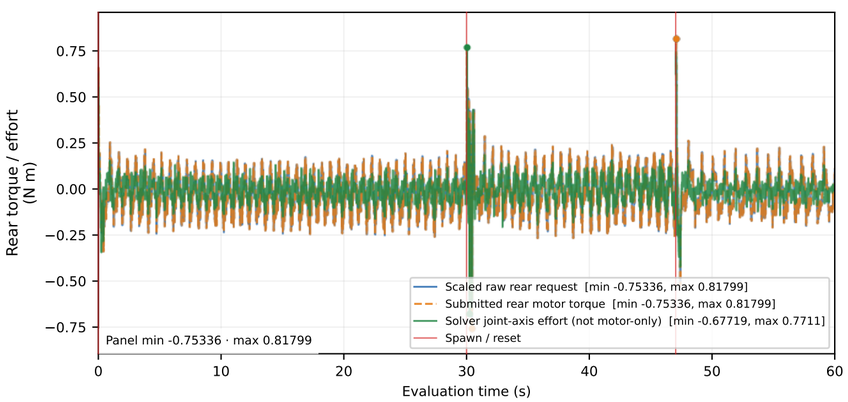
```
SAMPLED DR: env 34; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.12253, 1.19421, 1.07296, 1.08835]
  CoM XYZ mm [-4.21886, -1.91108, 0.617315]
  mu_s [0.666375, 0.700521, 0.637675, 0.633024]
  mu_d [0.519145, 0.641329, 0.637675, 0.568172]
  Kp [steer,rear] [9.69958, 0]; Kd [0.182268, 0]
  spawn roll 0.0739992 rad; rate 0.209117 rad/s
  gyro bias XYZ rad/s [-0.000150689, 0.00024818, -7.63476e-06]
  gravity bias YZ [-0.000274781, -0.000871272]
E2 [30.00, 30.72) s:
  mass x [1.12651, 1.17793, 0.985291, 0.969248]
  CoM XYZ mm [1.6075, 4.6953, -1.06634]
  mu_s [0.640016, 0.732187, 0.728342, 0.71155]
  mu_d [0.513141, 0.631664, 0.728342, 0.71155]
  Kp [steer,rear] [9.58907, 0]; Kd [0.209406, 0]
  spawn roll -0.0759934 rad; rate 0.197094 rad/s
  gyro bias XYZ rad/s [-0.000316628, -0.000359827, -0.000149901]
  gravity bias YZ [-0.00311456, 0.000597333]
E3 [30.72, 60.00) s:
  mass x [0.816499, 0.906458, 0.925283, 0.977816]
  CoM XYZ mm [-2.75219, -1.88555, 3.93467]
  mu_s [0.700521, 0.69316, 0.723429, 0.748289]
  mu_d [0.641329, 0.69316, 0.723429, 0.689499]
  Kp [steer,rear] [9.48051, 0]; Kd [0.20162, 0]
  spawn roll 0.0497188 rad; rate -0.136477 rad/s
  gyro bias XYZ rad/s [-0.000378376, 0.000175181, -7.20776e-05]
  gravity bias YZ [0.00183164, -0.0036954]
Six significant digits shown; full precision: sampled_dr_values.json.
```

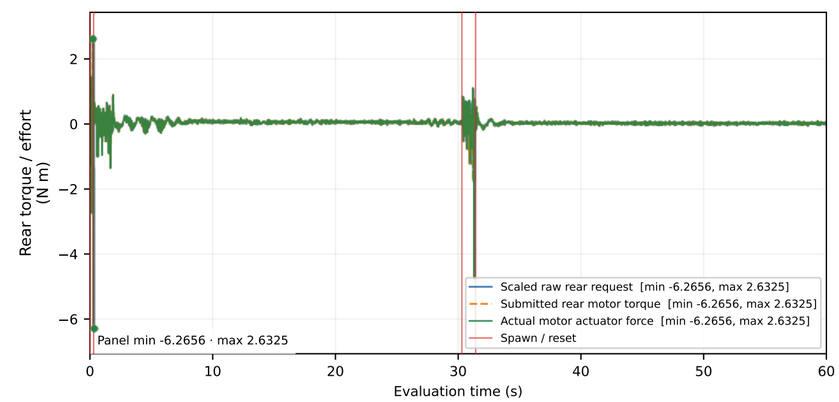
model_2500 · Demand versus actuation · env 35 · full 60 s

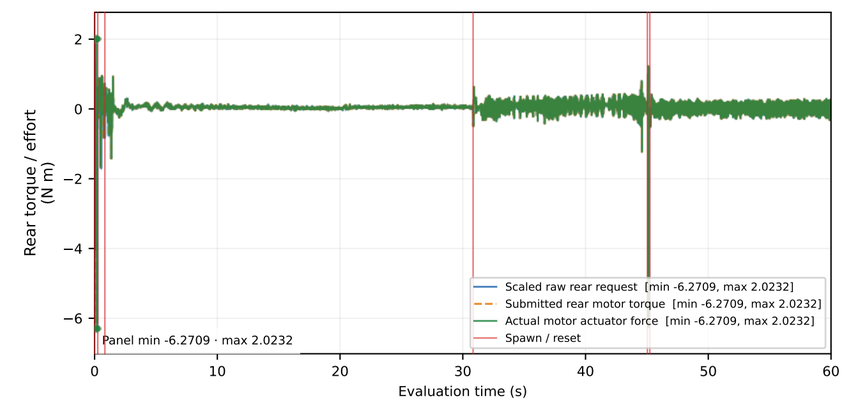












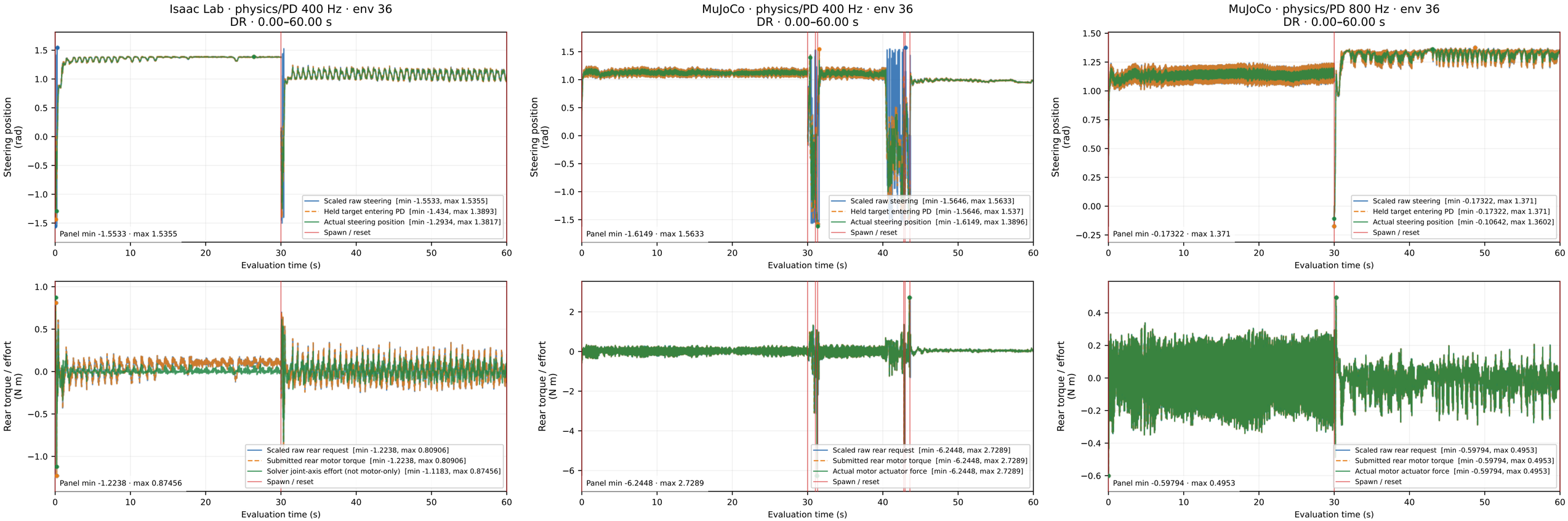
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 35; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.03099, 1.07698, 0.987322, 0.810236]
  CoM XYZ mm [1.15526, -1.85922, -3.17019]
  mu_s [0.751971, 0.532645, 0.774409, 0.547743]
  mu_d [0.541224, 0.532645, 0.561095, 0.547743]
  Kp [steer,rear] [10.4329, 0]; Kd [0.19945, 0]
  spawn roll 0.0616521 rad; rate 0.246993 rad/s
  gyro bias XYZ rad/s [0.000134506, -0.000247672, 0.000167419]
  gravity bias YZ [0.000144644, -0.00120618]
E2 [30.00, 47.04) s:
  mass x [1.16434, 1.18864, 0.980074, 1.1655]
  CoM XYZ mm [-0.537932, -1.72436, -0.138611]
  mu_s [0.659445, 0.547743, 0.785466, 0.774409]
  mu_d [0.547619, 0.547743, 0.52258, 0.561095]
  Kp [steer,rear] [9.36428, 0]; Kd [0.204561, 0]
  spawn roll -0.049918 rad; rate -0.074224 rad/s
  gyro bias XYZ rad/s [4.32197e-05, -0.000233409, 0.000143327]
  gravity bias YZ [-0.00169889, -0.00270366]
E3 [47.04, 60.00) s:
  mass x [0.946494, 0.976625, 1.12899, 0.829175]
  CoM XYZ mm [0.558808, 2.90988, 4.159]
  mu_s [0.576972, 0.532645, 0.751971, 0.533455]
  mu_d [0.576972, 0.532645, 0.541224, 0.533455]
  Kp [steer,rear] [10.5204, 0]; Kd [0.200146, 0]
  spawn roll 0.0970838 rad; rate 0.214616 rad/s
  gyro bias XYZ rad/s [0.000114234, -0.000117443, -0.000127933]
  gravity bias YZ [-0.000617411, -0.000349237]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 35; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.30) s:
  mass x [1.11437, 1.02047, 0.897555, 0.933781]
  CoM XYZ mm [-1.8131, -1.09846, 3.01301]
  mu_s [0.611239, 0.528253, 0.541939, 0.791209]
  mu_d [0.611239, 0.528253, 0.534359, 0.767936]
  Kp [steer,rear] [10.5826, 0]; Kd [0.204159, 0]
  spawn roll 0.0757045 rad; rate -0.169052 rad/s
  gyro bias XYZ rad/s [0.000179138, -0.000372661, -0.000247778]
  gravity bias YZ [0.00193906, -0.00125636]
E2 [0.30, 30.30) s:
  mass x [1.10358, 1.11516, 1.04214, 1.03538]
  CoM XYZ mm [-0.226786, 1.64802, 3.13249]
  mu_s [0.541939, 0.757579, 0.633024, 0.666375]
  mu_d [0.534359, 0.70921, 0.568172, 0.519145]
  Kp [steer,rear] [9.36067, 0]; Kd [0.192032, 0]
  spawn roll -0.0571343 rad; rate 0.00151967 rad/s
  gyro bias XYZ rad/s [0.000262581, -0.000138235, -8.50386e-05]
  gravity bias YZ [-0.00149944, -0.000276668]
E3 [30.30, 31.42) s:
  mass x [0.801902, 1.05794, 1.15258, 0.980549]
  CoM XYZ mm [2.70311, -3.33393, 2.11846]
  mu_s [0.568073, 0.611239, 0.559972, 0.748289]
  mu_d [0.568073, 0.611239, 0.502209, 0.689499]
  Kp [steer,rear] [10.8071, 0]; Kd [0.218447, 0]
  spawn roll -0.0246479 rad; rate -0.2484 rad/s
  gyro bias XYZ rad/s [0.000211986, 0.000154841, 0.000193949]
  gravity bias YZ [-0.00226276, 0.00128962]
E4 [31.42, 60.00) s:
  mass x [1.03001, 0.832794, 0.854795, 1.08168]
  CoM XYZ mm [-1.60549, 4.05211, 3.62333]
  mu_s [0.727426, 0.538434, 0.538976, 0.748289]
  mu_d [0.606358, 0.538434, 0.538976, 0.689499]
  Kp [steer,rear] [9.36101, 0]; Kd [0.214957, 0]
  spawn roll 0.0552535 rad; rate -0.00677227 rad/s
  gyro bias XYZ rad/s [0.000151423, 0.000256563, 0.000106564]
  gravity bias YZ [-0.000977961, 0.00130912]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 35; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.26) s:
  mass x [1.11437, 1.02047, 0.897555, 0.933781]
  CoM XYZ mm [-1.8131, -1.09846, 3.01301]
  mu_s [0.611239, 0.528253, 0.541939, 0.791209]
  mu_d [0.611239, 0.528253, 0.534359, 0.767936]
  Kp [steer,rear] [10.5826, 0]; Kd [0.204159, 0]
  spawn roll 0.0757045 rad; rate -0.169052 rad/s
  gyro bias XYZ rad/s [0.000179138, -0.000372661, -0.000247778]
  gravity bias YZ [0.00193906, -0.00125636]
E2 [0.26, 0.84) s:
  mass x [0.933735, 0.891164, 1.15416, 0.854002]
  CoM XYZ mm [3.52365, 4.83263, -0.209645]
  mu_s [0.71008, 0.633024, 0.640867, 0.538976]
  mu_d [0.59371, 0.568172, 0.556841, 0.538976]
  Kp [steer,rear] [10.2155, 0]; Kd [0.209338, 0]
  spawn roll -0.0365737 rad; rate 0.0966329 rad/s
  gyro bias XYZ rad/s [0.000403267, 8.54849e-05, -0.000120849]
  gravity bias YZ [0.000971239, 0.00115446]
E3 [0.84, 30.84) s:
  mass x [0.889133, 1.09597, 1.1635, 0.92522]
  CoM XYZ mm [0.101439, 3.38965, 4.61652]
  mu_s [0.757579, 0.640867, 0.616244, 0.700521]
  mu_d [0.70921, 0.556841, 0.586498, 0.641329]
  Kp [steer,rear] [10.2071, 0]; Kd [0.206588, 0]
  spawn roll -0.00866833 rad; rate -0.00740132 rad/s
  gyro bias XYZ rad/s [-0.000145499, 4.27028e-05, 0.000216237]
  gravity bias YZ [-0.000840456, -0.00225714]
E4 [30.84, 45.04) s:
  mass x [0.84311, 0.91119, 1.19725, 1.0504]
  CoM XYZ mm [4.53273, -3.95345, -2.3898]
  mu_s [0.616244, 0.757579, 0.732187, 0.538976]
  mu_d [0.586498, 0.70921, 0.631664, 0.538976]
  Kp [steer,rear] [10.1877, 0]; Kd [0.203099, 0]
  spawn roll 0.00957629 rad; rate 0.137029 rad/s
  gyro bias XYZ rad/s [-0.000176587, 0.000154702, -4.8365e-05]
  gravity bias YZ [0.000420363, -0.000894129]
E5 [45.04, 45.24) s:
  mass x [0.871442, 0.948927, 0.998415, 0.833973]
  CoM XYZ mm [-3.79589, -3.47431, -1.76624]
  mu_s [0.631146, 0.640016, 0.700521, 0.733515]
  mu_d [0.631146, 0.513141, 0.641329, 0.558392]
  Kp [steer,rear] [10.3672, 0]; Kd [0.218668, 0]
  spawn roll 0.0871196 rad; rate -0.196338 rad/s
  gyro bias XYZ rad/s [0.000544532, -5.09331e-05, 1.15576e-05]
  gravity bias YZ [0.00155484, -0.00326486]
E6 [45.24, 60.00) s:
  mass x [1.07334, 0.874021, 0.822333, 1.07059]
  CoM XYZ mm [2.44332, -1.95996, 1.66241]
  mu_s [0.538976, 0.559972, 0.728342, 0.568073]
  mu_d [0.538976, 0.502209, 0.728342, 0.568073]
  Kp [steer,rear] [10.1502, 0]; Kd [0.190974, 0]
  spawn roll -0.0383338 rad; rate 0.194893 rad/s
  gyro bias XYZ rad/s [9.6327e-05, -0.00023254, 0.000229655]
  gravity bias YZ [-0.00175814, 0.00257724]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 36 · full 60 s



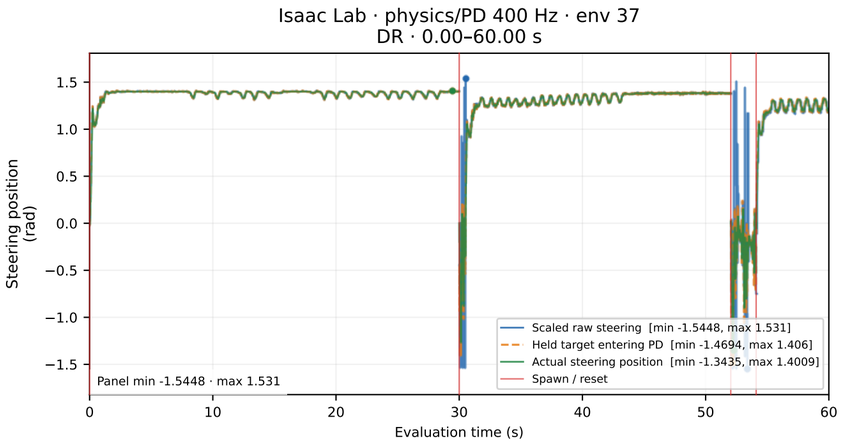
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

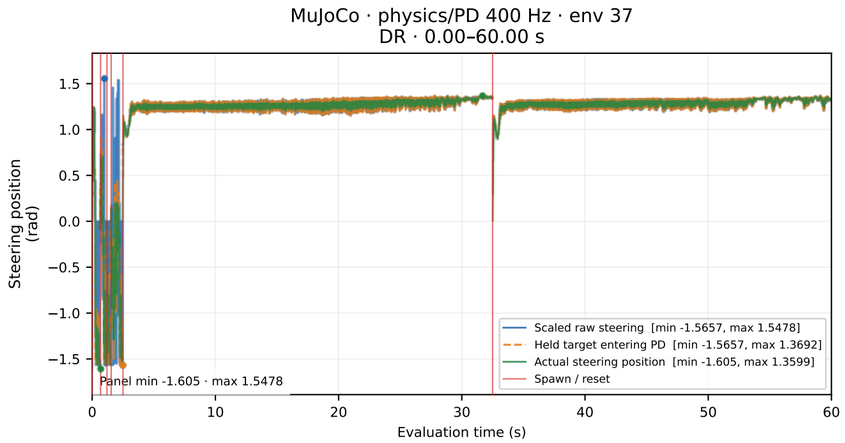
```
SAMPLED DR: env 36; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.889325, 0.823307, 1.18237, 1.18823]
  CoM XYZ mm [-0.652254, -3.03325, -0.461861]
  mu_s [0.794338, 0.579974, 0.533455, 0.764681]
  mu_d [0.526209, 0.579974, 0.533455, 0.764681]
  Kp [steer,rear] [10.0063, 0]; Kd [0.202994, 0]
  spawn roll -0.0838998 rad; rate -0.177664 rad/s
  gyro bias XYZ rad/s [0.000165333, -0.000268149, 0.000176489]
  gravity bias YZ [0.00087258, -0.000498517]
E2 [30.00, 60.00) s:
  mass x [1.11923, 0.87332, 1.10839, 1.15865]
  CoM XYZ mm [-1.55851, 0.599463, -3.3043]
  mu_s [0.787792, 0.632409, 0.703711, 0.598343]
  mu_d [0.617134, 0.589076, 0.703711, 0.598343]
  Kp [steer,rear] [9.84677, 0]; Kd [0.199817, 0]
  spawn roll -0.0626439 rad; rate 0.171068 rad/s
  gyro bias XYZ rad/s [3.1832e-05, 0.000210742, 0.000367422]
  gravity bias YZ [0.000256666, -0.00324526]
Six significant digits shown; full precision: sampled_dr_values.json.
```

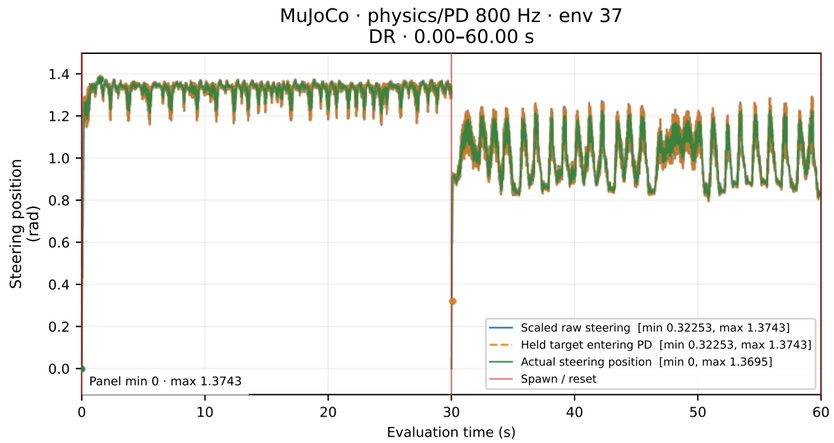
```
SAMPLED DR: env 36; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.967545, 1.16605, 1.00725, 1.13485]
  CoM XYZ mm [-1.27665, 0.695931, -1.20264]
  mu_s [0.538434, 0.727426, 0.631146, 0.538976]
  mu_d [0.538434, 0.606358, 0.631146, 0.538976]
  Kp [steer,rear] [10.3464, 0]; Kd [0.18993, 0]
  spawn roll -0.0134806 rad; rate 0.217073 rad/s
  gyro bias XYZ rad/s [0.000228825, 0.000338871, 0.000202232]
  gravity bias YZ [0.00100058, 0.00282381]
E2 [30.00, 31.04) s:
  mass x [1.19686, 0.970435, 1.07169, 0.896225]
  CoM XYZ mm [-2.67648, -4.26203, 4.45151]
  mu_s [0.728342, 0.611239, 0.748289, 0.669571]
  mu_d [0.728342, 0.611239, 0.689499, 0.669571]
  Kp [steer,rear] [9.8789, 0]; Kd [0.202617, 0]
  spawn roll -0.0216403 rad; rate -0.0927249 rad/s
  gyro bias XYZ rad/s [0.000198632, 0.000211382, -0.000252489]
  gravity bias YZ [-0.000844808, -0.000889235]
E3 [31.04, 31.34) s:
  mass x [1.04945, 0.939586, 1.1852, 0.835938]
  CoM XYZ mm [-0.0678102, 3.15345, 1.75922]
  mu_s [0.559972, 0.559972, 0.700521, 0.616244]
  mu_d [0.502209, 0.502209, 0.641329, 0.586498]
  Kp [steer,rear] [10.8011, 0]; Kd [0.211997, 0]
  spawn roll -0.0658707 rad; rate -0.0579494 rad/s
  gyro bias XYZ rad/s [-2.70543e-05, -0.000117642, -0.00013395]
  gravity bias YZ [-0.00245552, 0.00182166]
E4 [31.34, 42.78) s:
  mass x [0.996467, 0.876334, 0.981981, 1.04004]
  CoM XYZ mm [3.95249, 1.25703, 0.560635]
  mu_s [0.666375, 0.538976, 0.640016, 0.637675]
  mu_d [0.519145, 0.538976, 0.513141, 0.637675]
  Kp [steer,rear] [9.26179, 0]; Kd [0.181723, 0]
  spawn roll 0.0377449 rad; rate 0.0518503 rad/s
  gyro bias XYZ rad/s [2.28952e-05, 8.12965e-05, -6.94963e-05]
  gravity bias YZ [0.00093996, 0.000508482]
E5 [42.78, 42.96) s:
  mass x [0.935606, 0.852864, 1.14186, 1.16011]
  CoM XYZ mm [4.326, 0.230119, 4.46828]
  mu_s [0.700521, 0.541939, 0.637675, 0.791209]
  mu_d [0.641329, 0.534359, 0.637675, 0.767936]
  Kp [steer,rear] [9.06806, 0]; Kd [0.202557, 0]
  spawn roll 0.0745548 rad; rate 0.205111 rad/s
  gyro bias XYZ rad/s [0.000210011, 2.03846e-05, -0.000166352]
  gravity bias YZ [-0.0014714, 0.000873408]
E6 [42.96, 43.60) s:
  mass x [0.958102, 0.841826, 1.16035, 1.14905]
  CoM XYZ mm [4.85714, 2.82715, -4.6075]
  mu_s [0.559972, 0.546287, 0.631146, 0.633024]
  mu_d [0.502209, 0.546287, 0.631146, 0.568172]
  Kp [steer,rear] [10.0447, 0]; Kd [0.197069, 0]
  spawn roll 0.00466166 rad; rate 0.0758323 rad/s
  gyro bias XYZ rad/s [0.000512757, 0.000321531, -0.000107633]
  gravity bias YZ [-0.00185155, -0.000377457]
E7 [43.60, 60.00) s:
  mass x [1.10547, 1.18265, 0.971293, 1.12451]
  CoM XYZ mm [-4.75486, 1.14637, 1.48433]
  mu_s [0.559972, 0.749678, 0.727426, 0.704749]
  mu_d [0.502209, 0.741429, 0.606358, 0.541926]
  Kp [steer,rear] [10.0138, 0]; Kd [0.183276, 0]
  spawn roll 0.0676573 rad; rate -0.114212 rad/s
  gyro bias XYZ rad/s [-0.000322155, -0.000103443, -6.22804e-05]
  gravity bias YZ [0.000823562, 0.001472]
Six significant digits shown; full precision: sampled_dr_values.json.
```

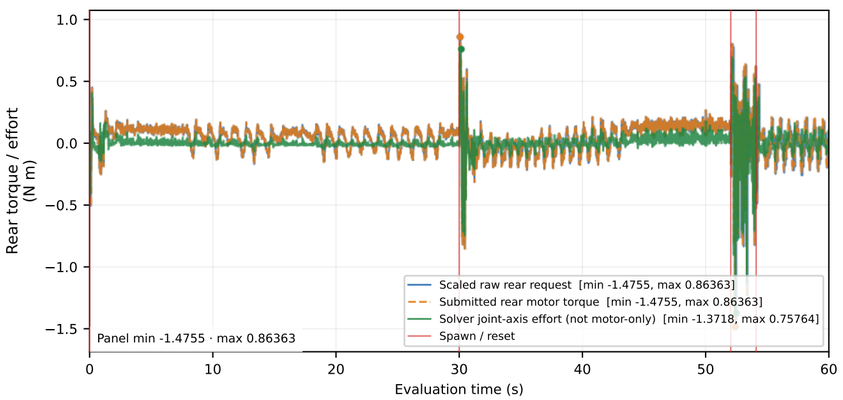
```
SAMPLED DR: env 36; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.967545, 1.16605, 1.00725, 1.13485]
  CoM XYZ mm [-1.27665, 0.695931, -1.20264]
  mu_s [0.538434, 0.727426, 0.631146, 0.538976]
  mu_d [0.538434, 0.606358, 0.631146, 0.538976]
  Kp [steer,rear] [10.3464, 0]; Kd [0.18993, 0]
  spawn roll -0.0134806 rad; rate 0.217073 rad/s
  gyro bias XYZ rad/s [0.000228825, 0.000338871, 0.000202232]
  gravity bias YZ [0.00100058, 0.00282381]
E2 [30.00, 60.00) s:
  mass x [1.19686, 0.970435, 1.07169, 0.896225]
  CoM XYZ mm [-2.67648, -4.26203, 4.45151]
  mu_s [0.728342, 0.611239, 0.748289, 0.669571]
  mu_d [0.728342, 0.611239, 0.689499, 0.669571]
  Kp [steer,rear] [9.8789, 0]; Kd [0.202617, 0]
  spawn roll -0.0216403 rad; rate -0.0927249 rad/s
  gyro bias XYZ rad/s [0.000198632, 0.000211382, -0.000252489]
  gravity bias YZ [-0.000844808, -0.000889235]
Six significant digits shown; full precision: sampled_dr_values.json.
```

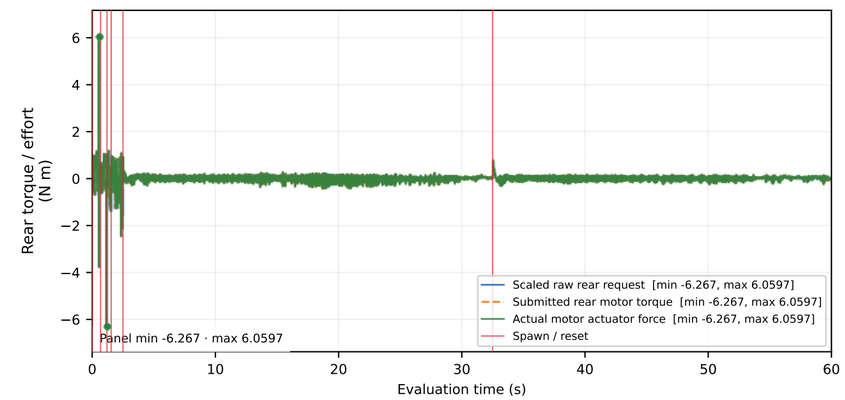
model_2500 · Demand versus actuation · env 37 · full 60 s

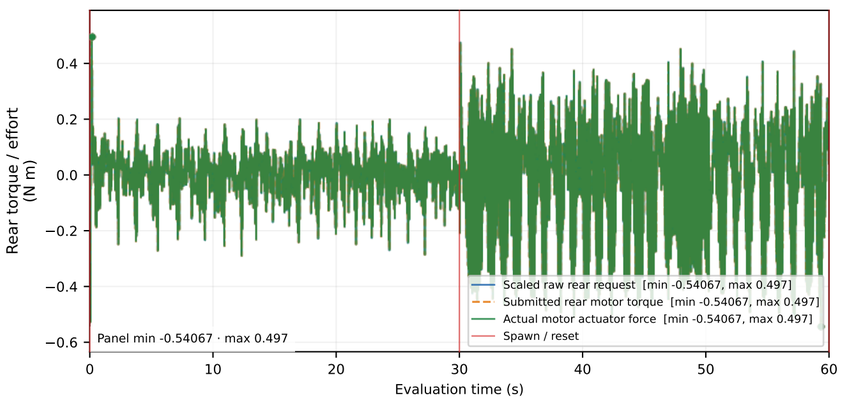












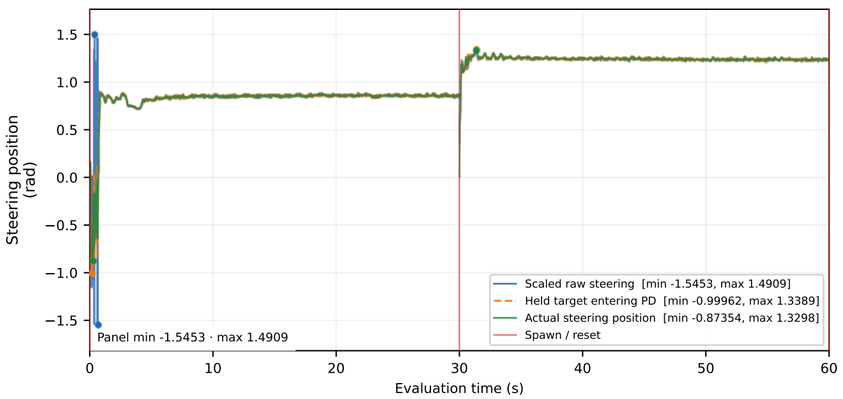
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

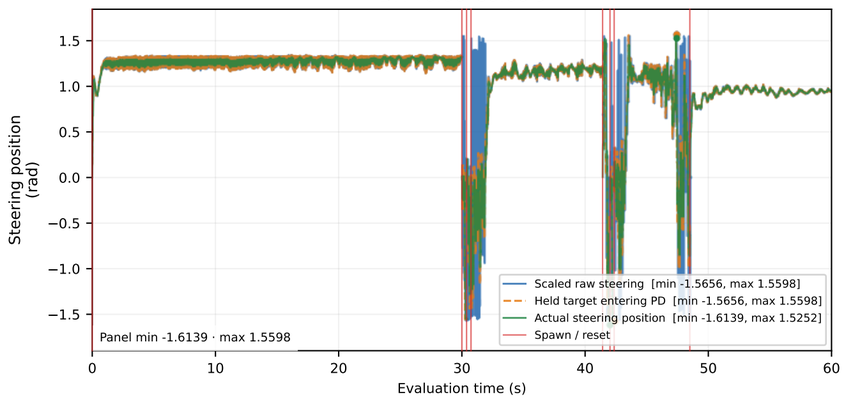
```
SAMPLED DR: env 37; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.06897, 0.86944, 1.04357, 1.07393]
  CoM XYZ mm [1.41931, -3.74473, -4.63834]
  mu_s [0.764681, 0.794338, 0.703711, 0.533455]
  mu_d [0.764681, 0.526209, 0.703711, 0.533455]
  Kp [steer,rear] [10.923, 0]; Kd [0.189328, 0]
  spawn roll -0.0258817 rad; rate -0.00667393 rad/s
  gyro bias XYZ rad/s [0.000218942, -0.00012712, -0.0001508]
  gravity bias YZ [0.00144611, -0.000632127]
E2 [30.00, 52.04) s:
  mass x [0.91121, 0.935715, 1.14339, 1.16952]
  CoM XYZ mm [-3.97393, -2.78473, -1.15326]
  mu_s [0.632409, 0.726175, 0.713129, 0.726175]
  mu_d [0.589076, 0.558574, 0.713129, 0.558574]
  Kp [steer,rear] [10.6151, 0]; Kd [0.211117, 0]
  spawn roll -0.0898541 rad; rate -0.0463745 rad/s
  gyro bias XYZ rad/s [-1.3226e-05, -8.08991e-06, -0.000160178]
  gravity bias YZ [0.00282522, -0.000587555]
E3 [52.04, 54.10) s:
  mass x [1.00566, 1.10026, 0.903106, 1.14291]
  CoM XYZ mm [0.992984, 3.46284, 4.27411]
  mu_s [0.673378, 0.533455, 0.659445, 0.533455]
  mu_d [0.673378, 0.533455, 0.547619, 0.533455]
  Kp [steer,rear] [9.40009, 0]; Kd [0.181659, 0]
  spawn roll -0.0688357 rad; rate 0.143508 rad/s
  gyro bias XYZ rad/s [9.38244e-05, -0.00012617, 0.000102573]
  gravity bias YZ [0.0019121, 0.000154381]
E4 [54.10, 60.00) s:
  mass x [1.12016, 0.933536, 0.905927, 1.11461]
  CoM XYZ mm [-0.808045, -1.85707, -0.963807]
  mu_s [0.632409, 0.602694, 0.632409, 0.602694]
  mu_d [0.589076, 0.602694, 0.589076, 0.602694]
  Kp [steer,rear] [9.15919, 0]; Kd [0.20601, 0]
  spawn roll -0.00417519 rad; rate -0.114425 rad/s
  gyro bias XYZ rad/s [0.000289598, -0.000254552, 0.00039431]
  gravity bias YZ [-0.00250523, -0.00284714]
Six significant digits shown; full precision: sampled_dr_values.json.
```

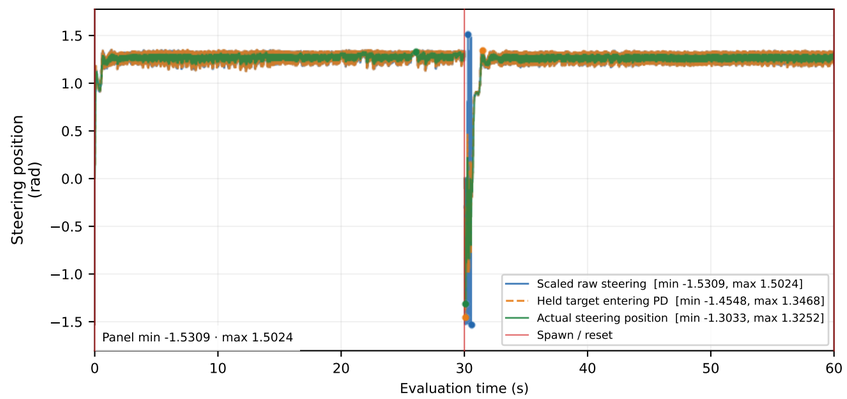
```
SAMPLED DR: env 37; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.68) s:
  mass x [1.16784, 0.828649, 1.16923, 0.814299]
  CoM XYZ mm [-4.12528, -3.74298, -0.222992]
  mu_s [0.749678, 0.791209, 0.616244, 0.728342]
  mu_d [0.741429, 0.767936, 0.586498, 0.728342]
  Kp [steer,rear] [10.2426, 0]; Kd [0.204752, 0]
  spawn roll -0.0450624 rad; rate 0.221879 rad/s
  gyro bias XYZ rad/s [0.000112079, -9.4086e-05, 0.00014188]
  gravity bias YZ [2.36296e-05, 0.00159927]
E2 [0.68, 1.20) s:
  mass x [1.04252, 1.02903, 0.811093, 0.871413]
  CoM XYZ mm [3.17014, -4.5089, 0.716271]
  mu_s [0.736077, 0.640867, 0.637675, 0.733515]
  mu_d [0.699455, 0.556841, 0.637675, 0.558392]
  Kp [steer,rear] [10.7549, 0]; Kd [0.216135, 0]
  spawn roll -0.0756035 rad; rate 0.107357 rad/s
  gyro bias XYZ rad/s [-6.14345e-05, -0.000103861, 0.000441419]
  gravity bias YZ [0.000703019, -0.000120722]
E3 [1.20, 1.54) s:
  mass x [0.842745, 1.10896, 1.10692, 1.01941]
  CoM XYZ mm [-4.18722, 4.37099, 4.20679]
  mu_s [0.597748, 0.640016, 0.568073, 0.727426]
  mu_d [0.597748, 0.513141, 0.568073, 0.606358]
  Kp [steer,rear] [9.37334, 0]; Kd [0.210042, 0]
  spawn roll -0.0998609 rad; rate 0.142483 rad/s
  gyro bias XYZ rad/s [3.11612e-05, -0.00021694, -0.00016323]
  gravity bias YZ [-0.00213008, -0.000319505]
E4 [1.54, 2.50) s:
  mass x [0.877247, 1.03203, 0.908988, 0.877389]
  CoM XYZ mm [-0.511892, -0.395673, 1.73933]
  mu_s [0.700521, 0.700521, 0.541939, 0.616244]
  mu_d [0.641329, 0.641329, 0.534359, 0.586498]
  Kp [steer,rear] [9.50982, 0]; Kd [0.188853, 0]
  spawn roll -0.0598529 rad; rate 0.138223 rad/s
  gyro bias XYZ rad/s [-0.000230918, -0.000146542, 7.09895e-05]
  gravity bias YZ [-0.000158763, -0.00332722]
E5 [2.50, 32.50) s:
  mass x [1.12477, 1.18193, 0.930487, 1.10288]
  CoM XYZ mm [-2.94233, -3.764, -0.874708]
  mu_s [0.727426, 0.748289, 0.611239, 0.704749]
  mu_d [0.606358, 0.689499, 0.611239, 0.541926]
  Kp [steer,rear] [9.46844, 0]; Kd [0.185555, 0]
  spawn roll 0.0677484 rad; rate 0.188911 rad/s
  gyro bias XYZ rad/s [5.97027e-05, -7.08922e-05, 8.33852e-05]
  gravity bias YZ [0.00199517, 0.00348367]
E6 [32.50, 60.00) s:
  mass x [1.17385, 1.18431, 1.02134, 1.13001]
  CoM XYZ mm [-2.76365, -3.50858, 0.862396]
  mu_s [0.611239, 0.700521, 0.728342, 0.732187]
  mu_d [0.611239, 0.641329, 0.728342, 0.631664]
  Kp [steer,rear] [10.734, 0]; Kd [0.19728, 0]
  spawn roll 0.0552036 rad; rate -0.142578 rad/s
  gyro bias XYZ rad/s [-0.000333262, -0.000305812, -0.000206869]
  gravity bias YZ [0.000398298, -0.00223014]
Six significant digits shown; full precision: sampled_dr_values.json.
```

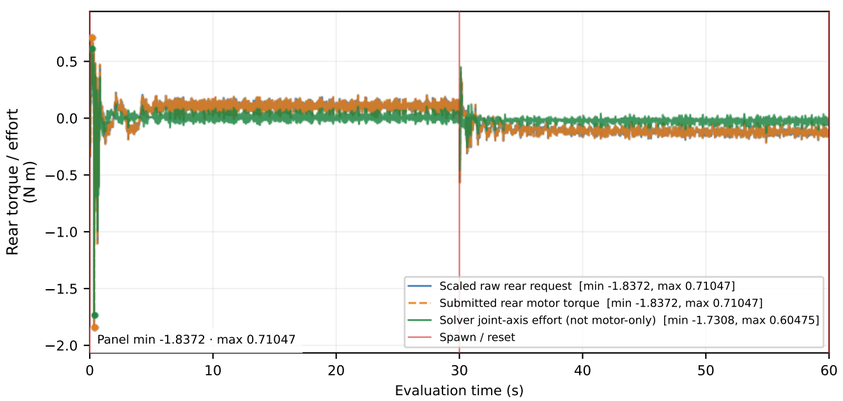
```
SAMPLED DR: env 37; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16784, 0.828649, 1.16923, 0.814299]
  CoM XYZ mm [-4.12528, -3.74298, -0.222992]
  mu_s [0.749678, 0.791209, 0.616244, 0.728342]
  mu_d [0.741429, 0.767936, 0.586498, 0.728342]
  Kp [steer,rear] [10.2426, 0]; Kd [0.204752, 0]
  spawn roll -0.0450624 rad; rate 0.221879 rad/s
  gyro bias XYZ rad/s [0.000112079, -9.4086e-05, 0.00014188]
  gravity bias YZ [2.36296e-05, 0.00159927]
E2 [30.00, 60.00) s:
  mass x [0.826023, 1.04052, 0.930369, 0.969218]
  CoM XYZ mm [1.75077, 2.31945, 3.03961]
  mu_s [0.541939, 0.71008, 0.637675, 0.597748]
  mu_d [0.534359, 0.59371, 0.637675, 0.597748]
  Kp [steer,rear] [10.4509, 0]; Kd [0.200802, 0]
  spawn roll 0.0432316 rad; rate -0.0721844 rad/s
  gyro bias XYZ rad/s [2.86129e-05, -0.00023106, 2.02665e-05]
  gravity bias YZ [0.00152038, 0.000273868]
Six significant digits shown; full precision: sampled_dr_values.json.
```

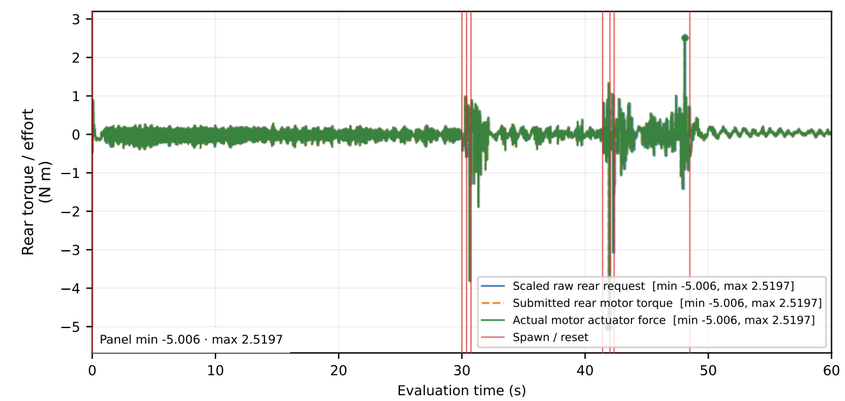
model_2500 · Demand versus actuation · env 38 · full 60 s

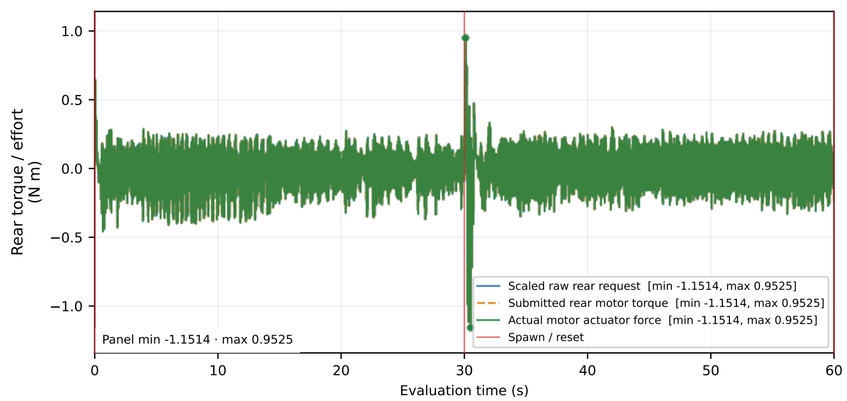












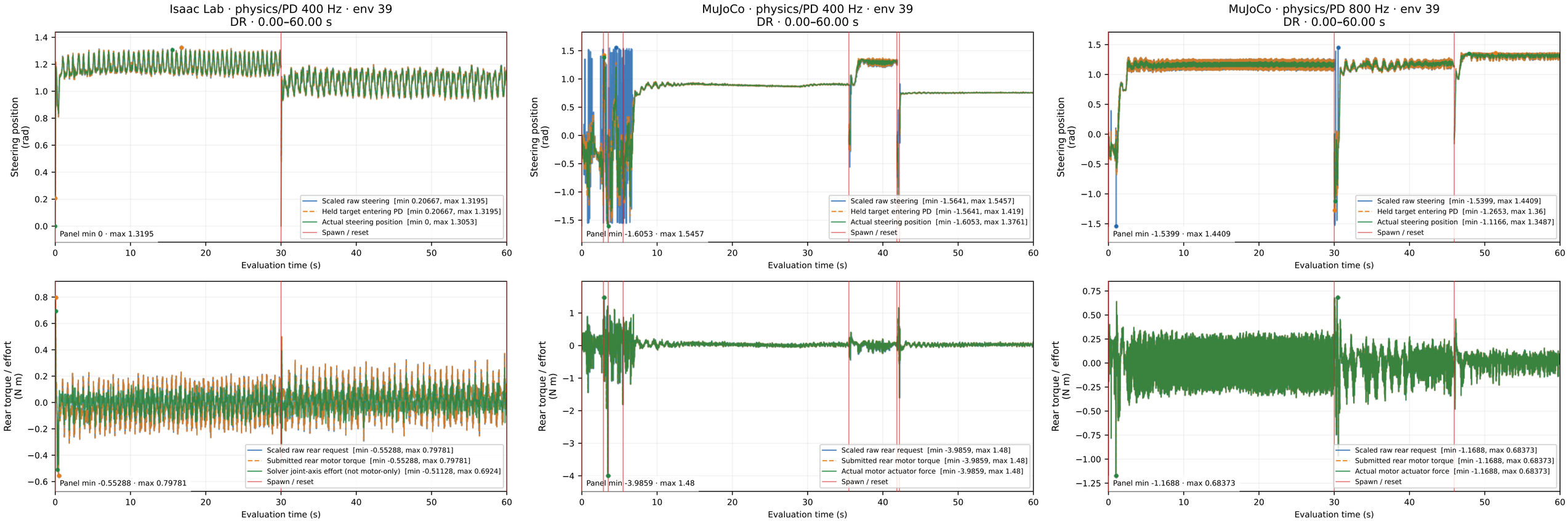
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 38; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.11762, 0.836914, 0.995436, 1.1308]
  CoM XYZ mm [3.84191, 4.0777, -4.95727]
  mu_s [0.794338, 0.647391, 0.673378, 0.531594]
  mu_d [0.526209, 0.647391, 0.673378, 0.531594]
  Kp [steer,rear] [10.4319, 0]; Kd [0.194748, 0]
  spawn roll -0.011885 rad; rate 0.0115525 rad/s
  gyro bias XYZ rad/s [-9.9337e-05, -9.97786e-05, -0.000188746]
  gravity bias YZ [-0.00060973, -0.00307423]
E2 [30.00, 60.00) s:
  mass x [0.948215, 0.951209, 0.903158, 1.15621]
  CoM XYZ mm [1.84169, -2.7413, 3.49542]
  mu_s [0.547743, 0.774409, 0.560535, 0.673378]
  mu_d [0.547743, 0.561095, 0.560535, 0.673378]
  Kp [steer,rear] [9.75346, 0]; Kd [0.181468, 0]
  spawn roll -0.0282039 rad; rate 0.14784 rad/s
  gyro bias XYZ rad/s [-3.54838e-05, -1.54418e-05, -8.73341e-05]
  gravity bias YZ [-0.00133313, 0.0021975]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 38; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.0946, 0.807368, 0.814653, 1.03788]
  CoM XYZ mm [0.857136, -2.76032, 1.38134]
  mu_s [0.736077, 0.704749, 0.633024, 0.616244]
  mu_d [0.699455, 0.541926, 0.568172, 0.586498]
  Kp [steer,rear] [9.30934, 0]; Kd [0.192084, 0]
  spawn roll 0.0494247 rad; rate -0.184587 rad/s
  gyro bias XYZ rad/s [-0.000134545, 0.000185694, -0.000102061]
  gravity bias YZ [-0.00183624, -0.000511672]
E2 [30.00, 30.38) s:
  mass x [0.890986, 1.19204, 0.890429, 0.852831]
  CoM XYZ mm [1.17917, -2.10881, 4.54459]
  mu_s [0.748289, 0.640016, 0.528253, 0.631146]
  mu_d [0.689499, 0.513141, 0.528253, 0.631146]
  Kp [steer,rear] [9.76925, 0]; Kd [0.187092, 0]
  spawn roll -0.0858828 rad; rate 0.0235947 rad/s
  gyro bias XYZ rad/s [-0.000119004, -7.4116e-05, -7.19028e-05]
  gravity bias YZ [0.00206062, 0.000455609]
E3 [30.38, 30.74) s:
  mass x [0.81892, 0.846206, 1.14164, 0.823848]
  CoM XYZ mm [-4.11037, 0.686048, 3.32889]
  mu_s [0.640867, 0.538434, 0.538434, 0.704749]
  mu_d [0.556841, 0.538434, 0.538434, 0.541926]
  Kp [steer,rear] [9.50962, 0]; Kd [0.202056, 0]
  spawn roll -0.0384562 rad; rate 0.0829481 rad/s
  gyro bias XYZ rad/s [-3.30915e-05, -0.00019653, -1.83674e-05]
  gravity bias YZ [0.000780591, 0.00202102]
E4 [30.74, 41.42) s:
  mass x [1.06578, 0.990997, 1.04596, 0.891202]
  CoM XYZ mm [-4.88459, -0.747627, -0.747745]
  mu_s [0.728342, 0.704749, 0.733515, 0.704749]
  mu_d [0.728342, 0.541926, 0.558392, 0.541926]
  Kp [steer,rear] [10.4303, 0]; Kd [0.194417, 0]
  spawn roll -0.0704001 rad; rate -0.197332 rad/s
  gyro bias XYZ rad/s [-2.40726e-05, 2.11562e-05, 0.000105931]
  gravity bias YZ [0.000769275, -0.000822098]
E5 [41.42, 42.02) s:
  mass x [1.12862, 0.860983, 0.910753, 0.814961]
  CoM XYZ mm [-2.81092, -2.96765, -3.40159]
  mu_s [0.748289, 0.546287, 0.538976, 0.723429]
  mu_d [0.689499, 0.546287, 0.538976, 0.723429]
  Kp [steer,rear] [10.7562, 0]; Kd [0.200965, 0]
  spawn roll 0.0476956 rad; rate 0.0818871 rad/s
  gyro bias XYZ rad/s [-0.000138082, 0.000215344, 0.000224316]
  gravity bias YZ [-0.00227307, 0.000391213]
E6 [42.02, 42.36) s:
  mass x [1.17562, 1.12714, 0.899412, 0.975255]
  CoM XYZ mm [-0.540265, -0.367724, -3.62225]
  mu_s [0.791209, 0.71008, 0.669571, 0.528253]
  mu_d [0.767936, 0.59371, 0.669571, 0.528253]
  Kp [steer,rear] [9.98523, 0]; Kd [0.210272, 0]
  spawn roll -0.0555902 rad; rate -0.107512 rad/s
  gyro bias XYZ rad/s [7.89347e-05, 0.000265749, 0.000297908]
  gravity bias YZ [-0.0043962, 0.00136375]
E7 [42.36, 48.50) s:
  mass x [0.863007, 1.19554, 0.975987, 0.931001]
  CoM XYZ mm [0.43362, -0.380969, -4.3105]
  mu_s [0.666375, 0.749678, 0.71008, 0.633024]
  mu_d [0.519145, 0.741429, 0.59371, 0.568172]
  Kp [steer,rear] [9.91488, 0]; Kd [0.197156, 0]
  spawn roll 0.0026603 rad; rate -0.145111 rad/s
  gyro bias XYZ rad/s [0.00036954, 0.000367114, -0.000111937]
  gravity bias YZ [-0.000991216, -0.00120392]
E8 [48.50, 60.00) s:
  mass x [1.00666, 1.18752, 0.84588, 0.946555]
  CoM XYZ mm [-4.85837, 1.58286, 3.49196]
  mu_s [0.538976, 0.748289, 0.728342, 0.546287]
  mu_d [0.538976, 0.689499, 0.728342, 0.546287]
  Kp [steer,rear] [10.0283, 0]; Kd [0.188856, 0]
  spawn roll -0.0358756 rad; rate 0.207634 rad/s
  gyro bias XYZ rad/s [-0.000185577, -9.61102e-05, -7.92296e-06]
  gravity bias YZ [-0.000677098, -0.000470997]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 38; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.0946, 0.807368, 0.814653, 1.03788]
  CoM XYZ mm [0.857136, -2.76032, 1.38134]
  mu_s [0.736077, 0.704749, 0.633024, 0.616244]
  mu_d [0.699455, 0.541926, 0.568172, 0.586498]
  Kp [steer,rear] [9.30934, 0]; Kd [0.192084, 0]
  spawn roll 0.0494247 rad; rate -0.184587 rad/s
  gyro bias XYZ rad/s [-0.000134545, 0.000185694, -0.000102061]
  gravity bias YZ [-0.00183624, -0.000511672]
E2 [30.00, 60.00) s:
  mass x [0.890986, 1.19204, 0.890429, 0.852831]
  CoM XYZ mm [1.17917, -2.10881, 4.54459]
  mu_s [0.748289, 0.640016, 0.528253, 0.631146]
  mu_d [0.689499, 0.513141, 0.528253, 0.631146]
  Kp [steer,rear] [9.76925, 0]; Kd [0.187092, 0]
  spawn roll -0.0858828 rad; rate 0.0235947 rad/s
  gyro bias XYZ rad/s [-0.000119004, -7.4116e-05, -7.19028e-05]
  gravity bias YZ [0.00206062, 0.000455609]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 39 · full 60 s



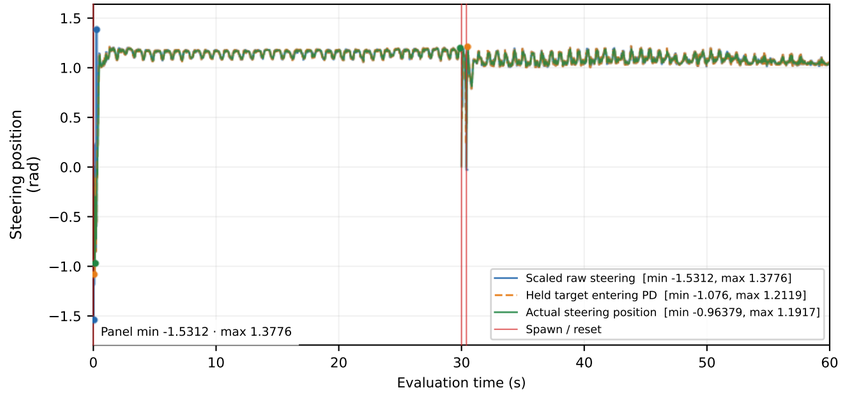
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

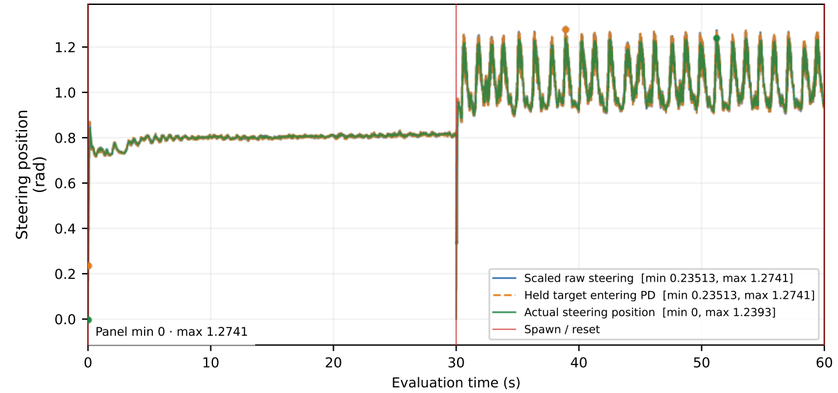
```
SAMPLED DR: env 39; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.18107, 0.976622, 1.16125, 0.85314]
  CoM XYZ mm [0.932112, -1.2884, 0.629038]
  mu_s [0.576972, 0.579974, 0.628821, 0.632409]
  mu_d [0.576972, 0.579974, 0.628821, 0.589076]
  Kp [steer,rear] [10.5127, 0]; Kd [0.189964, 0]
  spawn roll 0.0735025 rad; rate -0.229389 rad/s
  gyro bias XYZ rad/s [-7.55464e-05, -0.000265604, 0.000107245]
  gravity bias YZ [-0.00155328, -0.000994103]
E2 [30.00, 60.00) s:
  mass x [0.808221, 0.855749, 1.13878, 1.12369]
  CoM XYZ mm [-3.80968, 0.309244, 1.10134]
  mu_s [0.693205, 0.760821, 0.751971, 0.674963]
  mu_d [0.693205, 0.670315, 0.541224, 0.601294]
  Kp [steer,rear] [10.7521, 0]; Kd [0.189601, 0]
  spawn roll 0.0190763 rad; rate 0.0728261 rad/s
  gyro bias XYZ rad/s [2.52376e-07, 0.000105377, 0.000172258]
  gravity bias YZ [-0.00202497, -0.000776156]
Six significant digits shown; full precision: sampled_dr_values.json.
```

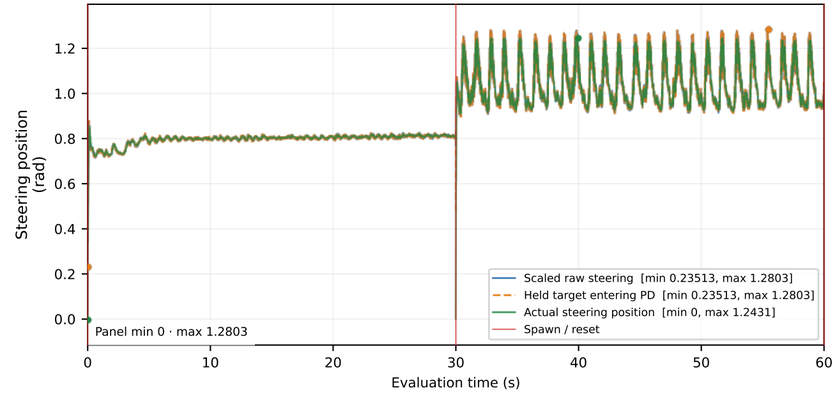
```
SAMPLED DR: env 39; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 2.86) s:
  mass x [0.973645, 0.884651, 1.16481, 1.04419]
  CoM XYZ mm [-0.554547, 0.030713, -0.346279]
  mu_s [0.736077, 0.728342, 0.640867, 0.666375]
  mu_d [0.699455, 0.728342, 0.556841, 0.519145]
  Kp [steer,rear] [9.09012, 0]; Kd [0.208688, 0]
  spawn roll -0.00991366 rad; rate -0.150883 rad/s
  gyro bias XYZ rad/s [-0.00036369, 8.10202e-05, 0.000139191]
  gravity bias YZ [0.000357033, 0.00113532]
E2 [2.86, 3.50) s:
  mass x [1.15148, 0.949854, 1.14238, 0.878367]
  CoM XYZ mm [4.51665, -2.24625, 4.09581]
  mu_s [0.559972, 0.69316, 0.704749, 0.71008]
  mu_d [0.502209, 0.69316, 0.541926, 0.59371]
  Kp [steer,rear] [10.3539, 0]; Kd [0.212476, 0]
  spawn roll 0.0796308 rad; rate 0.135644 rad/s
  gyro bias XYZ rad/s [-5.05106e-06, -0.000476714, -0.000228683]
  gravity bias YZ [0.00160196, 0.00495476]
E3 [3.50, 5.48) s:
  mass x [1.09439, 1.02436, 1.12288, 0.804516]
  CoM XYZ mm [-2.56357, -0.297027, 1.67533]
  mu_s [0.669571, 0.666375, 0.732187, 0.637675]
  mu_d [0.669571, 0.519145, 0.631664, 0.637675]
  Kp [steer,rear] [9.88587, 0]; Kd [0.184779, 0]
  spawn roll -0.0288558 rad; rate -0.0109164 rad/s
  gyro bias XYZ rad/s [8.3864e-05, 0.000309233, 0.000301804]
  gravity bias YZ [-0.00292473, -0.00174857]
E4 [5.48, 35.48) s:
  mass x [1.1337, 1.19379, 0.812131, 0.829387]
  CoM XYZ mm [4.27955, 3.12757, 1.81693]
  mu_s [0.704749, 0.640016, 0.568073, 0.704749]
  mu_d [0.541926, 0.513141, 0.568073, 0.541926]
  Kp [steer,rear] [10.579, 0]; Kd [0.182887, 0]
  spawn roll -0.0310055 rad; rate -0.235178 rad/s
  gyro bias XYZ rad/s [-0.000155919, -0.000381763, -0.000303682]
  gravity bias YZ [-0.000194632, 0.00101622]
E5 [35.48, 41.86) s:
  mass x [0.824867, 1.07223, 1.09368, 1.08353]
  CoM XYZ mm [-1.38669, -4.9189, -0.519183]
  mu_s [0.616244, 0.69316, 0.71155, 0.71155]
  mu_d [0.586498, 0.69316, 0.71155, 0.71155]
  Kp [steer,rear] [10.3698, 0]; Kd [0.182396, 0]
  spawn roll -0.0126218 rad; rate -0.140776 rad/s
  gyro bias XYZ rad/s [0.000145366, -0.000175, 0.000220368]
  gravity bias YZ [0.0015459, -6.49389e-05]
E6 [41.86, 42.20) s:
  mass x [0.857126, 1.01041, 0.928453, 1.01033]
  CoM XYZ mm [4.65481, -4.02962, 2.75011]
  mu_s [0.559972, 0.727426, 0.71008, 0.546287]
  mu_d [0.502209, 0.606358, 0.59371, 0.546287]
  Kp [steer,rear] [10.4123, 0]; Kd [0.199519, 0]
  spawn roll -0.0852302 rad; rate 0.0692869 rad/s
  gyro bias XYZ rad/s [5.6216e-05, -0.000164477, 6.30443e-06]
  gravity bias YZ [-0.00309446, -0.00183361]
E7 [42.20, 60.00) s:
  mass x [1.05785, 0.908579, 0.876384, 0.854862]
  CoM XYZ mm [-2.4568, 4.36928, -0.17605]
  mu_s [0.791209, 0.732187, 0.631146, 0.736077]
  mu_d [0.767936, 0.631664, 0.631146, 0.699455]
  Kp [steer,rear] [10.2709, 0]; Kd [0.180311, 0]
  spawn roll 0.0450392 rad; rate -0.187041 rad/s
  gyro bias XYZ rad/s [0.000214484, 0.000120642, 0.000102376]
  gravity bias YZ [-0.00178587, 0.00240232]
Six significant digits shown; full precision: sampled_dr_values.json.
```

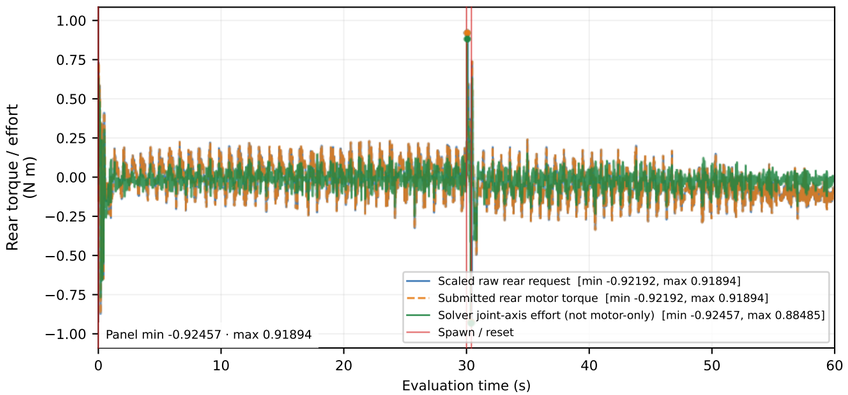
```
SAMPLED DR: env 39; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.973645, 0.884651, 1.16481, 1.04419]
  CoM XYZ mm [-0.554547, 0.030713, -0.346279]
  mu_s [0.736077, 0.728342, 0.640867, 0.666375]
  mu_d [0.699455, 0.728342, 0.556841, 0.519145]
  Kp [steer,rear] [9.09012, 0]; Kd [0.208688, 0]
  spawn roll -0.00991366 rad; rate -0.150883 rad/s
  gyro bias XYZ rad/s [-0.00036369, 8.10202e-05, 0.000139191]
  gravity bias YZ [0.000357033, 0.00113532]
E2 [30.00, 45.94) s:
  mass x [1.15842, 1.15223, 0.848549, 0.849509]
  CoM XYZ mm [-2.43123, -0.257908, 1.59667]
  mu_s [0.748289, 0.704749, 0.727426, 0.546287]
  mu_d [0.689499, 0.541926, 0.606358, 0.546287]
  Kp [steer,rear] [9.20614, 0]; Kd [0.215893, 0]
  spawn roll -0.0622703 rad; rate -0.0466344 rad/s
  gyro bias XYZ rad/s [0.000104251, 0.000172304, 0.000264019]
  gravity bias YZ [0.0030847, -0.00269552]
E3 [45.94, 60.00) s:
  mass x [1.1343, 1.00338, 1.0844, 1.03676]
  CoM XYZ mm [-2.87866, -4.3334, 0.3647467]
  mu_s [0.749678, 0.757579, 0.546287, 0.704749]
  mu_d [0.741429, 0.70921, 0.546287, 0.541926]
  Kp [steer,rear] [9.59958, 0]; Kd [0.206538, 0]
  spawn roll -0.0196737 rad; rate -0.0922485 rad/s
  gyro bias XYZ rad/s [-0.000267279, 0.000200594, 0.00017391]
  gravity bias YZ [-0.00165565, -0.00111979]
Six significant digits shown; full precision: sampled_dr_values.json.
```

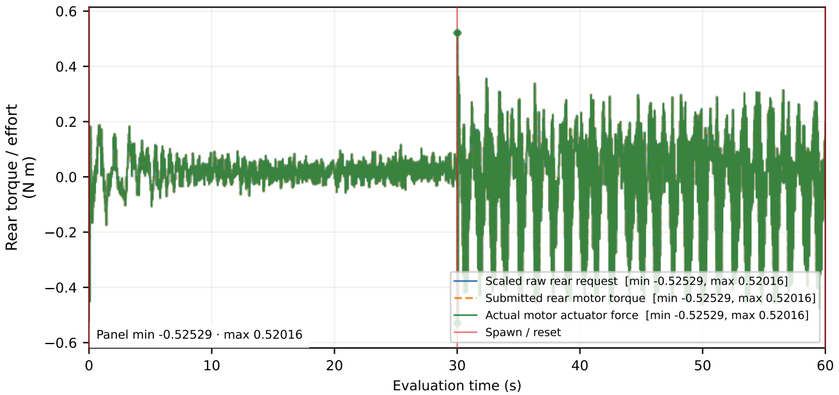
model_2500 · Demand versus actuation · env 40 · full 60 s

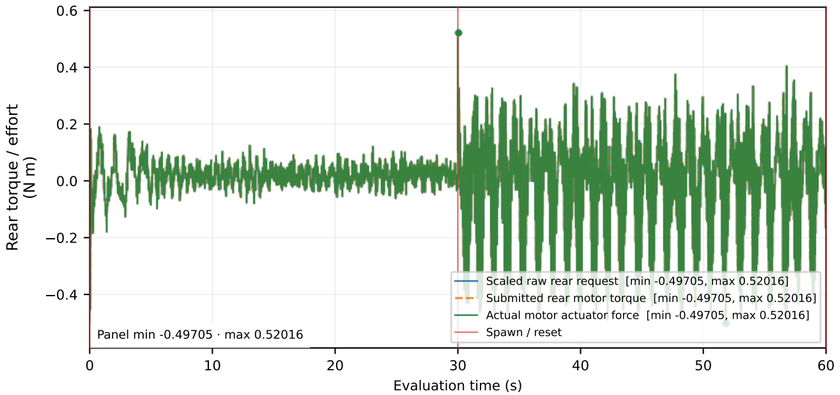












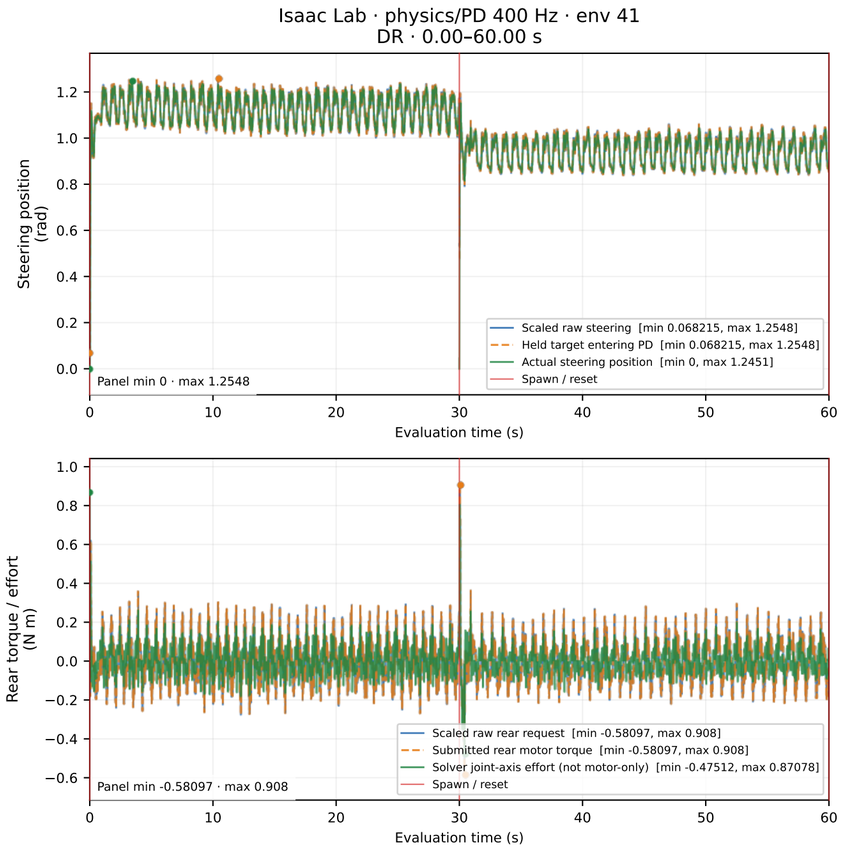
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

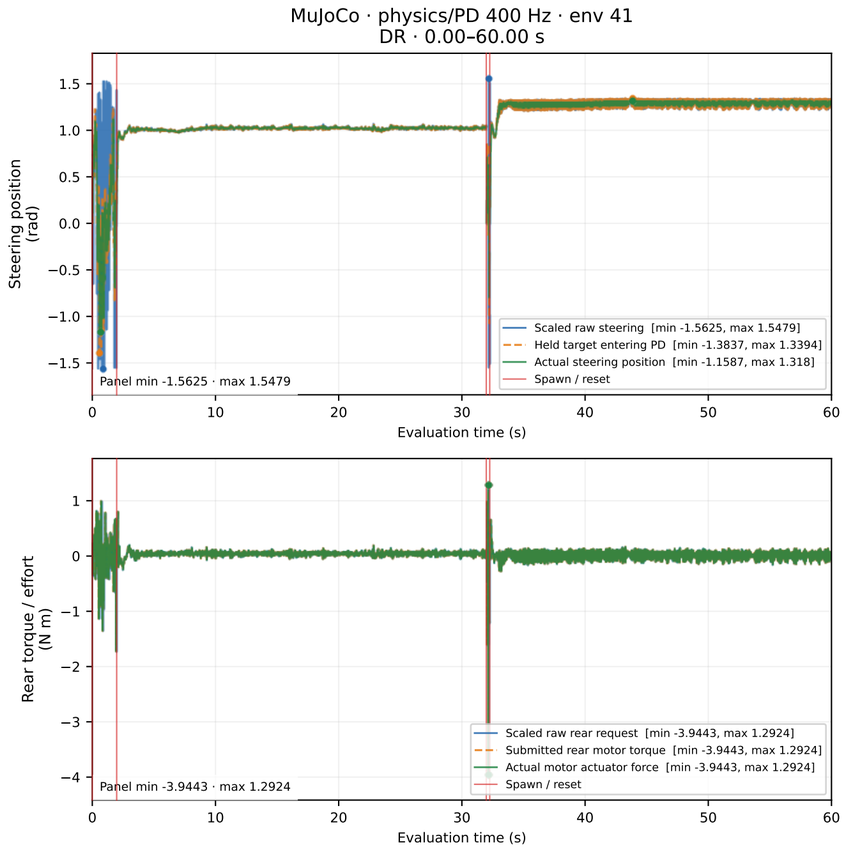
```
SAMPLED DR: env 40; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.04035, 1.04577, 0.979949, 1.08483]
  CoM XYZ mm [-0.0709146, -3.01796, 4.94215]
  mu_s [0.703711, 0.531594, 0.533455, 0.760821]
  mu_d [0.703711, 0.531594, 0.533455, 0.670315]
  Kp [steer,rear] [10.911, 0]; Kd [0.183236, 0]
  spawn roll -0.0712918 rad; rate 0.082492 rad/s
  gyro bias XYZ rad/s [-5.82015e-05, -0.000490921, -1.82122e-05]
  gravity bias YZ [0.000388784, 0.00204129]
E2 [30.00, 30.40) s:
  mass x [1.17788, 0.882942, 1.03983, 1.01883]
  CoM XYZ mm [2.16585, -4.89858, -4.63985]
  mu_s [0.764681, 0.774409, 0.760821, 0.632409]
  mu_d [0.764681, 0.561095, 0.670315, 0.589076]
  Kp [steer,rear] [10.5275, 0]; Kd [0.181964, 0]
  spawn roll 0.098748 rad; rate -0.121438 rad/s
  gyro bias XYZ rad/s [0.000241145, 0.000166553, -0.000184359]
  gravity bias YZ [0.00147458, -0.000643518]
E3 [30.40, 60.00) s:
  mass x [1.18893, 0.872782, 1.06795, 0.873044]
  CoM XYZ mm [4.9552, -0.0197962, 4.82924]
  mu_s [0.659445, 0.539956, 0.674963, 0.703711]
  mu_d [0.547619, 0.539956, 0.601294, 0.703711]
  Kp [steer,rear] [10.4112, 0]; Kd [0.200526, 0]
  spawn roll 0.069262 rad; rate -0.12872 rad/s
  gyro bias XYZ rad/s [-5.84255e-05, 0.000159831, -3.5917e-05]
  gravity bias YZ [0.000585921, -0.00253607]
Six significant digits shown; full precision: sampled_dr_values.json.
```

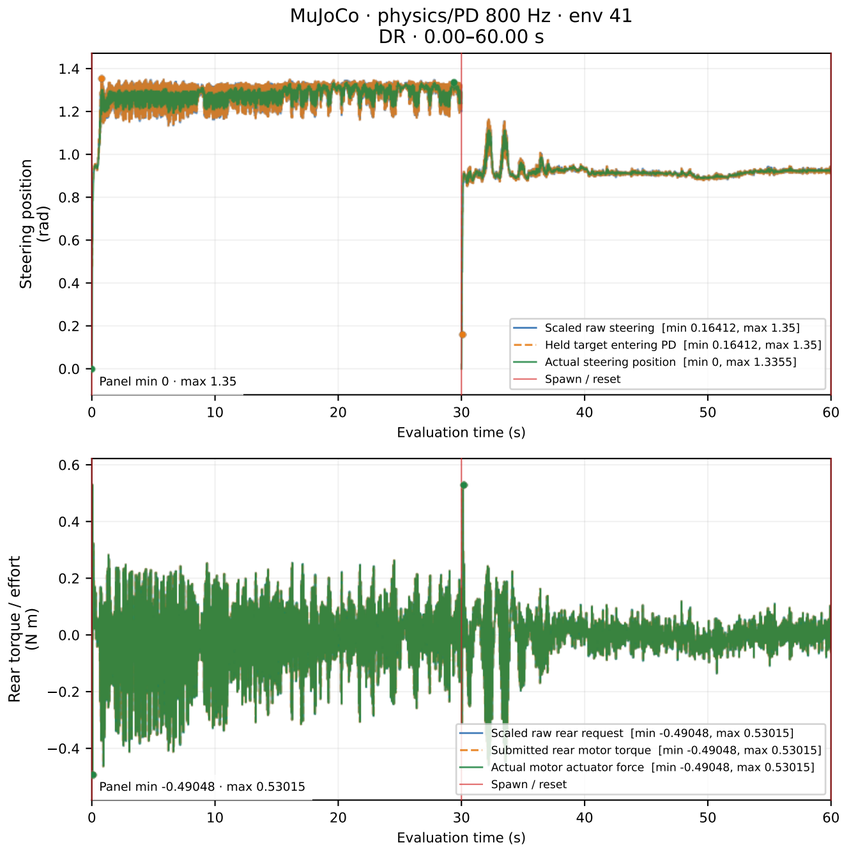
```
SAMPLED DR: env 40; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.19906, 1.14192, 0.887075, 1.19206]
  CoM XYZ mm [-0.91411, 4.39072, -2.76031]
  mu_s [0.757579, 0.748289, 0.666375, 0.700521]
  mu_d [0.70921, 0.689499, 0.519145, 0.641329]
  Kp [steer,rear] [9.28961, 0]; Kd [0.194382, 0]
  spawn roll 0.0207809 rad; rate -0.0459738 rad/s
  gyro bias XYZ rad/s [-0.000295331, 0.000226492, -0.000163432]
  gravity bias YZ [0.000325256, 0.000203647]
E2 [30.00, 60.00) s:
  mass x [0.888803, 1.02384, 0.93011, 1.12818]
  CoM XYZ mm [1.48571, 1.34925, 3.56072]
  mu_s [0.597748, 0.669571, 0.616244, 0.528253]
  mu_d [0.597748, 0.669571, 0.586498, 0.528253]
  Kp [steer,rear] [10.7976, 0]; Kd [0.180783, 0]
  spawn roll 0.0565158 rad; rate 0.209408 rad/s
  gyro bias XYZ rad/s [-5.45064e-05, -0.000171669, -0.000328325]
  gravity bias YZ [0.00244514, -0.000764611]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 40; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.19906, 1.14192, 0.887075, 1.19206]
  CoM XYZ mm [-0.91411, 4.39072, -2.76031]
  mu_s [0.757579, 0.748289, 0.666375, 0.700521]
  mu_d [0.70921, 0.689499, 0.519145, 0.641329]
  Kp [steer,rear] [9.28961, 0]; Kd [0.194382, 0]
  spawn roll 0.0207809 rad; rate -0.0459738 rad/s
  gyro bias XYZ rad/s [-0.000295331, 0.000226492, -0.000163432]
  gravity bias YZ [0.000325256, 0.000203647]
E2 [30.00, 60.00) s:
  mass x [0.888803, 1.02384, 0.93011, 1.12818]
  CoM XYZ mm [1.48571, 1.34925, 3.56072]
  mu_s [0.597748, 0.669571, 0.616244, 0.528253]
  mu_d [0.597748, 0.669571, 0.586498, 0.528253]
  Kp [steer,rear] [10.7976, 0]; Kd [0.180783, 0]
  spawn roll 0.0565158 rad; rate 0.209408 rad/s
  gyro bias XYZ rad/s [-5.45064e-05, -0.000171669, -0.000328325]
  gravity bias YZ [0.00244514, -0.000764611]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 41 · full 60 s







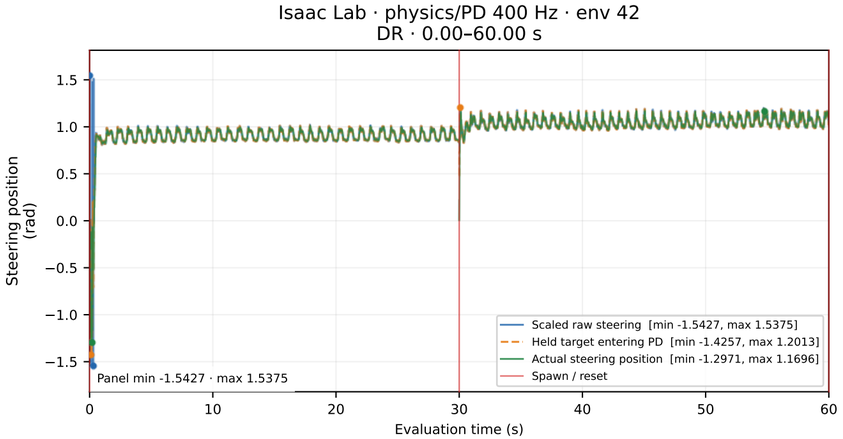
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

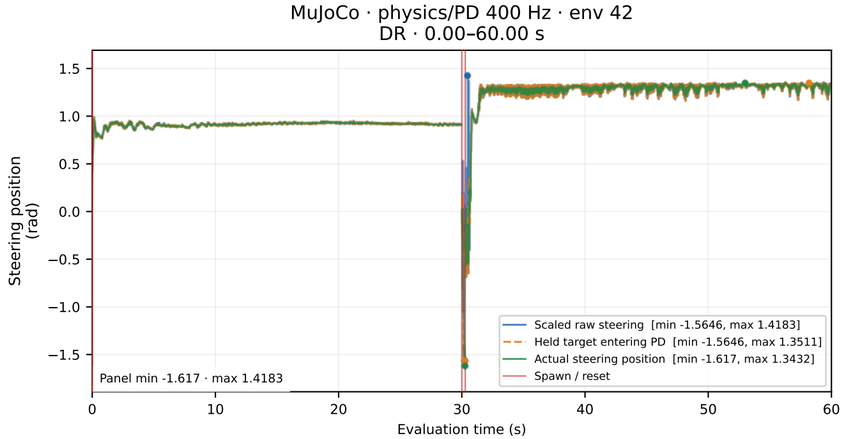
```
SAMPLED DR: env 41; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.02563, 0.983483, 1.08059, 0.972001]
  CoM XYZ mm [-3.28602, -0.157472, -3.32175]
  mu_s [0.764681, 0.693205, 0.559809, 0.674963]
  mu_d [0.764681, 0.693205, 0.559809, 0.601294]
  Kp [steer,rear] [10.9536, 0]; Kd [0.203215, 0]
  spawn roll 0.0304884 rad; rate -0.132846 rad/s
  gyro bias XYZ rad/s [4.11975e-05, -0.00018841, -8.61843e-05]
  gravity bias YZ [0.000214678, 0.00204717]
E2 [30.00, 60.00) s:
  mass x [0.975866, 1.08206, 0.843441, 1.14329]
  CoM XYZ mm [0.677288, 1.96078, -0.419378]
  mu_s [0.673378, 0.632409, 0.794338, 0.774409]
  mu_d [0.673378, 0.589076, 0.526209, 0.561095]
  Kp [steer,rear] [9.94711, 0]; Kd [0.207776, 0]
  spawn roll 0.0961761 rad; rate -0.169061 rad/s
  gyro bias XYZ rad/s [-3.30134e-06, -3.2802e-05, -0.000105036]
  gravity bias YZ [-0.000921459, -0.000930932]
Six significant digits shown; full precision: sampled_dr_values.json.
```

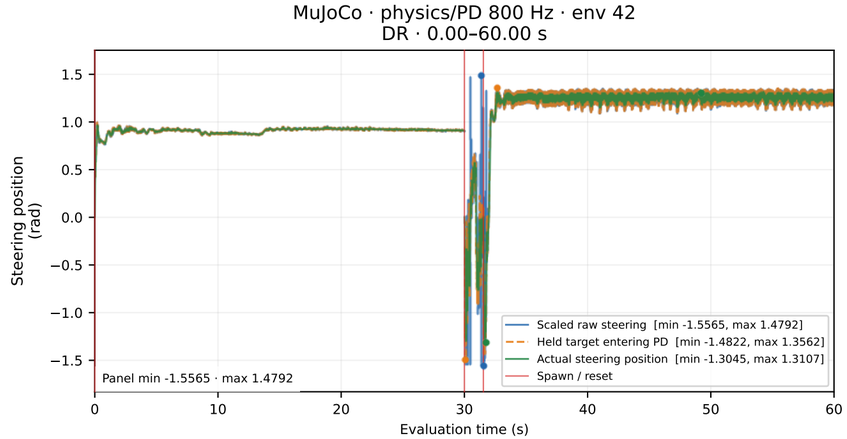
```
SAMPLED DR: env 41; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.98) s:
  mass x [0.926566, 0.817842, 1.05164, 0.922992]
  CoM XYZ mm [-1.5725, -2.52458, 3.87987]
  mu_s [0.640016, 0.640016, 0.666375, 0.791209]
  mu_d [0.513141, 0.513141, 0.519145, 0.767936]
  Kp [steer,rear] [10.1916, 0]; Kd [0.202588, 0]
  spawn roll 0.00910104 rad; rate 0.229638 rad/s
  gyro bias XYZ rad/s [-0.000103397, 0.00015139, -3.93015e-05]
  gravity bias YZ [0.00289244, 0.000194054]
E2 [1.98, 31.98) s:
  mass x [1.07076, 1.0927, 0.853399, 0.960507]
  CoM XYZ mm [-4.64046, 0.504941, 3.0578]
  mu_s [0.616244, 0.528253, 0.757579, 0.528253]
  mu_d [0.586498, 0.528253, 0.70921, 0.528253]
  Kp [steer,rear] [10.5888, 0]; Kd [0.197991, 0]
  spawn roll 0.0414456 rad; rate -0.120878 rad/s
  gyro bias XYZ rad/s [-0.00047001, -9.712e-05, -2.48645e-05]
  gravity bias YZ [-0.00115379, -0.000545909]
E3 [31.98, 32.26) s:
  mass x [0.876037, 0.815033, 0.989124, 1.15222]
  CoM XYZ mm [4.04001, 0.327695, 1.12065]
  mu_s [0.611239, 0.541939, 0.546287, 0.640867]
  mu_d [0.611239, 0.534359, 0.546287, 0.556841]
  Kp [steer,rear] [9.145, 0]; Kd [0.199156, 0]
  spawn roll 0.0656389 rad; rate -0.0234191 rad/s
  gyro bias XYZ rad/s [4.01662e-05, 0.000111145, -0.00016517]
  gravity bias YZ [-0.00235082, 0.00112893]
E4 [32.26, 60.00) s:
  mass x [1.16128, 1.118, 1.11372, 1.17773]
  CoM XYZ mm [-2.94626, -4.28865, -0.0523343]
  mu_s [0.733515, 0.528253, 0.568073, 0.749678]
  mu_d [0.558392, 0.528253, 0.568073, 0.741429]
  Kp [steer,rear] [9.90174, 0]; Kd [0.215294, 0]
  spawn roll 0.0237631 rad; rate -0.0780757 rad/s
  gyro bias XYZ rad/s [-7.63414e-05, 0.000195395, 0.000104616]
  gravity bias YZ [-0.000910715, -0.00288346]
Six significant digits shown; full precision: sampled_dr_values.json.
```

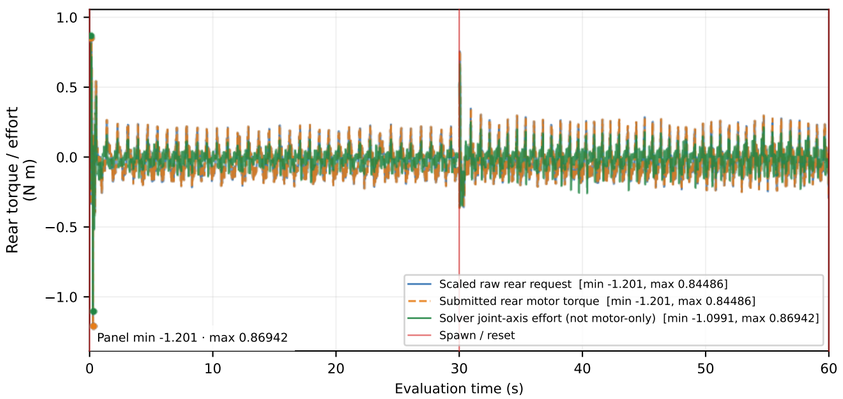
```
SAMPLED DR: env 41; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.926566, 0.817842, 1.05164, 0.922992]
  CoM XYZ mm [-1.5725, -2.52458, 3.87987]
  mu_s [0.640016, 0.640016, 0.666375, 0.791209]
  mu_d [0.513141, 0.513141, 0.519145, 0.767936]
  Kp [steer,rear] [10.1916, 0]; Kd [0.202588, 0]
  spawn roll 0.00910104 rad; rate 0.229638 rad/s
  gyro bias XYZ rad/s [-0.000103397, 0.00015139, -3.93015e-05]
  gravity bias YZ [0.00289244, 0.000194054]
E2 [30.00, 60.00) s:
  mass x [1.05303, 0.809968, 1.18286, 0.98333]
  CoM XYZ mm [3.12944, 2.71152, 3.2709]
  mu_s [0.757579, 0.736077, 0.736077, 0.541939]
  mu_d [0.70921, 0.699455, 0.699455, 0.534359]
  Kp [steer,rear] [9.98297, 0]; Kd [0.216082, 0]
  spawn roll 0.039654 rad; rate -0.122531 rad/s
  gyro bias XYZ rad/s [-0.000155446, -0.000121868, 0.000286776]
  gravity bias YZ [0.000968121, -0.000573735]
Six significant digits shown; full precision: sampled_dr_values.json.
```

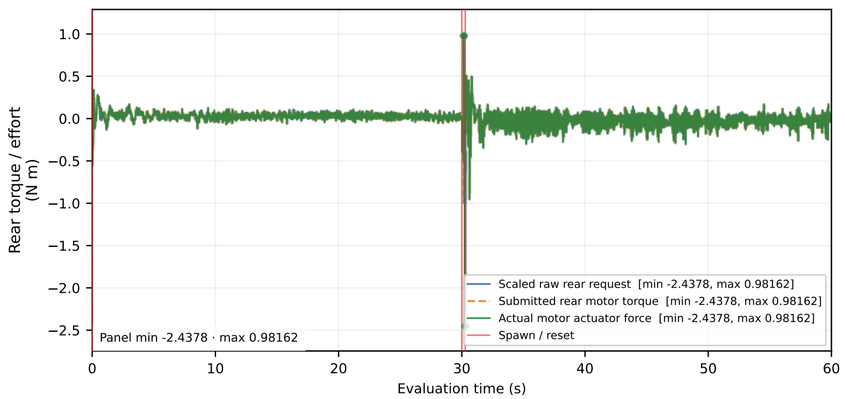
model_2500 · Demand versus actuation · env 42 · full 60 s

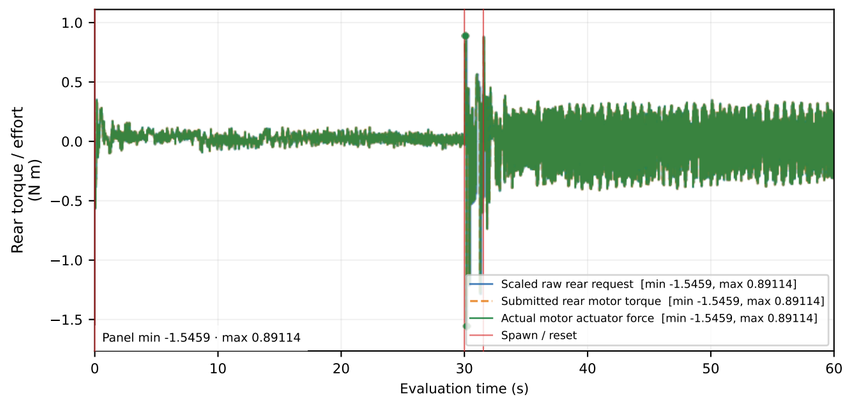












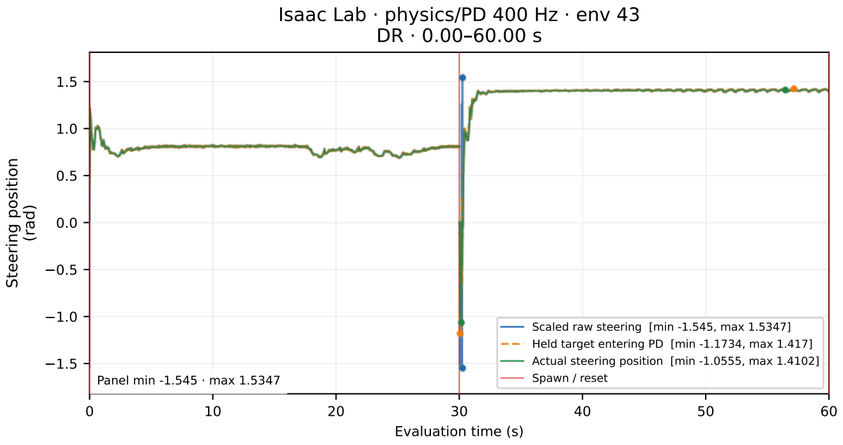
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

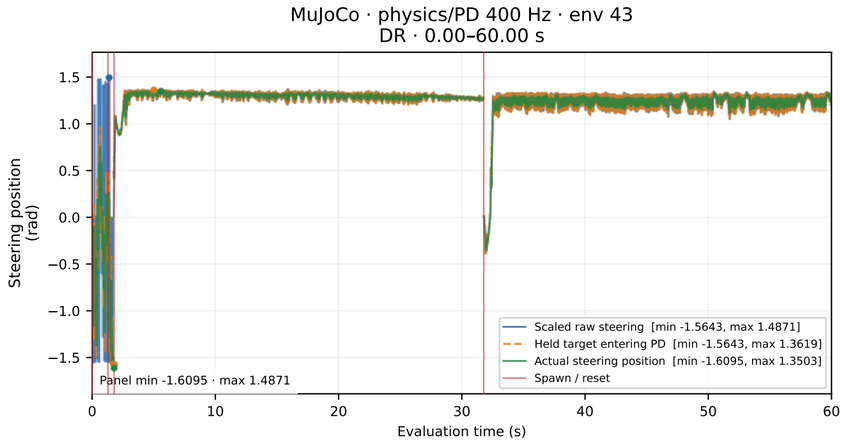
```
SAMPLED DR: env 42; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.00679, 1.17058, 1.15612, 1.14533]
  CoM XYZ mm [-3.16264, 2.93301, 3.42508]
  mu_s [0.560535, 0.713129, 0.703711, 0.598343]
  mu_d [0.560535, 0.713129, 0.703711, 0.598343]
  Kp [steer,rear] [9.3656, 0]; Kd [0.180648, 0]
  spawn roll -0.0355667 rad; rate -0.199113 rad/s
  gyro bias XYZ rad/s [7.18189e-05, 3.51286e-05, -4.70988e-05]
  gravity bias YZ [0.00497112, 0.000496189]
E2 [30.00, 60.00) s:
  mass x [1.11242, 1.10235, 0.950996, 0.941079]
  CoM XYZ mm [-1.10635, 0.569113, 2.59101]
  mu_s [0.785466, 0.764681, 0.673378, 0.693205]
  mu_d [0.52258, 0.764681, 0.673378, 0.693205]
  Kp [steer,rear] [10.383, 0]; Kd [0.183516, 0]
  spawn roll 0.0535442 rad; rate 0.212608 rad/s
  gyro bias XYZ rad/s [-0.000107776, -0.000431715, -2.87598e-05]
  gravity bias YZ [0.000301304, 0.000161024]
Six significant digits shown; full precision: sampled_dr_values.json.
```

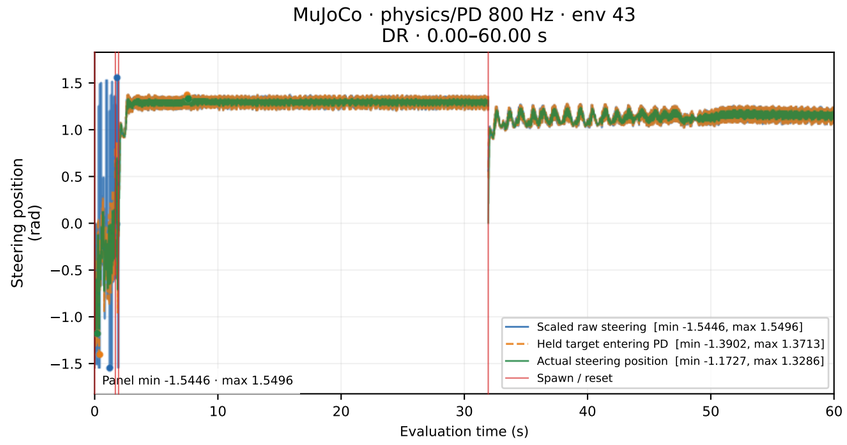
```
SAMPLED DR: env 42; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09284, 1.16008, 1.09972, 1.15357]
  CoM XYZ mm [-2.72544, 2.35069, 1.48796]
  mu_s [0.637675, 0.748289, 0.541939, 0.748289]
  mu_d [0.637675, 0.689499, 0.534359, 0.689499]
  Kp [steer,rear] [10.6795, 0]; Kd [0.195338, 0]
  spawn roll -0.0149055 rad; rate 0.134696 rad/s
  gyro bias XYZ rad/s [-0.000172796, -0.000281875, 0.000203764]
  gravity bias YZ [-0.00172276, -0.000162154]
E2 [30.00, 30.28) s:
  mass x [0.931213, 0.864553, 0.928057, 0.913452]
  CoM XYZ mm [-4.22538, 1.72408, -0.154362]
  mu_s [0.71008, 0.69316, 0.611239, 0.71155]
  mu_d [0.59371, 0.69316, 0.611239, 0.71155]
  Kp [steer,rear] [9.60562, 0]; Kd [0.212883, 0]
  spawn roll -0.0799131 rad; rate -0.0172209 rad/s
  gyro bias XYZ rad/s [0.000366462, 0.000184175, -0.000310861]
  gravity bias YZ [-0.00254312, -0.00370704]
E3 [30.28, 60.00) s:
  mass x [1.07123, 1.15427, 1.11965, 1.15157]
  CoM XYZ mm [-0.924762, -2.74748, 2.98176]
  mu_s [0.71008, 0.727426, 0.538976, 0.559972]
  mu_d [0.59371, 0.606358, 0.538976, 0.502209]
  Kp [steer,rear] [10.2255, 0]; Kd [0.213834, 0]
  spawn roll -0.0450423 rad; rate 0.0900405 rad/s
  gyro bias XYZ rad/s [-0.000205619, -2.14786e-05, -0.000159829]
  gravity bias YZ [0.0021959, -0.00092937]
Six significant digits shown; full precision: sampled_dr_values.json.
```

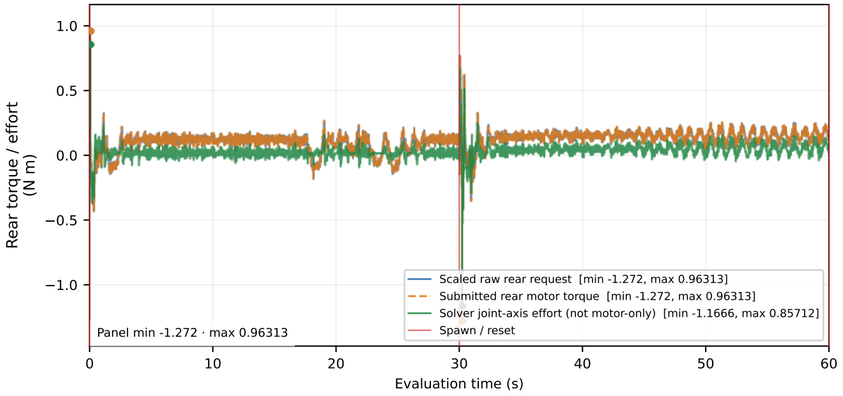
```
SAMPLED DR: env 42; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09284, 1.16008, 1.09972, 1.15357]
  CoM XYZ mm [-2.72544, 2.35069, 1.48796]
  mu_s [0.637675, 0.748289, 0.541939, 0.748289]
  mu_d [0.637675, 0.689499, 0.534359, 0.689499]
  Kp [steer,rear] [10.6795, 0]; Kd [0.195338, 0]
  spawn roll -0.0149055 rad; rate 0.134696 rad/s
  gyro bias XYZ rad/s [-0.000172796, -0.000281875, 0.000203764]
  gravity bias YZ [-0.00172276, -0.000162154]
E2 [30.00, 31.54) s:
  mass x [0.931213, 0.864553, 0.928057, 0.913452]
  CoM XYZ mm [-4.22538, 1.72408, -0.154362]
  mu_s [0.71008, 0.69316, 0.611239, 0.71155]
  mu_d [0.59371, 0.69316, 0.611239, 0.71155]
  Kp [steer,rear] [9.60562, 0]; Kd [0.212883, 0]
  spawn roll -0.0799131 rad; rate -0.0172209 rad/s
  gyro bias XYZ rad/s [0.000366462, 0.000184175, -0.000310861]
  gravity bias YZ [-0.00254312, -0.00370704]
E3 [31.54, 60.00) s:
  mass x [1.17508, 0.936711, 0.962443, 0.975015]
  CoM XYZ mm [3.56882, -1.68577, -3.84009]
  mu_s [0.727426, 0.69316, 0.736077, 0.732187]
  mu_d [0.606358, 0.69316, 0.699455, 0.631664]
  Kp [steer,rear] [10.618, 0]; Kd [0.208953, 0]
  spawn roll -0.0754326 rad; rate -0.180929 rad/s
  gyro bias XYZ rad/s [0.000251584, -0.000109471, 0.000467955]
  gravity bias YZ [0.000706018, 0.000487003]
Six significant digits shown; full precision: sampled_dr_values.json.
```

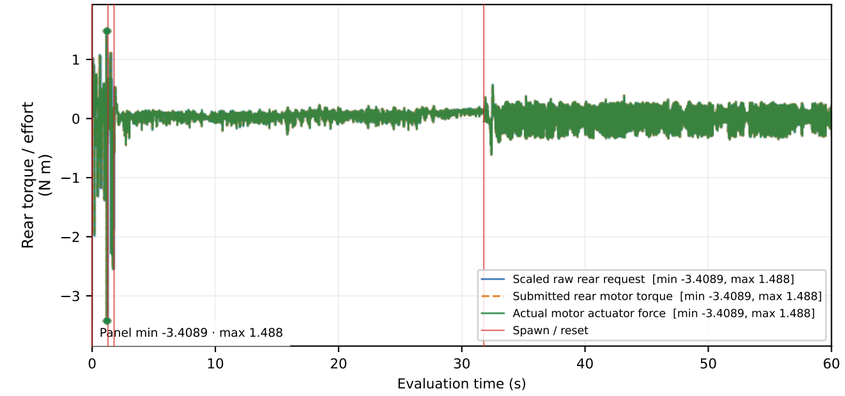
model_2500 · Demand versus actuation · env 43 · full 60 s

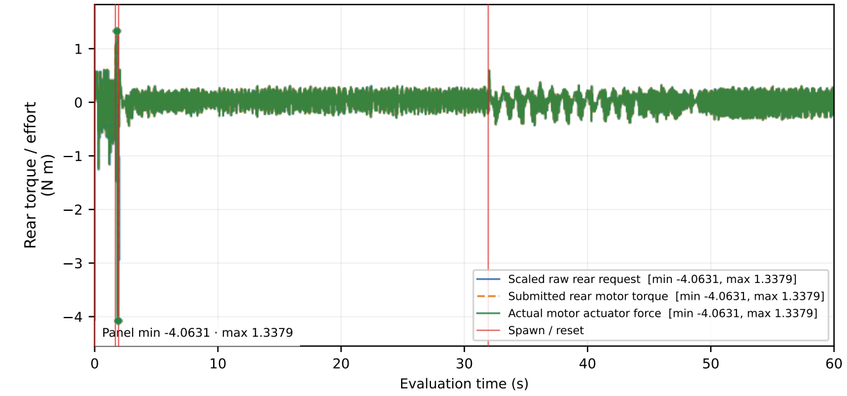












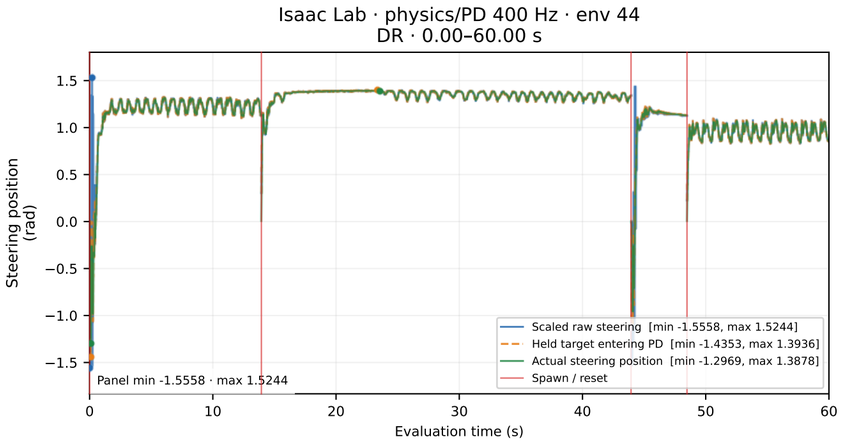
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

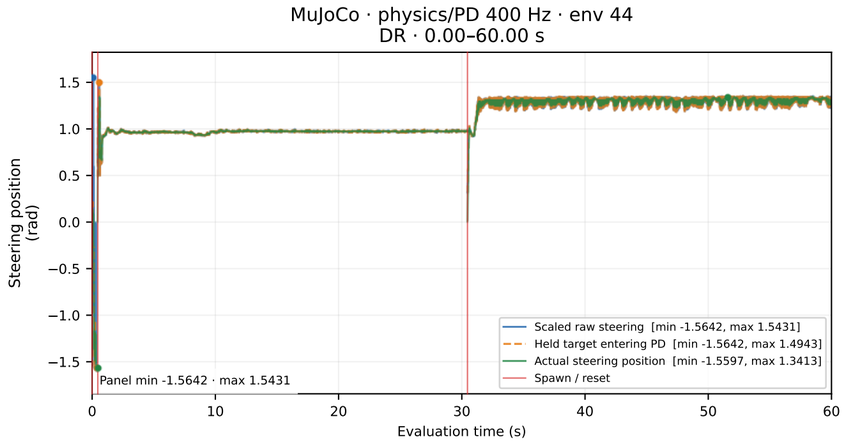
```
SAMPLED DR: env 43; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.10483, 0.886778, 1.00858, 1.14156]
  CoM XYZ mm [-4.88865, 3.97869, -1.99647]
  mu_s [0.751971, 0.751971, 0.560535, 0.531594]
  mu_d [0.541224, 0.541224, 0.560535, 0.531594]
  Kp [steer,rear] [9.30642, 0]; Kd [0.208974, 0]
  spawn roll 0.0878213 rad; rate 0.00262013 rad/s
  gyro bias XYZ rad/s [0.000470035, -0.000422879, 0.000217325]
  gravity bias YZ [-0.00177895, 0.00267846]
E2 [30.00, 60.00) s:
  mass x [1.16034, 0.871918, 0.825417, 0.964164]
  CoM XYZ mm [-2.4552, -4.01363, -4.01926]
  mu_s [0.560535, 0.576972, 0.560535, 0.726175]
  mu_d [0.560535, 0.576972, 0.560535, 0.558574]
  Kp [steer,rear] [10.7255, 0]; Kd [0.191452, 0]
  spawn roll -0.0755623 rad; rate 0.0821185 rad/s
  gyro bias XYZ rad/s [0.000369804, -0.000125499, -0.000205857]
  gravity bias YZ [-0.000956724, 0.00213195]
Six significant digits shown; full precision: sampled_dr_values.json.
```

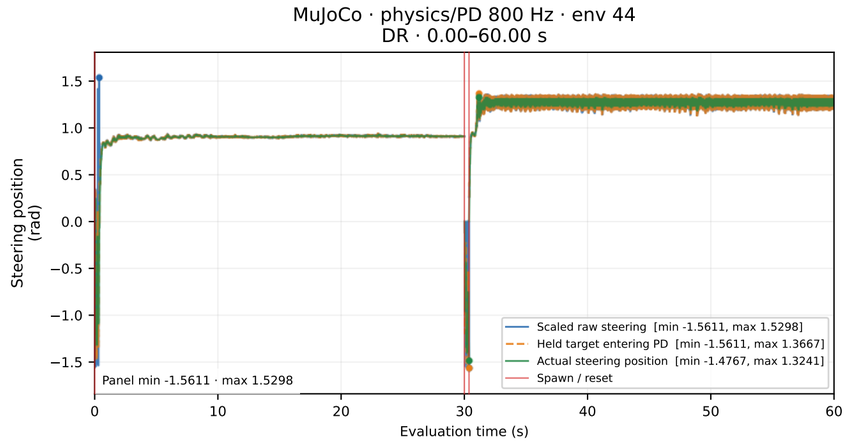
```
SAMPLED DR: env 43; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.28) s:
  mass x [0.817729, 0.988708, 0.985162, 0.991227]
  CoM XYZ mm [3.8026, 2.9309, -2.68711]
  mu_s [0.723429, 0.71008, 0.538976, 0.611239]
  mu_d [0.723429, 0.59371, 0.538976, 0.611239]
  Kp [steer,rear] [10.0508, 0]; Kd [0.200584, 0]
  spawn roll -0.050416 rad; rate -0.105342 rad/s
  gyro bias XYZ rad/s [0.000375649, -9.08079e-05, -0.000273867]
  gravity bias YZ [-0.00169986, -0.0021852]
E2 [1.28, 1.78) s:
  mass x [0.89827, 1.13738, 0.833986, 0.911589]
  CoM XYZ mm [3.19607, -1.52013, -4.4138]
  mu_s [0.728342, 0.791209, 0.748289, 0.723429]
  mu_d [0.728342, 0.767936, 0.689499, 0.723429]
  Kp [steer,rear] [9.44534, 0]; Kd [0.21778, 0]
  spawn roll -0.0843199 rad; rate 0.149925 rad/s
  gyro bias XYZ rad/s [0.000116279, 0.000368476, -0.000118591]
  gravity bias YZ [-0.000460961, 0.0007676697]
E3 [1.78, 31.78) s:
  mass x [1.10481, 0.887145, 0.816103, 1.12608]
  CoM XYZ mm [4.70513, -4.49192, 2.76855]
  mu_s [0.71155, 0.559972, 0.733515, 0.546287]
  mu_d [0.71155, 0.502209, 0.558392, 0.546287]
  Kp [steer,rear] [10.1804, 0]; Kd [0.20886, 0]
  spawn roll -0.0182207 rad; rate 0.169049 rad/s
  gyro bias XYZ rad/s [-4.77859e-05, -7.55718e-06, 0.000306216]
  gravity bias YZ [0.000457112, 0.00187042]
E4 [31.78, 60.00) s:
  mass x [0.958536, 0.89962, 1.00667, 0.894517]
  CoM XYZ mm [1.55971, -2.34448, -1.05464]
  mu_s [0.538434, 0.69316, 0.616244, 0.757579]
  mu_d [0.538434, 0.69316, 0.586498, 0.70921]
  Kp [steer,rear] [9.1567, 0]; Kd [0.186025, 0]
  spawn roll -0.0305574 rad; rate 0.0144263 rad/s
  gyro bias XYZ rad/s [-0.000241398, 0.00031125, -0.000228451]
  gravity bias YZ [-0.00162742, -0.00212006]
Six significant digits shown; full precision: sampled_dr_values.json.
```

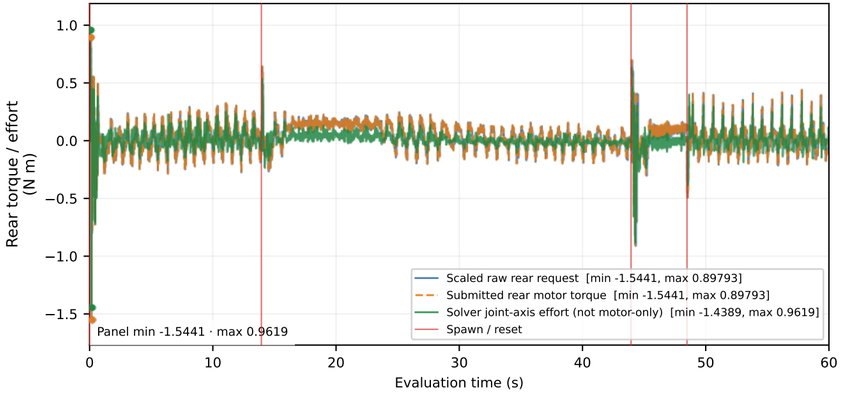
```
SAMPLED DR: env 43; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.68) s:
  mass x [0.817729, 0.988708, 0.985162, 0.991227]
  CoM XYZ mm [3.8026, 2.9309, -2.68711]
  mu_s [0.723429, 0.71008, 0.538976, 0.611239]
  mu_d [0.723429, 0.59371, 0.538976, 0.611239]
  Kp [steer,rear] [10.0508, 0]; Kd [0.200584, 0]
  spawn roll -0.050416 rad; rate -0.105342 rad/s
  gyro bias XYZ rad/s [0.000375649, -9.08079e-05, -0.000273867]
  gravity bias YZ [-0.00169986, -0.0021852]
E2 [1.68, 1.94) s:
  mass x [1.16563, 0.990179, 1.04739, 0.870516]
  CoM XYZ mm [2.19329, 1.80216, -4.02538]
  mu_s [0.728342, 0.538976, 0.704749, 0.669571]
  mu_d [0.728342, 0.538976, 0.541926, 0.669571]
  Kp [steer,rear] [10.6388, 0]; Kd [0.202311, 0]
  spawn roll 0.080898 rad; rate 0.0849563 rad/s
  gyro bias XYZ rad/s [0.00033814, -3.26728e-05, 5.41149e-05]
  gravity bias YZ [-0.000456631, -0.00314438]
E3 [1.94, 31.94) s:
  mass x [1.16411, 1.19538, 1.04018, 1.17439]
  CoM XYZ mm [4.28446, -3.79597, -2.24692]
  mu_s [0.749678, 0.757579, 0.637675, 0.616244]
  mu_d [0.741429, 0.70921, 0.637675, 0.586498]
  Kp [steer,rear] [9.82385, 0]; Kd [0.199662, 0]
  spawn roll 0.0131643 rad; rate -0.16866 rad/s
  gyro bias XYZ rad/s [-0.000417208, 0.000270835, -9.43824e-05]
  gravity bias YZ [9.76232e-05, 0.00206213]
E4 [31.94, 60.00) s:
  mass x [1.11138, 1.09319, 1.02976, 1.05999]
  CoM XYZ mm [-2.99738, -0.236625, -0.403381]
  mu_s [0.727426, 0.559972, 0.568073, 0.597748]
  mu_d [0.606358, 0.502209, 0.568073, 0.597748]
  Kp [steer,rear] [10.8315, 0]; Kd [0.212054, 0]
  spawn roll 0.0308381 rad; rate 0.0507097 rad/s
  gyro bias XYZ rad/s [-0.000151025, 1.28111e-05, -3.67526e-05]
  gravity bias YZ [-0.00213026, 0.00109776]
Six significant digits shown; full precision: sampled_dr_values.json.
```

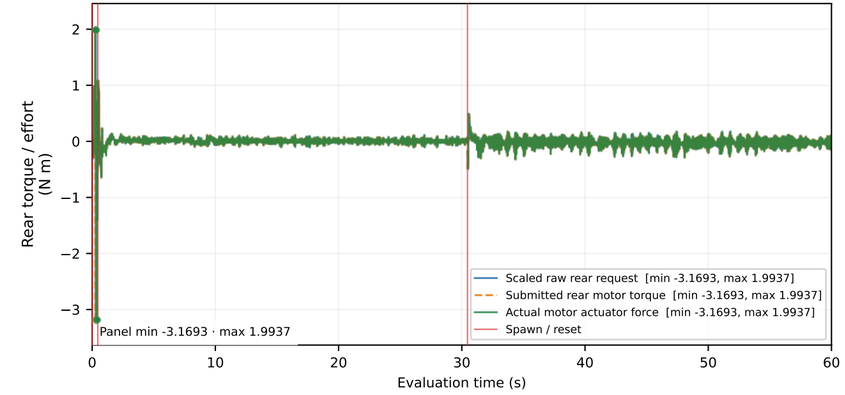
model_2500 · Demand versus actuation · env 44 · full 60 s

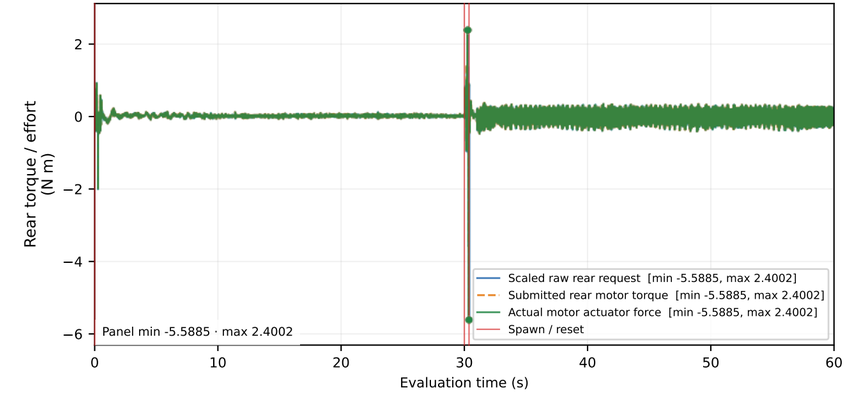












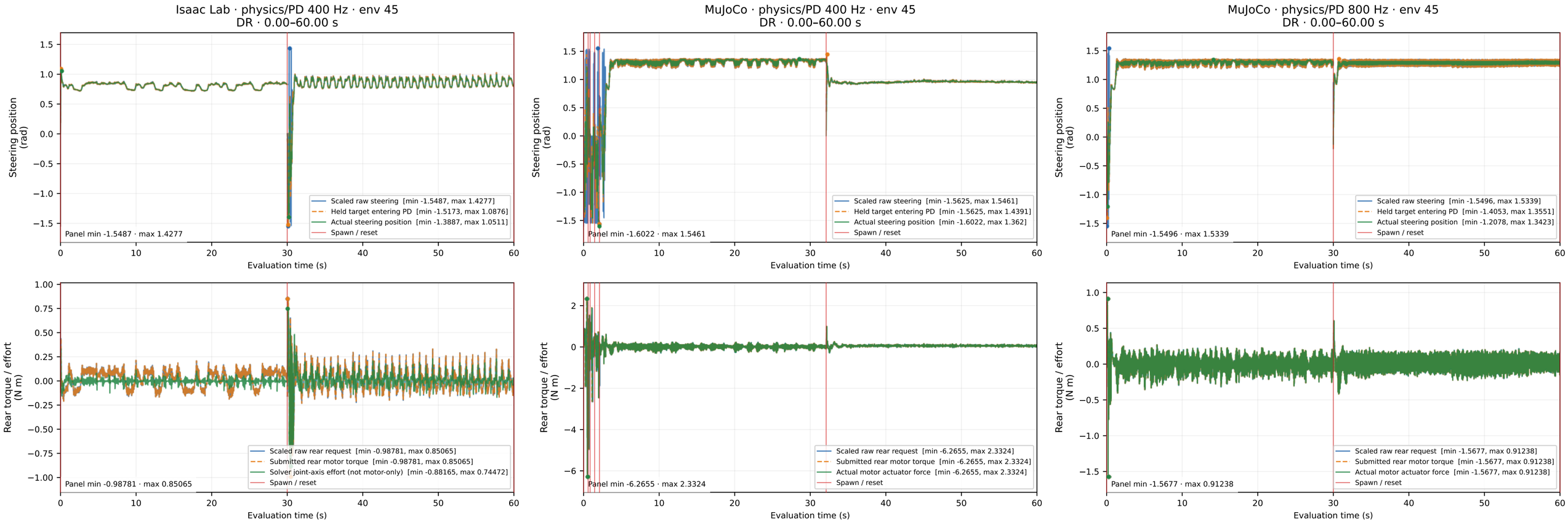
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 44; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 13.94) s:
  mass x [0.941381, 0.96656, 0.849268, 0.883645]
  CoM XYZ mm [-3.95469, -0.882261, -4.4395]
  mu_s [0.647391, 0.726175, 0.533455, 0.726175]
  mu_d [0.647391, 0.558574, 0.533455, 0.558574]
  Kp [steer,rear] [10.553, 0]; Kd [0.203251, 0]
  spawn roll -0.0903528 rad; rate -0.239559 rad/s
  gyro bias XYZ rad/s [-0.000238181, -9.13169e-05, -0.000286138]
  gravity bias YZ [0.00360941, -0.00105445]
E2 [13.94, 43.94) s:
  mass x [1.14842, 0.849827, 1.19386, 1.02724]
  CoM XYZ mm [-4.23548, -3.53206, -0.988275]
  mu_s [0.533455, 0.787792, 0.547743, 0.774409]
  mu_d [0.533455, 0.617134, 0.547743, 0.561095]
  Kp [steer,rear] [10.1177, 0]; Kd [0.192887, 0]
  spawn roll 0.0182498 rad; rate -0.0147221 rad/s
  gyro bias XYZ rad/s [-0.000116376, 0.000249171, 0.000123057]
  gravity bias YZ [1.07443e-05, 0.00194308]
E3 [43.94, 48.48) s:
  mass x [1.15761, 1.16175, 1.02536, 1.17213]
  CoM XYZ mm [-2.67453, -4.62616, 4.68497]
  mu_s [0.598343, 0.647391, 0.533455, 0.647391]
  mu_d [0.598343, 0.647391, 0.533455, 0.647391]
  Kp [steer,rear] [10.7051, 0]; Kd [0.203398, 0]
  spawn roll -0.0753268 rad; rate 0.106684 rad/s
  gyro bias XYZ rad/s [-0.000162301, -0.000414313, 7.95857e-05]
  gravity bias YZ [0.00263486, -0.00279752]
E4 [48.48, 60.00) s:
  mass x [1.18532, 1.03488, 1.18563, 1.19369]
  CoM XYZ mm [-2.79817, 1.75211, -3.49937]
  mu_s [0.774409, 0.579974, 0.726175, 0.674963]
  mu_d [0.561095, 0.579974, 0.558574, 0.601294]
  Kp [steer,rear] [9.84255, 0]; Kd [0.180638, 0]
  spawn roll -0.00713564 rad; rate 0.0294966 rad/s
  gyro bias XYZ rad/s [5.65455e-05, -0.000190306, -0.000184131]
  gravity bias YZ [5.0621e-05, -0.000538262]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 44; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.46) s:
  mass x [1.04618, 0.870142, 1.04623, 0.949018]
  CoM XYZ mm [-1.31651, 2.93645, 4.59471]
  mu_s [0.538434, 0.538434, 0.736077, 0.733515]
  mu_d [0.538434, 0.538434, 0.699455, 0.558392]
  Kp [steer,rear] [9.68445, 0]; Kd [0.190976, 0]
  spawn roll -0.0585563 rad; rate 0.157051 rad/s
  gyro bias XYZ rad/s [1.85104e-05, -0.000490205, 0.000237956]
  gravity bias YZ [0.00337487, -7.0869e-05]
E2 [0.46, 30.46) s:
  mass x [0.954605, 0.835809, 1.07741, 1.06988]
  CoM XYZ mm [-4.10035, 0.884504, 4.14273]
  mu_s [0.728342, 0.736077, 0.69316, 0.546287]
  mu_d [0.728342, 0.699455, 0.69316, 0.546287]
  Kp [steer,rear] [9.26279, 0]; Kd [0.187364, 0]
  spawn roll 0.0845821 rad; rate -0.0686739 rad/s
  gyro bias XYZ rad/s [-0.000148312, -4.68555e-05, 0.000189854]
  gravity bias YZ [-0.000816323, -0.00182729]
E3 [30.46, 60.00) s:
  mass x [0.990045, 1.03445, 0.878945, 1.18026]
  CoM XYZ mm [-1.52486, -3.60773, 3.41697]
  mu_s [0.631146, 0.757579, 0.749678, 0.749678]
  mu_d [0.631146, 0.70921, 0.741429, 0.741429]
  Kp [steer,rear] [9.51207, 0]; Kd [0.201351, 0]
  spawn roll -0.00503864 rad; rate 0.0751251 rad/s
  gyro bias XYZ rad/s [5.75694e-06, -0.000395627, -0.000485063]
  gravity bias YZ [0.00123184, -0.005065]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 44; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.04618, 0.870142, 1.04623, 0.949018]
  CoM XYZ mm [-1.31651, 2.93645, 4.59471]
  mu_s [0.538434, 0.538434, 0.736077, 0.733515]
  mu_d [0.538434, 0.538434, 0.699455, 0.558392]
  Kp [steer,rear] [9.68445, 0]; Kd [0.190976, 0]
  spawn roll -0.0585563 rad; rate 0.157051 rad/s
  gyro bias XYZ rad/s [1.85104e-05, -0.000490205, 0.000237956]
  gravity bias YZ [0.00337487, -7.0869e-05]
E2 [30.00, 30.38) s:
  mass x [0.948621, 1.00576, 1.15524, 1.08679]
  CoM XYZ mm [3.67924, 1.66306, 1.97143]
  mu_s [0.791209, 0.528253, 0.704749, 0.631146]
  mu_d [0.767936, 0.528253, 0.541926, 0.631146]
  Kp [steer,rear] [9.56033, 0]; Kd [0.181751, 0]
  spawn roll -0.0792774 rad; rate 0.0337022 rad/s
  gyro bias XYZ rad/s [3.46728e-05, -5.81238e-05, 0.000143752]
  gravity bias YZ [0.0014215, -5.02601e-05]
E3 [30.38, 60.00) s:
  mass x [0.909688, 1.1506, 0.868769, 0.976718]
  CoM XYZ mm [3.58117, -2.42407, -0.289591]
  mu_s [0.732187, 0.637675, 0.757579, 0.732187]
  mu_d [0.631664, 0.637675, 0.70921, 0.631664]
  Kp [steer,rear] [10.997, 0]; Kd [0.189741, 0]
  spawn roll 0.00442566 rad; rate 0.0122795 rad/s
  gyro bias XYZ rad/s [-0.000221453, -0.000250809, 5.67582e-05]
  gravity bias YZ [-0.00189388, -0.00247438]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 45 · full 60 s



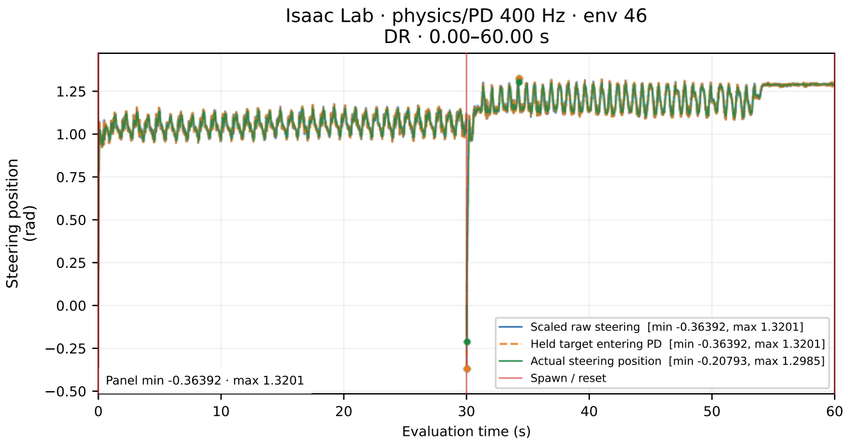
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

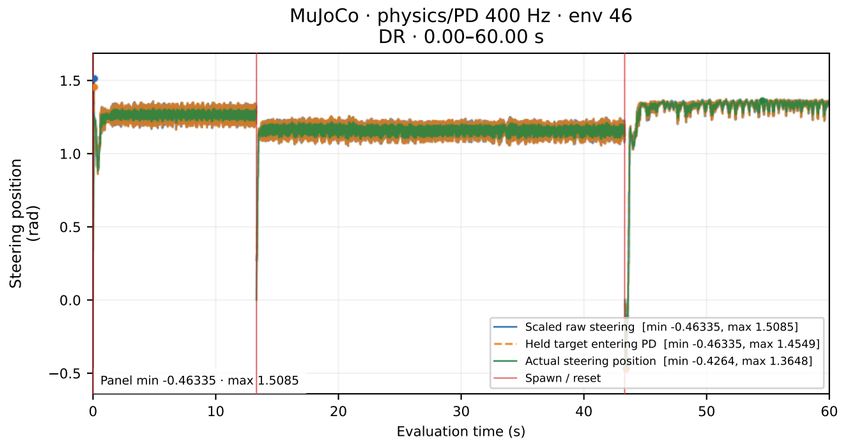
```
SAMPLED DR: env 45; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.91451, 1.0945, 1.03991, 1.15067]
  CoM XYZ mm [0.609383, 4.51562, 0.853941]
  mu_s [0.532645, 0.647391, 0.674963, 0.533455]
  mu_d [0.532645, 0.647391, 0.601294, 0.533455]
  Kp [steer,rear] [10.5839, 0]; Kd [0.20961, 0]
  spawn roll 0.0561155 rad; rate -0.163077 rad/s
  gyro bias XYZ rad/s [-9.15444e-05, -0.000168938, -1.55805e-05]
  gravity bias YZ [0.00347817, 0.00130226]
E2 [30.00, 60.00) s:
  mass x [1.15753, 0.834688, 1.14101, 1.08278]
  CoM XYZ mm [0.795841, 3.01766, -4.12476]
  mu_s [0.760821, 0.647391, 0.659445, 0.632409]
  mu_d [0.670315, 0.647391, 0.547619, 0.589076]
  Kp [steer,rear] [9.84092, 0]; Kd [0.197066, 0]
  spawn roll -0.0740835 rad; rate 0.093096 rad/s
  gyro bias XYZ rad/s [-0.00031712, 0.000120311, 0.000243353]
  gravity bias YZ [-0.00052785, -0.00170623]
Six significant digits shown; full precision: sampled_dr_values.json.
```

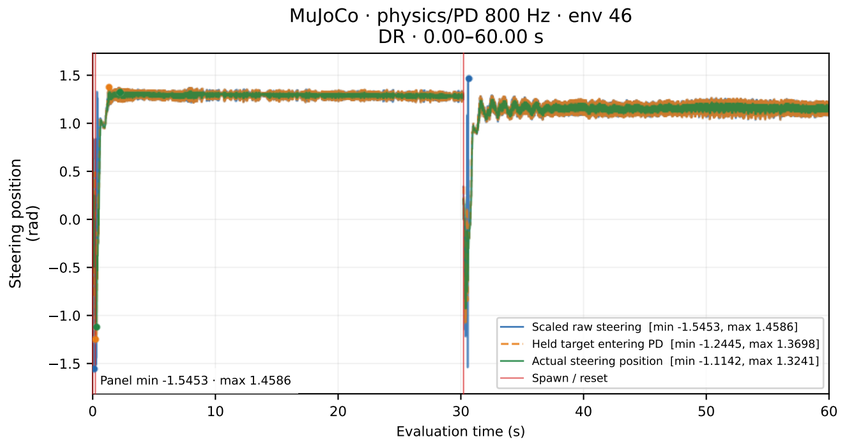
```
SAMPLED DR: env 45; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.60) s:
  mass x [1.0283, 1.11342, 0.868957, 0.992338]
  CoM XYZ mm [-0.719669, -3.11429, 0.391038]
  mu_s [0.616244, 0.597748, 0.736077, 0.616244]
  mu_d [0.586498, 0.597748, 0.699455, 0.586498]
  Kp [steer,rear] [9.60248, 0]; Kd [0.180136, 0]
  spawn roll -0.090286 rad; rate -0.0511712 rad/s
  gyro bias XYZ rad/s [0.000372074, 1.53988e-05, -0.000226969]
  gravity bias YZ [0.0011662, -0.00273945]
E2 [0.60, 0.86) s:
  mass x [0.84832, 1.17614, 0.915474, 0.899313]
  CoM XYZ mm [1.10572, -3.69804, 0.460751]
  mu_s [0.559972, 0.732187, 0.559972, 0.71155]
  mu_d [0.502209, 0.631664, 0.502209, 0.71155]
  Kp [steer,rear] [9.34733, 0]; Kd [0.19494, 0]
  spawn roll 0.0842819 rad; rate 0.176581 rad/s
  gyro bias XYZ rad/s [0.000186427, -9.71195e-05, -4.72349e-05]
  gravity bias YZ [0.00366361, 0.0010572]
E3 [0.86, 1.46) s:
  mass x [1.10776, 0.995629, 1.1679, 1.04241]
  CoM XYZ mm [-3.56775, 1.71396, -2.33411]
  mu_s [0.69316, 0.69316, 0.700521, 0.546287]
  mu_d [0.69316, 0.69316, 0.641329, 0.546287]
  Kp [steer,rear] [9.55618, 0]; Kd [0.185214, 0]
  spawn roll -0.0837655 rad; rate -0.190753 rad/s
  gyro bias XYZ rad/s [9.42402e-05, -0.000129821, -0.000217799]
  gravity bias YZ [-0.00055415, -0.00216323]
E4 [1.46, 2.10) s:
  mass x [0.936873, 0.87709, 0.844336, 0.859617]
  CoM XYZ mm [-0.365105, 2.8358, 4.34363]
  mu_s [0.541939, 0.757579, 0.791209, 0.757579]
  mu_d [0.534359, 0.70921, 0.767936, 0.70921]
  Kp [steer,rear] [9.17322, 0]; Kd [0.218362, 0]
  spawn roll -0.00954031 rad; rate 0.0414334 rad/s
  gyro bias XYZ rad/s [-0.000447509, -0.000180606, 8.89018e-05]
  gravity bias YZ [0.00342217, -6.01172e-05]
E5 [2.10, 32.10) s:
  mass x [0.928924, 1.12088, 1.17138, 0.908778]
  CoM XYZ mm [-0.303269, -4.07428, -0.365249]
  mu_s [0.640867, 0.732187, 0.541939, 0.791209]
  mu_d [0.556841, 0.631664, 0.534359, 0.767936]
  Kp [steer,rear] [10.7022, 0]; Kd [0.180027, 0]
  spawn roll 0.00338899 rad; rate 0.21835 rad/s
  gyro bias XYZ rad/s [5.8181e-05, 0.000126456, 9.24898e-05]
  gravity bias YZ [-0.00237553, -0.000891357]
E6 [32.10, 60.00) s:
  mass x [1.07428, 1.03262, 1.13889, 0.939128]
  CoM XYZ mm [-3.6004, 1.69268, -3.89672]
  mu_s [0.71008, 0.538434, 0.640867, 0.611239]
  mu_d [0.59371, 0.538434, 0.556841, 0.611239]
  Kp [steer,rear] [10.6163, 0]; Kd [0.192153, 0]
  spawn roll 0.085458 rad; rate -0.190884 rad/s
  gyro bias XYZ rad/s [-0.000152866, -0.000116929, -0.000136895]
  gravity bias YZ [-0.000396986, -0.0041253]
Six significant digits shown; full precision: sampled_dr_values.json.
```

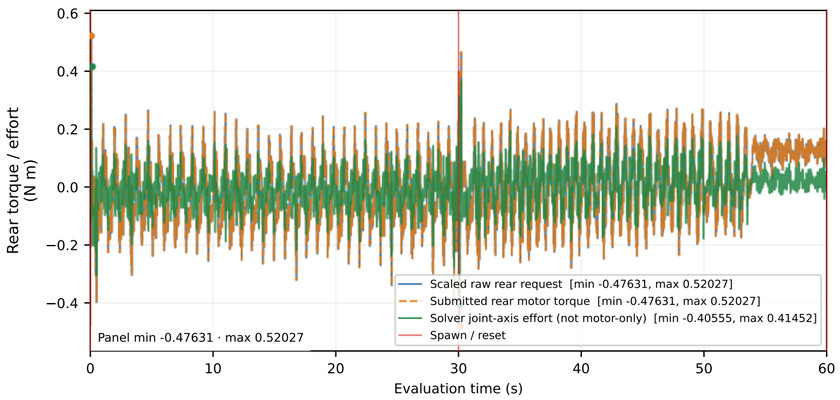
```
SAMPLED DR: env 45; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.0283, 1.11342, 0.868957, 0.992338]
  CoM XYZ mm [-0.719669, -3.11429, 0.391038]
  mu_s [0.616244, 0.597748, 0.736077, 0.616244]
  mu_d [0.586498, 0.597748, 0.699455, 0.586498]
  Kp [steer,rear] [9.60248, 0]; Kd [0.180136, 0]
  spawn roll -0.090286 rad; rate -0.0511712 rad/s
  gyro bias XYZ rad/s [0.000372074, 1.53988e-05, -0.000226969]
  gravity bias YZ [0.0011662, -0.00273945]
E2 [30.00, 60.00) s:
  mass x [0.979572, 1.07169, 1.03473, 1.18028]
  CoM XYZ mm [0.736894, -4.09673, 1.10124]
  mu_s [0.748289, 0.748289, 0.611239, 0.637675]
  mu_d [0.689499, 0.689499, 0.611239, 0.637675]
  Kp [steer,rear] [9.91425, 0]; Kd [0.191506, 0]
  spawn roll 0.0225475 rad; rate -0.237895 rad/s
  gyro bias XYZ rad/s [0.000177359, -0.000212698, 7.98098e-05]
  gravity bias YZ [-0.00389648, -0.000647599]
Six significant digits shown; full precision: sampled_dr_values.json.
```

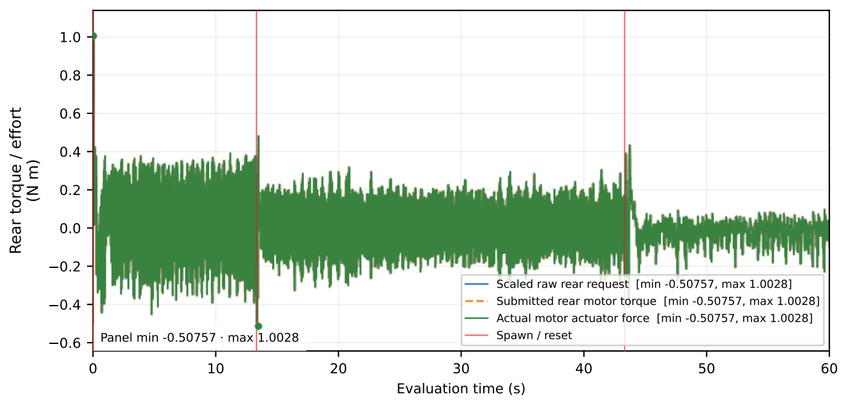
model_2500 · Demand versus actuation · env 46 · full 60 s

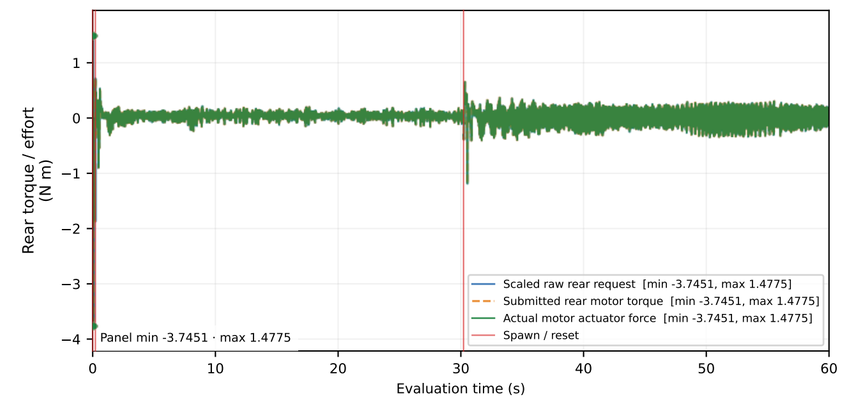












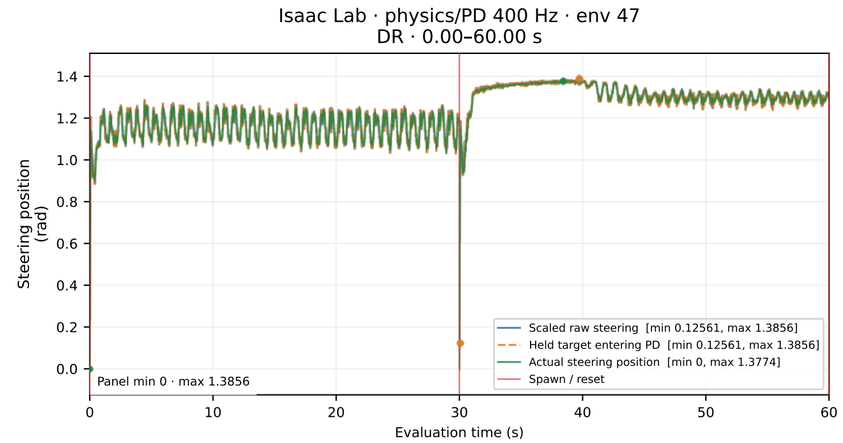
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

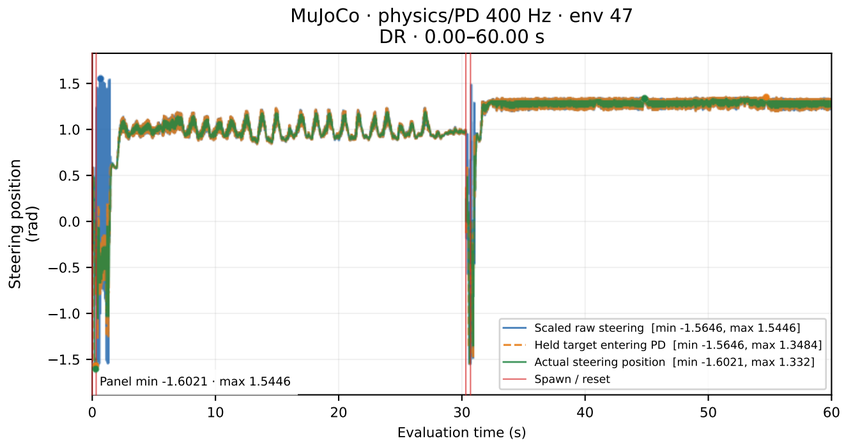
```
SAMPLED DR: env 46; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.813902, 0.964713, 1.09081, 0.926487]
  CoM XYZ mm [2.33594, 1.13692, 3.41862]
  mu_s [0.579974, 0.787792, 0.576972, 0.726175]
  mu_d [0.579974, 0.617134, 0.576972, 0.558574]
  Kp [steer,rear] [9.21021, 0]; Kd [0.214087, 0]
  spawn roll 0.0381843 rad; rate -0.107651 rad/s
  gyro bias XYZ rad/s [0.000202893, -0.000185471, -0.000147127]
  gravity bias YZ [0.00189576, 0.0012899]
E2 [30.00, 60.00) s:
  mass x [1.17386, 0.936098, 1.10321, 1.14728]
  CoM XYZ mm [-1.03818, -0.664905, 2.66431]
  mu_s [0.532645, 0.681711, 0.602694, 0.539956]
  mu_d [0.532645, 0.611756, 0.602694, 0.539956]
  Kp [steer,rear] [9.17165, 0]; Kd [0.212853, 0]
  spawn roll 0.00172687 rad; rate -0.246702 rad/s
  gyro bias XYZ rad/s [-5.22129e-05, 0.000184728, 0.00022664]
  gravity bias YZ [0.00270684, -0.00120501]
Six significant digits shown; full precision: sampled_dr_values.json.
```

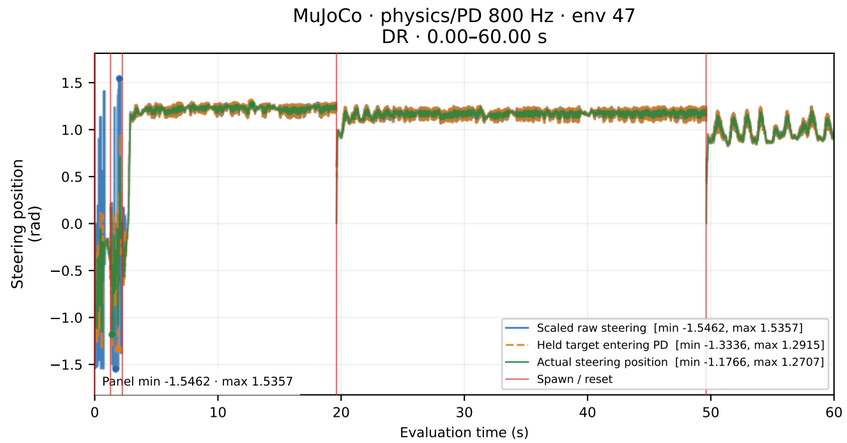
```
SAMPLED DR: env 46; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 13.32) s:
  mass x [0.882498, 1.08128, 1.0754, 1.07555]
  CoM XYZ mm [1.62438, -3.77742, -3.27171]
  mu_s [0.749678, 0.727426, 0.640016, 0.69316]
  mu_d [0.741429, 0.606358, 0.513141, 0.69316]
  Kp [steer,rear] [9.12117, 0]; Kd [0.207069, 0]
  spawn roll 0.0932981 rad; rate -0.0927966 rad/s
  gyro bias XYZ rad/s [0.000126401, 0.000390188, -0.000109424]
  gravity bias YZ [0.00110445, 0.00119378]
E2 [13.32, 43.32) s:
  mass x [1.01361, 0.802286, 0.990541, 1.13248]
  CoM XYZ mm [1.09206, 0.0480632, 1.10722]
  mu_s [0.732187, 0.637675, 0.69316, 0.538434]
  mu_d [0.631664, 0.637675, 0.69316, 0.538434]
  Kp [steer,rear] [10.5326, 0]; Kd [0.195451, 0]
  spawn roll -0.0183043 rad; rate 0.0863785 rad/s
  gyro bias XYZ rad/s [-5.88265e-05, 0.000388015, -0.000227037]
  gravity bias YZ [-0.000566167, 0.00130714]
E3 [43.32, 60.00) s:
  mass x [1.02491, 0.911508, 1.10754, 1.13636]
  CoM XYZ mm [-3.94382, -4.78781, 3.85707]
  mu_s [0.546287, 0.616244, 0.757579, 0.728342]
  mu_d [0.546287, 0.586498, 0.70921, 0.728342]
  Kp [steer,rear] [10.5, 0]; Kd [0.191548, 0]
  spawn roll -0.04388 rad; rate 0.0375422 rad/s
  gyro bias XYZ rad/s [-0.000171093, 0.000145314, 6.84799e-05]
  gravity bias YZ [-0.0004935, 0.00240581]
Six significant digits shown; full precision: sampled_dr_values.json.
```

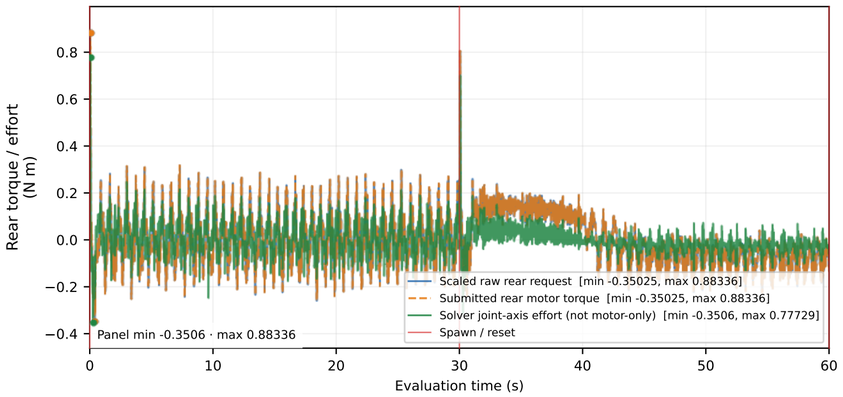
```
SAMPLED DR: env 46; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.22) s:
  mass x [0.882498, 1.08128, 1.0754, 1.07555]
  CoM XYZ mm [1.62438, -3.77742, -3.27171]
  mu_s [0.749678, 0.727426, 0.640016, 0.69316]
  mu_d [0.741429, 0.606358, 0.513141, 0.69316]
  Kp [steer,rear] [9.12117, 0]; Kd [0.207069, 0]
  spawn roll 0.0932981 rad; rate -0.0927966 rad/s
  gyro bias XYZ rad/s [0.000126401, 0.000390188, -0.000109424]
  gravity bias YZ [0.00110445, 0.00119378]
E2 [0.22, 30.22) s:
  mass x [0.885525, 1.18296, 1.07899, 0.9302]
  CoM XYZ mm [-2.16469, -4.08174, -1.2722]
  mu_s [0.732187, 0.640016, 0.69316, 0.640867]
  mu_d [0.631664, 0.513141, 0.69316, 0.556841]
  Kp [steer,rear] [9.38561, 0]; Kd [0.184505, 0]
  spawn roll -0.0790354 rad; rate 0.0140316 rad/s
  gyro bias XYZ rad/s [8.00441e-05, 5.28129e-05, -0.000146935]
  gravity bias YZ [0.00550037, 0.00103566]
E3 [30.22, 60.00) s:
  mass x [1.00945, 0.910747, 0.800438, 0.876145]
  CoM XYZ mm [-2.48635, -0.00111701, -0.126316]
  mu_s [0.728342, 0.723429, 0.568073, 0.538976]
  mu_d [0.728342, 0.723429, 0.568073, 0.538976]
  Kp [steer,rear] [9.47252, 0]; Kd [0.187993, 0]
  spawn roll -0.046428 rad; rate 0.187411 rad/s
  gyro bias XYZ rad/s [0.000288874, -0.000303692, 9.03436e-05]
  gravity bias YZ [0.000921866, 0.00245169]
Six significant digits shown; full precision: sampled_dr_values.json.
```

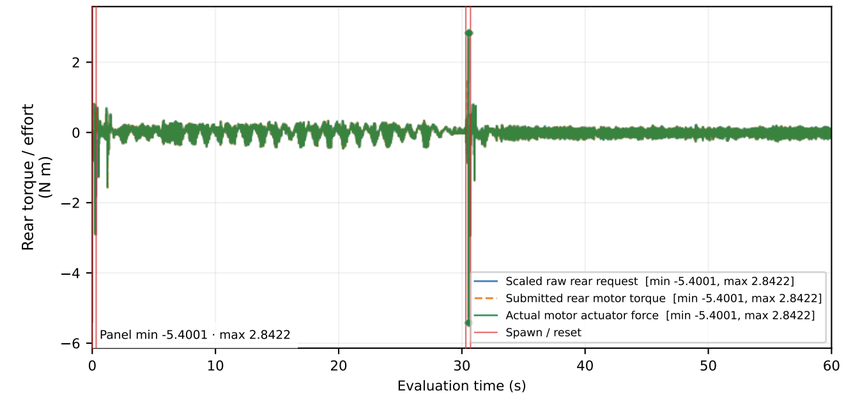
model_2500 · Demand versus actuation · env 47 · full 60 s

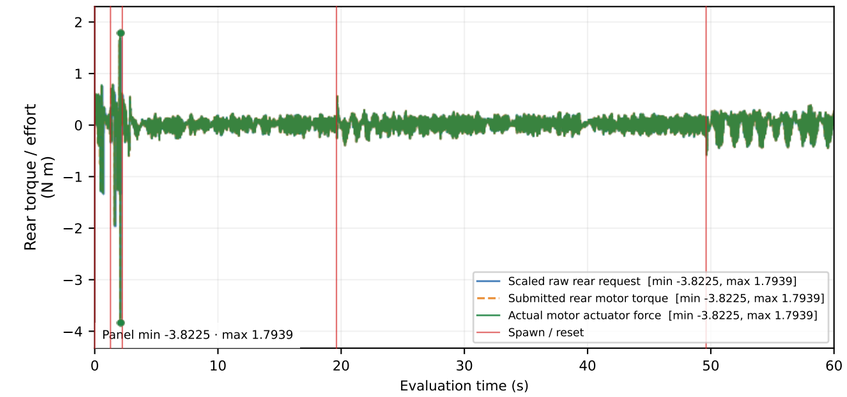












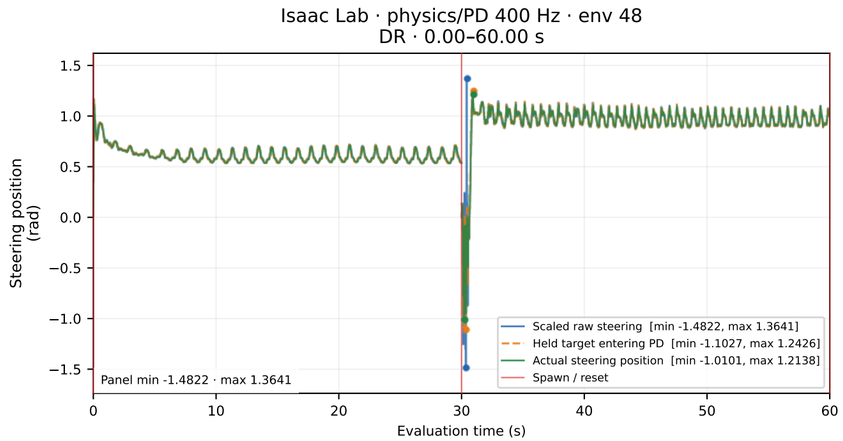
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

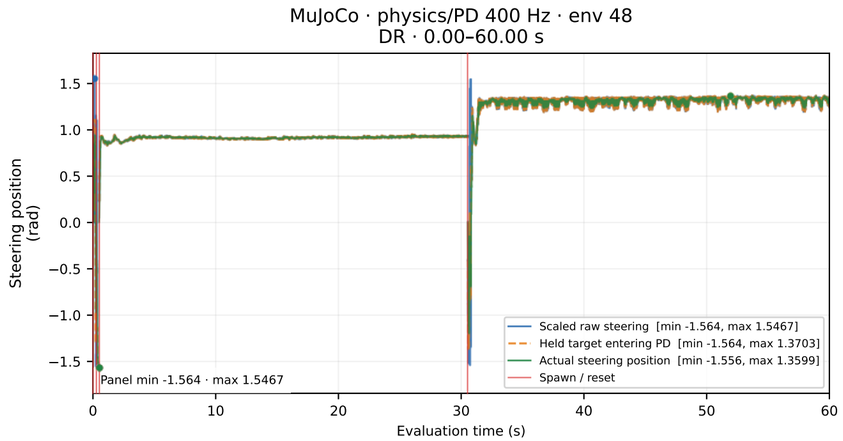
```
SAMPLED DR: env 47; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.03655, 1.17511, 1.11408, 0.877524]
  CoM XYZ mm [-4.44587, -0.567719, -2.63108]
  mu_s [0.760821, 0.726175, 0.560535, 0.726175]
  mu_d [0.670315, 0.558574, 0.560535, 0.558574]
  Kp [steer,rear] [10.6797, 0]; Kd [0.214648, 0]
  spawn roll 0.0742524 rad; rate -0.213024 rad/s
  gyro bias XYZ rad/s [-4.5567e-05, 0.000271351, -5.81968e-05]
  gravity bias YZ [0.00115448, -0.00144247]
E2 [30.00, 60.00) s:
  mass x [0.899864, 0.877441, 1.16899, 1.08508]
  CoM XYZ mm [-2.47404, -3.20068, -4.93082]
  mu_s [0.785466, 0.647391, 0.713129, 0.693205]
  mu_d [0.52258, 0.647391, 0.713129, 0.693205]
  Kp [steer,rear] [9.54417, 0]; Kd [0.213168, 0]
  spawn roll 0.0653071 rad; rate -0.227627 rad/s
  gyro bias XYZ rad/s [-5.883e-05, 0.000143332, 0.000143997]
  gravity bias YZ [0.000960929, -0.00292982]
Six significant digits shown; full precision: sampled_dr_values.json.
```

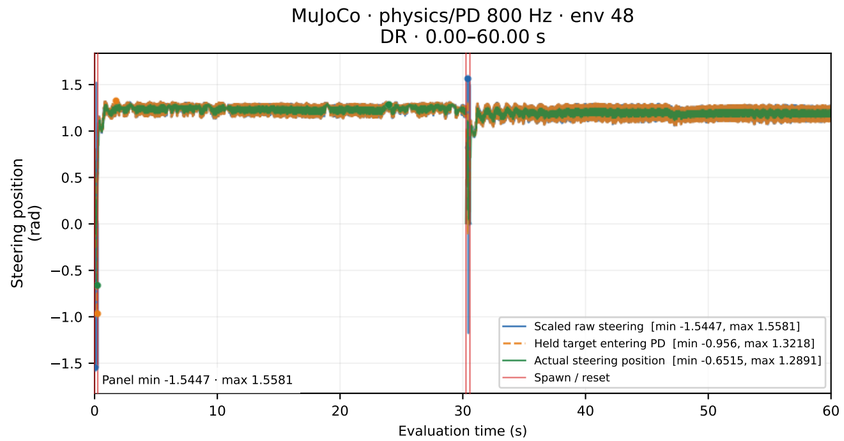
```
SAMPLED DR: env 47; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.32) s:
  mass x [1.03031, 1.10798, 1.09245, 0.807864]
  CoM XYZ mm [0.400709, 4.01135, 2.78235]
  mu_s [0.528253, 0.71008, 0.736077, 0.748289]
  mu_d [0.528253, 0.59371, 0.699455, 0.689499]
  Kp [steer,rear] [9.35461, 0]; Kd [0.21788, 0]
  spawn roll -0.0468968 rad; rate -0.0970046 rad/s
  gyro bias XYZ rad/s [3.01175e-06, -0.000120107, 6.87117e-05]
  gravity bias YZ [-0.00422143, -0.00194905]
E2 [0.32, 30.32) s:
  mass x [0.819334, 0.975526, 1.00746, 0.983347]
  CoM XYZ mm [-2.60108, 1.40927, 2.02062]
  mu_s [0.597748, 0.700521, 0.757579, 0.71008]
  mu_d [0.597748, 0.641329, 0.70921, 0.59371]
  Kp [steer,rear] [9.07265, 0]; Kd [0.185688, 0]
  spawn roll -0.0440747 rad; rate 0.0112541 rad/s
  gyro bias XYZ rad/s [3.78981e-05, 2.99002e-05, 8.45185e-05]
  gravity bias YZ [-0.00151106, 0.00107793]
E3 [30.32, 30.70) s:
  mass x [0.935038, 1.02052, 1.12262, 0.909411]
  CoM XYZ mm [1.74958, -3.78616, 4.51876]
  mu_s [0.704749, 0.69316, 0.597748, 0.538434]
  mu_d [0.541926, 0.69316, 0.597748, 0.538434]
  Kp [steer,rear] [10.458, 0]; Kd [0.211661, 0]
  spawn roll -0.0330262 rad; rate 0.160557 rad/s
  gyro bias XYZ rad/s [-0.000294888, -9.13743e-05, 0.000444933]
  gravity bias YZ [-6.74809e-05, 0.00103594]
E4 [30.70, 60.00) s:
  mass x [0.901264, 1.02977, 1.04103, 1.05278]
  CoM XYZ mm [-0.580672, -3.41664, 3.29062]
  mu_s [0.568073, 0.666375, 0.704749, 0.736077]
  mu_d [0.568073, 0.519145, 0.541926, 0.699455]
  Kp [steer,rear] [10.2127, 0]; Kd [0.186575, 0]
  spawn roll -0.0988457 rad; rate 0.195537 rad/s
  gyro bias XYZ rad/s [4.37454e-05, 0.000172272, 8.1821e-05]
  gravity bias YZ [0.00021133, 0.00109951]
Six significant digits shown; full precision: sampled_dr_values.json.
```

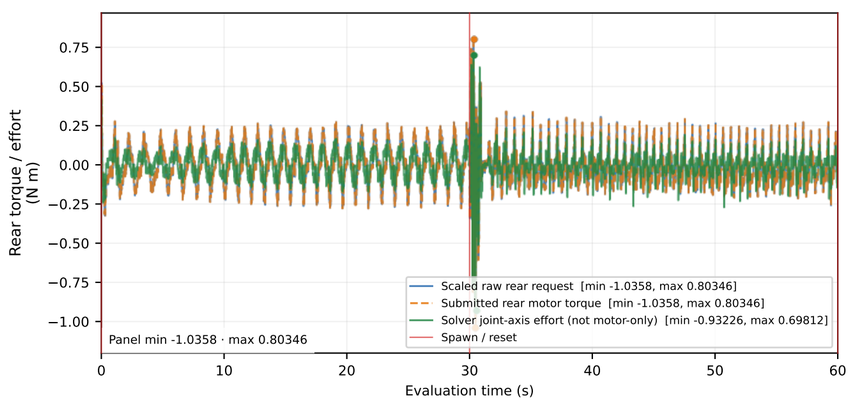
```
SAMPLED DR: env 47; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.28) s:
  mass x [1.03031, 1.10798, 1.09245, 0.807864]
  CoM XYZ mm [0.400709, 4.01135, 2.78235]
  mu_s [0.528253, 0.71008, 0.736077, 0.748289]
  mu_d [0.528253, 0.59371, 0.699455, 0.689499]
  Kp [steer,rear] [9.35461, 0]; Kd [0.21788, 0]
  spawn roll -0.0468968 rad; rate -0.0970046 rad/s
  gyro bias XYZ rad/s [3.01175e-06, -0.000120107, 6.87117e-05]
  gravity bias YZ [-0.00422143, -0.00194905]
E2 [1.28, 2.24) s:
  mass x [0.988684, 0.904768, 0.895446, 1.12629]
  CoM XYZ mm [-0.0644951, 2.06511, -3.90885]
  mu_s [0.71008, 0.71008, 0.69316, 0.538976]
  mu_d [0.59371, 0.59371, 0.69316, 0.538976]
  Kp [steer,rear] [10.5017, 0]; Kd [0.205067, 0]
  spawn roll -0.0737216 rad; rate 0.189624 rad/s
  gyro bias XYZ rad/s [-3.92332e-06, 0.000134535, -0.000308125]
  gravity bias YZ [4.32135e-05, -0.00068459]
E3 [2.24, 19.62) s:
  mass x [1.08582, 1.16874, 0.998601, 1.03475]
  CoM XYZ mm [-4.1558, -1.24468, 4.9313]
  mu_s [0.748289, 0.611239, 0.528253, 0.597748]
  mu_d [0.689499, 0.611239, 0.528253, 0.597748]
  Kp [steer,rear] [9.50513, 0]; Kd [0.195058, 0]
  spawn roll -0.0301954 rad; rate -0.0384992 rad/s
  gyro bias XYZ rad/s [2.39944e-06, -9.88794e-05, 0.000250088]
  gravity bias YZ [0.00208184, 0.000981116]
E4 [19.62, 49.62) s:
  mass x [1.1446, 1.15252, 1.16758, 1.04264]
  CoM XYZ mm [-4.53675, -0.501051, 2.79191]
  mu_s [0.71008, 0.757579, 0.538976, 0.728342]
  mu_d [0.59371, 0.70921, 0.538976, 0.728342]
  Kp [steer,rear] [9.44164, 0]; Kd [0.18992, 0]
  spawn roll 0.0247201 rad; rate 0.114003 rad/s
  gyro bias XYZ rad/s [-0.000271496, 0.000279174, 0.000242257]
  gravity bias YZ [0.0008539, -0.00111884]
E5 [49.62, 60.00) s:
  mass x [0.909109, 1.07095, 1.03261, 1.03408]
  CoM XYZ mm [-3.01552, 1.72354, -4.71103]
  mu_s [0.748289, 0.71155, 0.71008, 0.637675]
  mu_d [0.689499, 0.71155, 0.59371, 0.637675]
  Kp [steer,rear] [9.56325, 0]; Kd [0.208263, 0]
  spawn roll -0.00541428 rad; rate 0.220684 rad/s
  gyro bias XYZ rad/s [0.000146279, -5.36009e-05, 3.81239e-05]
  gravity bias YZ [-0.000445671, 0.000948283]
Six significant digits shown; full precision: sampled_dr_values.json.
```

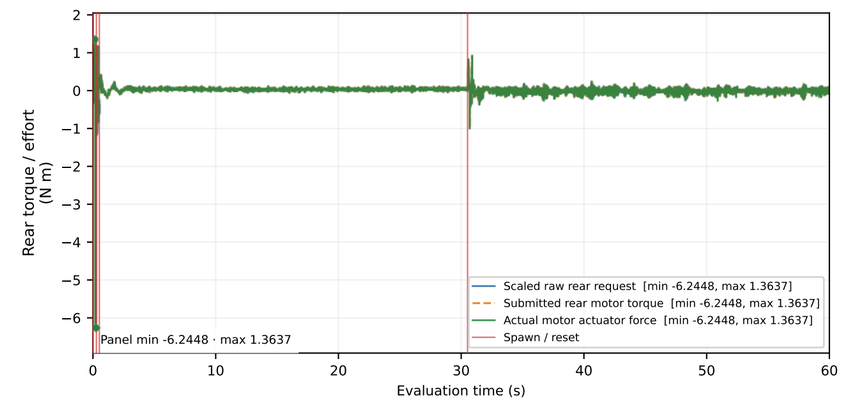
model_2500 · Demand versus actuation · env 48 · full 60 s

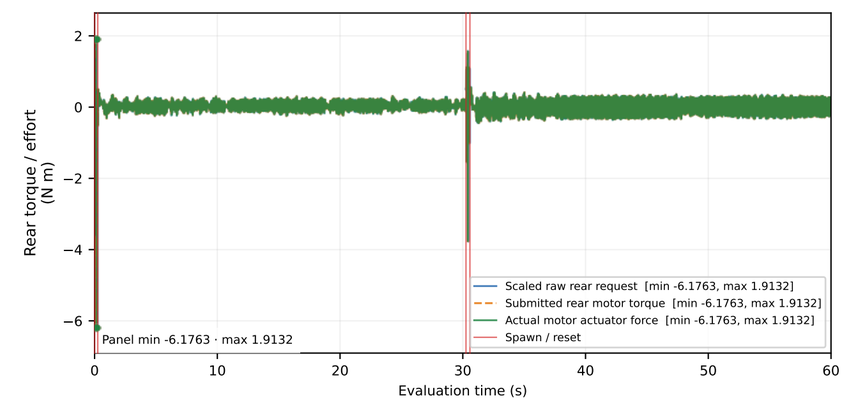












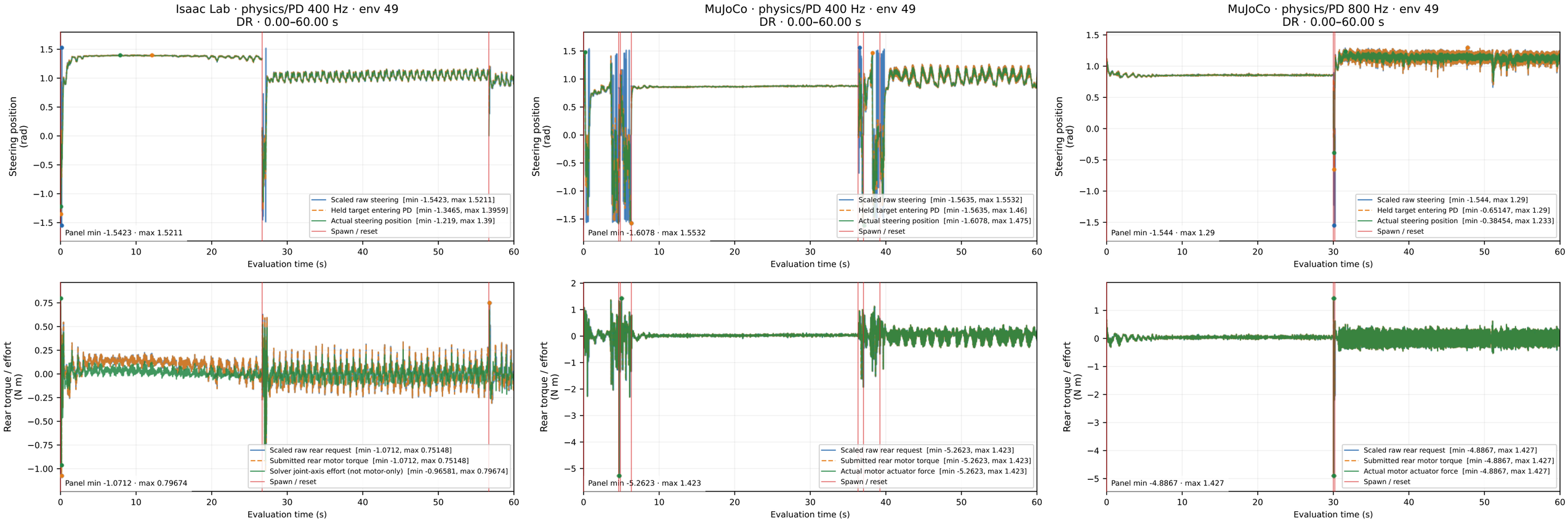
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 48; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.08753, 1.17826, 0.995849, 1.00857]
  CoM XYZ mm [-0.618026, 4.98509, 0.135079]
  mu_s [0.703711, 0.726175, 0.787792, 0.787792]
  mu_d [0.703711, 0.558574, 0.617134, 0.617134]
  Kp [steer,rear] [10.6269, 0]; Kd [0.208986, 0]
  spawn roll 0.0653718 rad; rate -0.159716 rad/s
  gyro bias XYZ rad/s [3.62393e-05, -9.55029e-05, -0.000273433]
  gravity bias YZ [-0.00302836, -0.00224948]
E2 [30.00, 60.00) s:
  mass x [1.14941, 1.15342, 1.11081, 0.896198]
  CoM XYZ mm [-2.72126, 1.15197, 1.6517]
  mu_s [0.579974, 0.693205, 0.547743, 0.693205]
  mu_d [0.579974, 0.693205, 0.547743, 0.693205]
  Kp [steer,rear] [10.6851, 0]; Kd [0.18368, 0]
  spawn roll -0.0243969 rad; rate 0.0606264 rad/s
  gyro bias XYZ rad/s [-0.000184089, -0.00030335, -0.000277833]
  gravity bias YZ [0.00169077, -0.00102462]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 48; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [0.948933, 1.07276, 1.00741, 0.96734]
  CoM XYZ mm [1.96312, -3.74609, 1.99845]
  mu_s [0.559972, 0.71155, 0.669571, 0.640016]
  mu_d [0.502209, 0.71155, 0.669571, 0.513141]
  Kp [steer,rear] [9.00075, 0]; Kd [0.213496, 0]
  spawn roll 0.0828675 rad; rate -0.189681 rad/s
  gyro bias XYZ rad/s [0.000127252, 0.000128662, -0.000120542]
  gravity bias YZ [0.00266815, -0.00159674]
E2 [0.28, 0.52) s:
  mass x [0.969414, 1.10477, 0.87968, 0.969751]
  CoM XYZ mm [-3.70687, 1.66721, -0.934434]
  mu_s [0.568073, 0.723429, 0.640867, 0.704749]
  mu_d [0.568073, 0.723429, 0.556841, 0.541926]
  Kp [steer,rear] [9.89196, 0]; Kd [0.213456, 0]
  spawn roll -0.0883098 rad; rate -0.0289943 rad/s
  gyro bias XYZ rad/s [6.7487e-05, -5.02594e-05, 0.000127656]
  gravity bias YZ [0.00475094, 0.0020392]
E3 [0.52, 30.52) s:
  mass x [1.10132, 0.988294, 1.07585, 1.10371]
  CoM XYZ mm [2.41057, 2.85973, 0.25693]
  mu_s [0.640016, 0.757579, 0.704749, 0.541939]
  mu_d [0.513141, 0.70921, 0.541926, 0.534359]
  Kp [steer,rear] [10.7175, 0]; Kd [0.205582, 0]
  spawn roll 0.0529119 rad; rate -0.148663 rad/s
  gyro bias XYZ rad/s [0.000160925, 0.000172416, 0.000212644]
  gravity bias YZ [0.000404312, -0.00192323]
E4 [30.52, 60.00) s:
  mass x [0.952696, 0.990547, 0.948737, 0.931427]
  CoM XYZ mm [-4.97849, -3.83979, -0.657533]
  mu_s [0.559972, 0.633024, 0.597748, 0.640016]
  mu_d [0.502209, 0.568172, 0.597748, 0.513141]
  Kp [steer,rear] [9.14485, 0]; Kd [0.191679, 0]
  spawn roll -0.0652092 rad; rate -0.0960493 rad/s
  gyro bias XYZ rad/s [0.000207544, 0.00013045, -0.000430529]
  gravity bias YZ [0.000374311, 0.000344802]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 48; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.26) s:
  mass x [0.948933, 1.07276, 1.00741, 0.96734]
  CoM XYZ mm [1.96312, -3.74609, 1.99845]
  mu_s [0.559972, 0.71155, 0.669571, 0.640016]
  mu_d [0.502209, 0.71155, 0.669571, 0.513141]
  Kp [steer,rear] [9.00075, 0]; Kd [0.213496, 0]
  spawn roll 0.0828675 rad; rate -0.189681 rad/s
  gyro bias XYZ rad/s [0.000127252, 0.000128662, -0.000120542]
  gravity bias YZ [0.00266815, -0.00159674]
E2 [0.26, 30.26) s:
  mass x [0.808865, 1.01398, 0.994135, 1.00275]
  CoM XYZ mm [-4.31078, -1.99309, 2.79664]
  mu_s [0.640867, 0.732187, 0.559972, 0.597748]
  mu_d [0.556841, 0.631664, 0.502209, 0.597748]
  Kp [steer,rear] [10.9784, 0]; Kd [0.196941, 0]
  spawn roll -0.00321518 rad; rate 0.139784 rad/s
  gyro bias XYZ rad/s [0.000153575, -0.000135664, 1.55938e-05]
  gravity bias YZ [-0.00113177, 0.000514633]
E3 [30.26, 30.58) s:
  mass x [1.08386, 0.853643, 1.16262, 0.861685]
  CoM XYZ mm [2.43457, -3.03032, 2.42355]
  mu_s [0.541939, 0.71155, 0.733515, 0.749678]
  mu_d [0.534359, 0.71155, 0.558392, 0.741429]
  Kp [steer,rear] [10.4606, 0]; Kd [0.216125, 0]
  spawn roll 0.0840548 rad; rate 0.0711848 rad/s
  gyro bias XYZ rad/s [-8.50104e-05, -0.000309878, -9.46143e-05]
  gravity bias YZ [-0.0036056, -0.000550167]
E4 [30.58, 60.00) s:
  mass x [0.9917, 1.00228, 1.00319, 0.839121]
  CoM XYZ mm [0.515675, 0.182563, -1.45984]
  mu_s [0.631146, 0.640016, 0.640867, 0.749678]
  mu_d [0.631146, 0.513141, 0.556841, 0.741429]
  Kp [steer,rear] [9.1803, 0]; Kd [0.206165, 0]
  spawn roll 0.0543008 rad; rate 0.124664 rad/s
  gyro bias XYZ rad/s [-0.00024219, -0.000192722, -1.61808e-05]
  gravity bias YZ [0.00465794, 0.0006785]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 49 · full 60 s



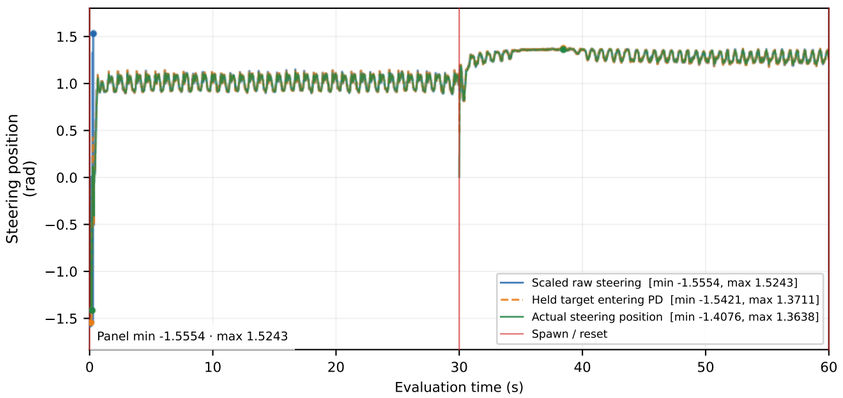
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

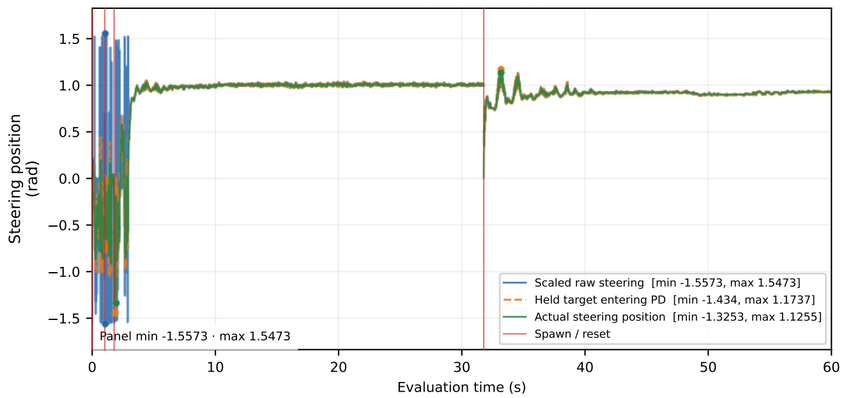
```
SAMPLED DR: env 49; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 26.68) s:
  mass x [1.16767, 1.02079, 1.07897, 0.814629]
  CoM XYZ mm [-3.47495, -3.38237, -3.22597]
  mu_s [0.547743, 0.598343, 0.774409, 0.760821]
  mu_d [0.547743, 0.598343, 0.561095, 0.670315]
  Kp [steer,rear] [10.5091, 0]; Kd [0.190662, 0]
  spawn roll -0.0669806 rad; rate -0.085151 rad/s
  gyro bias XYZ rad/s [0.000105286, 0.000217856, -4.12894e-06]
  gravity bias YZ [-0.000672248, 0.00182319]
E2 [26.68, 56.68) s:
  mass x [0.961654, 1.16869, 0.915261, 0.876968]
  CoM XYZ mm [3.00056, 0.714, 1.20407]
  mu_s [0.741039, 0.681711, 0.560535, 0.713129]
  mu_d [0.614405, 0.611756, 0.560535, 0.713129]
  Kp [steer,rear] [10.4558, 0]; Kd [0.180785, 0]
  spawn roll -0.0518287 rad; rate 0.136825 rad/s
  gyro bias XYZ rad/s [-0.000321651, -0.000150021, 0.000280684]
  gravity bias YZ [-0.00145164, 0.00206375]
E3 [56.68, 60.00) s:
  mass x [1.08793, 0.868082, 1.03885, 0.850176]
  CoM XYZ mm [-3.32075, 1.74008, 2.60136]
  mu_s [0.741039, 0.673378, 0.592046, 0.787792]
  mu_d [0.614405, 0.673378, 0.534947, 0.617134]
  Kp [steer,rear] [10.278, 0]; Kd [0.207587, 0]
  spawn roll 0.0610619 rad; rate -0.115938 rad/s
  gyro bias XYZ rad/s [-8.96865e-06, 0.00025883, 0.000231798]
  gravity bias YZ [0.00299946, -0.00336261]
Six significant digits shown; full precision: sampled_dr_values.json.
```

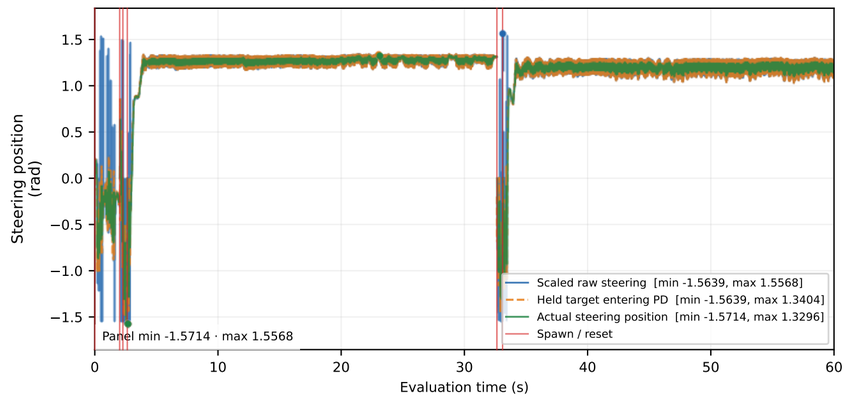
```
SAMPLED DR: env 49; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 4.64) s:
  mass x [0.913233, 1.1641, 1.12773, 0.91701]
  CoM XYZ mm [3.34854, 4.04615, -2.15763]
  mu_s [0.736077, 0.637675, 0.640016, 0.733515]
  mu_d [0.699455, 0.637675, 0.513141, 0.558392]
  Kp [steer,rear] [10.1476, 0]; Kd [0.213448, 0]
  spawn roll 0.0863189 rad; rate -0.0570891 rad/s
  gyro bias XYZ rad/s [0.000281771, -0.000166747, 0.000119675]
  gravity bias YZ [0.00174457, 0.00369669]
E2 [4.64, 4.86) s:
  mass x [1.19235, 1.08213, 1.02801, 1.05424]
  CoM XYZ mm [-1.72457, -4.12029, 2.35562]
  mu_s [0.541939, 0.546287, 0.736077, 0.637675]
  mu_d [0.534359, 0.546287, 0.699455, 0.637675]
  Kp [steer,rear] [9.4624, 0]; Kd [0.21421, 0]
  spawn roll 0.079729 rad; rate 0.119137 rad/s
  gyro bias XYZ rad/s [-0.000233366, -3.99513e-05, 0.000153254]
  gravity bias YZ [-0.00349055, 0.00278417]
E3 [4.86, 6.32) s:
  mass x [1.00931, 0.890937, 1.10744, 0.836926]
  CoM XYZ mm [4.30324, 4.22095, -2.89644]
  mu_s [0.568073, 0.704749, 0.637675, 0.616244]
  mu_d [0.568073, 0.541926, 0.637675, 0.586498]
  Kp [steer,rear] [10.8241, 0]; Kd [0.187122, 0]
  spawn roll 0.0697917 rad; rate 0.0812057 rad/s
  gyro bias XYZ rad/s [0.000127583, 5.33777e-05, -0.000312492]
  gravity bias YZ [0.000882079, 0.000244129]
E4 [6.32, 36.32) s:
  mass x [0.926608, 1.13568, 1.17766, 0.93652]
  CoM XYZ mm [2.38566, 3.78858, 2.70094]
  mu_s [0.71155, 0.631146, 0.733515, 0.732187]
  mu_d [0.71155, 0.631146, 0.558392, 0.631664]
  Kp [steer,rear] [9.62313, 0]; Kd [0.188571, 0]
  spawn roll 0.0609389 rad; rate 0.225905 rad/s
  gyro bias XYZ rad/s [0.000281571, -1.67543e-05, -8.8375e-05]
  gravity bias YZ [8.35985e-05, 0.00145331]
E5 [36.32, 37.04) s:
  mass x [0.839765, 1.11111, 0.801269, 0.816831]
  CoM XYZ mm [-4.50648, -3.85568, 3.59316]
  mu_s [0.559972, 0.749678, 0.757579, 0.640867]
  mu_d [0.502209, 0.741429, 0.70921, 0.556841]
  Kp [steer,rear] [9.09879, 0]; Kd [0.203822, 0]
  spawn roll 0.0164155 rad; rate -0.126637 rad/s
  gyro bias XYZ rad/s [-0.000320956, 6.99507e-05, -0.000274306]
  gravity bias YZ [0.00341725, 0.002529]
E6 [37.04, 39.22) s:
  mass x [1.03434, 1.08937, 1.01501, 0.951878]
  CoM XYZ mm [2.04634, -0.712359, -2.41383]
  mu_s [0.733515, 0.749678, 0.538976, 0.728342]
  mu_d [0.558392, 0.741429, 0.538976, 0.728342]
  Kp [steer,rear] [9.93877, 0]; Kd [0.208141, 0]
  spawn roll 0.0784524 rad; rate 0.166272 rad/s
  gyro bias XYZ rad/s [-0.000227385, -1.69427e-05, -0.000156127]
  gravity bias YZ [-0.000363731, -0.00107437]
E7 [39.22, 60.00) s:
  mass x [0.962941, 1.12214, 1.1465, 1.06223]
  CoM XYZ mm [0.831907, 1.17711, -2.0201]
  mu_s [0.728342, 0.597748, 0.704749, 0.546287]
  mu_d [0.728342, 0.597748, 0.541926, 0.546287]
  Kp [steer,rear] [9.31056, 0]; Kd [0.183019, 0]
  spawn roll -0.00838106 rad; rate -0.188766 rad/s
  gyro bias XYZ rad/s [0.000118329, -2.55333e-05, 0.000110444]
  gravity bias YZ [-0.00132805, 0.00111189]
Six significant digits shown; full precision: sampled_dr_values.json.
```

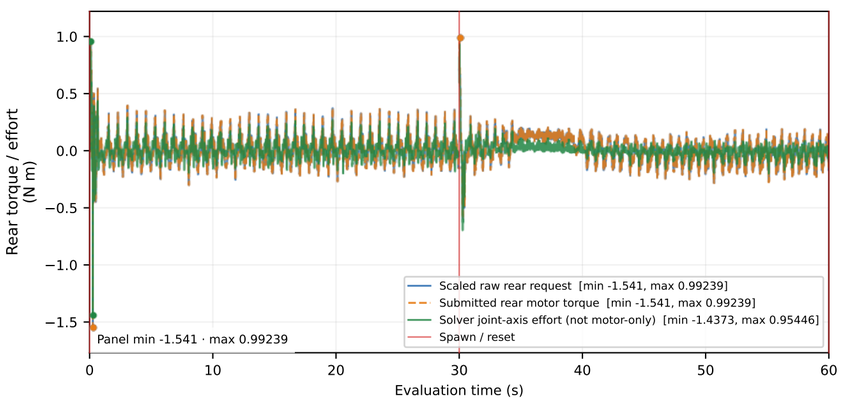
```
SAMPLED DR: env 49; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.913233, 1.1641, 1.12773, 0.91701]
  CoM XYZ mm [3.34854, 4.04615, -2.15763]
  mu_s [0.736077, 0.637675, 0.640016, 0.733515]
  mu_d [0.699455, 0.637675, 0.513141, 0.558392]
  Kp [steer,rear] [10.1476, 0]; Kd [0.213448, 0]
  spawn roll 0.0863189 rad; rate -0.0570891 rad/s
  gyro bias XYZ rad/s [0.000281771, -0.000166747, 0.000119675]
  gravity bias YZ [0.00174457, 0.00369669]
E2 [30.00, 30.20) s:
  mass x [1.05262, 1.19762, 0.845037, 0.981568]
  CoM XYZ mm [3.31378, -4.73576, -2.30997]
  mu_s [0.568073, 0.69316, 0.727426, 0.631146]
  mu_d [0.568073, 0.69316, 0.606358, 0.631146]
  Kp [steer,rear] [10.3401, 0]; Kd [0.216598, 0]
  spawn roll 0.0931377 rad; rate 0.193093 rad/s
  gyro bias XYZ rad/s [-0.000174902, -1.52907e-05, 5.03956e-05]
  gravity bias YZ [6.04822e-05, 0.00106008]
E3 [30.20, 60.00) s:
  mass x [0.903195, 0.901779, 1.18325, 0.942435]
  CoM XYZ mm [3.29046, 1.49534, -2.45754]
  mu_s [0.631146, 0.546287, 0.538434, 0.791209]
  mu_d [0.631146, 0.546287, 0.538434, 0.767936]
  Kp [steer,rear] [10.9304, 0]; Kd [0.211799, 0]
  spawn roll 0.0661296 rad; rate -0.0723594 rad/s
  gyro bias XYZ rad/s [1.84797e-06, 1.18338e-05, -0.00017462]
  gravity bias YZ [0.00631493, 0.00121862]
Six significant digits shown; full precision: sampled_dr_values.json.
```

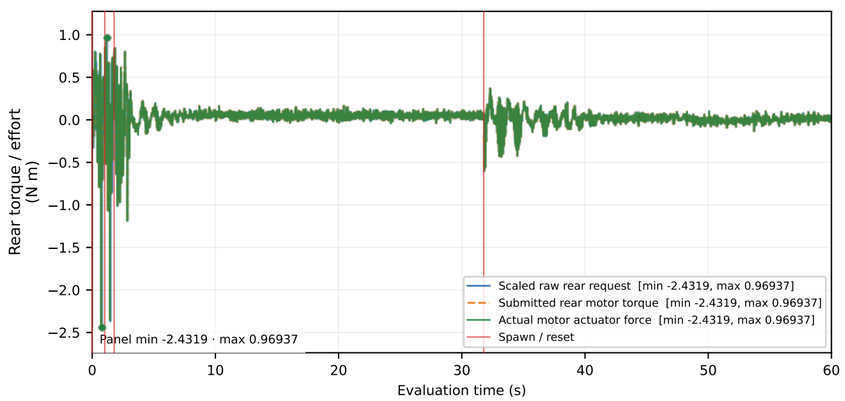
model_2500 · Demand versus actuation · env 50 · full 60 s

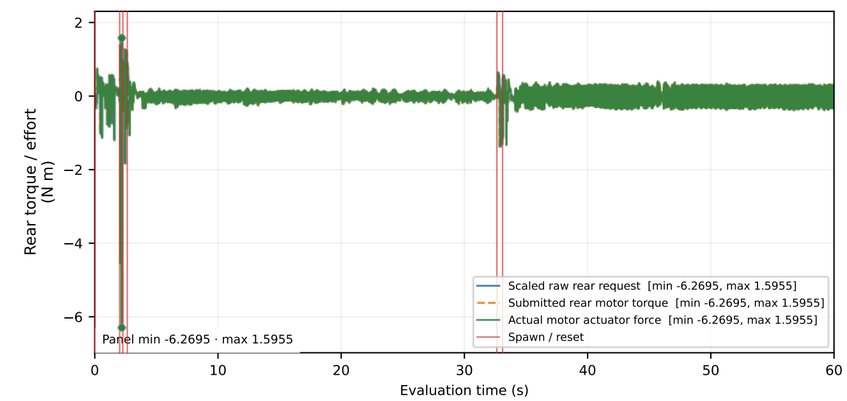












50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

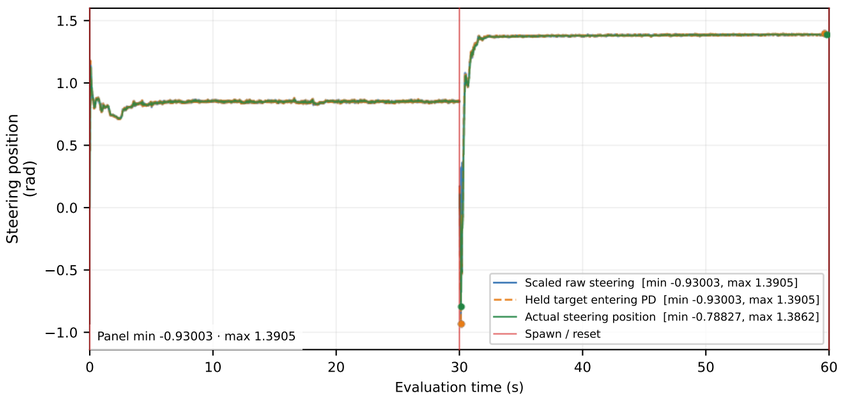
```
SAMPLED DR: env 50; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.15577, 0.969311, 0.993129, 0.980615]
  CoM XYZ mm [-2.54807, 1.30373, -2.17538]
  mu_s [0.751971, 0.764681, 0.764681, 0.693205]
  mu_d [0.541224, 0.764681, 0.764681, 0.693205]
  Kp [steer,rear] [10.8224, 0]; Kd [0.214074, 0]
  spawn roll -0.0884919 rad; rate -0.232642 rad/s
  gyro bias XYZ rad/s [0.000140697, 5.15835e-05, -0.000181121]
  gravity bias YZ [0.000233398, -0.000809242]
E2 [30.00, 60.00) s:
  mass x [1.18694, 0.806568, 0.949505, 0.879836]
  CoM XYZ mm [-3.80045, -2.67955, -3.76047]
  mu_s [0.785466, 0.532645, 0.760821, 0.760821]
  mu_d [0.52258, 0.532645, 0.670315, 0.670315]
  Kp [steer,rear] [10.3029, 0]; Kd [0.195031, 0]
  spawn roll 0.0895005 rad; rate 0.238192 rad/s
  gyro bias XYZ rad/s [-0.000127682, -0.000331672, 0.000413771]
  gravity bias YZ [-0.000251227, -0.00299106]
Six significant digits shown; full precision: sampled_dr_values.json.
```

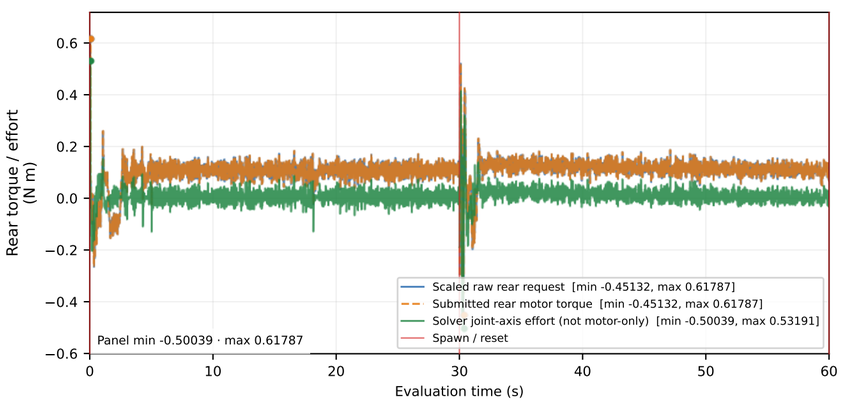
```
SAMPLED DR: env 50; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.02) s:
  mass x [1.04035, 0.830341, 0.940128, 0.933446]
  CoM XYZ mm [-4.62023, 3.57471, -4.64994]
  mu_s [0.631146, 0.732187, 0.597748, 0.631146]
  mu_d [0.631146, 0.631664, 0.597748, 0.631146]
  Kp [steer,rear] [9.08442, 0]; Kd [0.188041, 0]
  spawn roll 0.000687381 rad; rate -0.0492109 rad/s
  gyro bias XYZ rad/s [-4.09457e-05, -0.000282267, -0.00016859]
  gravity bias YZ [-0.000846323, -0.000773653]
E2 [1.02, 1.78) s:
  mass x [0.929787, 0.922862, 0.918625, 0.872758]
  CoM XYZ mm [2.83534, 1.67363, 0.0203466]
  mu_s [0.700521, 0.71155, 0.616244, 0.757579]
  mu_d [0.641329, 0.71155, 0.586498, 0.70921]
  Kp [steer,rear] [9.98066, 0]; Kd [0.195047, 0]
  spawn roll -0.0234883 rad; rate -0.223173 rad/s
  gyro bias XYZ rad/s [5.86512e-05, -0.000168819, 0.000113509]
  gravity bias YZ [0.000817212, -0.000122932]
E3 [1.78, 31.78) s:
  mass x [1.17386, 0.937473, 0.960115, 1.03836]
  CoM XYZ mm [-3.0154, 1.57505, 2.66425]
  mu_s [0.538976, 0.69316, 0.71008, 0.69316]
  mu_d [0.538976, 0.69316, 0.59371, 0.69316]
  Kp [steer,rear] [10.2724, 0]; Kd [0.180218, 0]
  spawn roll -0.0680822 rad; rate -0.181021 rad/s
  gyro bias XYZ rad/s [0.000169836, 6.95642e-05, 0.000131721]
  gravity bias YZ [0.000220411, 0.00153188]
E4 [31.78, 60.00) s:
  mass x [1.14809, 1.01048, 1.0009, 1.1621]
  CoM XYZ mm [4.30793, 2.27509, 0.90829]
  mu_s [0.69316, 0.640867, 0.791209, 0.749678]
  mu_d [0.69316, 0.556841, 0.767936, 0.741429]
  Kp [steer,rear] [9.29888, 0]; Kd [0.185282, 0]
  spawn roll -0.0135281 rad; rate 0.151796 rad/s
  gyro bias XYZ rad/s [0.000215499, 2.50955e-05, 0.000249005]
  gravity bias YZ [-0.0027542, -0.000682723]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 50; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 2.02) s:
  mass x [1.04035, 0.830341, 0.940128, 0.933446]
  CoM XYZ mm [-4.62023, 3.57471, -4.64994]
  mu_s [0.631146, 0.732187, 0.597748, 0.631146]
  mu_d [0.631146, 0.631664, 0.597748, 0.631146]
  Kp [steer,rear] [9.08442, 0]; Kd [0.188041, 0]
  spawn roll 0.000687381 rad; rate -0.0492109 rad/s
  gyro bias XYZ rad/s [-4.09457e-05, -0.000282267, -0.00016859]
  gravity bias YZ [-0.000846323, -0.000773653]
E2 [2.02, 2.28) s:
  mass x [1.15518, 0.810271, 0.828778, 1.07257]
  CoM XYZ mm [3.9666, 0.143293, -1.54585]
  mu_s [0.733515, 0.732187, 0.71155, 0.633024]
  mu_d [0.558392, 0.631664, 0.71155, 0.568172]
  Kp [steer,rear] [9.58315, 0]; Kd [0.202488, 0]
  spawn roll 0.0993319 rad; rate 0.00299045 rad/s
  gyro bias XYZ rad/s [-0.000284759, 4.29024e-05, -0.000236369]
  gravity bias YZ [-0.00225511, -0.0017409]
E3 [2.28, 2.64) s:
  mass x [0.809794, 0.840942, 0.932825, 1.19233]
  CoM XYZ mm [3.76137, 1.90411, 1.87976]
  mu_s [0.640016, 0.736077, 0.669571, 0.568073]
  mu_d [0.513141, 0.699455, 0.669571, 0.568073]
  Kp [steer,rear] [10.5795, 0]; Kd [0.185812, 0]
  spawn roll -0.0929147 rad; rate 0.0869282 rad/s
  gyro bias XYZ rad/s [0.000158452, -1.41046e-06, -5.29651e-05]
  gravity bias YZ [-0.00099225, -0.00054257]
E4 [2.64, 32.64) s:
  mass x [1.01306, 0.949765, 0.858973, 0.91838]
  CoM XYZ mm [-4.29277, -3.43135, 4.84826]
  mu_s [0.728342, 0.633024, 0.69316, 0.757579]
  mu_d [0.728342, 0.568172, 0.69316, 0.70921]
  Kp [steer,rear] [10.2756, 0]; Kd [0.206824, 0]
  spawn roll -0.0734395 rad; rate -0.222717 rad/s
  gyro bias XYZ rad/s [0.000155822, 0.000201897, 5.28504e-05]
  gravity bias YZ [-0.00252984, 0.000504211]
E5 [32.64, 33.10) s:
  mass x [0.950504, 0.884692, 0.910554, 1.06995]
  CoM XYZ mm [0.345044, 0.532505, 2.15373]
  mu_s [0.71008, 0.559972, 0.733515, 0.728342]
  mu_d [0.59371, 0.502209, 0.558392, 0.728342]
  Kp [steer,rear] [9.56995, 0]; Kd [0.201848, 0]
  spawn roll -0.0548667 rad; rate -0.0165733 rad/s
  gyro bias XYZ rad/s [-2.62859e-06, 9.29744e-05, -0.000120543]
  gravity bias YZ [0.000714771, -0.00144499]
E6 [33.10, 60.00) s:
  mass x [1.09967, 1.09061, 1.05979, 1.16773]
  CoM XYZ mm [3.92281, -0.491529, 0.452252]
  mu_s [0.791209, 0.541939, 0.732187, 0.749678]
  mu_d [0.767936, 0.534359, 0.631664, 0.741429]
  Kp [steer,rear] [10.32, 0]; Kd [0.195267, 0]
  spawn roll -0.0463786 rad; rate 0.00287988 rad/s
  gyro bias XYZ rad/s [-0.000251113, 0.000130964, 0.000205204]
  gravity bias YZ [-0.000343513, 0.0016595]
Six significant digits shown; full precision: sampled_dr_values.json.
```

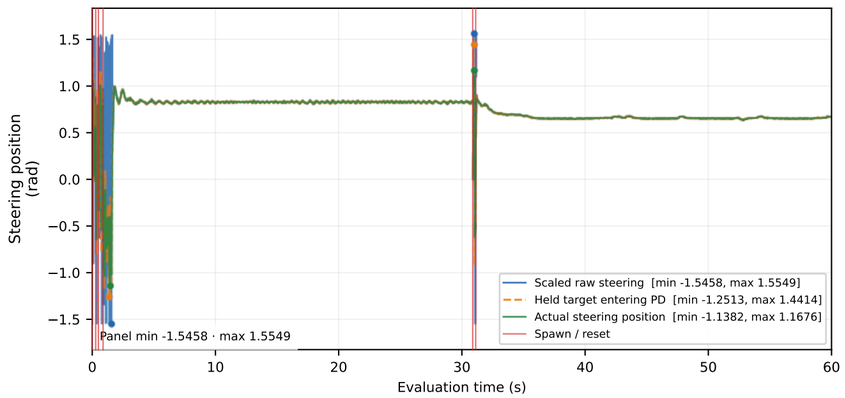
model_2500 · Demand versus actuation · env 51 · full 60 s

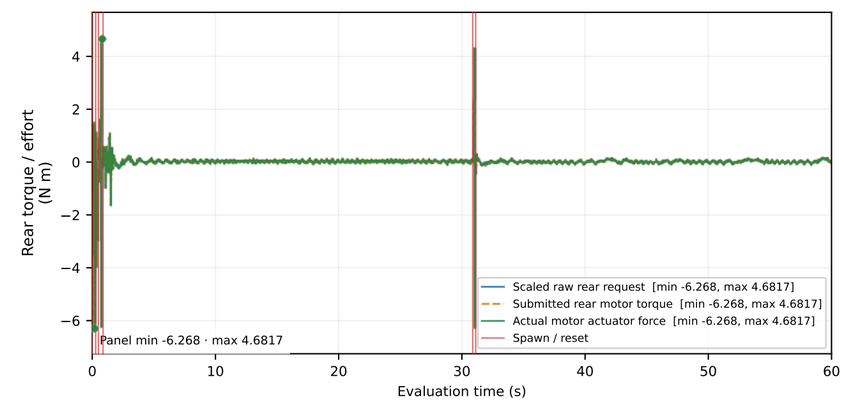
Isaac Lab · physics/PD 400 Hz · env 51
DR · 0.00–60.00 s



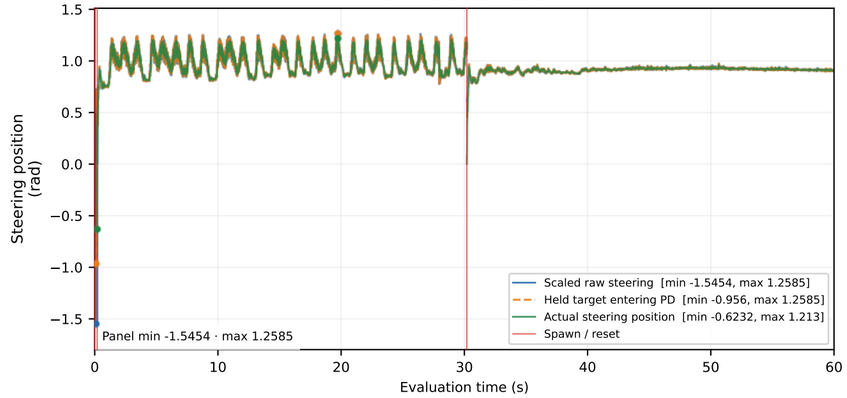


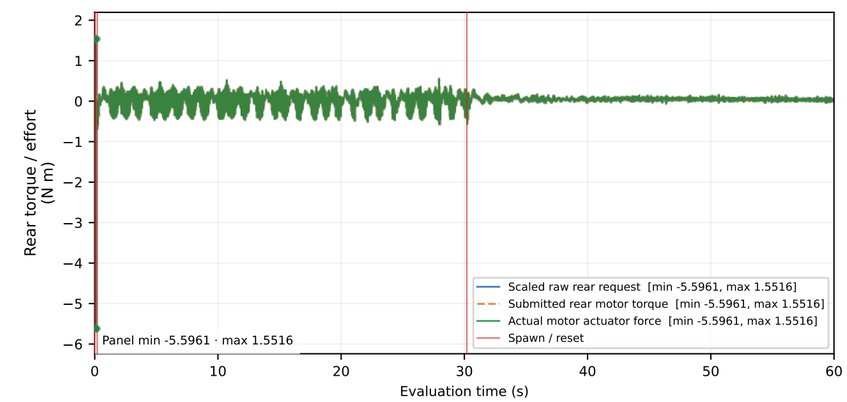
MuJoCo · physics/PD 400 Hz · env 51
DR · 0.00–60.00 s





MuJoCo · physics/PD 800 Hz · env 51
DR · 0.00–60.00 s





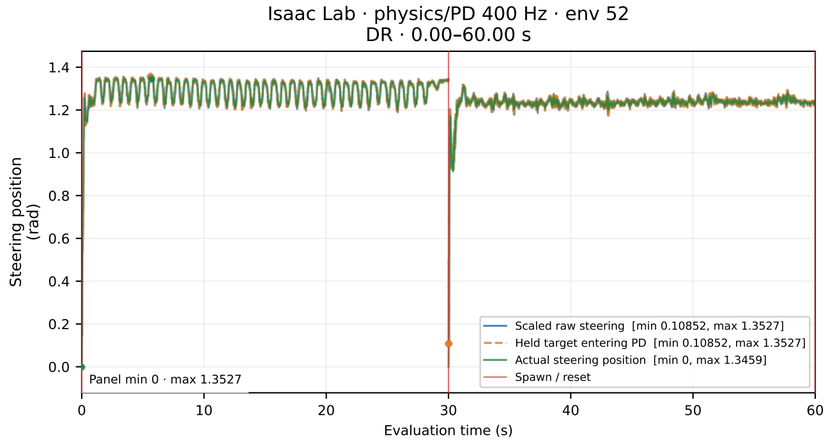
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

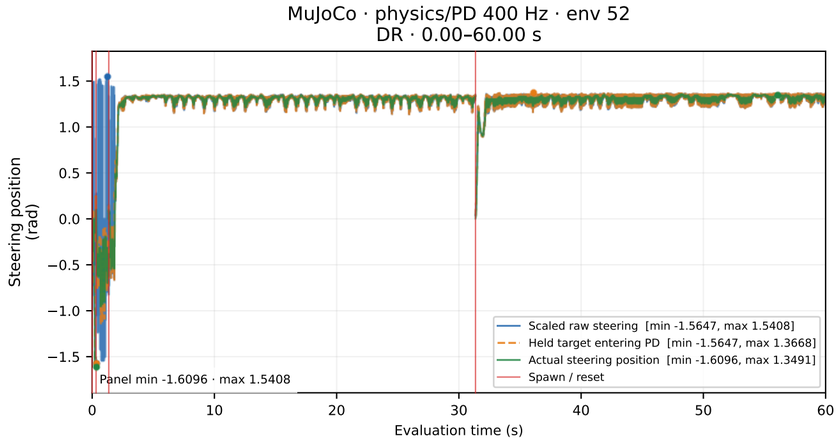
```
SAMPLED DR: env 51; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.87433, 1.19755, 0.963269, 0.971932]
  CoM XYZ mm [-1.10874, 4.205, -3.25464]
  mu_s [0.533455, 0.787792, 0.659445, 0.674963]
  mu_d [0.533455, 0.617134, 0.547619, 0.601294]
  Kp [steer,rear] [10.5429, 0]; Kd [0.189101, 0]
  spawn roll 0.0687147 rad; rate -0.105067 rad/s
  gyro bias XYZ rad/s [0.000170898, -5.42507e-05, 0.000132994]
  gravity bias YZ [0.00215066, -0.00110346]
E2 [30.00, 60.00) s:
  mass x [1.19788, 1.07216, 1.01968, 0.875061]
  CoM XYZ mm [2.23897, -4.09974, -4.60489]
  mu_s [0.533455, 0.598343, 0.751971, 0.632409]
  mu_d [0.533455, 0.598343, 0.541224, 0.589076]
  Kp [steer,rear] [10.859, 0]; Kd [0.219522, 0]
  spawn roll -0.0552671 rad; rate 0.156633 rad/s
  gyro bias XYZ rad/s [-8.94421e-05, 2.81277e-05, -0.000141213]
  gravity bias YZ [0.000624014, 0.00274033]
Six significant digits shown; full precision: sampled_dr_values.json.
```

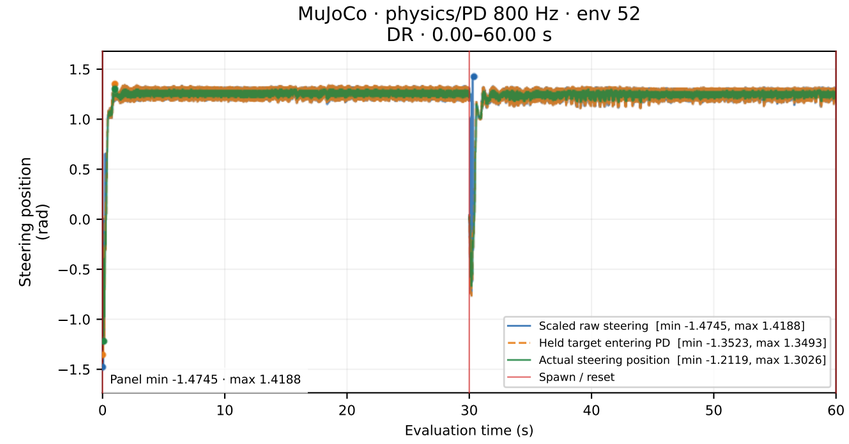
```
SAMPLED DR: env 51; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [1.1942, 1.01727, 0.983743, 0.973919]
  CoM XYZ mm [1.6316, -4.43244, 4.72033]
  mu_s [0.69316, 0.546287, 0.546287, 0.704749]
  mu_d [0.69316, 0.546287, 0.546287, 0.541926]
  Kp [steer,rear] [9.06551, 0]; Kd [0.21998, 0]
  spawn roll 0.0866621 rad; rate -0.22737 rad/s
  gyro bias XYZ rad/s [0.000142895, -0.000194485, -0.000107844]
  gravity bias YZ [0.00221892, 0.00273361]
E2 [0.28, 0.50) s:
  mass x [0.996803, 1.00338, 0.949832, 0.95773]
  CoM XYZ mm [-4.12259, -2.14861, -0.173559]
  mu_s [0.736077, 0.631146, 0.728342, 0.616244]
  mu_d [0.699455, 0.631146, 0.728342, 0.586498]
  Kp [steer,rear] [10.9851, 0]; Kd [0.182833, 0]
  spawn roll 0.0976148 rad; rate 0.101993 rad/s
  gyro bias XYZ rad/s [0.000105736, -0.000221656, -0.000179136]
  gravity bias YZ [-0.000576773, 0.000886982]
E3 [0.50, 0.88) s:
  mass x [1.05313, 1.11235, 0.994234, 0.868535]
  CoM XYZ mm [2.37178, -1.56825, -3.62737]
  mu_s [0.749678, 0.704749, 0.546287, 0.538434]
  mu_d [0.741429, 0.541926, 0.546287, 0.538434]
  Kp [steer,rear] [10.6829, 0]; Kd [0.218654, 0]
  spawn roll 0.0716195 rad; rate -0.117495 rad/s
  gyro bias XYZ rad/s [0.000107159, -9.19193e-05, 5.4211e-05]
  gravity bias YZ [0.00365437, 0.00200592]
E4 [0.88, 30.88) s:
  mass x [1.15023, 1.09454, 1.18487, 0.981418]
  CoM XYZ mm [1.46313, 4.18747, -2.39991]
  mu_s [0.640867, 0.633024, 0.640867, 0.611239]
  mu_d [0.556841, 0.568172, 0.556841, 0.611239]
  Kp [steer,rear] [9.64987, 0]; Kd [0.181477, 0]
  spawn roll -0.0252405 rad; rate 0.0123658 rad/s
  gyro bias XYZ rad/s [-1.4485e-06, 0.000125752, -4.57125e-05]
  gravity bias YZ [0.000808513, 0.00196315]
E5 [30.88, 31.12) s:
  mass x [1.05561, 0.952827, 0.80234, 0.958497]
  CoM XYZ mm [-2.01471, -3.74206, -4.28537]
  mu_s [0.633024, 0.71008, 0.669571, 0.597748]
  mu_d [0.568172, 0.59371, 0.669571, 0.597748]
  Kp [steer,rear] [9.2209, 0]; Kd [0.214402, 0]
  spawn roll 0.0690499 rad; rate -0.0117621 rad/s
  gyro bias XYZ rad/s [-4.94755e-05, -0.000266939, 0.000412373]
  gravity bias YZ [-0.00155495, -0.00130275]
E6 [31.12, 60.00) s:
  mass x [1.01532, 0.931921, 1.07484, 1.07977]
  CoM XYZ mm [-4.74891, 4.88739, -1.84585]
  mu_s [0.538976, 0.704749, 0.637675, 0.757579]
  mu_d [0.538976, 0.541926, 0.637675, 0.70921]
  Kp [steer,rear] [9.01141, 0]; Kd [0.190206, 0]
  spawn roll 0.0474584 rad; rate 0.0501261 rad/s
  gyro bias XYZ rad/s [-0.000107537, 0.000138557, -3.35874e-05]
  gravity bias YZ [-0.00247383, -0.00157511]
Six significant digits shown; full precision: sampled_dr_values.json.
```

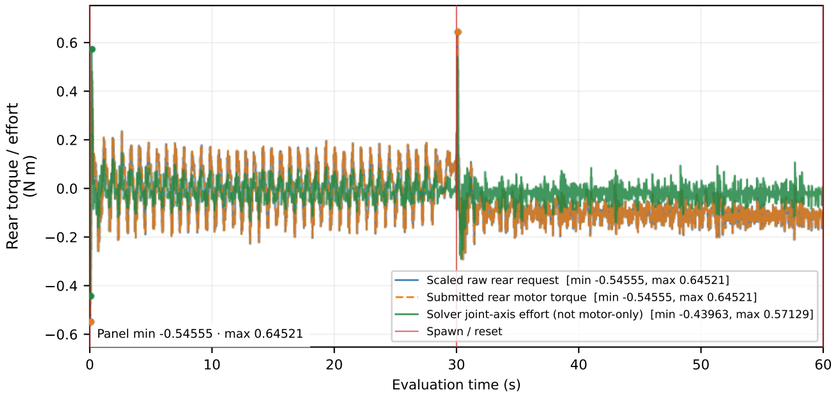
```
SAMPLED DR: env 51; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.20) s:
  mass x [1.1942, 1.01727, 0.983743, 0.973919]
  CoM XYZ mm [1.6316, -4.43244, 4.72033]
  mu_s [0.69316, 0.546287, 0.546287, 0.704749]
  mu_d [0.69316, 0.546287, 0.546287, 0.541926]
  Kp [steer,rear] [9.06551, 0]; Kd [0.21998, 0]
  spawn roll 0.0866621 rad; rate -0.22737 rad/s
  gyro bias XYZ rad/s [0.000142895, -0.000194485, -0.000107844]
  gravity bias YZ [0.00221892, 0.00273361]
E2 [0.20, 30.20) s:
  mass x [0.892959, 1.08917, 0.981928, 1.01878]
  CoM XYZ mm [3.27123, 2.47138, 2.76163]
  mu_s [0.611239, 0.732187, 0.700521, 0.728342]
  mu_d [0.611239, 0.631664, 0.641329, 0.728342]
  Kp [steer,rear] [10.3149, 0]; Kd [0.193014, 0]
  spawn roll -0.0296293 rad; rate 0.215667 rad/s
  gyro bias XYZ rad/s [0.000240257, 0.00035221, 0.000259583]
  gravity bias YZ [0.000904406, 0.0010964]
E3 [30.20, 60.00) s:
  mass x [0.971615, 0.966917, 0.991726, 0.828326]
  CoM XYZ mm [-3.34545, 2.40684, -1.81808]
  mu_s [0.748289, 0.732187, 0.704749, 0.528253]
  mu_d [0.689499, 0.631664, 0.541926, 0.528253]
  Kp [steer,rear] [9.8693, 0]; Kd [0.186607, 0]
  spawn roll -0.00592788 rad; rate 0.207095 rad/s
  gyro bias XYZ rad/s [-5.1523e-05, 0.0002882, 0.00029299]
  gravity bias YZ [0.000287504, -0.00250863]
Six significant digits shown; full precision: sampled_dr_values.json.
```

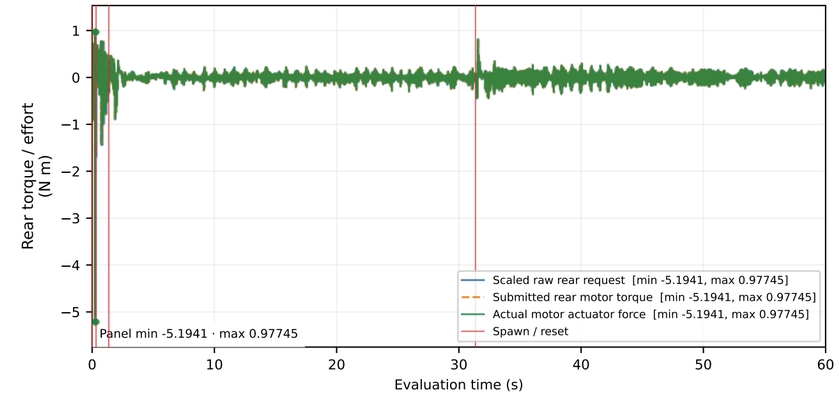
model_2500 · Demand versus actuation · env 52 · full 60 s

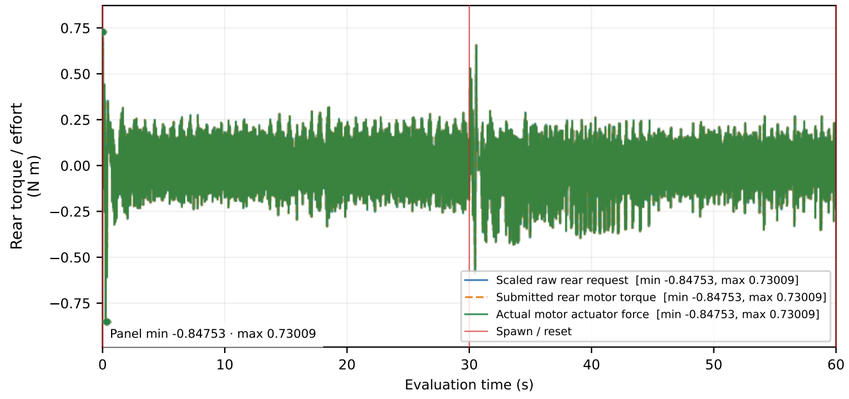












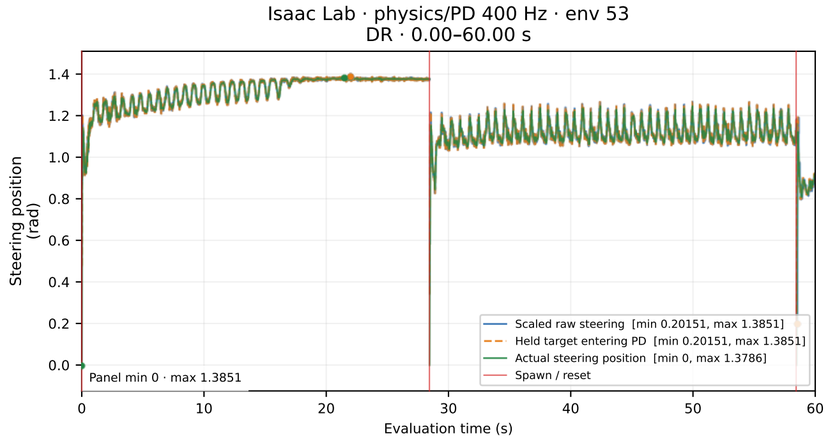
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

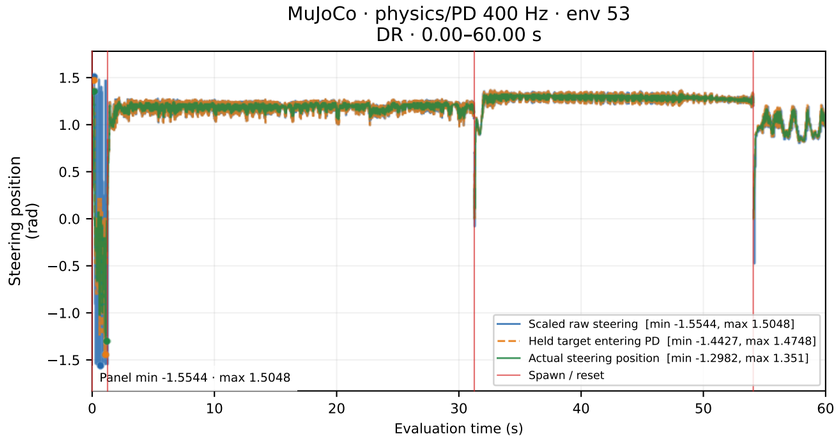
```
SAMPLED DR: env 52; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.01143, 0.921674, 0.904912, 0.988041]
  CoM XYZ mm [-1.64346, -2.42152, -3.33379]
  mu_s [0.532645, 0.751971, 0.628821, 0.751971]
  mu_d [0.532645, 0.541224, 0.628821, 0.541224]
  Kp [steer,rear] [9.6431, 0]; Kd [0.197862, 0]
  spawn roll -0.0318425 rad; rate 0.0760781 rad/s
  gyro bias XYZ rad/s [-8.25792e-05, -5.09338e-05, -0.000109599]
  gravity bias YZ [-0.0010959, -0.00106676]
E2 [30.00, 60.00) s:
  mass x [0.978785, 1.12238, 0.900565, 0.925982]
  CoM XYZ mm [4.98074, -3.23945, 1.40122]
  mu_s [0.751971, 0.559809, 0.628821, 0.713129]
  mu_d [0.541224, 0.559809, 0.628821, 0.713129]
  Kp [steer,rear] [10.7297, 0]; Kd [0.201636, 0]
  spawn roll 0.0360942 rad; rate -0.145563 rad/s
  gyro bias XYZ rad/s [-0.000104037, -4.83557e-05, 5.78648e-05]
  gravity bias YZ [-0.00165757, 5.49024e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

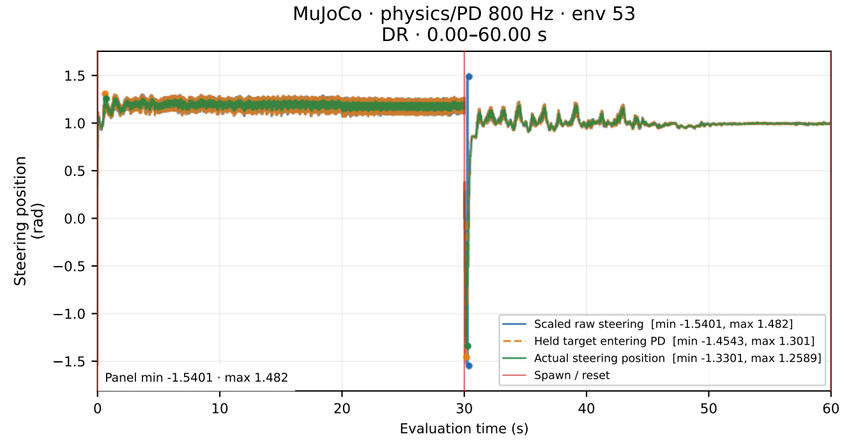
```
SAMPLED DR: env 52; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.32) s:
  mass x [0.895868, 1.11548, 1.08882, 0.818311]
  CoM XYZ mm [-1.70543, -2.37887, 0.642735]
  mu_s [0.69316, 0.637675, 0.640016, 0.748289]
  mu_d [0.69316, 0.637675, 0.513141, 0.689499]
  Kp [steer,rear] [10.9442, 0]; Kd [0.212329, 0]
  spawn roll -0.0656004 rad; rate -0.0835782 rad/s
  gyro bias XYZ rad/s [-4.7977e-05, 6.89662e-05, -9.91746e-05]
  gravity bias YZ [-0.000248333, -0.000571064]
E2 [0.32, 1.36) s:
  mass x [0.902628, 1.1733, 0.854954, 0.965253]
  CoM XYZ mm [0.700614, 4.80222, 4.26717]
  mu_s [0.640016, 0.732187, 0.700521, 0.666375]
  mu_d [0.513141, 0.631664, 0.641329, 0.519145]
  Kp [steer,rear] [10.0627, 0]; Kd [0.197855, 0]
  spawn roll 0.0103001 rad; rate -0.208955 rad/s
  gyro bias XYZ rad/s [9.74961e-05, 0.000102484, 0.000323916]
  gravity bias YZ [0.0003707, 0.0020176]
E3 [1.36, 31.36) s:
  mass x [1.19162, 1.19224, 1.00279, 1.1625]
  CoM XYZ mm [-1.02146, -2.66206, -1.10231]
  mu_s [0.669571, 0.559972, 0.637675, 0.71155]
  mu_d [0.669571, 0.502209, 0.637675, 0.71155]
  Kp [steer,rear] [9.80641, 0]; Kd [0.200827, 0]
  spawn roll -0.0404535 rad; rate 0.114395 rad/s
  gyro bias XYZ rad/s [-4.05207e-05, -2.45261e-06, -3.14968e-05]
  gravity bias YZ [0.00122725, -0.000385151]
E4 [31.36, 60.00) s:
  mass x [0.996757, 0.962115, 1.19327, 0.997593]
  CoM XYZ mm [3.8777, -2.88413, -0.403296]
  mu_s [0.723429, 0.757579, 0.700521, 0.528253]
  mu_d [0.723429, 0.70921, 0.641329, 0.528253]
  Kp [steer,rear] [9.35724, 0]; Kd [0.208195, 0]
  spawn roll -0.0178964 rad; rate 0.006131 rad/s
  gyro bias XYZ rad/s [0.000110999, -0.000215221, -9.27952e-05]
  gravity bias YZ [0.00187409, -6.57911e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

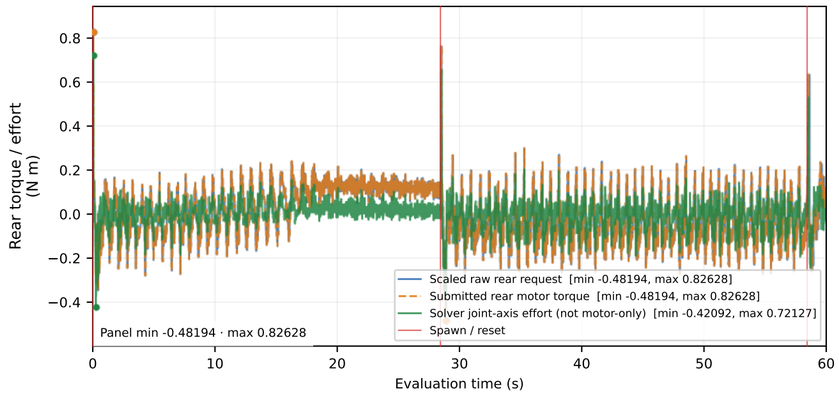
```
SAMPLED DR: env 52; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.895868, 1.11548, 1.08882, 0.818311]
  CoM XYZ mm [-1.70543, -2.37887, 0.642735]
  mu_s [0.69316, 0.637675, 0.640016, 0.748289]
  mu_d [0.69316, 0.637675, 0.513141, 0.689499]
  Kp [steer,rear] [10.9442, 0]; Kd [0.212329, 0]
  spawn roll -0.0656004 rad; rate -0.0835782 rad/s
  gyro bias XYZ rad/s [-4.7977e-05, 6.89662e-05, -9.91746e-05]
  gravity bias YZ [-0.000248333, -0.000571064]
E2 [30.00, 60.00) s:
  mass x [1.13978, 0.995018, 0.97675, 0.876827]
  CoM XYZ mm [1.76979, -1.93484, 3.17605]
  mu_s [0.597748, 0.637675, 0.633024, 0.611239]
  mu_d [0.597748, 0.637675, 0.568172, 0.611239]
  Kp [steer,rear] [10.201, 0]; Kd [0.201236, 0]
  spawn roll -0.0432077 rad; rate 0.0697385 rad/s
  gyro bias XYZ rad/s [-0.000407284, 0.000293345, 0.000263711]
  gravity bias YZ [-0.000910457, 0.000317174]
Six significant digits shown; full precision: sampled_dr_values.json.
```

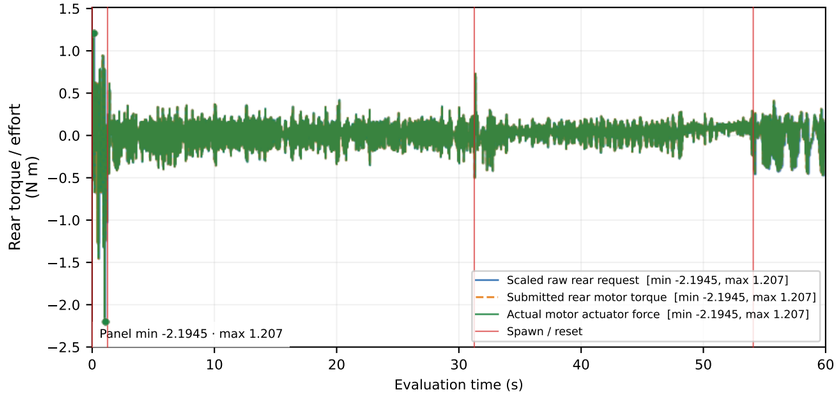
model_2500 · Demand versus actuation · env 53 · full 60 s

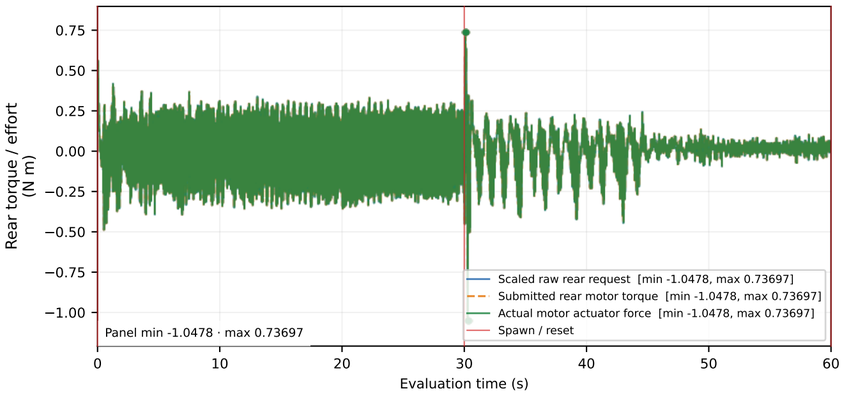












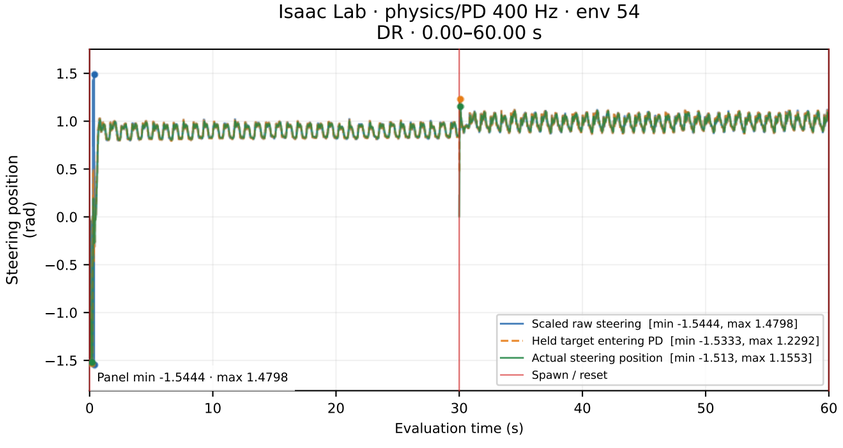
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

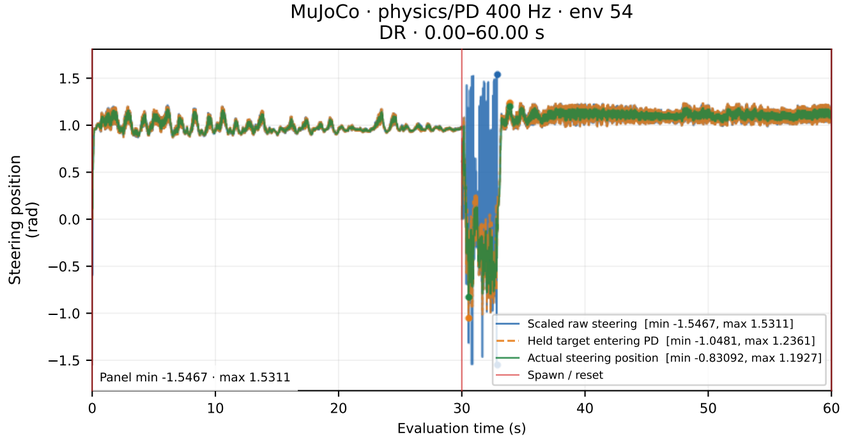
```
SAMPLED DR: env 53; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 28.44) s:
  mass x [0.937191, 1.19344, 0.855895, 1.14525]
  CoM XYZ mm [-1.96986, -2.81425, 2.27009]
  mu_s [0.774409, 0.539956, 0.774409, 0.785466]
  mu_d [0.561095, 0.539956, 0.561095, 0.52258]
  Kp [steer,rear] [9.10881, 0]; Kd [0.200427, 0]
  spawn roll 0.0774201 rad; rate 0.199795 rad/s
  gyro bias XYZ rad/s [-2.1234e-05, 7.10412e-05, 0.000266473]
  gravity bias YZ [0.00113377, 0.00341815]
E2 [28.44, 58.44) s:
  mass x [1.08785, 0.933505, 0.894214, 0.904942]
  CoM XYZ mm [2.82797, -0.615619, 0.79608]
  mu_s [0.531594, 0.774409, 0.693205, 0.531594]
  mu_d [0.531594, 0.561095, 0.693205, 0.531594]
  Kp [steer,rear] [9.15879, 0]; Kd [0.188207, 0]
  spawn roll 0.065437 rad; rate -0.16356 rad/s
  gyro bias XYZ rad/s [4.0565e-05, -0.000219733, 6.90915e-06]
  gravity bias YZ [-0.00324229, -0.000688782]
E3 [58.44, 60.00) s:
  mass x [1.02962, 1.16839, 1.18665, 0.912459]
  CoM XYZ mm [-4.35281, 3.14168, -1.23222]
  mu_s [0.741039, 0.659445, 0.713129, 0.647391]
  mu_d [0.614405, 0.547619, 0.713129, 0.647391]
  Kp [steer,rear] [10.5804, 0]; Kd [0.211518, 0]
  spawn roll 0.0621697 rad; rate -0.189606 rad/s
  gyro bias XYZ rad/s [0.000247501, 3.50317e-05, 0.000232372]
  gravity bias YZ [0.00033728, 0.00111157]
Six significant digits shown; full precision: sampled_dr_values.json.
```

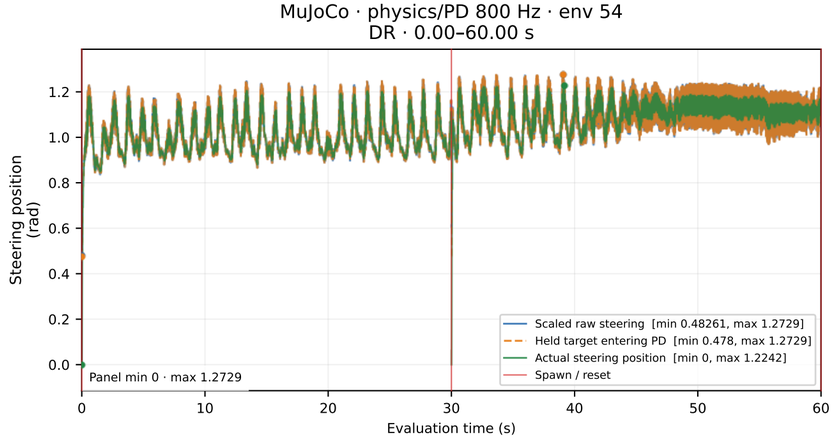
```
SAMPLED DR: env 53; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.26) s:
  mass x [1.01658, 1.16532, 0.890464, 0.952267]
  CoM XYZ mm [-0.896499, -0.23245, -3.38234]
  mu_s [0.616244, 0.704749, 0.791209, 0.559972]
  mu_d [0.586498, 0.541926, 0.767936, 0.502209]
  Kp [steer,rear] [9.83361, 0]; Kd [0.199999, 0]
  spawn roll 0.0480047 rad; rate 0.0813598 rad/s
  gyro bias XYZ rad/s [-1.68078e-05, -0.000542628, 0.000220491]
  gravity bias YZ [0.00110247, 0.000413204]
E2 [1.26, 31.26) s:
  mass x [0.854132, 1.05172, 0.994098, 1.09104]
  CoM XYZ mm [-1.3453, -0.772613, 3.85896]
  mu_s [0.568073, 0.723429, 0.611239, 0.541939]
  mu_d [0.568073, 0.723429, 0.611239, 0.534359]
  Kp [steer,rear] [10.1132, 0]; Kd [0.21786, 0]
  spawn roll -9.27329e-05 rad; rate -0.0141128 rad/s
  gyro bias XYZ rad/s [6.8371e-05, -7.18617e-05, 0.000189908]
  gravity bias YZ [-0.00102904, -0.000470824]
E3 [31.26, 54.08) s:
  mass x [0.922679, 0.983297, 0.956886, 1.0459]
  CoM XYZ mm [4.99604, -4.88986, 3.77766]
  mu_s [0.559972, 0.704749, 0.568073, 0.538976]
  mu_d [0.502209, 0.541926, 0.568073, 0.538976]
  Kp [steer,rear] [10.7575, 0]; Kd [0.205237, 0]
  spawn roll 0.0373654 rad; rate 0.115207 rad/s
  gyro bias XYZ rad/s [0.000234425, 6.02528e-05, -0.000494244]
  gravity bias YZ [-0.00335598, 0.00117259]
E4 [54.08, 60.00) s:
  mass x [0.953553, 1.12825, 1.05418, 0.953149]
  CoM XYZ mm [3.20866, 2.32445, 2.39002]
  mu_s [0.538434, 0.736077, 0.631146, 0.568073]
  mu_d [0.538434, 0.699455, 0.631146, 0.568073]
  Kp [steer,rear] [10.0327, 0]; Kd [0.184673, 0]
  spawn roll 0.0341392 rad; rate 0.122202 rad/s
  gyro bias XYZ rad/s [-0.000102103, -0.000161652, -0.00031826]
  gravity bias YZ [0.00161921, 4.34852e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

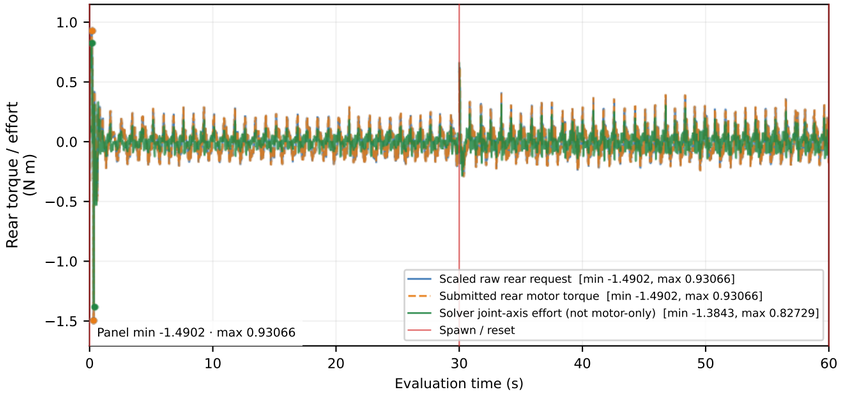
```
SAMPLED DR: env 53; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.01658, 1.16532, 0.890464, 0.952267]
  CoM XYZ mm [-0.896499, -0.23245, -3.38234]
  mu_s [0.616244, 0.704749, 0.791209, 0.559972]
  mu_d [0.586498, 0.541926, 0.767936, 0.502209]
  Kp [steer,rear] [9.83361, 0]; Kd [0.199999, 0]
  spawn roll 0.0480047 rad; rate 0.0813598 rad/s
  gyro bias XYZ rad/s [-1.68078e-05, -0.000542628, 0.000220491]
  gravity bias YZ [0.00110247, 0.000413204]
E2 [30.00, 60.00) s:
  mass x [1.15391, 1.11127, 0.962362, 1.17442]
  CoM XYZ mm [-4.7896, 0.566434, 2.75297]
  mu_s [0.728342, 0.69316, 0.528253, 0.71155]
  mu_d [0.728342, 0.69316, 0.528253, 0.71155]
  Kp [steer,rear] [10.9894, 0]; Kd [0.204184, 0]
  spawn roll -0.0670953 rad; rate 0.171254 rad/s
  gyro bias XYZ rad/s [-0.000121232, -5.29606e-05, 0.000156749]
  gravity bias YZ [-0.000861907, -0.0036078]
Six significant digits shown; full precision: sampled_dr_values.json.
```

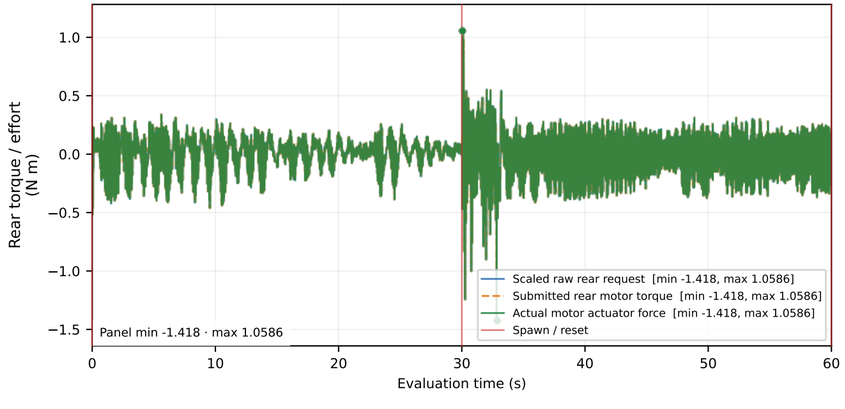
model_2500 · Demand versus actuation · env 54 · full 60 s

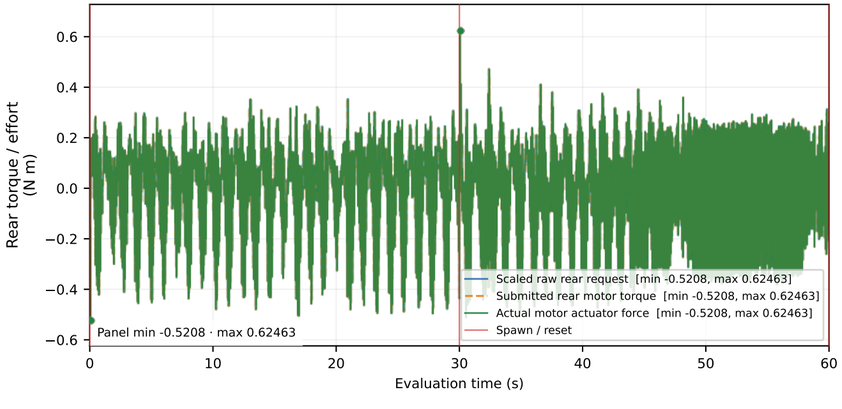












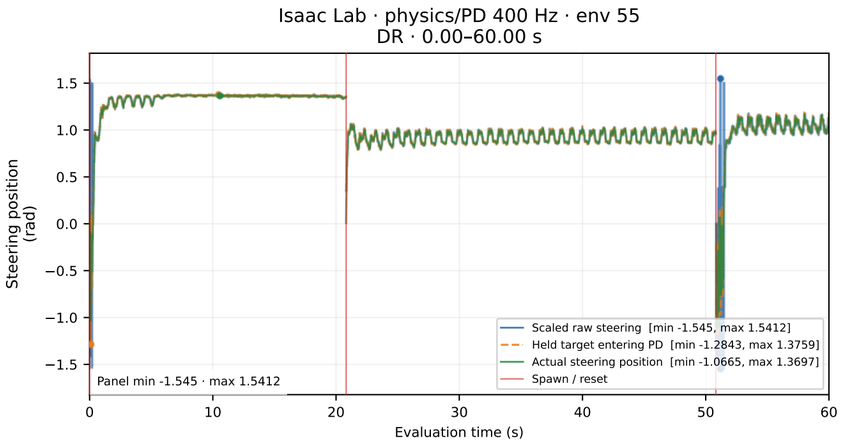
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

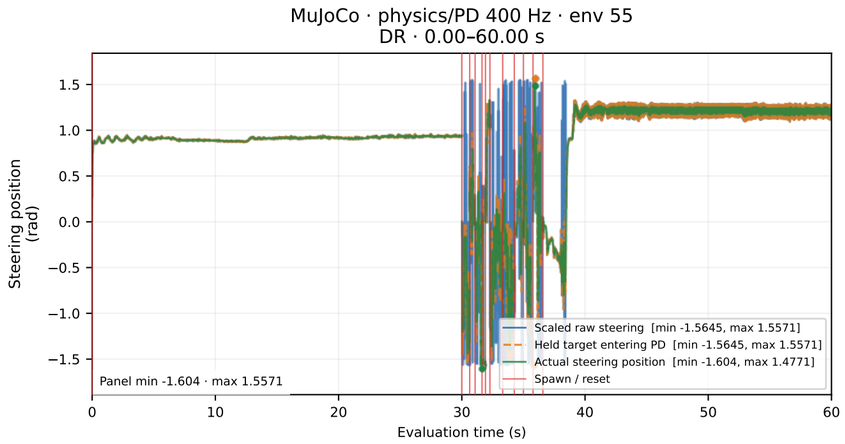
```
SAMPLED DR: env 54; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.10934, 1.19379, 0.825476, 1.17256]
  CoM XYZ mm [2.97143, 2.55803, 3.43199]
  mu_s [0.560535, 0.703711, 0.726175, 0.681711]
  mu_d [0.560535, 0.703711, 0.558574, 0.611756]
  Kp [steer,rear] [10.0082, 0]; Kd [0.186213, 0]
  spawn roll -0.0964281 rad; rate 0.181293 rad/s
  gyro bias XYZ rad/s [-0.000101903, -9.48728e-05, 6.15729e-05]
  gravity bias YZ [-0.000767394, 0.000607192]
E2 [30.00, 60.00) s:
  mass x [0.921562, 0.915659, 0.964448, 0.957485]
  CoM XYZ mm [-0.282958, 0.988118, -3.6774]
  mu_s [0.751971, 0.584424, 0.547743, 0.726175]
  mu_d [0.541224, 0.584424, 0.547743, 0.558574]
  Kp [steer,rear] [9.502, 0]; Kd [0.182184, 0]
  spawn roll 0.0621586 rad; rate 0.244119 rad/s
  gyro bias XYZ rad/s [-0.000509188, -3.44135e-05, 4.87007e-06]
  gravity bias YZ [-0.0040903, -0.000794371]
Six significant digits shown; full precision: sampled_dr_values.json.
```

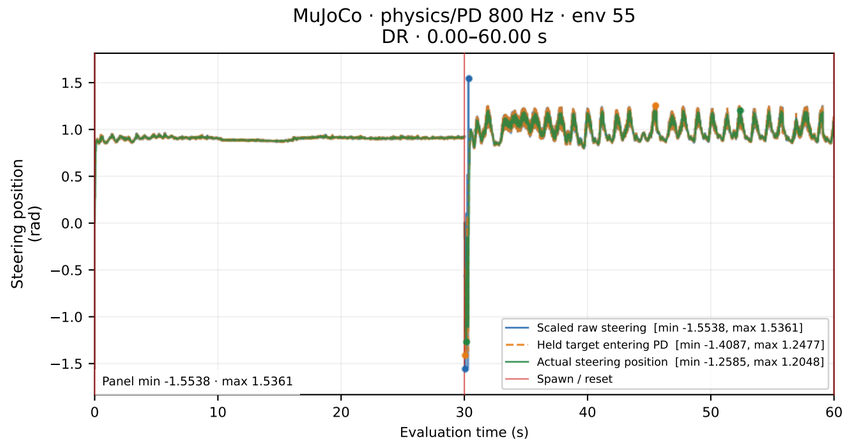
```
SAMPLED DR: env 54; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09167, 1.09356, 1.01932, 0.995919]
  CoM XYZ mm [0.318897, 1.84907, -4.07307]
  mu_s [0.666375, 0.700521, 0.723429, 0.541939]
  mu_d [0.519145, 0.641329, 0.723429, 0.534359]
  Kp [steer,rear] [9.46515, 0]; Kd [0.187392, 0]
  spawn roll 0.0116196 rad; rate 0.0807588 rad/s
  gyro bias XYZ rad/s [6.48641e-05, -8.75067e-05, 0.000224428]
  gravity bias YZ [0.0010855, -0.00040928]
E2 [30.00, 60.00) s:
  mass x [1.09167, 0.850294, 1.03901, 1.13524]
  CoM XYZ mm [2.66514, 1.43688, 1.52467]
  mu_s [0.732187, 0.669571, 0.559972, 0.597748]
  mu_d [0.631664, 0.669571, 0.502209, 0.597748]
  Kp [steer,rear] [9.70921, 0]; Kd [0.203106, 0]
  spawn roll 0.0644941 rad; rate -0.00531168 rad/s
  gyro bias XYZ rad/s [0.000128906, 0.000211222, -0.000220637]
  gravity bias YZ [0.00353561, 0.000974373]
Six significant digits shown; full precision: sampled_dr_values.json.
```

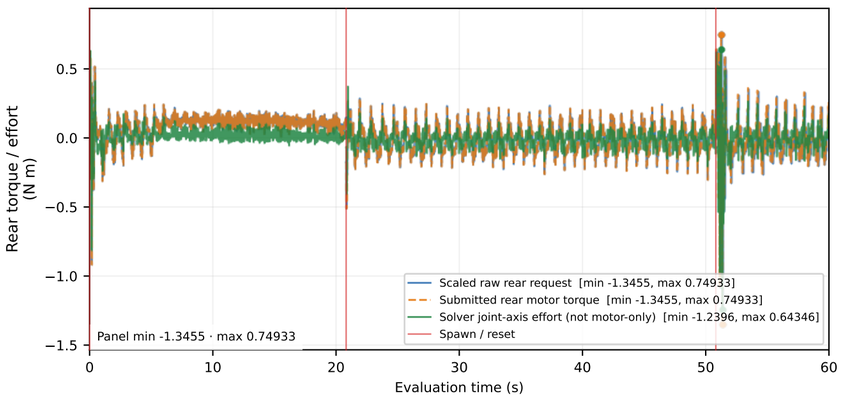
```
SAMPLED DR: env 54; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09167, 1.09356, 1.01932, 0.995919]
  CoM XYZ mm [0.318897, 1.84907, -4.07307]
  mu_s [0.666375, 0.700521, 0.723429, 0.541939]
  mu_d [0.519145, 0.641329, 0.723429, 0.534359]
  Kp [steer,rear] [9.46515, 0]; Kd [0.187392, 0]
  spawn roll 0.0116196 rad; rate 0.0807588 rad/s
  gyro bias XYZ rad/s [6.48641e-05, -8.75067e-05, 0.000224428]
  gravity bias YZ [0.0010855, -0.00040928]
E2 [30.00, 60.00) s:
  mass x [1.09167, 0.850294, 1.03901, 1.13524]
  CoM XYZ mm [2.66514, 1.43688, 1.52467]
  mu_s [0.732187, 0.669571, 0.559972, 0.597748]
  mu_d [0.631664, 0.669571, 0.502209, 0.597748]
  Kp [steer,rear] [9.70921, 0]; Kd [0.203106, 0]
  spawn roll 0.0644941 rad; rate -0.00531168 rad/s
  gyro bias XYZ rad/s [0.000128906, 0.000211222, -0.000220637]
  gravity bias YZ [0.00353561, 0.000974373]
Six significant digits shown; full precision: sampled_dr_values.json.
```

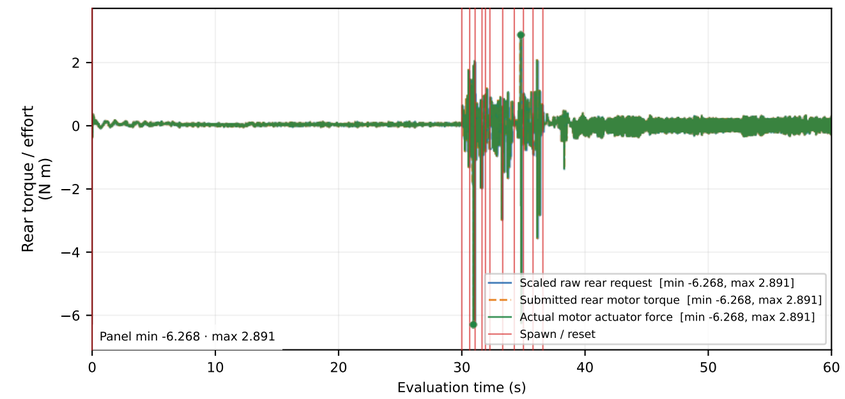
model_2500 · Demand versus actuation · env 55 · full 60 s

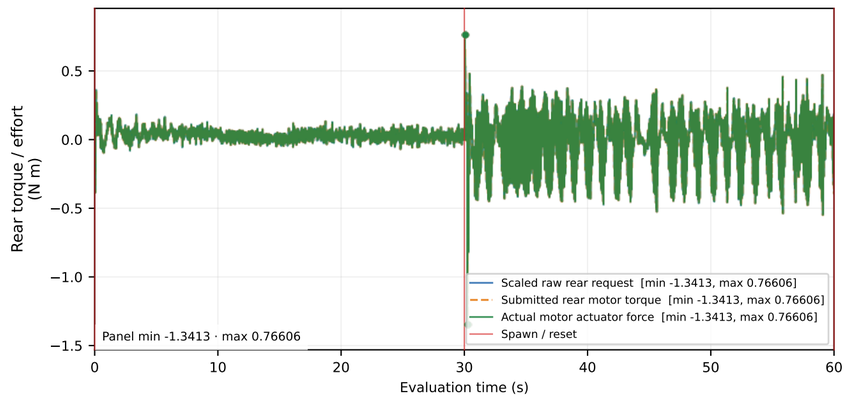












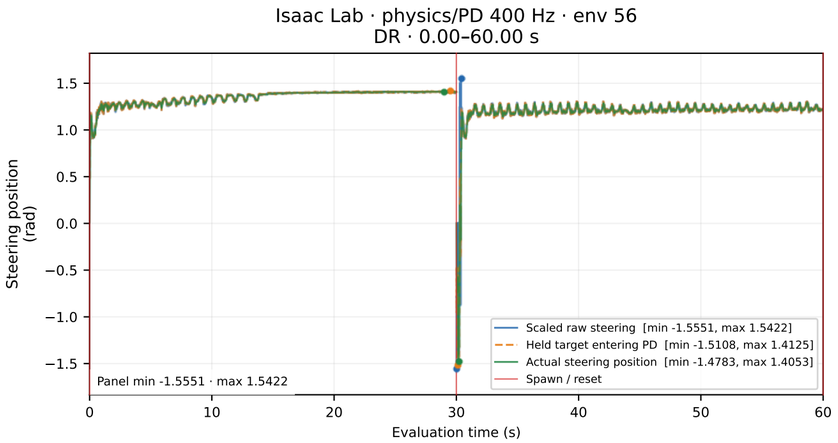
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

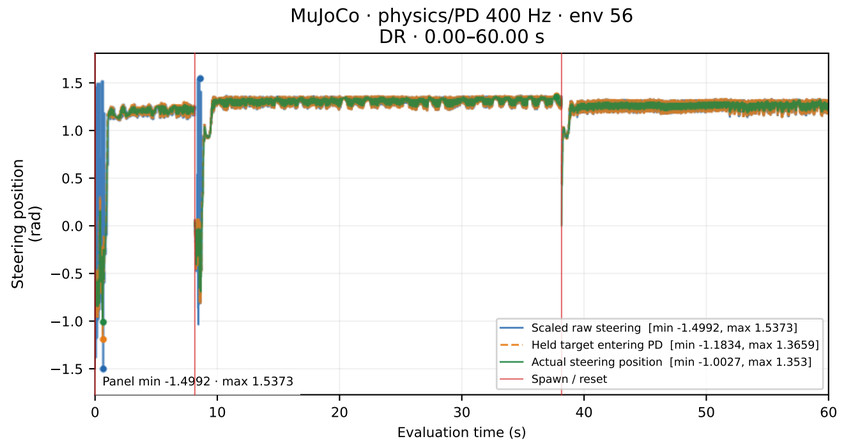
```
SAMPLED DR: env 55; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 20.82) s:
  mass x [0.963485, 1.07764, 1.18683, 0.820207]
  CoM XYZ mm [-3.90576, -2.59108, 0.0577271]
  mu_s [0.628821, 0.560535, 0.760821, 0.647391]
  mu_d [0.628821, 0.560535, 0.670315, 0.647391]
  Kp [steer,rear] [10.2764, 0]; Kd [0.200364, 0]
  spawn roll -0.0387032 rad; rate -0.189371 rad/s
  gyro bias XYZ rad/s [-2.12844e-05, 3.17022e-05, 0.000155771]
  gravity bias YZ [0.00176665, 0.00100496]
E2 [20.82, 50.82) s:
  mass x [1.03942, 1.09104, 1.02408, 0.815617]
  CoM XYZ mm [4.00533, 2.65287, 1.77476]
  mu_s [0.693205, 0.787792, 0.559809, 0.794338]
  mu_d [0.693205, 0.617134, 0.559809, 0.526209]
  Kp [steer,rear] [10.527, 0]; Kd [0.210141, 0]
  spawn roll 0.00214958 rad; rate 0.066284 rad/s
  gyro bias XYZ rad/s [-8.82509e-05, -1.80504e-05, 6.60728e-05]
  gravity bias YZ [0.00286012, 0.00110588]
E3 [50.82, 60.00) s:
  mass x [1.01917, 0.803289, 1.02937, 0.877224]
  CoM XYZ mm [0.233695, 0.62497, -1.17226]
  mu_s [0.531594, 0.713129, 0.592046, 0.659445]
  mu_d [0.531594, 0.713129, 0.534947, 0.547619]
  Kp [steer,rear] [10.654, 0]; Kd [0.187429, 0]
  spawn roll -0.0647806 rad; rate -0.00877151 rad/s
  gyro bias XYZ rad/s [-0.000406793, 0.000384208, -2.96344e-05]
  gravity bias YZ [-0.00204948, -0.00215862]
Six significant digits shown; full precision: sampled_dr_values.json.
```

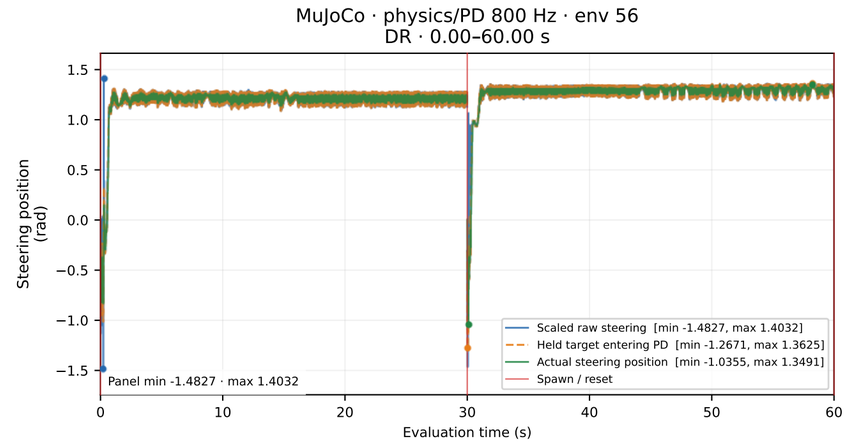
```
SAMPLED DR: env 55; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.06971, 0.972924, 1.00779, 1.19138]
  CoM XYZ mm [-3.20054, 2.46581, -4.66681]
  mu_s [0.71155, 0.757579, 0.633024, 0.791209]
  mu_d [0.71155, 0.70921, 0.568172, 0.767936]
  Kp [steer,rear] [10.3741, 0]; Kd [0.186761, 0]
  spawn roll 0.02939 rad; rate -0.0619636 rad/s
  gyro bias XYZ rad/s [-0.000345929, 6.10222e-05, 0.000149293]
  gravity bias YZ [0.000140127, -0.000249329]
E2 [30.00, 30.64) s:
  mass x [0.950851, 1.15439, 1.06268, 1.03878]
  CoM XYZ mm [1.79913, 1.45826, 3.13366]
  mu_s [0.541939, 0.633024, 0.611239, 0.640016]
  mu_d [0.534359, 0.568172, 0.611239, 0.513141]
  Kp [steer,rear] [10.104, 0]; Kd [0.200072, 0]
  spawn roll -0.0578729 rad; rate -0.213384 rad/s
  gyro bias XYZ rad/s [-0.000286517, 0.000241971, 0.000139544]
  gravity bias YZ [-0.00335196, 0.003988]
E3 [30.64, 31.08) s:
  mass x [0.841558, 0.811055, 0.850415, 0.838444]
  CoM XYZ mm [0.964484, -2.95318, 3.32143]
  mu_s [0.723429, 0.704749, 0.640867, 0.541939]
  mu_d [0.723429, 0.541926, 0.556841, 0.534359]
  Kp [steer,rear] [9.83051, 0]; Kd [0.185102, 0]
  spawn roll -0.0238694 rad; rate 0.238069 rad/s
  gyro bias XYZ rad/s [0.000296954, 1.99077e-05, -5.1428e-05]
  gravity bias YZ [0.00185396, 0.00271763]
E4 [31.08, 31.64) s:
  mass x [0.977562, 0.939718, 1.10087, 0.859839]
  CoM XYZ mm [-0.713412, -2.41105, 3.34113]
  mu_s [0.669571, 0.611239, 0.700521, 0.631146]
  mu_d [0.669571, 0.611239, 0.641329, 0.631146]
  Kp [steer,rear] [10.5322, 0]; Kd [0.219664, 0]
  spawn roll -0.0492317 rad; rate 0.0183658 rad/s
  gyro bias XYZ rad/s [-0.000102469, -0.000191119, -0.000104485]
  gravity bias YZ [-0.00264815, -0.00138903]
E5 [31.64, 31.92) s:
  mass x [0.807522, 0.950731, 0.913443, 0.939251]
  CoM XYZ mm [3.64069, 3.87714, 0.785652]
  mu_s [0.640867, 0.723429, 0.704749, 0.538976]
  mu_d [0.556841, 0.723429, 0.541926, 0.538976]
  Kp [steer,rear] [9.35143, 0]; Kd [0.215904, 0]
  spawn roll -0.0752084 rad; rate 0.00748958 rad/s
  gyro bias XYZ rad/s [0.000333094, 1.15888e-05, -3.8052e-05]
  gravity bias YZ [-0.00305434, -0.000285825]
E6 [31.92, 32.28) s:
  mass x [0.915522, 1.19818, 1.06993, 0.841503]
  CoM XYZ mm [-3.59426, -3.46838, 2.35408]
  mu_s [0.71008, 0.733515, 0.611239, 0.748289]
  mu_d [0.59371, 0.558392, 0.611239, 0.689499]
  Kp [steer,rear] [9.17657, 0]; Kd [0.204401, 0]
  spawn roll 0.0669363 rad; rate 0.246136 rad/s
  gyro bias XYZ rad/s [2.15728e-05, -0.000658089, -0.000151426]
  gravity bias YZ [-0.000504504, -0.00157344]
E7 [32.28, 33.32) s:
  mass x [1.11837, 1.09435, 0.803739, 0.918133]
  CoM XYZ mm [-4.62211, 3.42452, -4.20362]
  mu_s [0.71008, 0.791209, 0.700521, 0.791209]
  mu_d [0.59371, 0.767936, 0.641329, 0.767936]
  Kp [steer,rear] [9.1785, 0]; Kd [0.192983, 0]
  spawn roll -0.0799881 rad; rate -0.0901051 rad/s
  gyro bias XYZ rad/s [-0.000159882, 2.21676e-05, 0.000189237]
  gravity bias YZ [0.000174635, -0.000361986]
E8 [33.32, 34.26) s:
  mass x [1.18228, 0.893188, 1.11623, 0.916708]
  CoM XYZ mm [3.98366, 0.741484, -3.22077]
  mu_s [0.559972, 0.538434, 0.749678, 0.732187]
  mu_d [0.502209, 0.538434, 0.741429, 0.631664]
  Kp [steer,rear] [9.21255, 0]; Kd [0.210853, 0]
  spawn roll -0.0273076 rad; rate -0.0692347 rad/s
  gyro bias XYZ rad/s [-6.51231e-05, 0.000370741, -0.000267846]
  gravity bias YZ [0.00228716, -0.00299861]
E9 [34.26, 35.00) s:
  mass x [1.12591, 0.900137, 0.954882, 0.933317]
  CoM XYZ mm [4.09999, -3.45071, -0.619425]
  mu_s [0.757579, 0.733515, 0.640016, 0.749678]
  mu_d [0.70921, 0.558392, 0.513141, 0.741429]
  Kp [steer,rear] [10.0242, 0]; Kd [0.20782, 0]
  spawn roll -0.0288042 rad; rate -0.0110942 rad/s
  gyro bias XYZ rad/s [-0.000205313, 0.000134744, 1.35886e-05]
  gravity bias YZ [0.00112404, 0.00168542]
E10 [35.00, 35.78) s:
  mass x [1.13111, 1.17678, 0.979955, 0.806021]
  CoM XYZ mm [1.04925, 2.58589, -1.10245]
  mu_s [0.568073, 0.791209, 0.69316, 0.640867]
  mu_d [0.568073, 0.767936, 0.69316, 0.556841]
  Kp [steer,rear] [9.52077, 0]; Kd [0.190062, 0]
  spawn roll 0.00935767 rad; rate -0.217517 rad/s
  gyro bias XYZ rad/s [0.000296809, -0.000175332, -0.000138796]
  gravity bias YZ [-0.00145292, 0.0010066]
E11 [35.78, 36.58) s:
  mass x [0.968944, 0.916349, 0.881028, 0.899013]
  CoM XYZ mm [-1.30586, -3.42872, -0.650544]
  mu_s [0.748289, 0.69316, 0.728342, 0.666375]
  mu_d [0.689499, 0.69316, 0.728342, 0.519145]
  Kp [steer,rear] [10.9664, 0]; Kd [0.215449, 0]
  spawn roll -0.00875553 rad; rate 0.0956034 rad/s
  gyro bias XYZ rad/s [-0.000132225, -1.92237e-05, -1.94566e-05]
  gravity bias YZ [0.000422119, 0.00267768]
E12 [36.58, 60.00) s:
  mass x [0.938605, 0.838997, 1.12104, 0.999597]
  CoM XYZ mm [-2.19752, -1.34488, -1.573]
  mu_s [0.700521, 0.723429, 0.631146, 0.669571]
  mu_d [0.641329, 0.723429, 0.631146, 0.669571]
  Kp [steer,rear] [9.02062, 0]; Kd [0.200193, 0]
  spawn roll -0.013736 rad; rate -0.118537 rad/s
  gyro bias XYZ rad/s [-0.000175155, -0.00023395, -5.02368e-05]
  gravity bias YZ [-0.00209707, 0.00430118]
Six significant digits shown; full precision: sampled_dr_values.json.
```

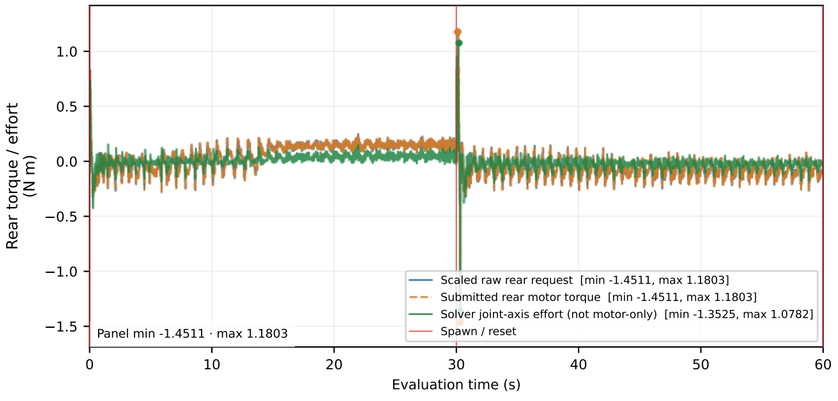
```
SAMPLED DR: env 55; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.06971, 0.972924, 1.00779, 1.19138]
  CoM XYZ mm [-3.20054, 2.46581, -4.66681]
  mu_s [0.71155, 0.757579, 0.633024, 0.791209]
  mu_d [0.71155, 0.70921, 0.568172, 0.767936]
  Kp [steer,rear] [10.3741, 0]; Kd [0.186761, 0]
  spawn roll 0.02939 rad; rate -0.0619636 rad/s
  gyro bias XYZ rad/s [-0.000345929, 6.10222e-05, 0.000149293]
  gravity bias YZ [0.000140127, -0.000249329]
E2 [30.00, 60.00) s:
  mass x [0.950851, 1.15439, 1.06268, 1.03878]
  CoM XYZ mm [1.79913, 1.45826, 3.13366]
  mu_s [0.541939, 0.633024, 0.611239, 0.640016]
  mu_d [0.534359, 0.568172, 0.611239, 0.513141]
  Kp [steer,rear] [10.104, 0]; Kd [0.200072, 0]
  spawn roll -0.0578729 rad; rate -0.213384 rad/s
  gyro bias XYZ rad/s [-0.000286517, 0.000241971, 0.000139544]
  gravity bias YZ [-0.00335196, 0.003988]
Six significant digits shown; full precision: sampled_dr_values.json.
```

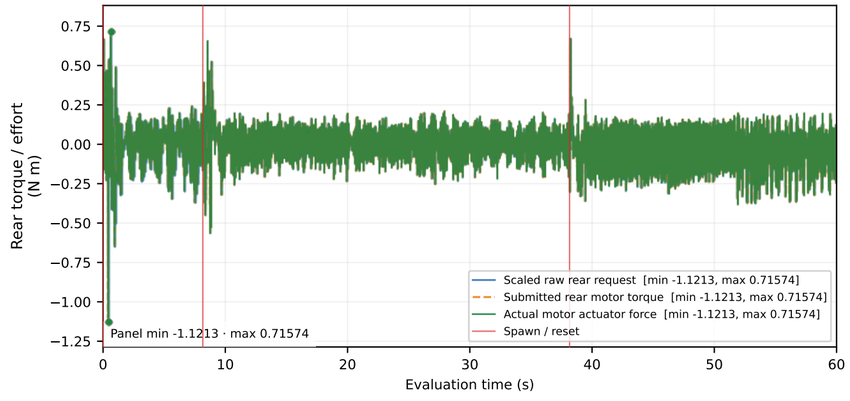
model_2500 · Demand versus actuation · env 56 · full 60 s

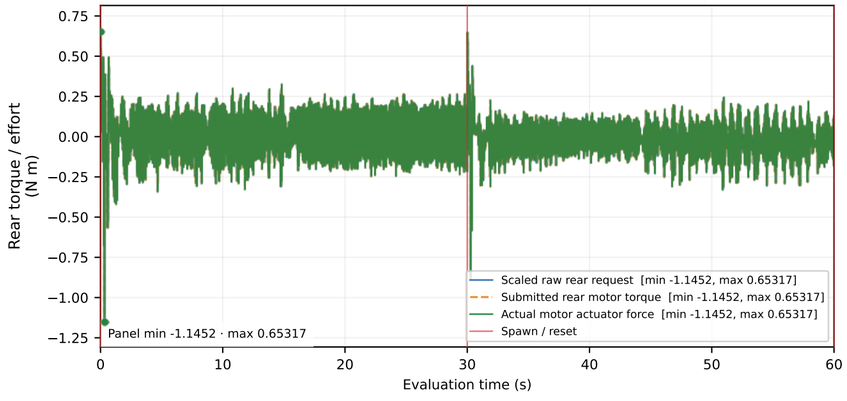












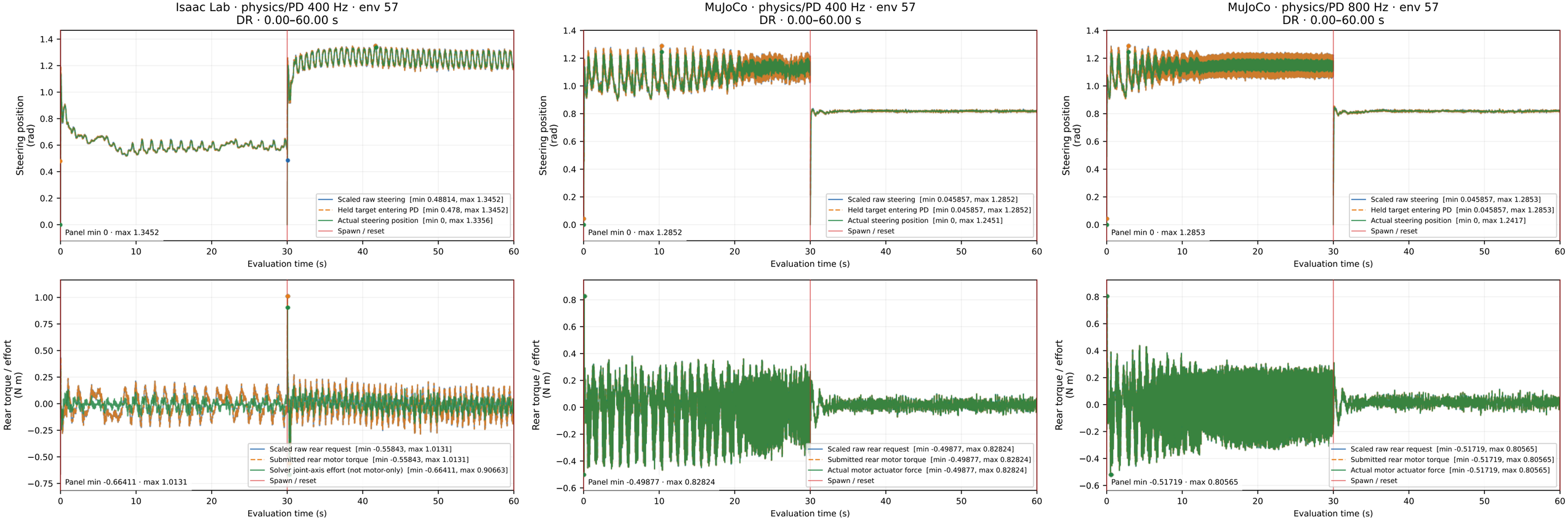
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 56; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.978548, 0.933041, 0.929648, 0.983315]
  CoM XYZ mm [-3.25596, -4.31841, 2.93314]
  mu_s [0.787792, 0.628821, 0.598343, 0.785466]
  mu_d [0.617134, 0.628821, 0.598343, 0.52258]
  Kp [steer,rear] [9.19854, 0]; Kd [0.208574, 0]
  spawn roll 0.0423985 rad; rate -0.00768098 rad/s
  gyro bias XYZ rad/s [-0.000145776, -0.000196373, 5.05469e-05]
  gravity bias YZ [-0.00159637, 0.00238998]
E2 [30.00, 60.00) s:
  mass x [0.994449, 0.849617, 0.945306, 1.1639]
  CoM XYZ mm [-3.41831, -2.02272, 4.95651]
  mu_s [0.681711, 0.632409, 0.751971, 0.632409]
  mu_d [0.611756, 0.589076, 0.541224, 0.589076]
  Kp [steer,rear] [10.3618, 0]; Kd [0.195038, 0]
  spawn roll -0.0946161 rad; rate -0.231065 rad/s
  gyro bias XYZ rad/s [0.00018137, 0.000240971, 9.6531e-05]
  gravity bias YZ [0.00487678, -0.00370653]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 56; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 8.16) s:
  mass x [1.17746, 1.14053, 0.82888, 0.916452]
  CoM XYZ mm [-4.13249, -1.50093, -2.20324]
  mu_s [0.538976, 0.757579, 0.559972, 0.748289]
  mu_d [0.538976, 0.70921, 0.502209, 0.689499]
  Kp [steer,rear] [9.78832, 0]; Kd [0.18094, 0]
  spawn roll -0.0446931 rad; rate -0.236748 rad/s
  gyro bias XYZ rad/s [-1.33536e-05, -0.000180171, 9.54982e-05]
  gravity bias YZ [-0.00148982, 0.00237804]
E2 [8.16, 38.16) s:
  mass x [0.856354, 0.811058, 1.04212, 1.0589]
  CoM XYZ mm [0.588333, -3.77611, -0.66241]
  mu_s [0.640016, 0.597748, 0.631146, 0.559972]
  mu_d [0.513141, 0.597748, 0.631146, 0.502209]
  Kp [steer,rear] [10.7806, 0]; Kd [0.183061, 0]
  spawn roll -0.0436161 rad; rate 0.0759602 rad/s
  gyro bias XYZ rad/s [-4.65311e-05, -0.000153393, 7.44427e-05]
  gravity bias YZ [-0.00122513, -0.00191494]
E3 [38.16, 60.00) s:
  mass x [1.13284, 0.807767, 1.00117, 1.19138]
  CoM XYZ mm [-0.178846, -2.61705, 3.94731]
  mu_s [0.700521, 0.666375, 0.71155, 0.666375]
  mu_d [0.641329, 0.519145, 0.71155, 0.519145]
  Kp [steer,rear] [9.39402, 0]; Kd [0.182916, 0]
  spawn roll 0.0244649 rad; rate 0.0222008 rad/s
  gyro bias XYZ rad/s [-0.000224125, 0.000416155, 0.000182462]
  gravity bias YZ [-0.00161299, -0.00305434]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 56; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.17746, 1.14053, 0.82888, 0.916452]
  CoM XYZ mm [-4.13249, -1.50093, -2.20324]
  mu_s [0.538976, 0.757579, 0.559972, 0.748289]
  mu_d [0.538976, 0.70921, 0.502209, 0.689499]
  Kp [steer,rear] [9.78832, 0]; Kd [0.18094, 0]
  spawn roll -0.0446931 rad; rate -0.236748 rad/s
  gyro bias XYZ rad/s [-1.33536e-05, -0.000180171, 9.54982e-05]
  gravity bias YZ [-0.00148982, 0.00237804]
E2 [30.00, 60.00) s:
  mass x [1.08147, 1.04442, 1.08994, 1.11074]
  CoM XYZ mm [-3.25819, -4.05595, 1.49957]
  mu_s [0.666375, 0.631146, 0.633024, 0.727426]
  mu_d [0.519145, 0.631146, 0.568172, 0.606358]
  Kp [steer,rear] [10.354, 0]; Kd [0.193614, 0]
  spawn roll -0.064361 rad; rate -0.0590807 rad/s
  gyro bias XYZ rad/s [0.000253148, 0.000173571, -0.000125265]
  gravity bias YZ [-0.00262255, -0.00076504]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 57 · full 60 s



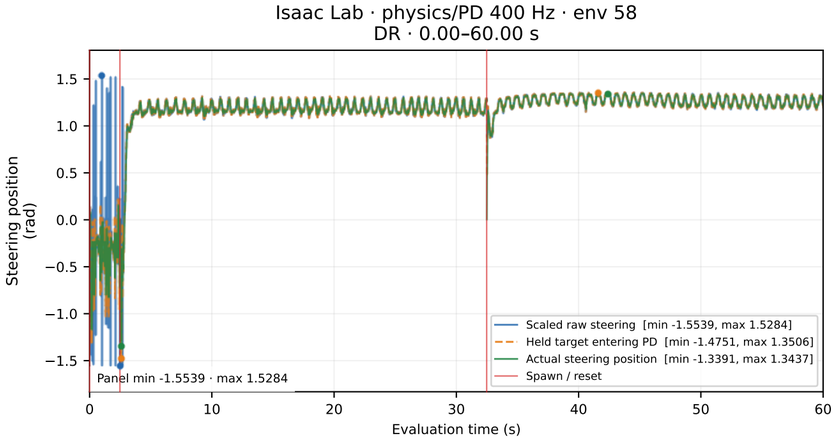
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

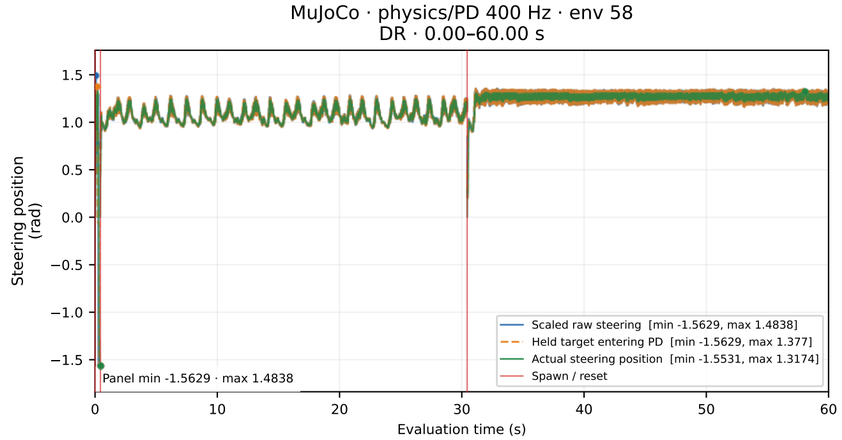
```
SAMPLED DR: env 57; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.03432, 0.881207, 1.02932, 1.18585]
  CoM XYZ mm [-0.983864, 4.94792, -4.15209]
  mu_s [0.681711, 0.584424, 0.584424, 0.764681]
  mu_d [0.611756, 0.584424, 0.584424, 0.764681]
  Kp [steer,rear] [9.40944, 0]; Kd [0.182484, 0]
  spawn roll 0.0620867 rad; rate -0.00318035 rad/s
  gyro bias XYZ rad/s [0.000308029, 0.000200594, 0.000217675]
  gravity bias YZ [-0.00138079, 0.0029808]
E2 [30.00, 60.00) s:
  mass x [0.854298, 1.007, 1.06667, 0.956546]
  CoM XYZ mm [-3.08383, -2.12793, -2.92704]
  mu_s [0.794338, 0.531594, 0.787792, 0.713129]
  mu_d [0.526209, 0.531594, 0.617134, 0.713129]
  Kp [steer,rear] [9.88376, 0]; Kd [0.20941, 0]
  spawn roll 0.091352 rad; rate -0.174524 rad/s
  gyro bias XYZ rad/s [-3.3195e-05, -5.78408e-05, -5.78026e-05]
  gravity bias YZ [-0.00101303, -0.00231498]
Six significant digits shown; full precision: sampled_dr_values.json.
```

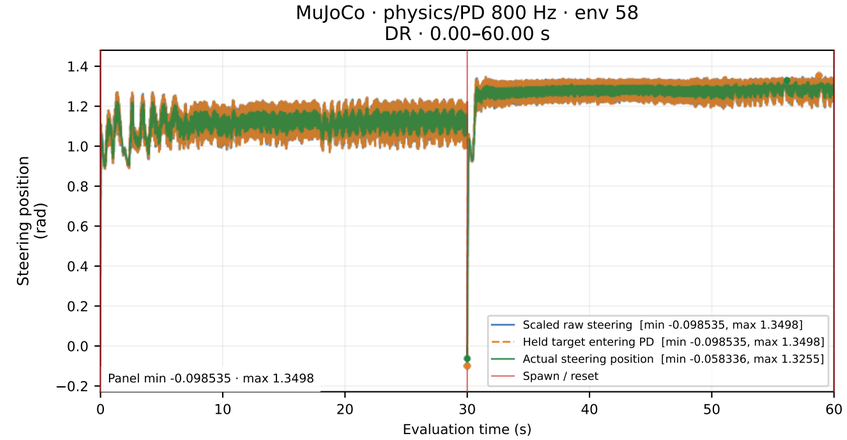
```
SAMPLED DR: env 57; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.00241, 1.02604, 1.00477, 1.18887]
  CoM XYZ mm [1.14903, 0.682835, -2.13213]
  mu_s [0.700521, 0.640867, 0.791209, 0.546287]
  mu_d [0.641329, 0.556841, 0.767936, 0.546287]
  Kp [steer,rear] [10.2201, 0]; Kd [0.189835, 0]
  spawn roll 0.0549298 rad; rate -0.229716 rad/s
  gyro bias XYZ rad/s [5.2601e-05, 6.57036e-05, 0.000187049]
  gravity bias YZ [-0.00175523, -9.17922e-05]
E2 [30.00, 60.00) s:
  mass x [1.08541, 0.924102, 1.11539, 1.05289]
  CoM XYZ mm [-3.29912, 4.5428, 1.20135]
  mu_s [0.559972, 0.71155, 0.640016, 0.631146]
  mu_d [0.502209, 0.71155, 0.513141, 0.631146]
  Kp [steer,rear] [10.8444, 0]; Kd [0.210072, 0]
  spawn roll 0.0469243 rad; rate -0.131273 rad/s
  gyro bias XYZ rad/s [-0.000102709, 0.00027766, -1.37505e-05]
  gravity bias YZ [0.00161779, 0.00214689]
Six significant digits shown; full precision: sampled_dr_values.json.
```

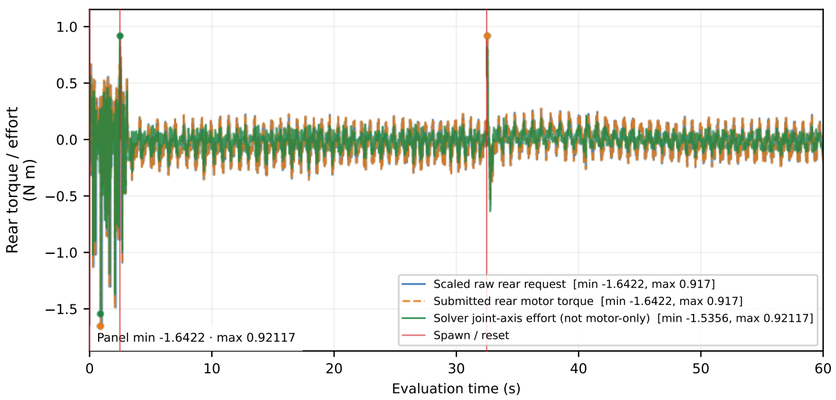
```
SAMPLED DR: env 57; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.00241, 1.02604, 1.00477, 1.18887]
  CoM XYZ mm [1.14903, 0.682835, -2.13213]
  mu_s [0.700521, 0.640867, 0.791209, 0.546287]
  mu_d [0.641329, 0.556841, 0.767936, 0.546287]
  Kp [steer,rear] [10.2201, 0]; Kd [0.189835, 0]
  spawn roll 0.0549298 rad; rate -0.229716 rad/s
  gyro bias XYZ rad/s [5.2601e-05, 6.57036e-05, 0.000187049]
  gravity bias YZ [-0.00175523, -9.17922e-05]
E2 [30.00, 60.00) s:
  mass x [1.08541, 0.924102, 1.11539, 1.05289]
  CoM XYZ mm [-3.29912, 4.5428, 1.20135]
  mu_s [0.559972, 0.71155, 0.640016, 0.631146]
  mu_d [0.502209, 0.71155, 0.513141, 0.631146]
  Kp [steer,rear] [10.8444, 0]; Kd [0.210072, 0]
  spawn roll 0.0469243 rad; rate -0.131273 rad/s
  gyro bias XYZ rad/s [-0.000102709, 0.00027766, -1.37505e-05]
  gravity bias YZ [0.00161779, 0.00214689]
Six significant digits shown; full precision: sampled_dr_values.json.
```

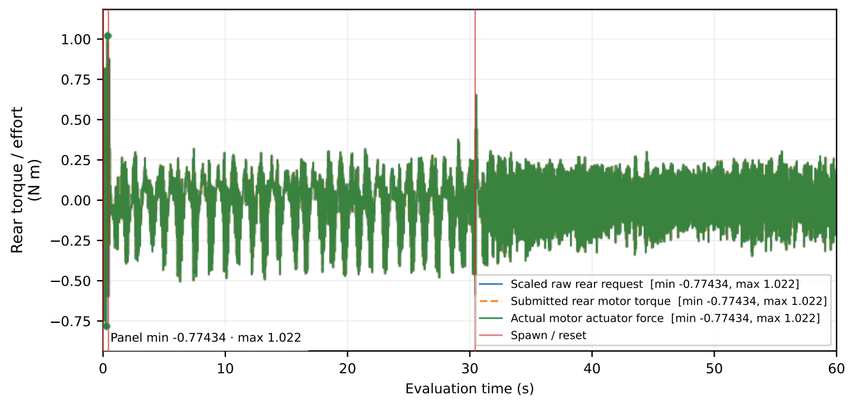
model_2500 · Demand versus actuation · env 58 · full 60 s

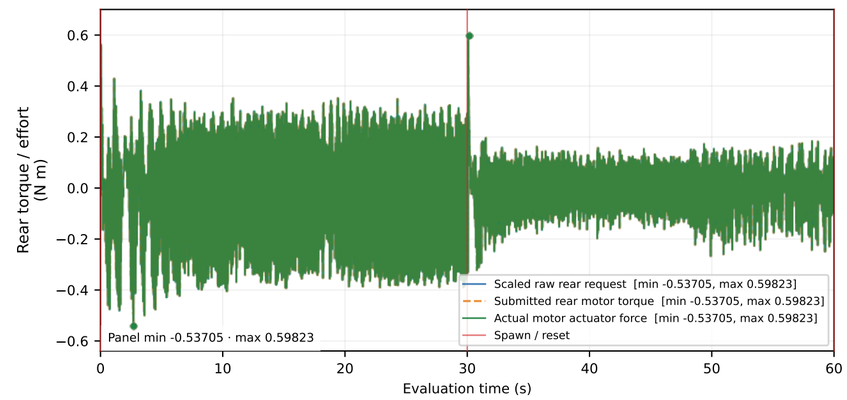












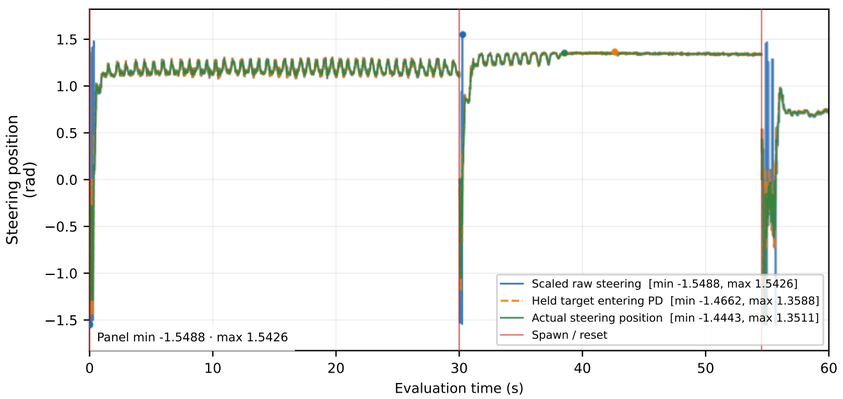
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

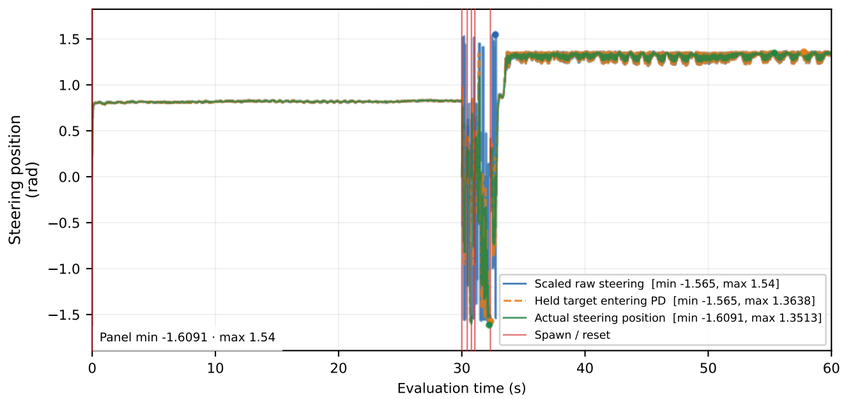
```
SAMPLED DR: env 58; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 2.48) s:
  mass x [1.18769, 1.06116, 1.01873, 0.96856]
  CoM XYZ mm [4.84164, 4.93649, -1.37092]
  mu_s [0.531594, 0.547743, 0.703711, 0.628821]
  mu_d [0.531594, 0.547743, 0.703711, 0.628821]
  Kp [steer,rear] [10.3875, 0]; Kd [0.217806, 0]
  spawn roll -0.0226035 rad; rate 0.0441529 rad/s
  gyro bias XYZ rad/s [0.000287679, 0.000107617, -1.96523e-06]
  gravity bias YZ [-0.00102433, -0.00207661]
E2 [2.48, 32.48) s:
  mass x [1.18104, 0.895951, 0.89566, 1.1683]
  CoM XYZ mm [1.12556, -1.30424, 3.61978]
  mu_s [0.673378, 0.681711, 0.602694, 0.647391]
  mu_d [0.673378, 0.611756, 0.602694, 0.647391]
  Kp [steer,rear] [9.22221, 0]; Kd [0.186914, 0]
  spawn roll -0.0732774 rad; rate -0.210146 rad/s
  gyro bias XYZ rad/s [0.000170324, -0.000262011, 0.000160037]
  gravity bias YZ [-0.00118923, 0.000579164]
E3 [32.48, 60.00) s:
  mass x [1.05184, 0.975285, 0.958412, 0.853304]
  CoM XYZ mm [-4.75869, -2.26922, -2.61918]
  mu_s [0.659445, 0.751971, 0.681711, 0.584424]
  mu_d [0.547619, 0.541224, 0.611756, 0.584424]
  Kp [steer,rear] [9.22393, 0]; Kd [0.1872, 0]
  spawn roll 0.0823566 rad; rate 0.196909 rad/s
  gyro bias XYZ rad/s [9.81637e-05, -0.000120994, 3.03913e-05]
  gravity bias YZ [3.49841e-06, 0.00121471]
Six significant digits shown; full precision: sampled_dr_values.json.
```

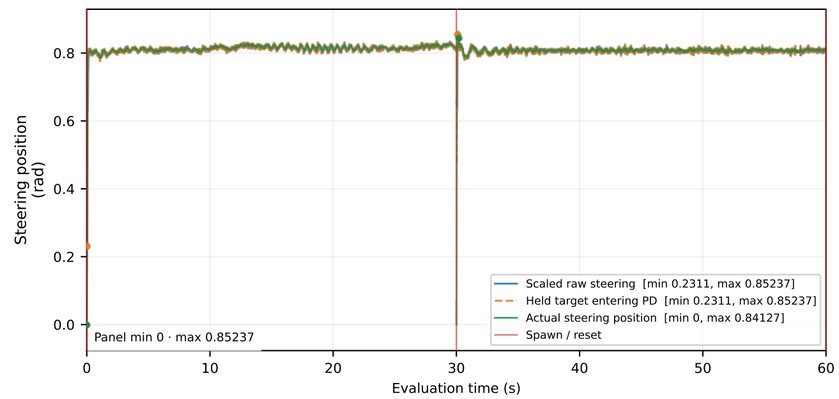
```
SAMPLED DR: env 58; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.44) s:
  mass x [1.13399, 1.03862, 0.915545, 0.817181]
  CoM XYZ mm [4.73654, 0.964717, 2.90263]
  mu_s [0.541939, 0.541939, 0.732187, 0.749678]
  mu_d [0.534359, 0.534359, 0.631664, 0.741429]
  Kp [steer,rear] [9.37998, 0]; Kd [0.19139, 0]
  spawn roll 0.0542045 rad; rate -0.00291911 rad/s
  gyro bias XYZ rad/s [0.000350847, -0.000267596, 6.51149e-05]
  gravity bias YZ [-0.00137824, -3.96436e-05]
E2 [0.44, 30.44) s:
  mass x [1.18822, 0.970738, 0.849511, 1.0935]
  CoM XYZ mm [3.21299, 0.60756, -0.358027]
  mu_s [0.538976, 0.633024, 0.559972, 0.700521]
  mu_d [0.538976, 0.568172, 0.502209, 0.641329]
  Kp [steer,rear] [10.7819, 0]; Kd [0.185642, 0]
  spawn roll 0.0740205 rad; rate 0.0585414 rad/s
  gyro bias XYZ rad/s [0.000159138, -0.000206941, -0.00018457]
  gravity bias YZ [0.00128601, -0.00104469]
E3 [30.44, 60.00) s:
  mass x [0.937507, 0.860351, 1.10632, 1.16687]
  CoM XYZ mm [2.00202, -2.7896, 0.884465]
  mu_s [0.723429, 0.727426, 0.637675, 0.631146]
  mu_d [0.723429, 0.606358, 0.637675, 0.631146]
  Kp [steer,rear] [9.60298, 0]; Kd [0.20182, 0]
  spawn roll 0.0333651 rad; rate 0.233172 rad/s
  gyro bias XYZ rad/s [-5.82319e-05, -0.000232986, 0.00010145]
  gravity bias YZ [-0.00382904, 0.000744485]
Six significant digits shown; full precision: sampled_dr_values.json.
```

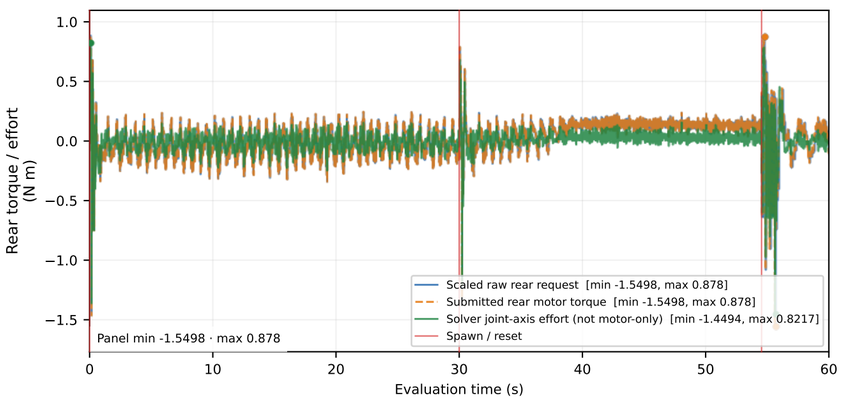
```
SAMPLED DR: env 58; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.13399, 1.03862, 0.915545, 0.817181]
  CoM XYZ mm [4.73654, 0.964717, 2.90263]
  mu_s [0.541939, 0.541939, 0.732187, 0.749678]
  mu_d [0.534359, 0.534359, 0.631664, 0.741429]
  Kp [steer,rear] [9.37998, 0]; Kd [0.19139, 0]
  spawn roll 0.0542045 rad; rate -0.00291911 rad/s
  gyro bias XYZ rad/s [0.000350847, -0.000267596, 6.51149e-05]
  gravity bias YZ [-0.00137824, -3.96436e-05]
E2 [30.00, 60.00) s:
  mass x [0.95846, 1.07212, 0.934933, 1.07707]
  CoM XYZ mm [-2.64052, -4.04246, 3.64696]
  mu_s [0.71008, 0.637675, 0.559972, 0.666375]
  mu_d [0.59371, 0.637675, 0.502209, 0.519145]
  Kp [steer,rear] [9.89135, 0]; Kd [0.209232, 0]
  spawn roll 0.00937321 rad; rate -0.152752 rad/s
  gyro bias XYZ rad/s [-5.94112e-06, -0.000154327, 1.45148e-05]
  gravity bias YZ [-0.0028596, -0.00716214]
Six significant digits shown; full precision: sampled_dr_values.json.
```

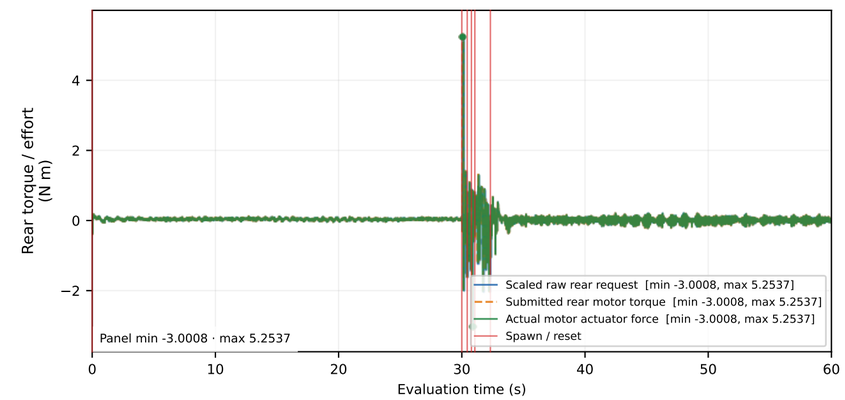
model_2500 · Demand versus actuation · env 59 · full 60 s

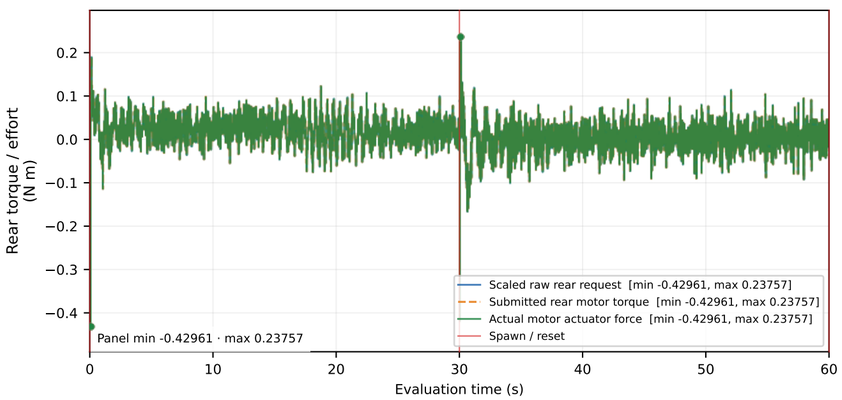












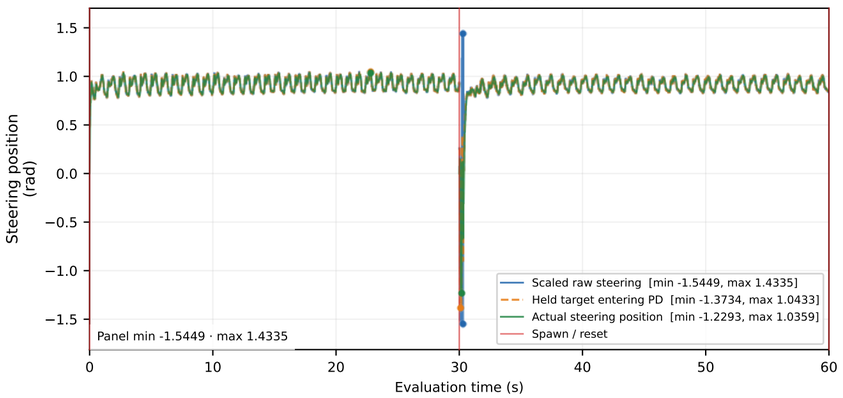
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

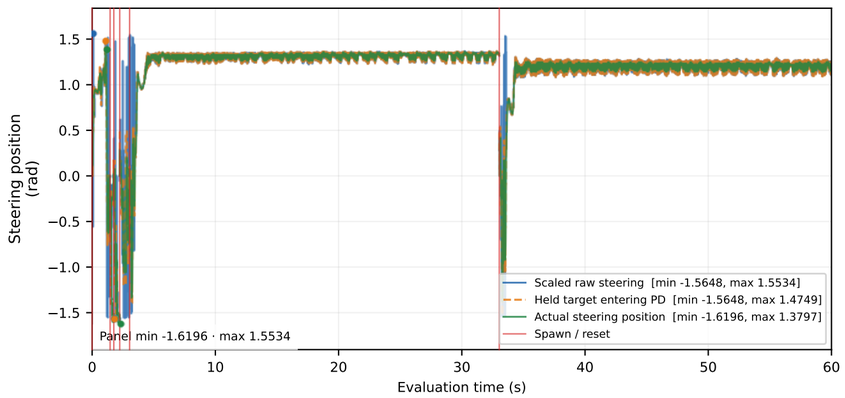
```
SAMPLED DR: env 59; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.925089, 0.860388, 1.05125, 0.952876]
  CoM XYZ mm [-4.30211, -1.42686, 3.08357]
  mu_s [0.584424, 0.741039, 0.576972, 0.647391]
  mu_d [0.584424, 0.614405, 0.576972, 0.647391]
  Kp [steer,rear] [10.6092, 0]; Kd [0.218709, 0]
  spawn roll -0.0933618 rad; rate -0.0975262 rad/s
  gyro bias XYZ rad/s [0.000129794, 0.000176123, -0.000140878]
  gravity bias YZ [-0.00166934, -0.00157474]
E2 [30.00, 54.54) s:
  mass x [0.851888, 1.17, 1.1751, 0.853816]
  CoM XYZ mm [-4.69279, -2.17381, -1.16156]
  mu_s [0.592046, 0.576972, 0.726175, 0.539956]
  mu_d [0.534947, 0.576972, 0.558574, 0.539956]
  Kp [steer,rear] [10.4138, 0]; Kd [0.188609, 0]
  spawn roll -0.0725461 rad; rate 0.0849203 rad/s
  gyro bias XYZ rad/s [-0.000238301, 0.000195208, 1.21025e-05]
  gravity bias YZ [0.00193001, 0.00228548]
E3 [54.54, 60.00) s:
  mass x [1.16187, 0.929548, 1.05204, 1.11966]
  CoM XYZ mm [-4.33715, 4.26448, 0.252157]
  mu_s [0.579974, 0.774409, 0.674963, 0.751971]
  mu_d [0.579974, 0.561095, 0.601294, 0.541224]
  Kp [steer,rear] [10.5803, 0]; Kd [0.203757, 0]
  spawn roll -0.0341186 rad; rate 0.134961 rad/s
  gyro bias XYZ rad/s [-0.000160216, 0.000208338, -9.83976e-05]
  gravity bias YZ [-0.000390665, 0.00344456]
Six significant digits shown; full precision: sampled_dr_values.json.
```

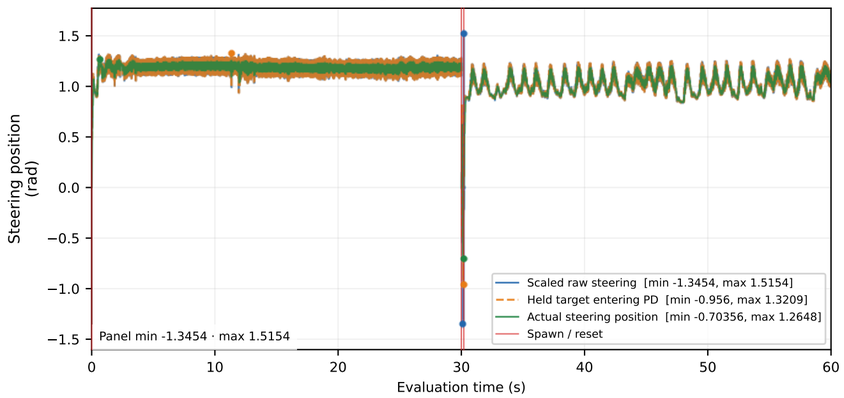
```
SAMPLED DR: env 59; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.17741, 0.943768, 1.11392, 1.03651]
  CoM XYZ mm [-2.05671, 4.22726, 3.69332]
  mu_s [0.538434, 0.791209, 0.528253, 0.669571]
  mu_d [0.538434, 0.767936, 0.528253, 0.669571]
  Kp [steer,rear] [9.44905, 0]; Kd [0.207236, 0]
  spawn roll 0.0303946 rad; rate -0.0764858 rad/s
  gyro bias XYZ rad/s [-6.64184e-05, -8.74588e-05, -0.000345704]
  gravity bias YZ [-0.000220678, 0.00328691]
E2 [30.00, 30.44) s:
  mass x [0.849147, 0.951963, 1.03885, 0.821198]
  CoM XYZ mm [3.0137, 4.69018, 3.77614]
  mu_s [0.611239, 0.723429, 0.559972, 0.546287]
  mu_d [0.611239, 0.723429, 0.502209, 0.546287]
  Kp [steer,rear] [9.52311, 0]; Kd [0.192271, 0]
  spawn roll 0.0423355 rad; rate 0.080037 rad/s
  gyro bias XYZ rad/s [0.000245426, -0.000110354, -4.2951e-05]
  gravity bias YZ [-0.000782561, -0.00261552]
E3 [30.44, 30.78) s:
  mass x [0.976334, 0.801364, 0.805555, 0.923883]
  CoM XYZ mm [1.72599, 4.78828, 4.04951]
  mu_s [0.732187, 0.666375, 0.728342, 0.559972]
  mu_d [0.631664, 0.519145, 0.728342, 0.502209]
  Kp [steer,rear] [9.45019, 0]; Kd [0.213617, 0]
  spawn roll -0.0546403 rad; rate 0.241166 rad/s
  gyro bias XYZ rad/s [0.000162156, -3.30168e-05, 0.000239966]
  gravity bias YZ [-0.00083723, 0.00312913]
E4 [30.78, 31.06) s:
  mass x [1.14196, 1.09368, 1.03432, 0.988288]
  CoM XYZ mm [-4.41055, -2.45491, -1.67236]
  mu_s [0.669571, 0.568073, 0.541939, 0.616244]
  mu_d [0.669571, 0.568073, 0.534359, 0.586498]
  Kp [steer,rear] [10.1747, 0]; Kd [0.187399, 0]
  spawn roll 0.0736781 rad; rate 0.088248 rad/s
  gyro bias XYZ rad/s [4.33695e-05, 0.000223197, -0.000219348]
  gravity bias YZ [0.00175509, 2.87761e-05]
E5 [31.06, 32.32) s:
  mass x [1.19659, 1.19068, 0.85643, 0.854471]
  CoM XYZ mm [-0.689385, 1.85821, 0.480161]
  mu_s [0.71155, 0.640867, 0.69316, 0.69316]
  mu_d [0.71155, 0.556841, 0.69316, 0.69316]
  Kp [steer,rear] [9.59194, 0]; Kd [0.20081, 0]
  spawn roll 0.0393102 rad; rate -0.175448 rad/s
  gyro bias XYZ rad/s [-0.000181485, 0.000140642, 0.00026491]
  gravity bias YZ [-0.00412691, -0.0017282]
E6 [32.32, 60.00) s:
  mass x [0.8593, 1.07949, 1.09295, 1.18038]
  CoM XYZ mm [-1.27312, -3.5976, -0.983021]
  mu_s [0.704749, 0.723429, 0.723429, 0.633024]
  mu_d [0.541926, 0.723429, 0.723429, 0.568172]
  Kp [steer,rear] [9.92248, 0]; Kd [0.192594, 0]
  spawn roll -0.0607455 rad; rate 0.186155 rad/s
  gyro bias XYZ rad/s [-0.000243478, -0.000226265, 0.000304189]
  gravity bias YZ [0.00200882, 0.000407443]
Six significant digits shown; full precision: sampled_dr_values.json.
```

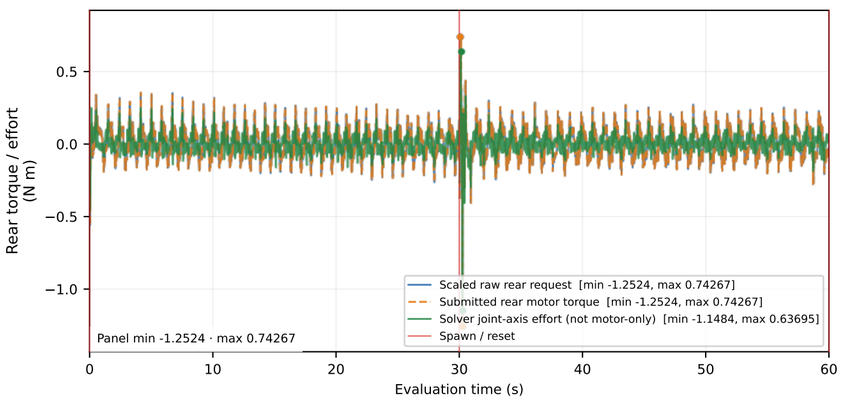
```
SAMPLED DR: env 59; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.17741, 0.943768, 1.11392, 1.03651]
  CoM XYZ mm [-2.05671, 4.22726, 3.69332]
  mu_s [0.538434, 0.791209, 0.528253, 0.669571]
  mu_d [0.538434, 0.767936, 0.528253, 0.669571]
  Kp [steer,rear] [9.44905, 0]; Kd [0.207236, 0]
  spawn roll 0.0303946 rad; rate -0.0764858 rad/s
  gyro bias XYZ rad/s [-6.64184e-05, -8.74588e-05, -0.000345704]
  gravity bias YZ [-0.000220678, 0.00328691]
E2 [30.00, 60.00) s:
  mass x [0.849147, 0.951963, 1.03885, 0.821198]
  CoM XYZ mm [3.0137, 4.69018, 3.77614]
  mu_s [0.611239, 0.723429, 0.559972, 0.546287]
  mu_d [0.611239, 0.723429, 0.502209, 0.546287]
  Kp [steer,rear] [9.52311, 0]; Kd [0.192271, 0]
  spawn roll 0.0423355 rad; rate 0.080037 rad/s
  gyro bias XYZ rad/s [0.000245426, -0.000110354, -4.2951e-05]
  gravity bias YZ [-0.000782561, -0.00261552]
Six significant digits shown; full precision: sampled_dr_values.json.
```

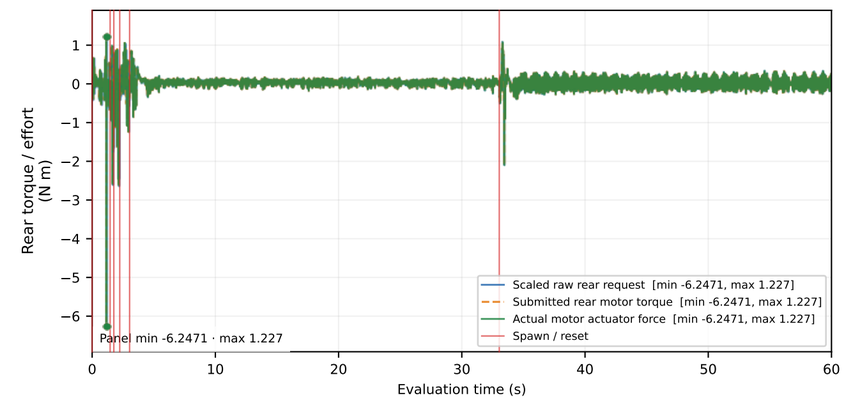
model_2500 · Demand versus actuation · env 60 · full 60 s

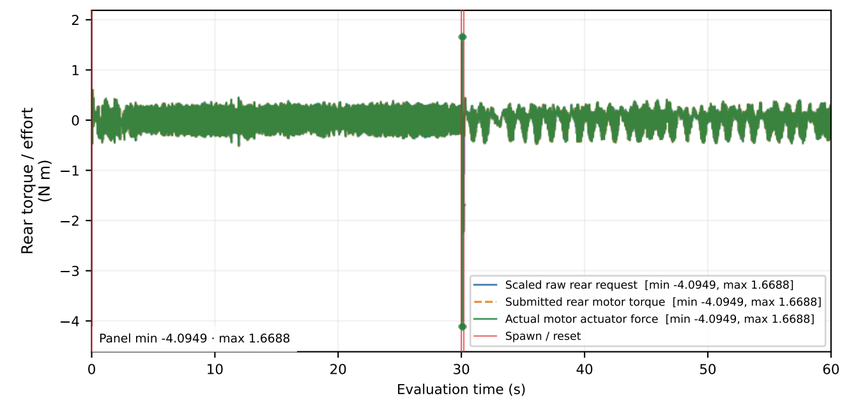












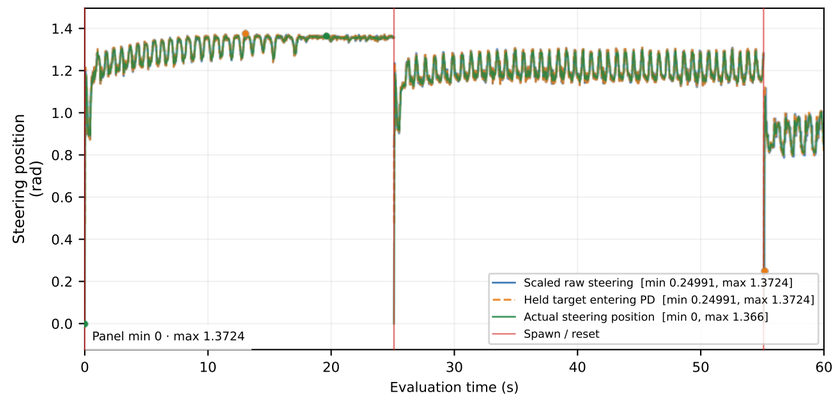
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

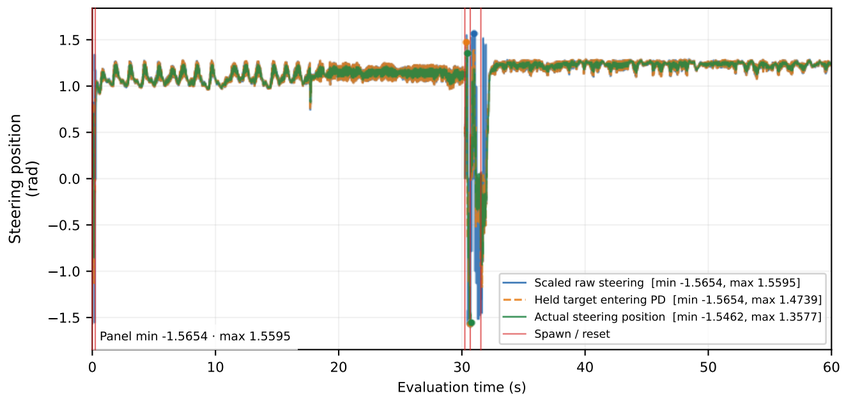
```
SAMPLED DR: env 60; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.900482, 1.10467, 0.935847, 1.18135]
  CoM XYZ mm [-3.37732, 2.48318, 4.58559]
  mu_s [0.547743, 0.703711, 0.539956, 0.693205]
  mu_d [0.547743, 0.703711, 0.539956, 0.693205]
  Kp [steer,rear] [10.9047, 0]; Kd [0.182122, 0]
  spawn roll -0.00452483 rad; rate 0.138263 rad/s
  gyro bias XYZ rad/s [0.000207397, 1.01754e-05, -0.000280071]
  gravity bias YZ [0.00226908, -0.000901314]
E2 [30.00, 60.00) s:
  mass x [0.800306, 1.12363, 0.945733, 0.825305]
  CoM XYZ mm [4.75881, 2.31119, 3.9894]
  mu_s [0.533455, 0.539956, 0.602694, 0.531594]
  mu_d [0.533455, 0.539956, 0.602694, 0.531594]
  Kp [steer,rear] [9.1609, 0]; Kd [0.207236, 0]
  spawn roll -0.076851 rad; rate 0.24644 rad/s
  gyro bias XYZ rad/s [0.000419029, 0.000123381, -0.000165383]
  gravity bias YZ [-0.000913082, 0.00265855]
Six significant digits shown; full precision: sampled_dr_values.json.
```

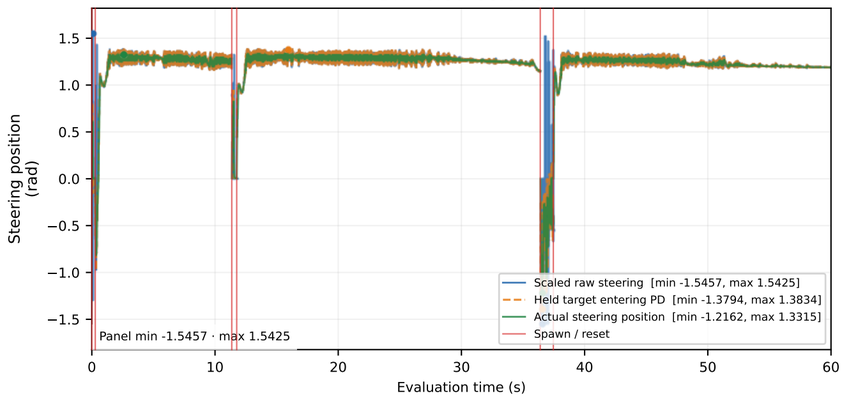
```
SAMPLED DR: env 60; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.46) s:
  mass x [0.863996, 1.03438, 1.12953, 0.918904]
  CoM XYZ mm [1.849, -0.361158, 3.91357]
  mu_s [0.559972, 0.538976, 0.568073, 0.669571]
  mu_d [0.502209, 0.538976, 0.568073, 0.669571]
  Kp [steer,rear] [9.48504, 0]; Kd [0.193822, 0]
  spawn roll 0.0180142 rad; rate 0.112822 rad/s
  gyro bias XYZ rad/s [-0.000120723, -7.37988e-05, 0.000199168]
  gravity bias YZ [-0.000972708, -0.00177809]
E2 [1.46, 1.76) s:
  mass x [0.917893, 1.07381, 0.843499, 0.825065]
  CoM XYZ mm [-2.4307, 0.779571, 4.62948]
  mu_s [0.538976, 0.727426, 0.640016, 0.528253]
  mu_d [0.538976, 0.606358, 0.513141, 0.528253]
  Kp [steer,rear] [10.581, 0]; Kd [0.194328, 0]
  spawn roll -0.0303228 rad; rate -0.212432 rad/s
  gyro bias XYZ rad/s [-0.000131211, -0.000202156, -0.000439974]
  gravity bias YZ [-0.00121586, -0.000284463]
E3 [1.76, 2.24) s:
  mass x [1.1042, 1.1535, 0.911097, 0.820985]
  CoM XYZ mm [-1.70354, -1.84323, -2.08665]
  mu_s [0.669571, 0.631146, 0.71008, 0.633024]
  mu_d [0.669571, 0.631146, 0.59371, 0.568172]
  Kp [steer,rear] [10.4149, 0]; Kd [0.204249, 0]
  spawn roll -0.0748894 rad; rate -0.0321481 rad/s
  gyro bias XYZ rad/s [7.76e-05, 0.000126532, 0.000184117]
  gravity bias YZ [-0.00145927, -0.00195968]
E4 [2.24, 3.04) s:
  mass x [0.874368, 1.03209, 0.827202, 1.11763]
  CoM XYZ mm [1.8891, 2.49642, -1.97656]
  mu_s [0.616244, 0.640016, 0.538976, 0.633024]
  mu_d [0.586498, 0.513141, 0.538976, 0.568172]
  Kp [steer,rear] [9.23719, 0]; Kd [0.219542, 0]
  spawn roll -0.0172836 rad; rate 0.236369 rad/s
  gyro bias XYZ rad/s [2.82859e-06, -4.40666e-05, -0.000137117]
  gravity bias YZ [0.00154996, -0.00150121]
E5 [3.04, 33.04) s:
  mass x [1.17026, 0.851044, 1.05302, 1.05592]
  CoM XYZ mm [-1.43378, -4.5833, -2.38461]
  mu_s [0.666375, 0.728342, 0.723429, 0.538976]
  mu_d [0.519145, 0.728342, 0.723429, 0.538976]
  Kp [steer,rear] [10.7656, 0]; Kd [0.18096, 0]
  spawn roll -0.092104 rad; rate 0.219224 rad/s
  gyro bias XYZ rad/s [-0.000116992, -0.000102171, -0.000155466]
  gravity bias YZ [-0.0035581, 0.00140918]
E6 [33.04, 60.00) s:
  mass x [0.989861, 1.10855, 1.03997, 1.10907]
  CoM XYZ mm [2.49538, -0.314775, 1.84938]
  mu_s [0.700521, 0.71008, 0.700521, 0.727426]
  mu_d [0.641329, 0.59371, 0.641329, 0.606358]
  Kp [steer,rear] [9.94775, 0]; Kd [0.187628, 0]
  spawn roll -0.0658516 rad; rate 0.232003 rad/s
  gyro bias XYZ rad/s [-0.0003429, -3.6408e-05, -9.68509e-05]
  gravity bias YZ [0.00214218, -0.000116128]
Six significant digits shown; full precision: sampled_dr_values.json.
```

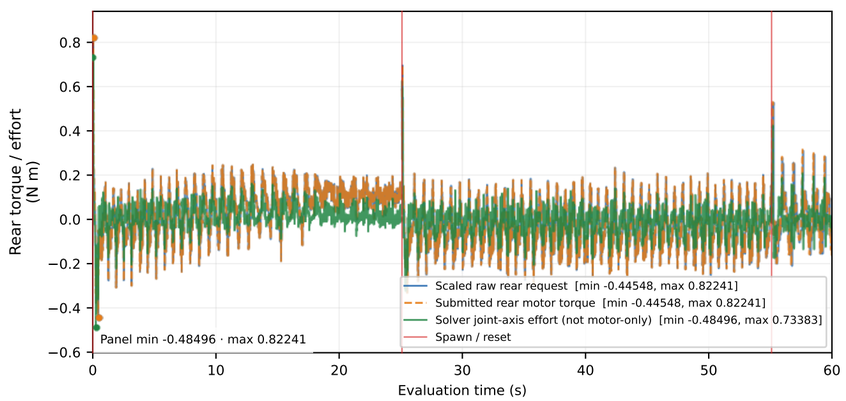
```
SAMPLED DR: env 60; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.863996, 1.03438, 1.12953, 0.918904]
  CoM XYZ mm [1.849, -0.361158, 3.91357]
  mu_s [0.559972, 0.538976, 0.568073, 0.669571]
  mu_d [0.502209, 0.538976, 0.568073, 0.669571]
  Kp [steer,rear] [9.48504, 0]; Kd [0.193822, 0]
  spawn roll 0.0180142 rad; rate 0.112822 rad/s
  gyro bias XYZ rad/s [-0.000120723, -7.37988e-05, 0.000199168]
  gravity bias YZ [-0.000972708, -0.00177809]
E2 [30.00, 30.20) s:
  mass x [0.904864, 1.15044, 1.1004, 1.0186]
  CoM XYZ mm [0.1525, -3.43279, -3.12837]
  mu_s [0.791209, 0.637675, 0.559972, 0.538976]
  mu_d [0.767936, 0.637675, 0.502209, 0.538976]
  Kp [steer,rear] [9.24463, 0]; Kd [0.18092, 0]
  spawn roll 0.0903796 rad; rate -0.21792 rad/s
  gyro bias XYZ rad/s [-9.52816e-05, 1.26954e-05, 1.48451e-07]
  gravity bias YZ [-0.00200569, -0.00128185]
E3 [30.20, 60.00) s:
  mass x [1.00374, 0.883257, 1.08934, 0.883413]
  CoM XYZ mm [-1.56465, 1.33965, -4.50909]
  mu_s [0.71155, 0.791209, 0.640867, 0.640867]
  mu_d [0.71155, 0.767936, 0.556841, 0.556841]
  Kp [steer,rear] [9.87806, 0]; Kd [0.192029, 0]
  spawn roll 0.0277374 rad; rate -0.123222 rad/s
  gyro bias XYZ rad/s [7.64843e-05, 0.000117377, -9.60973e-05]
  gravity bias YZ [-0.002893, 0.00274328]
Six significant digits shown; full precision: sampled_dr_values.json.
```

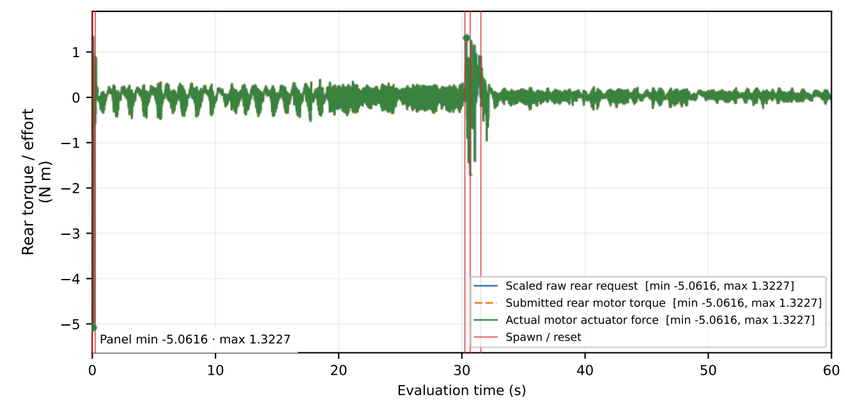
model_2500 · Demand versus actuation · env 61 · full 60 s

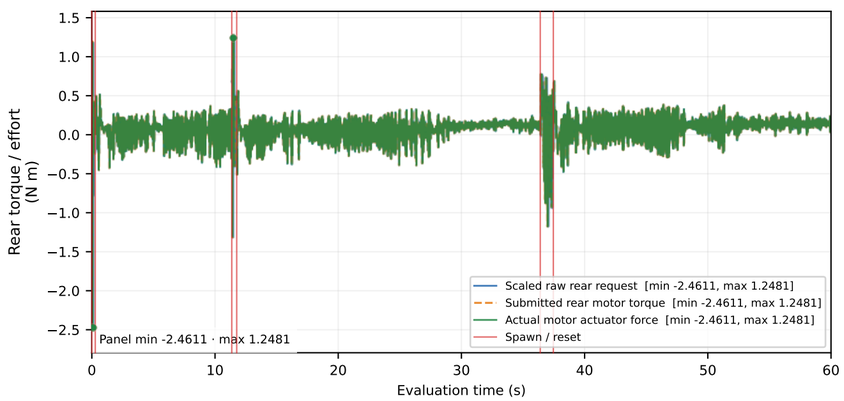












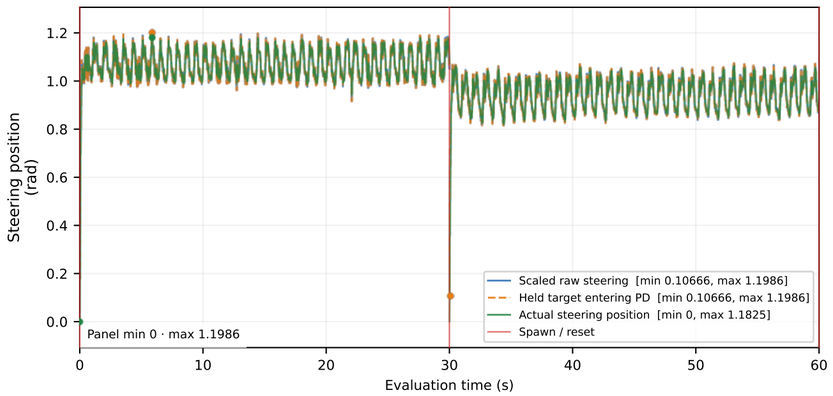
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

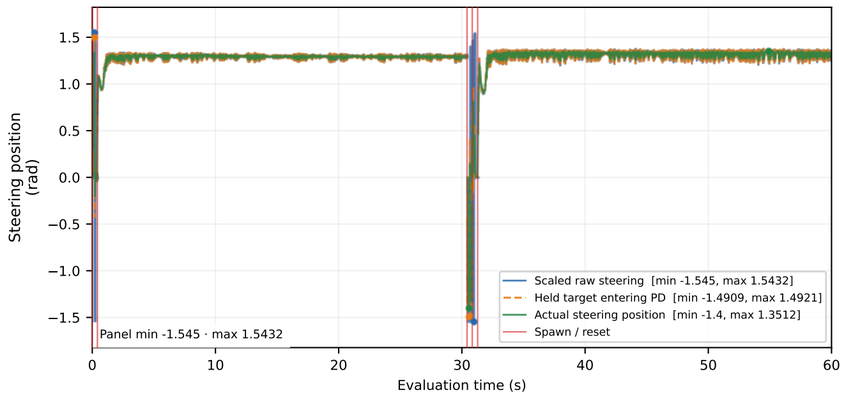
```
SAMPLED DR: env 61; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 25.10) s:
  mass x [1.04593, 1.00667, 0.908218, 0.97081]
  CoM XYZ mm [-3.09935, -2.72854, 2.47374]
  mu_s [0.532645, 0.576972, 0.681711, 0.693205]
  mu_d [0.532645, 0.576972, 0.611756, 0.693205]
  Kp [steer,rear] [10.6263, 0]; Kd [0.202767, 0]
  spawn roll 0.0778073 rad; rate 0.198054 rad/s
  gyro bias XYZ rad/s [0.000370692, -0.000214087, 0.000211219]
  gravity bias YZ [0.000401441, -0.00307435]
E2 [25.10, 55.10) s:
  mass x [0.987953, 1.03808, 0.810178, 1.14461]
  CoM XYZ mm [2.43007, -0.791136, -0.0987351]
  mu_s [0.787792, 0.533455, 0.785466, 0.592046]
  mu_d [0.617134, 0.533455, 0.52258, 0.534947]
  Kp [steer,rear] [9.74718, 0]; Kd [0.192989, 0]
  spawn roll 0.0754349 rad; rate 0.18036 rad/s
  gyro bias XYZ rad/s [-0.000316195, 0.000214386, -6.07899e-05]
  gravity bias YZ [0.00331178, -0.00432885]
E3 [55.10, 60.00) s:
  mass x [0.98811, 1.0164, 1.02224, 1.03655]
  CoM XYZ mm [-3.74888, 2.32713, 1.3504]
  mu_s [0.785466, 0.751971, 0.598343, 0.598343]
  mu_d [0.52258, 0.541224, 0.598343, 0.598343]
  Kp [steer,rear] [9.23459, 0]; Kd [0.208641, 0]
  spawn roll 0.0454457 rad; rate -0.126205 rad/s
  gyro bias XYZ rad/s [0.000118363, 0.000193584, 6.91042e-05]
  gravity bias YZ [-0.00215871, -0.000615227]
Six significant digits shown; full precision: sampled_dr_values.json.
```

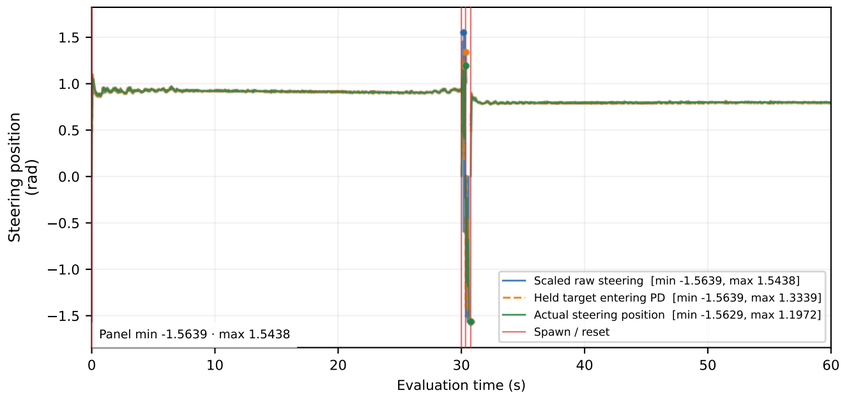
```
SAMPLED DR: env 61; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.24) s:
  mass x [0.925073, 0.889727, 0.872804, 1.14407]
  CoM XYZ mm [3.57542, -3.96438, -2.1091]
  mu_s [0.538976, 0.71008, 0.568073, 0.749678]
  mu_d [0.538976, 0.59371, 0.568073, 0.741429]
  Kp [steer,rear] [9.87736, 0]; Kd [0.209599, 0]
  spawn roll 0.0844232 rad; rate 0.0744198 rad/s
  gyro bias XYZ rad/s [0.000148847, -0.000162669, 0.000397757]
  gravity bias YZ [-4.9864e-05, -0.000578105]
E2 [0.24, 30.24) s:
  mass x [1.05433, 1.16247, 0.914729, 0.93478]
  CoM XYZ mm [-0.741662, 0.0209448, -0.15809]
  mu_s [0.732187, 0.723429, 0.748289, 0.736077]
  mu_d [0.631664, 0.723429, 0.689499, 0.699455]
  Kp [steer,rear] [9.65851, 0]; Kd [0.187563, 0]
  spawn roll 0.0733542 rad; rate 0.128861 rad/s
  gyro bias XYZ rad/s [-0.000183336, -0.000173914, 0.000224317]
  gravity bias YZ [0.00101996, -0.00164359]
E3 [30.24, 30.68) s:
  mass x [0.920135, 0.923472, 1.10074, 0.844097]
  CoM XYZ mm [-4.73483, 4.79922, -4.08565]
  mu_s [0.733515, 0.640867, 0.704749, 0.611239]
  mu_d [0.558392, 0.556841, 0.541926, 0.611239]
  Kp [steer,rear] [9.80817, 0]; Kd [0.210265, 0]
  spawn roll 0.0999161 rad; rate -0.15168 rad/s
  gyro bias XYZ rad/s [1.77661e-05, 0.000285011, -8.55378e-05]
  gravity bias YZ [0.000725614, -0.00029702]
E4 [30.68, 31.54) s:
  mass x [0.800301, 1.07805, 0.995308, 0.853685]
  CoM XYZ mm [-2.49548, -3.73241, -4.03448]
  mu_s [0.732187, 0.736077, 0.568073, 0.700521]
  mu_d [0.631664, 0.699455, 0.568073, 0.641329]
  Kp [steer,rear] [10.8423, 0]; Kd [0.206484, 0]
  spawn roll 0.0193762 rad; rate -0.0981116 rad/s
  gyro bias XYZ rad/s [-7.14101e-06, -1.94999e-05, -0.000164435]
  gravity bias YZ [0.00283172, 0.00119335]
E5 [31.54, 60.00) s:
  mass x [1.15984, 1.16839, 0.875318, 1.0121]
  CoM XYZ mm [-4.58364, -1.56581, 0.789727]
  mu_s [0.723429, 0.541939, 0.727426, 0.611239]
  mu_d [0.723429, 0.534359, 0.606358, 0.611239]
  Kp [steer,rear] [9.18896, 0]; Kd [0.196851, 0]
  spawn roll -0.0508898 rad; rate 0.101941 rad/s
  gyro bias XYZ rad/s [0.000356061, 1.52802e-05, -2.66949e-05]
  gravity bias YZ [-0.000579852, 2.96125e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

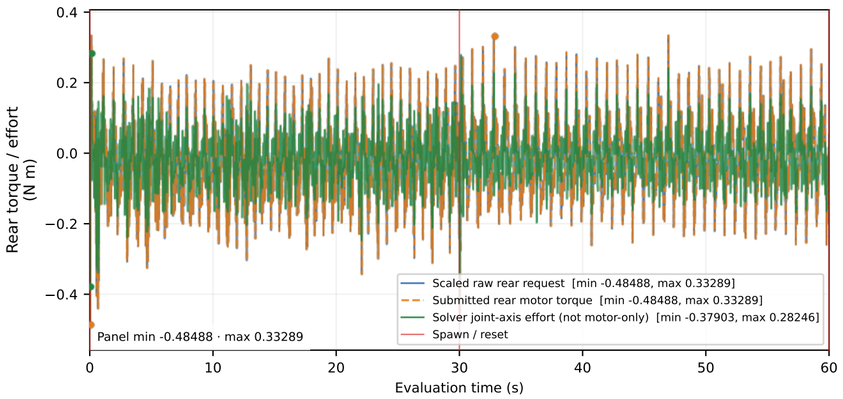
```
SAMPLED DR: env 61; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [0.925073, 0.889727, 0.872804, 1.14407]
  CoM XYZ mm [3.57542, -3.96438, -2.1091]
  mu_s [0.538976, 0.71008, 0.568073, 0.749678]
  mu_d [0.538976, 0.59371, 0.568073, 0.741429]
  Kp [steer,rear] [9.87736, 0]; Kd [0.209599, 0]
  spawn roll 0.0844232 rad; rate 0.0744198 rad/s
  gyro bias XYZ rad/s [0.000148847, -0.000162669, 0.000397757]
  gravity bias YZ [-4.9864e-05, -0.000578105]
E2 [0.28, 11.36) s:
  mass x [0.84446, 1.05807, 1.1099, 0.900103]
  CoM XYZ mm [0.461385, -4.53228, -0.61191]
  mu_s [0.71155, 0.732187, 0.538434, 0.69316]
  mu_d [0.71155, 0.631664, 0.538434, 0.69316]
  Kp [steer,rear] [10.2815, 0]; Kd [0.21241, 0]
  spawn roll -0.0540162 rad; rate -0.00341233 rad/s
  gyro bias XYZ rad/s [5.52317e-05, 0.000117328, -0.000106188]
  gravity bias YZ [0.0036445, 0.00104761]
E3 [11.36, 11.76) s:
  mass x [0.976073, 0.952244, 1.14694, 0.947814]
  CoM XYZ mm [-4.80501, 2.48823, 3.20485]
  mu_s [0.736077, 0.732187, 0.69316, 0.640016]
  mu_d [0.699455, 0.631664, 0.69316, 0.513141]
  Kp [steer,rear] [10.6694, 0]; Kd [0.18178, 0]
  spawn roll 0.0879026 rad; rate 0.217445 rad/s
  gyro bias XYZ rad/s [1.17678e-05, -0.000241729, 8.43254e-05]
  gravity bias YZ [0.00272294, 0.00291104]
E4 [11.76, 36.40) s:
  mass x [1.00683, 1.04085, 0.836539, 0.859487]
  CoM XYZ mm [1.19432, -4.86366, -1.23454]
  mu_s [0.69316, 0.640867, 0.736077, 0.732187]
  mu_d [0.69316, 0.556841, 0.699455, 0.631664]
  Kp [steer,rear] [9.68612, 0]; Kd [0.214206, 0]
  spawn roll -0.00215132 rad; rate 0.102432 rad/s
  gyro bias XYZ rad/s [-0.000253889, 0.000169451, -2.6825e-05]
  gravity bias YZ [-0.00186643, 0.00197135]
E5 [36.40, 37.46) s:
  mass x [1.19775, 1.03569, 1.05299, 0.845648]
  CoM XYZ mm [-0.448106, 3.76588, 4.11845]
  mu_s [0.71155, 0.640016, 0.733515, 0.640016]
  mu_d [0.71155, 0.513141, 0.558392, 0.513141]
  Kp [steer,rear] [9.41897, 0]; Kd [0.201391, 0]
  spawn roll -0.0667015 rad; rate -0.0125078 rad/s
  gyro bias XYZ rad/s [0.000240755, -0.000445043, -0.000135583]
  gravity bias YZ [-0.000442733, -0.00151677]
E6 [37.46, 60.00) s:
  mass x [1.1405, 0.954225, 0.936967, 0.98647]
  CoM XYZ mm [3.66501, -3.95416, -4.22501]
  mu_s [0.611239, 0.748289, 0.749678, 0.637675]
  mu_d [0.611239, 0.689499, 0.741429, 0.637675]
  Kp [steer,rear] [9.36258, 0]; Kd [0.216975, 0]
  spawn roll 0.0713233 rad; rate -0.235914 rad/s
  gyro bias XYZ rad/s [-4.11503e-05, -0.000301463, 0.000178844]
  gravity bias YZ [-0.000992252, -0.00333554]
Six significant digits shown; full precision: sampled_dr_values.json.
```

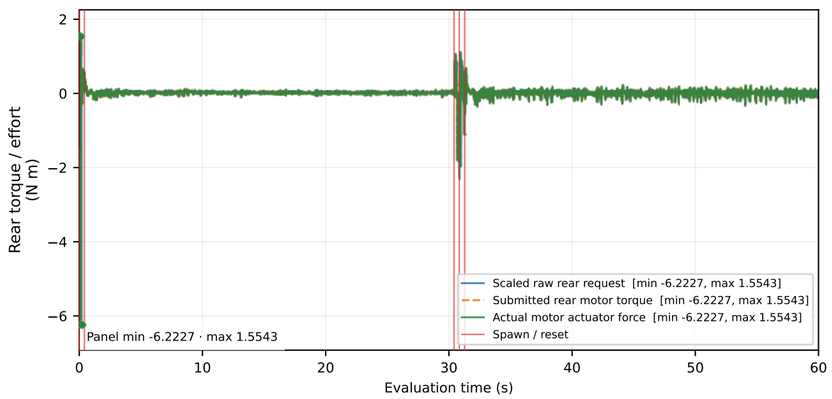
model_2500 · Demand versus actuation · env 62 · full 60 s

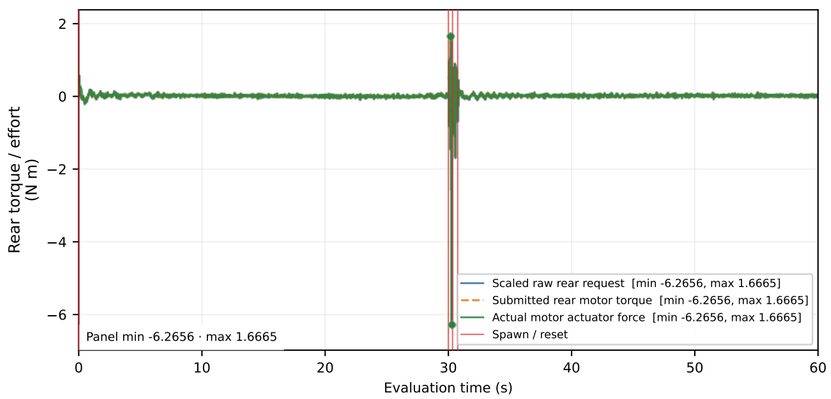












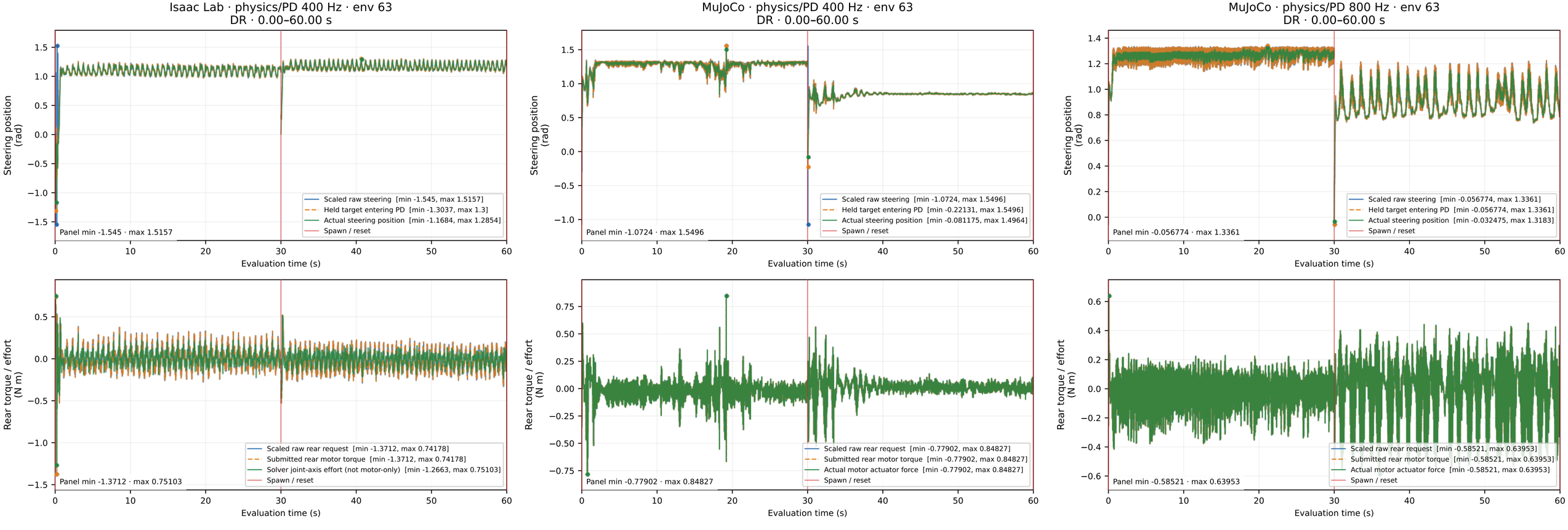
50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 62; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.01205, 0.910727, 1.1656, 1.1771]
  CoM XYZ mm [2.69511, 0.666026, -1.98537]
  mu_s [0.647391, 0.751971, 0.579974, 0.647391]
  mu_d [0.647391, 0.541224, 0.579974, 0.647391]
  Kp [steer,rear] [10.7399, 0]; Kd [0.186251, 0]
  spawn roll 0.00838556 rad; rate -0.0323896 rad/s
  gyro bias XYZ rad/s [0.000244845, -4.33467e-05, -9.10838e-05]
  gravity bias YZ [-0.00124117, -0.000246043]
E2 [30.00, 60.00) s:
  mass x [0.96663, 0.928456, 0.920087, 1.10597]
  CoM XYZ mm [0.830397, 2.0506, 0.429973]
  mu_s [0.531594, 0.794338, 0.532645, 0.774409]
  mu_d [0.531594, 0.526209, 0.532645, 0.561095]
  Kp [steer,rear] [9.85539, 0]; Kd [0.191222, 0]
  spawn roll 0.0110493 rad; rate -0.0630774 rad/s
  gyro bias XYZ rad/s [0.000312513, 4.54513e-05, -4.47451e-06]
  gravity bias YZ [0.000620301, -0.000547595]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 62; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.42) s:
  mass x [1.13543, 1.07686, 0.886436, 0.850165]
  CoM XYZ mm [-1.24931, 2.53889, 2.35188]
  mu_s [0.791209, 0.640016, 0.669571, 0.631146]
  mu_d [0.767936, 0.513141, 0.669571, 0.631146]
  Kp [steer,rear] [10.6484, 0]; Kd [0.201532, 0]
  spawn roll 0.059666 rad; rate 0.0300787 rad/s
  gyro bias XYZ rad/s [0.000176113, 0.000152281, -0.000194907]
  gravity bias YZ [0.00362988, 0.000554935]
E2 [0.42, 30.42) s:
  mass x [1.02572, 1.03521, 0.926834, 1.18918]
  CoM XYZ mm [-4.81969, -4.60651, 1.93796]
  mu_s [0.541939, 0.559972, 0.669571, 0.736077]
  mu_d [0.534359, 0.502209, 0.669571, 0.699455]
  Kp [steer,rear] [10.4723, 0]; Kd [0.205074, 0]
  spawn roll 0.0246067 rad; rate 0.244275 rad/s
  gyro bias XYZ rad/s [-0.000401951, 0.000148931, 6.53948e-06]
  gravity bias YZ [0.00162322, 0.00193128]
E3 [30.42, 30.84) s:
  mass x [1.10759, 1.09162, 0.873657, 1.17475]
  CoM XYZ mm [0.415086, -2.26386, -0.216222]
  mu_s [0.633024, 0.616244, 0.568073, 0.69316]
  mu_d [0.568172, 0.586498, 0.568073, 0.69316]
  Kp [steer,rear] [9.83548, 0]; Kd [0.215189, 0]
  spawn roll -0.0990009 rad; rate -0.0385385 rad/s
  gyro bias XYZ rad/s [-4.38903e-05, -5.06665e-06, 0.000156636]
  gravity bias YZ [-0.00315005, -0.000378556]
E4 [30.84, 31.28) s:
  mass x [1.01935, 0.854325, 0.956056, 0.900375]
  CoM XYZ mm [-4.14943, 1.34356, 3.07301]
  mu_s [0.748289, 0.538434, 0.631146, 0.637675]
  mu_d [0.689499, 0.538434, 0.631146, 0.637675]
  Kp [steer,rear] [9.42288, 0]; Kd [0.186403, 0]
  spawn roll 0.0655409 rad; rate 0.0403173 rad/s
  gyro bias XYZ rad/s [-0.000120343, -9.64318e-05, 0.000125775]
  gravity bias YZ [-0.00197305, -0.00103767]
E5 [31.28, 60.00) s:
  mass x [0.850267, 1.19756, 1.19215, 0.985976]
  CoM XYZ mm [0.438962, -4.99748, 3.52757]
  mu_s [0.71155, 0.541939, 0.733515, 0.69316]
  mu_d [0.71155, 0.534359, 0.558392, 0.69316]
  Kp [steer,rear] [10.2673, 0]; Kd [0.197007, 0]
  spawn roll -0.0176868 rad; rate 0.159026 rad/s
  gyro bias XYZ rad/s [-5.37826e-05, 2.1177e-05, -0.000322962]
  gravity bias YZ [-0.00347019, 0.000245845]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 62; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.13543, 1.07686, 0.886436, 0.850165]
  CoM XYZ mm [-1.24931, 2.53889, 2.35188]
  mu_s [0.791209, 0.640016, 0.669571, 0.631146]
  mu_d [0.767936, 0.513141, 0.669571, 0.631146]
  Kp [steer,rear] [10.6484, 0]; Kd [0.201532, 0]
  spawn roll 0.059666 rad; rate 0.0300787 rad/s
  gyro bias XYZ rad/s [0.000176113, 0.000152281, -0.000194907]
  gravity bias YZ [0.00362988, 0.000554935]
E2 [30.00, 30.34) s:
  mass x [1.10435, 0.832656, 1.02369, 0.801257]
  CoM XYZ mm [2.57136, -3.06342, -1.87806]
  mu_s [0.71008, 0.633024, 0.704749, 0.538434]
  mu_d [0.59371, 0.568172, 0.541926, 0.538434]
  Kp [steer,rear] [9.83748, 0]; Kd [0.197791, 0]
  spawn roll 0.0647167 rad; rate 0.14349 rad/s
  gyro bias XYZ rad/s [-7.07498e-06, 0.000204651, -0.000314145]
  gravity bias YZ [-0.00081528, -0.00145384]
E3 [30.34, 30.76) s:
  mass x [0.84416, 1.00113, 1.17612, 0.844773]
  CoM XYZ mm [0.435414, 1.86383, -0.135084]
  mu_s [0.732187, 0.538976, 0.749678, 0.538976]
  mu_d [0.631664, 0.538976, 0.741429, 0.538976]
  Kp [steer,rear] [9.90804, 0]; Kd [0.198313, 0]
  spawn roll -0.0646306 rad; rate -0.116687 rad/s
  gyro bias XYZ rad/s [0.000238552, 0.000172162, 0.000378569]
  gravity bias YZ [-0.00113772, -0.00426276]
E4 [30.76, 60.00) s:
  mass x [0.843686, 0.984973, 0.877347, 0.823193]
  CoM XYZ mm [-3.60021, 4.60623, 0.829765]
  mu_s [0.69316, 0.546287, 0.791209, 0.546287]
  mu_d [0.69316, 0.546287, 0.767936, 0.546287]
  Kp [steer,rear] [10.8657, 0]; Kd [0.214676, 0]
  spawn roll 0.0416617 rad; rate 0.0262887 rad/s
  gyro bias XYZ rad/s [0.000278631, 0.00010735, -0.000126949]
  gravity bias YZ [0.00130734, 0.00124375]
Six significant digits shown; full precision: sampled_dr_values.json.
```

model_2500 · Demand versus actuation · env 63 · full 60 s



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 63; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.18675, 1.05393, 1.00472, 0.870532]
  CoM XYZ mm [-2.55463, 0.144061, -2.76555]
  mu_s [0.531594, 0.764681, 0.632409, 0.751971]
  mu_d [0.531594, 0.764681, 0.589076, 0.541224]
  Kp [steer,rear] [9.7267, 0]; Kd [0.202504, 0]
  spawn roll -0.0648638 rad; rate -0.066083 rad/s
  gyro bias XYZ rad/s [-7.52822e-05, 0.000125875, 3.08003e-05]
  gravity bias YZ [-0.000469823, 0.00191541]
E2 [30.00, 60.00) s:
  mass x [1.15717, 0.96587, 0.985375, 0.849986]
  CoM XYZ mm [2.38675, -1.09744, 4.74863]
  mu_s [0.533455, 0.681711, 0.592046, 0.559809]
  mu_d [0.533455, 0.611756, 0.534947, 0.559809]
  Kp [steer,rear] [10.6957, 0]; Kd [0.205494, 0]
  spawn roll -0.0318545 rad; rate 0.135924 rad/s
  gyro bias XYZ rad/s [-0.000113761, -6.63928e-05, 2.04461e-06]
  gravity bias YZ [-0.00269847, 0.00169603]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 63; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05528, 1.19658, 0.838692, 1.00268]
  CoM XYZ mm [-1.03561, -2.93984, 3.83369]
  mu_s [0.538434, 0.640867, 0.538976, 0.568073]
  mu_d [0.538434, 0.556841, 0.538976, 0.568073]
  Kp [steer,rear] [9.57072, 0]; Kd [0.209347, 0]
  spawn roll 0.0203002 rad; rate 0.143305 rad/s
  gyro bias XYZ rad/s [-3.16734e-05, -0.000187012, -2.90293e-05]
  gravity bias YZ [-0.0023683, -0.00234085]
E2 [30.00, 60.00) s:
  mass x [0.945349, 1.19401, 0.994917, 0.962857]
  CoM XYZ mm [4.57202, 3.35095, -4.27227]
  mu_s [0.546287, 0.733515, 0.748289, 0.727426]
  mu_d [0.546287, 0.558392, 0.689499, 0.606358]
  Kp [steer,rear] [9.64075, 0]; Kd [0.200368, 0]
  spawn roll 0.014617 rad; rate -0.146467 rad/s
  gyro bias XYZ rad/s [3.55681e-05, -0.000131989, 0.000156822]
  gravity bias YZ [-0.00224773, 0.000255713]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 63; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05528, 1.19658, 0.838692, 1.00268]
  CoM XYZ mm [-1.03561, -2.93984, 3.83369]
  mu_s [0.538434, 0.640867, 0.538976, 0.568073]
  mu_d [0.538434, 0.556841, 0.538976, 0.568073]
  Kp [steer,rear] [9.57072, 0]; Kd [0.209347, 0]
  spawn roll 0.0203002 rad; rate 0.143305 rad/s
  gyro bias XYZ rad/s [-3.16734e-05, -0.000187012, -2.90293e-05]
  gravity bias YZ [-0.0023683, -0.00234085]
E2 [30.00, 60.00) s:
  mass x [0.945349, 1.19401, 0.994917, 0.962857]
  CoM XYZ mm [4.57202, 3.35095, -4.27227]
  mu_s [0.546287, 0.733515, 0.748289, 0.727426]
  mu_d [0.546287, 0.558392, 0.689499, 0.606358]
  Kp [steer,rear] [9.64075, 0]; Kd [0.200368, 0]
  spawn roll 0.014617 rad; rate -0.146467 rad/s
  gyro bias XYZ rad/s [3.55681e-05, -0.000131989, 0.000156822]
  gravity bias YZ [-0.00224773, 0.000255713]
Six significant digits shown; full precision: sampled_dr_values.json.
```